# PG No. 41　$C_3(c)$　3　[ trigonal ] (jml basis)

bra: $= \langle \frac{1}{2}, \frac{1}{2}; s|\,, \langle \frac{1}{2}, -\frac{1}{2}; s|$
ket: $= |\frac{1}{2}, \frac{1}{2}; s\rangle\,, |\frac{1}{2}, -\frac{1}{2}; s\rangle$

Table 1: (s,s) block.

| No. | multipole | matrix |
|---|---|---|
| 1 | symmetry | $1$ |
| | $\mathbb{Q}_0^{(a)}(A)$ | $\begin{bmatrix} \frac{\sqrt{2}}{2} & 0 \\ 0 & \frac{\sqrt{2}}{2} \end{bmatrix}$ |
| 2 | symmetry | $z$ |
| | $\mathbb{M}_1^{(1,-1;a)}(A)$ | $\begin{bmatrix} \frac{\sqrt{2}}{2} & 0 \\ 0 & -\frac{\sqrt{2}}{2} \end{bmatrix}$ |
| 3 | symmetry | $\frac{\sqrt{2}x}{2} + \frac{\sqrt{2}iy}{2}$ |
| | $\mathbb{M}_1^{(1,-1;a)}(E^{(a)})$ | $\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$ |
| 4 | symmetry | $\frac{\sqrt{2}x}{2} - \frac{\sqrt{2}iy}{2}$ |
| | $\mathbb{M}_1^{(1,-1;a)}(E^{(b)})$ | $\begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$ |

bra: $= \langle \frac{1}{2}, \frac{1}{2}; s|\,, \langle \frac{1}{2}, -\frac{1}{2}; s|$
ket: $= |\frac{1}{2}, \frac{1}{2}; p\rangle\,, |\frac{1}{2}, -\frac{1}{2}; p\rangle\,, |\frac{3}{2}, \frac{3}{2}; p\rangle\,, |\frac{3}{2}, \frac{1}{2}; p\rangle\,, |\frac{3}{2}, -\frac{1}{2}; p\rangle\,, |\frac{3}{2}, -\frac{3}{2}; p\rangle$

Table 2: (s,p) block.

| No. | multipole | matrix |
|---|---|---|
| 5 | symmetry | $z$ |
| | $\mathbb{Q}_1^{(a)}(A)$ | $\begin{bmatrix} -\frac{\sqrt{3}}{6} & 0 & 0 & \frac{\sqrt{6}}{6} & 0 & 0 \\ 0 & \frac{\sqrt{3}}{6} & 0 & 0 & \frac{\sqrt{6}}{6} & 0 \end{bmatrix}$ |
| 6 | symmetry | $\frac{\sqrt{2}x}{2} + \frac{\sqrt{2}iy}{2}$ |
| | $\mathbb{Q}_1^{(a)}(E^{(a)})$ | $\begin{bmatrix} 0 & -\frac{\sqrt{6}}{6} & 0 & 0 & \frac{\sqrt{3}}{6} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{2} \end{bmatrix}$ |
| 7 | symmetry | $\frac{\sqrt{2}x}{2} - \frac{\sqrt{2}iy}{2}$ |

*continued ...*

Table 2

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_1^{(a)}(E^{(b)})$ | $\begin{bmatrix} 0 & 0 & -\frac{1}{2} & 0 & 0 & 0 \\ -\frac{\sqrt{6}}{6} & 0 & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 \end{bmatrix}$ |
| 8 | symmetry | $z$ |
| | $\mathbb{Q}_1^{(1,0;a)}(A)$ | $\begin{bmatrix} \frac{\sqrt{6}}{6} & 0 & 0 & \frac{\sqrt{3}}{6} & 0 & 0 \\ 0 & -\frac{\sqrt{6}}{6} & 0 & 0 & \frac{\sqrt{3}}{6} & 0 \end{bmatrix}$ |
| 9 | symmetry | $\frac{\sqrt{2}x}{2} + \frac{\sqrt{2}iy}{2}$ |
| | $\mathbb{Q}_1^{(1,0;a)}(E^{(a)})$ | $\begin{bmatrix} 0 & \frac{\sqrt{3}}{3} & 0 & 0 & \frac{\sqrt{6}}{12} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{4} \end{bmatrix}$ |
| 10 | symmetry | $\frac{\sqrt{2}x}{2} - \frac{\sqrt{2}iy}{2}$ |
| | $\mathbb{Q}_1^{(1,0;a)}(E^{(b)})$ | $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{2}}{4} & 0 & 0 & 0 \\ \frac{\sqrt{3}}{3} & 0 & 0 & -\frac{\sqrt{6}}{12} & 0 & 0 \end{bmatrix}$ |
| 11 | symmetry | $-\frac{x^2}{2} - \frac{y^2}{2} + z^2$ |
| | $\mathbb{G}_2^{(1,-1;a)}(A)$ | $\begin{bmatrix} 0 & 0 & 0 & \frac{i}{2} & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{i}{2} & 0 \end{bmatrix}$ |
| 12 | symmetry | $\frac{\sqrt{6}xz}{2} + \frac{\sqrt{6}iyz}{2}$ |
| | $\mathbb{G}_2^{(1,-1;a)}(E^{(a)},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{4} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} \end{bmatrix}$ |
| 13 | symmetry | $\frac{\sqrt{6}x^2}{4} - \frac{\sqrt{6}ixy}{2} - \frac{\sqrt{6}y^2}{4}$ |
| | $\mathbb{G}_2^{(1,-1;a)}(E^{(a)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{2}i}{2} & 0 & 0 & 0 \end{bmatrix}$ |
| 14 | symmetry | $\frac{\sqrt{6}xz}{2} - \frac{\sqrt{6}iyz}{2}$ |
| | $\mathbb{G}_2^{(1,-1;a)}(E^{(b)},1)$ | $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{6}i}{4} & 0 & 0 \end{bmatrix}$ |
| 15 | symmetry | $\frac{\sqrt{6}x^2}{4} + \frac{\sqrt{6}ixy}{2} - \frac{\sqrt{6}y^2}{4}$ |
| | $\mathbb{G}_2^{(1,-1;a)}(E^{(b)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{2} \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 16 | symmetry | 1 |

*continued ...*

Table 2

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_0^{(1,1;a)}(A)$ | $\begin{bmatrix} -\frac{i}{2} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{i}{2} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 17 | symmetry | $z$ |
| | $\mathbb{T}_1^{(a)}(A)$ | $\begin{bmatrix} -\frac{\sqrt{3}i}{6} & 0 & 0 & \frac{\sqrt{6}i}{6} & 0 & 0 \\ 0 & \frac{\sqrt{3}i}{6} & 0 & 0 & \frac{\sqrt{6}i}{6} & 0 \end{bmatrix}$ |
| 18 | symmetry | $\frac{\sqrt{2}x}{2} + \frac{\sqrt{2}iy}{2}$ |
| | $\mathbb{T}_1^{(a)}(E^{(a)})$ | $\begin{bmatrix} 0 & -\frac{\sqrt{6}i}{6} & 0 & 0 & \frac{\sqrt{3}i}{6} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{i}{2} \end{bmatrix}$ |
| 19 | symmetry | $\frac{\sqrt{2}x}{2} - \frac{\sqrt{2}iy}{2}$ |
| | $\mathbb{T}_1^{(a)}(E^{(b)})$ | $\begin{bmatrix} 0 & 0 & -\frac{i}{2} & 0 & 0 & 0 \\ -\frac{\sqrt{6}i}{6} & 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 \end{bmatrix}$ |
| 20 | symmetry | $z$ |
| | $\mathbb{T}_1^{(1,0;a)}(A)$ | $\begin{bmatrix} -\frac{\sqrt{6}i}{6} & 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 \\ 0 & \frac{\sqrt{6}i}{6} & 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 \end{bmatrix}$ |
| 21 | symmetry | $\frac{\sqrt{2}x}{2} + \frac{\sqrt{2}iy}{2}$ |
| | $\mathbb{T}_1^{(1,0;a)}(E^{(a)})$ | $\begin{bmatrix} 0 & -\frac{\sqrt{3}i}{3} & 0 & 0 & -\frac{\sqrt{6}i}{12} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} \end{bmatrix}$ |
| 22 | symmetry | $\frac{\sqrt{2}x}{2} - \frac{\sqrt{2}iy}{2}$ |
| | $\mathbb{T}_1^{(1,0;a)}(E^{(b)})$ | $\begin{bmatrix} 0 & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & 0 \\ -\frac{\sqrt{3}i}{3} & 0 & 0 & \frac{\sqrt{6}i}{12} & 0 & 0 \end{bmatrix}$ |
| 23 | symmetry | $-\frac{x^2}{2} - \frac{y^2}{2} + z^2$ |
| | $\mathbb{M}_2^{(1,-1;a)}(A)$ | $\begin{bmatrix} 0 & 0 & 0 & \frac{1}{2} & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{1}{2} & 0 \end{bmatrix}$ |
| 24 | symmetry | $\frac{\sqrt{6}xz}{2} + \frac{\sqrt{6}iyz}{2}$ |
| | $\mathbb{M}_2^{(1,-1;a)}(E^{(a)},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{4} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{4} \end{bmatrix}$ |
| 25 | symmetry | $\frac{\sqrt{6}x^2}{4} - \frac{\sqrt{6}ixy}{2} - \frac{\sqrt{6}y^2}{4}$ |

*continued ...*

Table 2

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_2^{(1,-1;a)}(E^{(a)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{2}}{2} & 0 & 0 & 0 \end{bmatrix}$ |
| 26 | symmetry | $\frac{\sqrt{6}xz}{2} - \frac{\sqrt{6}iyz}{2}$ |
| | $\mathbb{M}_2^{(1,-1;a)}(E^{(b)},1)$ | $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{2}}{4} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{6}}{4} & 0 & 0 \end{bmatrix}$ |
| 27 | symmetry | $\frac{\sqrt{6}x^2}{4} + \frac{\sqrt{6}ixy}{2} - \frac{\sqrt{6}y^2}{4}$ |
| | $\mathbb{M}_2^{(1,-1;a)}(E^{(b)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{2} \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 28 | symmetry | 1 |
| | $\mathbb{M}_0^{(1,1;a)}(A)$ | $\begin{bmatrix} -\frac{1}{2} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{1}{2} & 0 & 0 & 0 & 0 \end{bmatrix}$ |

bra: $= \langle \frac{1}{2}, \frac{1}{2}; s|, \langle \frac{1}{2}, -\frac{1}{2}; s|$

ket: $= |\frac{3}{2}, \frac{3}{2}; d\rangle, |\frac{3}{2}, \frac{1}{2}; d\rangle, |\frac{3}{2}, -\frac{1}{2}; d\rangle, |\frac{3}{2}, -\frac{3}{2}; d\rangle, |\frac{5}{2}, \frac{5}{2}; d\rangle, |\frac{5}{2}, \frac{3}{2}; d\rangle, |\frac{5}{2}, \frac{1}{2}; d\rangle, |\frac{5}{2}, -\frac{1}{2}; d\rangle, |\frac{5}{2}, -\frac{3}{2}; d\rangle, |\frac{5}{2}, -\frac{5}{2}; d\rangle$

Table 3: (s,d) block.

| No. | multipole | matrix |
|---|---|---|
| 29 | symmetry | $-\frac{x^2}{2} - \frac{y^2}{2} + z^2$ |
| | $\mathbb{Q}_2^{(a)}(A)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{10}}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{10} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{10}}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{10} & 0 & 0 \end{bmatrix}$ |
| 30 | symmetry | $\frac{\sqrt{6}xz}{2} + \frac{\sqrt{6}iyz}{2}$ |
| | $\mathbb{Q}_2^{(a)}(E^{(a)},1)$ | $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{15}}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{10} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{5}}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{5} & 0 \end{bmatrix}$ |
| 31 | symmetry | $\frac{\sqrt{6}x^2}{4} - \frac{\sqrt{6}ixy}{2} - \frac{\sqrt{6}y^2}{4}$ |
| | $\mathbb{Q}_2^{(a)}(E^{(a)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{5}}{5} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{10} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 32 | symmetry | $\frac{\sqrt{6}xz}{2} - \frac{\sqrt{6}iyz}{2}$ |

*continued ...*

Table 3

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_2^{(a)}(E^{(b)},1)$ | $\begin{bmatrix} \frac{\sqrt{5}}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{5} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{15}}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{10} & 0 & 0 & 0 \end{bmatrix}$ |
| 33 | symmetry | $\frac{\sqrt{6}x^2}{4} + \frac{\sqrt{6}ixy}{2} - \frac{\sqrt{6}y^2}{4}$ |
| | $\mathbb{Q}_2^{(a)}(E^{(b)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{5}}{5} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{10} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} \end{bmatrix}$ |
| 34 | symmetry | $-\frac{x^2}{2} - \frac{y^2}{2} + z^2$ |
| | $\mathbb{Q}_2^{(1,0;a)}(A)$ | $\begin{bmatrix} 0 & \frac{\sqrt{15}}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{10} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{15}}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{10} & 0 & 0 \end{bmatrix}$ |
| 35 | symmetry | $\frac{\sqrt{6}xz}{2} + \frac{\sqrt{6}iyz}{2}$ |
| | $\mathbb{Q}_2^{(1,0;a)}(E^{(a)},1)$ | $\begin{bmatrix} 0 & 0 & \frac{3\sqrt{10}}{20} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{15} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{30}}{20} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{15} & 0 \end{bmatrix}$ |
| 36 | symmetry | $\frac{\sqrt{6}x^2}{4} - \frac{\sqrt{6}ixy}{2} - \frac{\sqrt{6}y^2}{4}$ |
| | $\mathbb{Q}_2^{(1,0;a)}(E^{(a)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{6} & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{30}}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{30} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 37 | symmetry | $\frac{\sqrt{6}xz}{2} - \frac{\sqrt{6}iyz}{2}$ |
| | $\mathbb{Q}_2^{(1,0;a)}(E^{(b)},1)$ | $\begin{bmatrix} -\frac{\sqrt{30}}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{15} & 0 & 0 & 0 & 0 \\ 0 & \frac{3\sqrt{10}}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{15} & 0 & 0 & 0 \end{bmatrix}$ |
| 38 | symmetry | $\frac{\sqrt{6}x^2}{4} + \frac{\sqrt{6}ixy}{2} - \frac{\sqrt{6}y^2}{4}$ |
| | $\mathbb{Q}_2^{(1,0;a)}(E^{(b)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & \frac{\sqrt{30}}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{30} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{6} \end{bmatrix}$ |
| 39 | symmetry | $-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$ |
| | $\mathbb{G}_3^{(1,-1;a)}(A,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{2} & 0 & 0 \end{bmatrix}$ |
| 40 | symmetry | $\frac{3\sqrt{10}x^2y}{4} - \frac{\sqrt{10}y^3}{4}$ |
| | $\mathbb{G}_3^{(1,-1;a)}(A,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} \\ 0 & 0 & 0 & 0 & -\frac{1}{2} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 41 | symmetry | $\frac{\sqrt{10}x^3}{4} - \frac{3\sqrt{10}xy^2}{4}$ |

*continued ...*

Table 3

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_3^{(1,-1;a)}(A,3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{2} \\ 0 & 0 & 0 & 0 & \frac{i}{2} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 42 | symmetry | $-\frac{\sqrt{3}x^3}{4} - \frac{\sqrt{3}ix^2y}{4} - \frac{\sqrt{3}xy^2}{4} + \sqrt{3}xz^2 - \frac{\sqrt{3}iy^3}{4} + \sqrt{3}iyz^2$ |
| | $\mathbb{G}_3^{(1,-1;a)}(E^{(a)},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{3} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{6} & 0 \end{bmatrix}$ |
| 43 | symmetry | $\frac{\sqrt{30}x^2z}{4} - \frac{\sqrt{30}ixyz}{2} - \frac{\sqrt{30}y^2z}{4}$ |
| | $\mathbb{G}_3^{(1,-1;a)}(E^{(a)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{6} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 44 | symmetry | $-\frac{\sqrt{3}x^3}{4} + \frac{\sqrt{3}ix^2y}{4} - \frac{\sqrt{3}xy^2}{4} + \sqrt{3}xz^2 + \frac{\sqrt{3}iy^3}{4} - \sqrt{3}iyz^2$ |
| | $\mathbb{G}_3^{(1,-1;a)}(E^{(b)},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{6} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{3} & 0 & 0 & 0 \end{bmatrix}$ |
| 45 | symmetry | $\frac{\sqrt{30}x^2z}{4} + \frac{\sqrt{30}ixyz}{2} - \frac{\sqrt{30}y^2z}{4}$ |
| | $\mathbb{G}_3^{(1,-1;a)}(E^{(b)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{6} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} \end{bmatrix}$ |
| 46 | symmetry | $z$ |
| | $\mathbb{G}_1^{(1,1;a)}(A)$ | $\begin{bmatrix} 0 & -\frac{i}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{i}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 47 | symmetry | $\frac{\sqrt{2}x}{2} + \frac{\sqrt{2}iy}{2}$ |
| | $\mathbb{G}_1^{(1,1;a)}(E^{(a)})$ | $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{6}i}{4} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 48 | symmetry | $\frac{\sqrt{2}x}{2} - \frac{\sqrt{2}iy}{2}$ |
| | $\mathbb{G}_1^{(1,1;a)}(E^{(b)})$ | $\begin{bmatrix} \frac{\sqrt{6}i}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 49 | symmetry | $-\frac{x^2}{2} - \frac{y^2}{2} + z^2$ |
| | $\mathbb{T}_2^{(a)}(A)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{10}i}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{10} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{10}i}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{10} & 0 & 0 \end{bmatrix}$ |
| 50 | symmetry | $\frac{\sqrt{6}xz}{2} + \frac{\sqrt{6}iyz}{2}$ |

*continued ...*

Table 3

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_2^{(a)}(E^{(a)},1)$ | $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{15}i}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{10} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{5} & 0 \end{bmatrix}$ |
| 51 | symmetry | $\frac{\sqrt{6}x^2}{4} - \frac{\sqrt{6}ixy}{2} - \frac{\sqrt{6}y^2}{4}$ |
| | $\mathbb{T}_2^{(a)}(E^{(a)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & \frac{i}{2} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{5}i}{5} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 52 | symmetry | $\frac{\sqrt{6}xz}{2} - \frac{\sqrt{6}iyz}{2}$ |
| | $\mathbb{T}_2^{(a)}(E^{(b)},1)$ | $\begin{bmatrix} \frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{5} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{15}i}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{10} & 0 & 0 & 0 \end{bmatrix}$ |
| 53 | symmetry | $\frac{\sqrt{6}x^2}{4} + \frac{\sqrt{6}ixy}{2} - \frac{\sqrt{6}y^2}{4}$ |
| | $\mathbb{T}_2^{(a)}(E^{(b)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{5}i}{5} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{10} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{2} \end{bmatrix}$ |
| 54 | symmetry | $-\frac{x^2}{2} - \frac{y^2}{2} + z^2$ |
| | $\mathbb{T}_2^{(1,0;a)}(A)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{15}i}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{10} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{15}i}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{10} & 0 & 0 \end{bmatrix}$ |
| 55 | symmetry | $\frac{\sqrt{6}xz}{2} + \frac{\sqrt{6}iyz}{2}$ |
| | $\mathbb{T}_2^{(1,0;a)}(E^{(a)},1)$ | $\begin{bmatrix} 0 & 0 & -\frac{3\sqrt{10}i}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{15} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{30}i}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{15} & 0 \end{bmatrix}$ |
| 56 | symmetry | $\frac{\sqrt{6}x^2}{4} - \frac{\sqrt{6}ixy}{2} - \frac{\sqrt{6}y^2}{4}$ |
| | $\mathbb{T}_2^{(1,0;a)}(E^{(a)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{6} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{30}i}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{30} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 57 | symmetry | $\frac{\sqrt{6}xz}{2} - \frac{\sqrt{6}iyz}{2}$ |
| | $\mathbb{T}_2^{(1,0;a)}(E^{(b)},1)$ | $\begin{bmatrix} \frac{\sqrt{30}i}{20} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{15} & 0 & 0 & 0 & 0 \\ 0 & -\frac{3\sqrt{10}i}{20} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{15} & 0 & 0 & 0 \end{bmatrix}$ |
| 58 | symmetry | $\frac{\sqrt{6}x^2}{4} + \frac{\sqrt{6}ixy}{2} - \frac{\sqrt{6}y^2}{4}$ |
| | $\mathbb{T}_2^{(1,0;a)}(E^{(b)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{30}i}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{30} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{6} \end{bmatrix}$ |
| 59 | symmetry | $-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$ |

*continued ...*

Table 3

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_3^{(1,-1;a)}(A,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{2} & 0 & 0 \end{bmatrix}$ |
| 60 | symmetry | $\frac{3\sqrt{10}x^2y}{4} - \frac{\sqrt{10}y^3}{4}$ |
| | $\mathbb{M}_3^{(1,-1;a)}(A,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{2} \\ 0 & 0 & 0 & 0 & \frac{i}{2} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 61 | symmetry | $\frac{\sqrt{10}x^3}{4} - \frac{3\sqrt{10}xy^2}{4}$ |
| | $\mathbb{M}_3^{(1,-1;a)}(A,3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} \\ 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 62 | symmetry | $-\frac{\sqrt{3}x^3}{4} - \frac{\sqrt{3}ix^2y}{4} - \frac{\sqrt{3}xy^2}{4} + \sqrt{3}xz^2 - \frac{\sqrt{3}iy^3}{4} + \sqrt{3}iyz^2$ |
| | $\mathbb{M}_3^{(1,-1;a)}(E^{(a)},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{3} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{6} & 0 \end{bmatrix}$ |
| 63 | symmetry | $\frac{\sqrt{30}x^2z}{4} - \frac{\sqrt{30}ixyz}{2} - \frac{\sqrt{30}y^2z}{4}$ |
| | $\mathbb{M}_3^{(1,-1;a)}(E^{(a)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{6} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 64 | symmetry | $-\frac{\sqrt{3}x^3}{4} + \frac{\sqrt{3}ix^2y}{4} - \frac{\sqrt{3}xy^2}{4} + \sqrt{3}xz^2 + \frac{\sqrt{3}iy^3}{4} - \sqrt{3}iyz^2$ |
| | $\mathbb{M}_3^{(1,-1;a)}(E^{(b)},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{6} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{3} & 0 & 0 & 0 \end{bmatrix}$ |
| 65 | symmetry | $\frac{\sqrt{30}x^2z}{4} + \frac{\sqrt{30}ixyz}{2} - \frac{\sqrt{30}y^2z}{4}$ |
| | $\mathbb{M}_3^{(1,-1;a)}(E^{(b)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{6} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{6} \end{bmatrix}$ |
| 66 | symmetry | $z$ |
| | $\mathbb{M}_1^{(1,1;a)}(A)$ | $\begin{bmatrix} 0 & -\frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 67 | symmetry | $\frac{\sqrt{2}x}{2} + \frac{\sqrt{2}iy}{2}$ |
| | $\mathbb{M}_1^{(1,1;a)}(E^{(a)})$ | $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{6}}{4} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 68 | symmetry | $\frac{\sqrt{2}x}{2} - \frac{\sqrt{2}iy}{2}$ |

*continued ...*

Table 3

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_1^{(1,1;a)}(E^{(b)})$ | $\begin{bmatrix} \frac{\sqrt{6}}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |

bra: $= \langle \frac{1}{2}, \frac{1}{2}; s| \, , \langle \frac{1}{2}, -\frac{1}{2}; s|$

ket: $= |\frac{5}{2}, \frac{5}{2}; f\rangle \, , |\frac{5}{2}, \frac{3}{2}; f\rangle \, , |\frac{5}{2}, \frac{1}{2}; f\rangle \, , |\frac{5}{2}, -\frac{1}{2}; f\rangle \, , |\frac{5}{2}, -\frac{3}{2}; f\rangle \, , |\frac{5}{2}, -\frac{5}{2}; f\rangle \, , |\frac{7}{2}, \frac{7}{2}; f\rangle \, , |\frac{7}{2}, \frac{5}{2}; f\rangle \, , |\frac{7}{2}, \frac{3}{2}; f\rangle \, , |\frac{7}{2}, \frac{1}{2}; f\rangle \, , |\frac{7}{2}, -\frac{1}{2}; f\rangle \, , |\frac{7}{2}, -\frac{3}{2}; f\rangle \, , |\frac{7}{2}, -\frac{5}{2}; f\rangle \, , |\frac{7}{2}, -\frac{7}{2}; f\rangle$

Table 4: (s,f) block.

| No. | multipole | matrix |
|---|---|---|
| 69 | symmetry | $-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$ |
| | $\mathbb{Q}_3^{(a)}(A,1)$ | $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{21}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{7} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{21}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{7} & 0 & 0 & 0 \end{bmatrix}$ |
| 70 | symmetry | $\frac{3\sqrt{10}x^2y}{4} - \frac{\sqrt{10}y^3}{4}$ |
| | $\mathbb{Q}_3^{(a)}(A,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{14} & -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{28} & 0 \\ -\frac{\sqrt{21}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{28} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} \end{bmatrix}$ |
| 71 | symmetry | $\frac{\sqrt{10}x^3}{4} - \frac{3\sqrt{10}xy^2}{4}$ |
| | $\mathbb{Q}_3^{(a)}(A,3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{14} & -\frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{28} & 0 \\ -\frac{\sqrt{21}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{28} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{4} \end{bmatrix}$ |
| 72 | symmetry | $-\frac{\sqrt{3}x^3}{4} - \frac{\sqrt{3}ix^2y}{4} - \frac{\sqrt{3}xy^2}{4} + \sqrt{3}xz^2 - \frac{\sqrt{3}iy^3}{4} + \sqrt{3}iyz^2$ |
| | $\mathbb{Q}_3^{(a)}(E^{(a)},1)$ | $\begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{7}}{7} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{14} & 0 & 0 \end{bmatrix}$ |
| 73 | symmetry | $\frac{\sqrt{30}x^2z}{4} - \frac{\sqrt{30}ixyz}{2} - \frac{\sqrt{30}y^2z}{4}$ |
| | $\mathbb{Q}_3^{(a)}(E^{(a)},2)$ | $\begin{bmatrix} -\frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{35}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 74 | symmetry | $-\frac{\sqrt{3}x^3}{4} + \frac{\sqrt{3}ix^2y}{4} - \frac{\sqrt{3}xy^2}{4} + \sqrt{3}xz^2 + \frac{\sqrt{3}iy^3}{4} - \sqrt{3}iyz^2$ |
| | $\mathbb{Q}_3^{(a)}(E^{(b)},1)$ | $\begin{bmatrix} 0 & \frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{7}}{7} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{14} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 75 | symmetry | $\frac{\sqrt{30}x^2z}{4} + \frac{\sqrt{30}ixyz}{2} - \frac{\sqrt{30}y^2z}{4}$ |

*continued ...*

Table 4

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_3^{(a)}(E^{(b)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{14} & 0 \end{bmatrix}$ |
| 76 | symmetry | $-\frac{3x^2z}{2}-\frac{3y^2z}{2}+z^3$ |
| | $\mathbb{Q}_3^{(1,0;a)}(A,1)$ | $\begin{bmatrix} 0 & 0 & \frac{\sqrt{7}}{7} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{7}}{7} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{14} & 0 & 0 & 0 \end{bmatrix}$ |
| 77 | symmetry | $\frac{3\sqrt{10}x^2y}{4}-\frac{\sqrt{10}y^3}{4}$ |
| | $\mathbb{Q}_3^{(1,0;a)}(A,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{7} & -\frac{\sqrt{6}i}{8} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{56} & 0 \\ \frac{\sqrt{7}i}{7} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{56} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{8} \end{bmatrix}$ |
| 78 | symmetry | $\frac{\sqrt{10}x^3}{4}-\frac{3\sqrt{10}xy^2}{4}$ |
| | $\mathbb{Q}_3^{(1,0;a)}(A,3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{7} & -\frac{\sqrt{6}}{8} & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{56} & 0 \\ \frac{\sqrt{7}}{7} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{56} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{8} \end{bmatrix}$ |
| 79 | symmetry | $-\frac{\sqrt{3}x^3}{4}-\frac{\sqrt{3}ix^2y}{4}-\frac{\sqrt{3}xy^2}{4}+\sqrt{3}xz^2-\frac{\sqrt{3}iy^3}{4}+\sqrt{3}iyz^2$ |
| | $\mathbb{Q}_3^{(1,0;a)}(E^{(a)},1)$ | $\begin{bmatrix} 0 & 0 & 0 & \frac{2\sqrt{21}}{21} & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{7}}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{21} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{28} & 0 & 0 \end{bmatrix}$ |
| 80 | symmetry | $\frac{\sqrt{30}x^2z}{4}-\frac{\sqrt{30}ixyz}{2}-\frac{\sqrt{30}y^2z}{4}$ |
| | $\mathbb{Q}_3^{(1,0;a)}(E^{(a)},2)$ | $\begin{bmatrix} \frac{\sqrt{21}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{14}}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{105}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{28} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 81 | symmetry | $-\frac{\sqrt{3}x^3}{4}+\frac{\sqrt{3}ix^2y}{4}-\frac{\sqrt{3}xy^2}{4}+\sqrt{3}xz^2+\frac{\sqrt{3}iy^3}{4}-\sqrt{3}iyz^2$ |
| | $\mathbb{Q}_3^{(1,0;a)}(E^{(b)},1)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{42}}{21} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{2\sqrt{21}}{21} & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{7}}{28} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 82 | symmetry | $\frac{\sqrt{30}x^2z}{4}+\frac{\sqrt{30}ixyz}{2}-\frac{\sqrt{30}y^2z}{4}$ |
| | $\mathbb{Q}_3^{(1,0;a)}(E^{(b)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{14}}{28} & 0 \end{bmatrix}$ |
| 83 | symmetry | $\frac{3x^4}{8}+\frac{3x^2y^2}{4}-3x^2z^2+\frac{3y^4}{8}-3y^2z^2+z^4$ |
| | $\mathbb{G}_4^{(1,-1;a)}(A,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{2} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{2} & 0 & 0 & 0 \end{bmatrix}$ |
| 84 | symmetry | $\frac{\sqrt{70}x^3z}{4}-\frac{3\sqrt{70}xy^2z}{4}$ |

*continued ...*

Table 4

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_4^{(1,-1;a)}(A,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{8} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{8} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{8} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{8} \end{bmatrix}$ |
| 85 | symmetry | $\frac{3\sqrt{70}x^2yz}{4} - \frac{\sqrt{70}y^3z}{4}$ |
| | $\mathbb{G}_4^{(1,-1;a)}(A,3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{8} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{8} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{8} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{8} \end{bmatrix}$ |
| 86 | symmetry | $-\frac{3\sqrt{5}x^3z}{4} - \frac{3\sqrt{5}ix^2yz}{4} - \frac{3\sqrt{5}xy^2z}{4} + \sqrt{5}xz^3 - \frac{3\sqrt{5}iy^3z}{4} + \sqrt{5}iyz^3$ |
| | $\mathbb{G}_4^{(1,-1;a)}(E^{(a)},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{4} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{4} & 0 & 0 \end{bmatrix}$ |
| 87 | symmetry | $\frac{\sqrt{70}x^4}{16} + \frac{\sqrt{70}ix^3y}{4} - \frac{3\sqrt{70}x^2y^2}{8} - \frac{\sqrt{70}ixy^3}{4} + \frac{\sqrt{70}y^4}{16}$ |
| | $\mathbb{G}_4^{(1,-1;a)}(E^{(a)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{2} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 88 | symmetry | $-\frac{\sqrt{10}x^4}{8} + \frac{\sqrt{10}ix^3y}{4} + \frac{3\sqrt{10}x^2z^2}{4} + \frac{\sqrt{10}ixy^3}{4} - \frac{3\sqrt{10}ixyz^2}{2} + \frac{\sqrt{10}y^4}{8} - \frac{3\sqrt{10}y^2z^2}{4}$ |
| | $\mathbb{G}_4^{(1,-1;a)}(E^{(a)},3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{4} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 89 | symmetry | $-\frac{3\sqrt{5}x^3z}{4} + \frac{3\sqrt{5}ix^2yz}{4} - \frac{3\sqrt{5}xy^2z}{4} + \sqrt{5}xz^3 + \frac{3\sqrt{5}iy^3z}{4} - \sqrt{5}iyz^3$ |
| | $\mathbb{G}_4^{(1,-1;a)}(E^{(b)},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{4} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{4} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 90 | symmetry | $\frac{\sqrt{70}x^4}{16} - \frac{\sqrt{70}ix^3y}{4} - \frac{3\sqrt{70}x^2y^2}{8} + \frac{\sqrt{70}ixy^3}{4} + \frac{\sqrt{70}y^4}{16}$ |
| | $\mathbb{G}_4^{(1,-1;a)}(E^{(b)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 91 | symmetry | $-\frac{\sqrt{10}x^4}{8} - \frac{\sqrt{10}ix^3y}{4} + \frac{3\sqrt{10}x^2z^2}{4} - \frac{\sqrt{10}ixy^3}{4} + \frac{3\sqrt{10}ixyz^2}{2} + \frac{\sqrt{10}y^4}{8} - \frac{3\sqrt{10}y^2z^2}{4}$ |
| | $\mathbb{G}_4^{(1,-1;a)}(E^{(b)},3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{4} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 \end{bmatrix}$ |
| 92 | symmetry | $-\frac{x^2}{2} - \frac{y^2}{2} + z^2$ |
| | $\mathbb{G}_2^{(1,1;a)}(A)$ | $\begin{bmatrix} 0 & 0 & -\frac{i}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{i}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 93 | symmetry | $\frac{\sqrt{6}xz}{2} + \frac{\sqrt{6}iyz}{2}$ |

*continued ...*

Table 4

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_2^{(1,1;a)}(E^{(a)},1)$ | $\begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{6}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{3} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 94 | symmetry | $\frac{\sqrt{6}x^2}{4} - \frac{\sqrt{6}ixy}{2} - \frac{\sqrt{6}y^2}{4}$ |
| | $\mathbb{G}_2^{(1,1;a)}(E^{(a)},2)$ | $\begin{bmatrix} -\frac{\sqrt{15}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 95 | symmetry | $\frac{\sqrt{6}xz}{2} - \frac{\sqrt{6}iyz}{2}$ |
| | $\mathbb{G}_2^{(1,1;a)}(E^{(b)},1)$ | $\begin{bmatrix} 0 & \frac{\sqrt{3}i}{3} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{6}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 96 | symmetry | $\frac{\sqrt{6}x^2}{4} + \frac{\sqrt{6}ixy}{2} - \frac{\sqrt{6}y^2}{4}$ |
| | $\mathbb{G}_2^{(1,1;a)}(E^{(b)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 97 | symmetry | $-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$ |
| | $\mathbb{T}_3^{(a)}(A,1)$ | $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{21}i}{14} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{7} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{21}i}{14} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{7} & 0 & 0 & 0 \end{bmatrix}$ |
| 98 | symmetry | $\frac{3\sqrt{10}x^2y}{4} - \frac{\sqrt{10}y^3}{4}$ |
| | $\mathbb{T}_3^{(a)}(A,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{14} & \frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{28} & 0 \\ \frac{\sqrt{21}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{4} \end{bmatrix}$ |
| 99 | symmetry | $\frac{\sqrt{10}x^3}{4} - \frac{3\sqrt{10}xy^2}{4}$ |
| | $\mathbb{T}_3^{(a)}(A,3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{14} & -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{28} & 0 \\ -\frac{\sqrt{21}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{4} \end{bmatrix}$ |
| 100 | symmetry | $-\frac{\sqrt{3}x^3}{4} - \frac{\sqrt{3}ix^2y}{4} - \frac{\sqrt{3}xy^2}{4} + \sqrt{3}xz^2 - \frac{\sqrt{3}iy^3}{4} + \sqrt{3}iyz^2$ |
| | $\mathbb{T}_3^{(a)}(E^{(a)},1)$ | $\begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{7}i}{7} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{14} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{14} & 0 & 0 \end{bmatrix}$ |
| 101 | symmetry | $\frac{\sqrt{30}x^2z}{4} - \frac{\sqrt{30}ixyz}{2} - \frac{\sqrt{30}y^2z}{4}$ |
| | $\mathbb{T}_3^{(a)}(E^{(a)},2)$ | $\begin{bmatrix} -\frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{14} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{35}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{14} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 102 | symmetry | $-\frac{\sqrt{3}x^3}{4} + \frac{\sqrt{3}ix^2y}{4} - \frac{\sqrt{3}xy^2}{4} + \sqrt{3}xz^2 + \frac{\sqrt{3}iy^3}{4} - \sqrt{3}iyz^2$ |

*continued ...*

Table 4

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_3^{(a)}(E^{(b)},1)$ | $\begin{bmatrix} 0 & \frac{\sqrt{14}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{14} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{7}i}{7} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{14} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 103 | symmetry | $\frac{\sqrt{30}x^2z}{4}+\frac{\sqrt{30}ixyz}{2}-\frac{\sqrt{30}y^2z}{4}$ |
| | $\mathbb{T}_3^{(a)}(E^{(b)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{14} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{14} & 0 \end{bmatrix}$ |
| 104 | symmetry | $-\frac{3x^2z}{2}-\frac{3y^2z}{2}+z^3$ |
| | $\mathbb{T}_3^{(1,0;a)}(A,1)$ | $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{7}i}{7} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{7}i}{7} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{14} & 0 & 0 & 0 \end{bmatrix}$ |
| 105 | symmetry | $\frac{3\sqrt{10}x^2y}{4}-\frac{\sqrt{10}y^3}{4}$ |
| | $\mathbb{T}_3^{(1,0;a)}(A,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{7} & -\frac{\sqrt{6}}{8} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{56} & 0 \\ \frac{\sqrt{7}}{7} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{56} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{8} \end{bmatrix}$ |
| 106 | symmetry | $\frac{\sqrt{10}x^3}{4}-\frac{3\sqrt{10}xy^2}{4}$ |
| | $\mathbb{T}_3^{(1,0;a)}(A,3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{7} & \frac{\sqrt{6}i}{8} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{56} & 0 \\ -\frac{\sqrt{7}i}{7} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{56} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{8} \end{bmatrix}$ |
| 107 | symmetry | $-\frac{\sqrt{3}x^3}{4}-\frac{\sqrt{3}ix^2y}{4}-\frac{\sqrt{3}xy^2}{4}+\sqrt{3}xz^2-\frac{\sqrt{3}iy^3}{4}+\sqrt{3}iyz^2$ |
| | $\mathbb{T}_3^{(1,0;a)}(E^{(a)},1)$ | $\begin{bmatrix} 0 & 0 & 0 & -\frac{2\sqrt{21}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{7}i}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{28} & 0 & 0 \end{bmatrix}$ |
| 108 | symmetry | $\frac{\sqrt{30}x^2z}{4}-\frac{\sqrt{30}ixyz}{2}-\frac{\sqrt{30}y^2z}{4}$ |
| | $\mathbb{T}_3^{(1,0;a)}(E^{(a)},2)$ | $\begin{bmatrix} -\frac{\sqrt{21}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{14}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{105}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{28} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 109 | symmetry | $-\frac{\sqrt{3}x^3}{4}+\frac{\sqrt{3}ix^2y}{4}-\frac{\sqrt{3}xy^2}{4}+\sqrt{3}xz^2+\frac{\sqrt{3}iy^3}{4}-\sqrt{3}iyz^2$ |
| | $\mathbb{T}_3^{(1,0;a)}(E^{(b)},1)$ | $\begin{bmatrix} 0 & \frac{\sqrt{42}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{2\sqrt{21}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{7}i}{28} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 110 | symmetry | $\frac{\sqrt{30}x^2z}{4}+\frac{\sqrt{30}ixyz}{2}-\frac{\sqrt{30}y^2z}{4}$ |
| | $\mathbb{T}_3^{(1,0;a)}(E^{(b)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{14}i}{28} & 0 \end{bmatrix}$ |
| 111 | symmetry | $\frac{3x^4}{8}+\frac{3x^2y^2}{4}-3x^2z^2+\frac{3y^4}{8}-3y^2z^2+z^4$ |

*continued ...*

Table 4

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_4^{(1,-1;a)}(A,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{2} & 0 & 0 & 0 \end{bmatrix}$ |
| 112 | symmetry | $\frac{\sqrt{70}x^3z}{4} - \frac{3\sqrt{70}xy^2z}{4}$ |
| | $\mathbb{M}_4^{(1,-1;a)}(A,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{8} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{8} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{8} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{8} \end{bmatrix}$ |
| 113 | symmetry | $\frac{3\sqrt{70}x^2yz}{4} - \frac{\sqrt{70}y^3z}{4}$ |
| | $\mathbb{M}_4^{(1,-1;a)}(A,3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{8} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{8} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{8} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{8} \end{bmatrix}$ |
| 114 | symmetry | $-\frac{3\sqrt{5}x^3z}{4} - \frac{3\sqrt{5}ix^2yz}{4} - \frac{3\sqrt{5}xy^2z}{4} + \sqrt{5}xz^3 - \frac{3\sqrt{5}iy^3z}{4} + \sqrt{5}iyz^3$ |
| | $\mathbb{M}_4^{(1,-1;a)}(E^{(a)},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{4} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{4} & 0 & 0 \end{bmatrix}$ |
| 115 | symmetry | $\frac{\sqrt{70}x^4}{16} + \frac{\sqrt{70}ix^3y}{4} - \frac{3\sqrt{70}x^2y^2}{8} - \frac{\sqrt{70}ixy^3}{4} + \frac{\sqrt{70}y^4}{16}$ |
| | $\mathbb{M}_4^{(1,-1;a)}(E^{(a)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{2} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 116 | symmetry | $-\frac{\sqrt{10}x^4}{8} + \frac{\sqrt{10}ix^3y}{4} + \frac{3\sqrt{10}x^2z^2}{4} + \frac{\sqrt{10}ixy^3}{4} - \frac{3\sqrt{10}ixyz^2}{2} + \frac{\sqrt{10}y^4}{8} - \frac{3\sqrt{10}y^2z^2}{4}$ |
| | $\mathbb{M}_4^{(1,-1;a)}(E^{(a)},3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{4} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 117 | symmetry | $-\frac{3\sqrt{5}x^3z}{4} + \frac{3\sqrt{5}ix^2yz}{4} - \frac{3\sqrt{5}xy^2z}{4} + \sqrt{5}xz^3 + \frac{3\sqrt{5}iy^3z}{4} - \sqrt{5}iyz^3$ |
| | $\mathbb{M}_4^{(1,-1;a)}(E^{(b)},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{4} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{4} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 118 | symmetry | $\frac{\sqrt{70}x^4}{16} - \frac{\sqrt{70}ix^3y}{4} - \frac{3\sqrt{70}x^2y^2}{8} + \frac{\sqrt{70}ixy^3}{4} + \frac{\sqrt{70}y^4}{16}$ |
| | $\mathbb{M}_4^{(1,-1;a)}(E^{(b)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 119 | symmetry | $-\frac{\sqrt{10}x^4}{8} - \frac{\sqrt{10}ix^3y}{4} + \frac{3\sqrt{10}x^2z^2}{4} - \frac{\sqrt{10}ixy^3}{4} + \frac{3\sqrt{10}ixyz^2}{2} + \frac{\sqrt{10}y^4}{8} - \frac{3\sqrt{10}y^2z^2}{4}$ |
| | $\mathbb{M}_4^{(1,-1;a)}(E^{(b)},3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{4} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{4} & 0 \end{bmatrix}$ |
| 120 | symmetry | $-\frac{x^2}{2} - \frac{y^2}{2} + z^2$ |

*continued ...*

Table 4

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_2^{(1,1;a)}(A)$ | $\begin{bmatrix} 0 & 0 & -\frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 121 | symmetry | $\frac{\sqrt{6}xz}{2} + \frac{\sqrt{6}iyz}{2}$ |
| | $\mathbb{M}_2^{(1,1;a)}(E^{(a)},1)$ | $\begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{6}}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{3} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 122 | symmetry | $\frac{\sqrt{6}x^2}{4} - \frac{\sqrt{6}ixy}{2} - \frac{\sqrt{6}y^2}{4}$ |
| | $\mathbb{M}_2^{(1,1;a)}(E^{(a)},2)$ | $\begin{bmatrix} -\frac{\sqrt{15}}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 123 | symmetry | $\frac{\sqrt{6}xz}{2} - \frac{\sqrt{6}iyz}{2}$ |
| | $\mathbb{M}_2^{(1,1;a)}(E^{(b)},1)$ | $\begin{bmatrix} 0 & \frac{\sqrt{3}}{3} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{6}}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 124 | symmetry | $\frac{\sqrt{6}x^2}{4} + \frac{\sqrt{6}ixy}{2} - \frac{\sqrt{6}y^2}{4}$ |
| | $\mathbb{M}_2^{(1,1;a)}(E^{(b)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |

bra: $= \langle \frac{1}{2}, \frac{1}{2}; p| \, , \langle \frac{1}{2}, -\frac{1}{2}; p| \, , \langle \frac{3}{2}, \frac{3}{2}; p| \, , \langle \frac{3}{2}, \frac{1}{2}; p| \, , \langle \frac{3}{2}, -\frac{1}{2}; p| \, , \langle \frac{3}{2}, -\frac{3}{2}; p|$

ket: $= |\frac{1}{2}, \frac{1}{2}; p\rangle \, , |\frac{1}{2}, -\frac{1}{2}; p\rangle \, , |\frac{3}{2}, \frac{3}{2}; p\rangle \, , |\frac{3}{2}, \frac{1}{2}; p\rangle \, , |\frac{3}{2}, -\frac{1}{2}; p\rangle \, , |\frac{3}{2}, -\frac{3}{2}; p\rangle$

Table 5: (p,p) block.

| No. | multipole | matrix |
|---|---|---|
| 125 | symmetry | 1 |
| | $\mathbb{Q}_0^{(a)}(A)$ | $\begin{bmatrix} \frac{\sqrt{6}}{6} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{6}}{6} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{6}}{6} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{6}}{6} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{6} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{6} \end{bmatrix}$ |
| 126 | symmetry | $-\frac{x^2}{2} - \frac{y^2}{2} + z^2$ |

*continued ...*

Table 5

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_2^{(a)}(A)$ | $\begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{6}}{6} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{6} & 0 \\ 0 & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & 0 \\ -\frac{\sqrt{6}}{6} & 0 & 0 & \frac{\sqrt{3}}{6} & 0 & 0 \\ 0 & \frac{\sqrt{6}}{6} & 0 & 0 & \frac{\sqrt{3}}{6} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{6} \end{bmatrix}$ |
| 127 | symmetry | $\frac{\sqrt{6}xz}{2} + \frac{\sqrt{6}iyz}{2}$ |
| | $\mathbb{Q}_2^{(a)}(E^{(a)},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{1}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{6} \\ \frac{\sqrt{3}}{6} & 0 & 0 & -\frac{\sqrt{6}}{6} & 0 & 0 \\ 0 & -\frac{1}{2} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{6} \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 128 | symmetry | $\frac{\sqrt{6}x^2}{4} - \frac{\sqrt{6}ixy}{2} - \frac{\sqrt{6}y^2}{4}$ |
| | $\mathbb{Q}_2^{(a)}(E^{(a)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{3}}{3} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{6}}{6} & 0 & 0 & 0 \\ -\frac{\sqrt{3}}{3} & 0 & 0 & -\frac{\sqrt{6}}{6} & 0 & 0 \end{bmatrix}$ |
| 129 | symmetry | $\frac{\sqrt{6}xz}{2} - \frac{\sqrt{6}iyz}{2}$ |
| | $\mathbb{Q}_2^{(a)}(E^{(b)},1)$ | $\begin{bmatrix} 0 & 0 & \frac{\sqrt{3}}{6} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{1}{2} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{6}}{6} & 0 & 0 & 0 \\ -\frac{1}{2} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{3}}{6} & 0 & 0 & \frac{\sqrt{6}}{6} & 0 \end{bmatrix}$ |
| 130 | symmetry | $\frac{\sqrt{6}x^2}{4} + \frac{\sqrt{6}ixy}{2} - \frac{\sqrt{6}y^2}{4}$ |

*continued ...*

Table 5

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_2^{(a)}(E^{(b)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{3} \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{3}}{3} & 0 & 0 & -\frac{\sqrt{6}}{6} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{6} \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 131 | symmetry | $-\frac{x^2}{2}-\frac{y^2}{2}+z^2$ |
| | $\mathbb{Q}_2^{(1,-1;a)}(A)$ | $\begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{6} & 0 \\ 0 & 0 & \frac{\sqrt{6}}{6} & 0 & 0 & 0 \\ -\frac{\sqrt{3}}{6} & 0 & 0 & -\frac{\sqrt{6}}{6} & 0 & 0 \\ 0 & \frac{\sqrt{3}}{6} & 0 & 0 & -\frac{\sqrt{6}}{6} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{6} \end{bmatrix}$ |
| 132 | symmetry | $\frac{\sqrt{6}xz}{2}+\frac{\sqrt{6}iyz}{2}$ |
| | $\mathbb{Q}_2^{(1,-1;a)}(E^{(a)},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{4} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{12} \\ \frac{\sqrt{6}}{12} & 0 & 0 & \frac{\sqrt{3}}{3} & 0 & 0 \\ 0 & -\frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{3} \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 133 | symmetry | $\frac{\sqrt{6}x^2}{4}-\frac{\sqrt{6}ixy}{2}-\frac{\sqrt{6}y^2}{4}$ |
| | $\mathbb{Q}_2^{(1,-1;a)}(E^{(a)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{6}}{6} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{3}}{3} & 0 & 0 & 0 \\ -\frac{\sqrt{6}}{6} & 0 & 0 & \frac{\sqrt{3}}{3} & 0 & 0 \end{bmatrix}$ |
| 134 | symmetry | $\frac{\sqrt{6}xz}{2}-\frac{\sqrt{6}iyz}{2}$ |

*continued ...*

Table 5

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_2^{(1,-1;a)}(E^{(b)},1)$ | $\begin{bmatrix} 0 & 0 & \frac{\sqrt{6}}{12} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{2}}{4} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{3}}{3} & 0 & 0 & 0 \\ -\frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{6}}{12} & 0 & 0 & -\frac{\sqrt{3}}{3} & 0 \end{bmatrix}$ |
| 135 | symmetry | $\frac{\sqrt{6}x^2}{4}+\frac{\sqrt{6}ixy}{2}-\frac{\sqrt{6}y^2}{4}$ |
| | $\mathbb{Q}_2^{(1,-1;a)}(E^{(b)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{6} \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{6}}{6} & 0 & 0 & \frac{\sqrt{3}}{3} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{3} \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 136 | symmetry | 1 |
| | $\mathbb{Q}_0^{(1,1;a)}(A)$ | $\begin{bmatrix} -\frac{\sqrt{3}}{3} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{3}}{3} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{3}}{6} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{3}}{6} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{6} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{6} \end{bmatrix}$ |
| 137 | symmetry | $z$ |
| | $\mathbb{G}_1^{(1,0;a)}(A)$ | $\begin{bmatrix} 0 & 0 & 0 & -\frac{i}{2} & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{i}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{i}{2} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{i}{2} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 138 | symmetry | $\frac{\sqrt{2}x}{2}+\frac{\sqrt{2}iy}{2}$ |

*continued ...*

Table 5

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_1^{(1,0;a)}(E^{(a)})$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{4} \\ -\frac{\sqrt{6}i}{4} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 139 | symmetry | $\frac{\sqrt{2}x}{2} - \frac{\sqrt{2}iy}{2}$ |
| | $\mathbb{G}_1^{(1,0;a)}(E^{(b)})$ | $\begin{bmatrix} 0 & 0 & \frac{\sqrt{6}i}{4} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{6}i}{4} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 140 | symmetry | $-\frac{x^2}{2} - \frac{y^2}{2} + z^2$ |
| | $\mathbb{T}_2^{(1,0;a)}(A)$ | $\begin{bmatrix} 0 & 0 & 0 & \frac{i}{2} & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{i}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{i}{2} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{i}{2} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 141 | symmetry | $\frac{\sqrt{6}xz}{2} + \frac{\sqrt{6}iyz}{2}$ |
| | $\mathbb{T}_2^{(1,0;a)}(E^{(a)},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{4} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} \\ \frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{6}i}{4} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 142 | symmetry | $\frac{\sqrt{6}x^2}{4} - \frac{\sqrt{6}ixy}{2} - \frac{\sqrt{6}y^2}{4}$ |

*continued ...*

Table 5

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_2^{(1,0;a)}(E^{(a)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{2}i}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{2}i}{2} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 143 | symmetry | $\frac{\sqrt{6}xz}{2} - \frac{\sqrt{6}iyz}{2}$ |
| | $\mathbb{T}_2^{(1,0;a)}(E^{(b)},1)$ | $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{6}i}{4} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{6}i}{4} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 144 | symmetry | $\frac{\sqrt{6}x^2}{4} + \frac{\sqrt{6}ixy}{2} - \frac{\sqrt{6}y^2}{4}$ |
| | $\mathbb{T}_2^{(1,0;a)}(E^{(b)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{2} \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{2}i}{2} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 145 | symmetry | $z$ |
| | $\mathbb{M}_1^{(a)}(A)$ | $\begin{bmatrix} \frac{1}{3} & 0 & 0 & \frac{\sqrt{2}}{6} & 0 & 0 \\ 0 & -\frac{1}{3} & 0 & 0 & \frac{\sqrt{2}}{6} & 0 \\ 0 & 0 & \frac{1}{2} & 0 & 0 & 0 \\ \frac{\sqrt{2}}{6} & 0 & 0 & \frac{1}{6} & 0 & 0 \\ 0 & \frac{\sqrt{2}}{6} & 0 & 0 & -\frac{1}{6} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{1}{2} \end{bmatrix}$ |
| 146 | symmetry | $\frac{\sqrt{2}x}{2} + \frac{\sqrt{2}iy}{2}$ |

*continued ...*

Table 5

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_1^{(a)}(E^{(a)})$ | $\begin{bmatrix} 0 & \frac{\sqrt{2}}{3} & 0 & 0 & \frac{1}{6} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{6} \\ -\frac{\sqrt{3}}{6} & 0 & 0 & \frac{\sqrt{6}}{6} & 0 & 0 \\ 0 & -\frac{1}{6} & 0 & 0 & \frac{\sqrt{2}}{3} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{6} \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 147 | symmetry | $\frac{\sqrt{2}x}{2} - \frac{\sqrt{2}iy}{2}$ |
| | $\mathbb{M}_1^{(a)}(E^{(b)})$ | $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & 0 \\ \frac{\sqrt{2}}{3} & 0 & 0 & -\frac{1}{6} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{6}}{6} & 0 & 0 & 0 \\ \frac{1}{6} & 0 & 0 & \frac{\sqrt{2}}{3} & 0 & 0 \\ 0 & \frac{\sqrt{3}}{6} & 0 & 0 & \frac{\sqrt{6}}{6} & 0 \end{bmatrix}$ |
| 148 | symmetry | $z$ |
| | $\mathbb{M}_1^{(1,-1;a)}(A)$ | $\begin{bmatrix} -\frac{\sqrt{6}}{18} & 0 & 0 & -\frac{2\sqrt{3}}{9} & 0 & 0 \\ 0 & \frac{\sqrt{6}}{18} & 0 & 0 & -\frac{2\sqrt{3}}{9} & 0 \\ 0 & 0 & \frac{\sqrt{6}}{6} & 0 & 0 & 0 \\ -\frac{2\sqrt{3}}{9} & 0 & 0 & \frac{\sqrt{6}}{18} & 0 & 0 \\ 0 & -\frac{2\sqrt{3}}{9} & 0 & 0 & -\frac{\sqrt{6}}{18} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{6} \end{bmatrix}$ |
| 149 | symmetry | $\frac{\sqrt{2}x}{2} + \frac{\sqrt{2}iy}{2}$ |
| | $\mathbb{M}_1^{(1,-1;a)}(E^{(a)})$ | $\begin{bmatrix} 0 & -\frac{\sqrt{3}}{9} & 0 & 0 & -\frac{\sqrt{6}}{9} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{3} \\ \frac{\sqrt{2}}{3} & 0 & 0 & \frac{1}{3} & 0 & 0 \\ 0 & \frac{\sqrt{6}}{9} & 0 & 0 & \frac{2\sqrt{3}}{9} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{3} \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 150 | symmetry | $\frac{\sqrt{2}x}{2} - \frac{\sqrt{2}iy}{2}$ |

*continued ...*

Table 5

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_1^{(1,-1;a)}(E^{(b)})$ | $\begin{bmatrix} 0 & 0 & \frac{\sqrt{2}}{3} & 0 & 0 & 0 \\ -\frac{\sqrt{3}}{9} & 0 & 0 & \frac{\sqrt{6}}{9} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{1}{3} & 0 & 0 & 0 \\ -\frac{\sqrt{6}}{9} & 0 & 0 & \frac{2\sqrt{3}}{9} & 0 & 0 \\ 0 & -\frac{\sqrt{2}}{3} & 0 & 0 & \frac{1}{3} & 0 \end{bmatrix}$ |
| 151 | symmetry | $-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$ |
| | $\mathbb{M}_3^{(1,-1;a)}(A,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{3\sqrt{5}}{10} & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{3\sqrt{5}}{10} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{10} \end{bmatrix}$ |
| 152 | symmetry | $\frac{3\sqrt{10}x^2y}{4} - \frac{\sqrt{10}y^3}{4}$ |
| | $\mathbb{M}_3^{(1,-1;a)}(A,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{2} \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{2}i}{2} & 0 & 0 & 0 \end{bmatrix}$ |
| 153 | symmetry | $\frac{\sqrt{10}x^3}{4} - \frac{3\sqrt{10}xy^2}{4}$ |
| | $\mathbb{M}_3^{(1,-1;a)}(A,3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{2} \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{2}}{2} & 0 & 0 & 0 \end{bmatrix}$ |
| 154 | symmetry | $-\frac{\sqrt{3}x^3}{4} - \frac{\sqrt{3}ix^2y}{4} - \frac{\sqrt{3}xy^2}{4} + \sqrt{3}xz^2 - \frac{\sqrt{3}iy^3}{4} + \sqrt{3}iyz^2$ |

*continued ...*

Table 5

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_3^{(1,-1;a)}(E^{(a)},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{5}}{5} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{5} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{5} \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 155 | symmetry | $\frac{\sqrt{30}x^2z}{4} - \frac{\sqrt{30}ixyz}{2} - \frac{\sqrt{30}y^2z}{4}$ |
| | $\mathbb{M}_3^{(1,-1;a)}(E^{(a)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{2}}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{2}}{2} & 0 & 0 \end{bmatrix}$ |
| 156 | symmetry | $-\frac{\sqrt{3}x^3}{4} + \frac{\sqrt{3}ix^2y}{4} - \frac{\sqrt{3}xy^2}{4} + \sqrt{3}xz^2 + \frac{\sqrt{3}iy^3}{4} - \sqrt{3}iyz^2$ |
| | $\mathbb{M}_3^{(1,-1;a)}(E^{(b)},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{5}}{5} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{15}}{5} & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{5} & 0 \end{bmatrix}$ |
| 157 | symmetry | $\frac{\sqrt{30}x^2z}{4} + \frac{\sqrt{30}ixyz}{2} - \frac{\sqrt{30}y^2z}{4}$ |
| | $\mathbb{M}_3^{(1,-1;a)}(E^{(b)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{2} \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 158 | symmetry | $z$ |

*continued ...*

Table 5

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_1^{(1,1;a)}(A)$ | $\begin{bmatrix} \frac{\sqrt{30}}{9} & 0 & 0 & -\frac{\sqrt{15}}{18} & 0 & 0 \\ 0 & -\frac{\sqrt{30}}{9} & 0 & 0 & -\frac{\sqrt{15}}{18} & 0 \\ 0 & 0 & -\frac{\sqrt{30}}{30} & 0 & 0 & 0 \\ -\frac{\sqrt{15}}{18} & 0 & 0 & -\frac{\sqrt{30}}{90} & 0 & 0 \\ 0 & -\frac{\sqrt{15}}{18} & 0 & 0 & \frac{\sqrt{30}}{90} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{30} \end{bmatrix}$ |
| 159 | symmetry | $\frac{\sqrt{2}x}{2} + \frac{\sqrt{2}iy}{2}$ |
| | $\mathbb{M}_1^{(1,1;a)}(E^{(a)})$ | $\begin{bmatrix} 0 & \frac{2\sqrt{15}}{9} & 0 & 0 & -\frac{\sqrt{30}}{36} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{12} \\ \frac{\sqrt{10}}{12} & 0 & 0 & -\frac{\sqrt{5}}{15} & 0 & 0 \\ 0 & \frac{\sqrt{30}}{36} & 0 & 0 & -\frac{2\sqrt{15}}{45} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{15} \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 160 | symmetry | $\frac{\sqrt{2}x}{2} - \frac{\sqrt{2}iy}{2}$ |
| | $\mathbb{M}_1^{(1,1;a)}(E^{(b)})$ | $\begin{bmatrix} 0 & 0 & \frac{\sqrt{10}}{12} & 0 & 0 & 0 \\ \frac{2\sqrt{15}}{9} & 0 & 0 & \frac{\sqrt{30}}{36} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{5}}{15} & 0 & 0 & 0 \\ -\frac{\sqrt{30}}{36} & 0 & 0 & -\frac{2\sqrt{15}}{45} & 0 & 0 \\ 0 & -\frac{\sqrt{10}}{12} & 0 & 0 & -\frac{\sqrt{5}}{15} & 0 \end{bmatrix}$ |

bra: $= \langle \frac{1}{2}, \frac{1}{2}; p|\,, \langle \frac{1}{2}, -\frac{1}{2}; p|\,, \langle \frac{3}{2}, \frac{3}{2}; p|\,, \langle \frac{3}{2}, \frac{1}{2}; p|\,, \langle \frac{3}{2}, -\frac{1}{2}; p|\,, \langle \frac{3}{2}, -\frac{3}{2}; p|$

ket: $= |\frac{3}{2}, \frac{3}{2}; d\rangle\,, |\frac{3}{2}, \frac{1}{2}; d\rangle\,, |\frac{3}{2}, -\frac{1}{2}; d\rangle\,, |\frac{3}{2}, -\frac{3}{2}; d\rangle\,, |\frac{5}{2}, \frac{5}{2}; d\rangle\,, |\frac{5}{2}, \frac{3}{2}; d\rangle\,, |\frac{5}{2}, \frac{1}{2}; d\rangle\,, |\frac{5}{2}, -\frac{1}{2}; d\rangle\,, |\frac{5}{2}, -\frac{3}{2}; d\rangle\,, |\frac{5}{2}, -\frac{5}{2}; d\rangle$

Table 6: (p,d) block.

| No. | multipole | matrix |
|---|---|---|
| 161 | symmetry | $z$ |

*continued ...*

Table 6

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_1^{(a)}(A)$ | $\begin{bmatrix} 0 & \frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{6}}{20} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{10} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{6}}{60} & 0 & 0 & 0 & 0 & \frac{3}{10} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{6}}{60} & 0 & 0 & 0 & 0 & \frac{3}{10} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{6}}{20} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{10} & 0 \end{bmatrix}$ |
| 162 | symmetry | $\frac{\sqrt{2}x}{2} + \frac{\sqrt{2}iy}{2}$ |
| | $\mathbb{Q}_1^{(a)}(E^{(a)})$ | $\begin{bmatrix} 0 & 0 & \frac{\sqrt{6}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{1}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{20} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{3}}{15} & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}}{20} & 0 & 0 \\ 0 & 0 & 0 & -\frac{1}{10} & 0 & 0 & 0 & 0 & \frac{3}{10} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{10} \end{bmatrix}$ |
| 163 | symmetry | $\frac{\sqrt{2}x}{2} - \frac{\sqrt{2}iy}{2}$ |
| | $\mathbb{Q}_1^{(a)}(E^{(b)})$ | $\begin{bmatrix} -\frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{6}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{10} & 0 & 0 & 0 & 0 & 0 \\ -\frac{1}{10} & 0 & 0 & 0 & 0 & -\frac{3}{10} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{3}}{15} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}}{20} & 0 & 0 & 0 \\ 0 & 0 & -\frac{1}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{20} & 0 & 0 \end{bmatrix}$ |
| 164 | symmetry | $-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$ |
| | $\mathbb{Q}_3^{(a)}(A,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{6} & 0 & 0 \\ \frac{1}{10} & 0 & 0 & 0 & 0 & -\frac{1}{5} & 0 & 0 & 0 & 0 \\ 0 & -\frac{3}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{15} & 0 & 0 & 0 \\ 0 & 0 & \frac{3}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{15} & 0 & 0 \\ 0 & 0 & 0 & -\frac{1}{10} & 0 & 0 & 0 & 0 & -\frac{1}{5} & 0 \end{bmatrix}$ |
| 165 | symmetry | $\frac{3\sqrt{10}x^2y}{4} - \frac{\sqrt{10}y^3}{4}$ |

*continued ...*

Table 6

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_3^{(a)}(A,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{6} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{10}i}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{20} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{12} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{12} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{10}i}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{20} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 166 | symmetry | $\frac{\sqrt{10}x^3}{4} - \frac{3\sqrt{10}xy^2}{4}$ |
| | $\mathbb{Q}_3^{(a)}(A,3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{6} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{10}}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{20} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{12} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{12} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{10}}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{20} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 167 | symmetry | $-\frac{\sqrt{3}x^3}{4} - \frac{\sqrt{3}ix^2y}{4} - \frac{\sqrt{3}xy^2}{4} + \sqrt{3}xz^2 - \frac{\sqrt{3}iy^3}{4} + \sqrt{3}iyz^2$ |
| | $\mathbb{Q}_3^{(a)}(E^{(a)},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{3} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{6} & 0 \\ 0 & \frac{1}{5} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{10} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{3}}{5} & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{30} & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{5} & 0 & 0 & 0 & 0 & \frac{7}{30} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{30} \end{bmatrix}$ |
| 168 | symmetry | $\frac{\sqrt{30}x^2z}{4} - \frac{\sqrt{30}ixyz}{2} - \frac{\sqrt{30}y^2z}{4}$ |
| | $\mathbb{Q}_3^{(a)}(E^{(a)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{1}{6} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{6} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{6} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{10}}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{30} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{10}}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{15} & 0 & 0 & 0 \end{bmatrix}$ |
| 169 | symmetry | $-\frac{\sqrt{3}x^3}{4} + \frac{\sqrt{3}ix^2y}{4} - \frac{\sqrt{3}xy^2}{4} + \sqrt{3}xz^2 + \frac{\sqrt{3}iy^3}{4} - \sqrt{3}iyz^2$ |

*continued ...*

Table 6

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_3^{(a)}(E^{(b)},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{6} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{3} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{30} & 0 & 0 & 0 & 0 & 0 \\ \frac{1}{5} & 0 & 0 & 0 & 0 & -\frac{7}{30} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{3}}{5} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{30} & 0 & 0 & 0 \\ 0 & 0 & \frac{1}{5} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{10} & 0 & 0 \end{bmatrix}$ |
| 170 | symmetry | $\frac{\sqrt{30}x^2z}{4} + \frac{\sqrt{30}ixyz}{2} - \frac{\sqrt{30}y^2z}{4}$ |
| | $\mathbb{Q}_3^{(a)}(E^{(b)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{6} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{6} \\ 0 & 0 & \frac{\sqrt{10}}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{15} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{10}}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{30} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{6} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 171 | symmetry | $-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$ |
| | $\mathbb{Q}_3^{(1,-1;a)}(A,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{6} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{6} & 0 & 0 \\ \frac{\sqrt{6}}{60} & 0 & 0 & 0 & 0 & \frac{2\sqrt{6}}{15} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{6}}{20} & 0 & 0 & 0 & 0 & -\frac{4}{15} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{6}}{20} & 0 & 0 & 0 & 0 & -\frac{4}{15} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{6}}{60} & 0 & 0 & 0 & 0 & \frac{2\sqrt{6}}{15} & 0 \end{bmatrix}$ |
| 172 | symmetry | $\frac{3\sqrt{10}x^2y}{4} - \frac{\sqrt{10}y^3}{4}$ |
| | $\mathbb{Q}_3^{(1,-1;a)}(A,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{6} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{6} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{15}i}{30} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{15} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{3} \\ 0 & 0 & 0 & 0 & -\frac{i}{3} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{15}i}{30} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{15} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 173 | symmetry | $\frac{\sqrt{10}x^3}{4} - \frac{3\sqrt{10}xy^2}{4}$ |

*continued ...*

Table 6

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_3^{(1,-1;a)}(A,3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{6} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{6} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{15}}{30} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{15} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{3} \\ 0 & 0 & 0 & 0 & -\frac{1}{3} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{15}}{30} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{15} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 174 | symmetry | $-\frac{\sqrt{3}x^3}{4} - \frac{\sqrt{3}ix^2y}{4} - \frac{\sqrt{3}xy^2}{4} + \sqrt{3}xz^2 - \frac{\sqrt{3}iy^3}{4} + \sqrt{3}iyz^2$ |
| | $\mathbb{Q}_3^{(1,-1;a)}(E^{(a)},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{9} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{9} & 0 \\ 0 & \frac{\sqrt{6}}{30} & 0 & 0 & 0 & 0 & \frac{2}{5} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{2}}{10} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{3}}{45} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{6}}{30} & 0 & 0 & 0 & 0 & -\frac{7\sqrt{6}}{45} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{15} \end{bmatrix}$ |
| 175 | symmetry | $\frac{\sqrt{30}x^2z}{4} - \frac{\sqrt{30}ixyz}{2} - \frac{\sqrt{30}y^2z}{4}$ |
| | $\mathbb{Q}_3^{(1,-1;a)}(E^{(a)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{18} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{18} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{2\sqrt{3}}{9} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{15}}{30} & 0 & 0 & 0 & 0 & \frac{2\sqrt{15}}{45} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{15}}{30} & 0 & 0 & 0 & 0 & \frac{2\sqrt{10}}{15} & 0 & 0 & 0 \end{bmatrix}$ |
| 176 | symmetry | $-\frac{\sqrt{3}x^3}{4} + \frac{\sqrt{3}ix^2y}{4} - \frac{\sqrt{3}xy^2}{4} + \sqrt{3}xz^2 + \frac{\sqrt{3}iy^3}{4} - \sqrt{3}iyz^2$ |
| | $\mathbb{Q}_3^{(1,-1;a)}(E^{(b)},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{9} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{9} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{15} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{6}}{30} & 0 & 0 & 0 & 0 & \frac{7\sqrt{6}}{45} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{2}}{10} & 0 & 0 & 0 & 0 & \frac{2\sqrt{3}}{45} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{6}}{30} & 0 & 0 & 0 & 0 & -\frac{2}{5} & 0 & 0 \end{bmatrix}$ |
| 177 | symmetry | $\frac{\sqrt{30}x^2z}{4} + \frac{\sqrt{30}ixyz}{2} - \frac{\sqrt{30}y^2z}{4}$ |

*continued ...*

Table 6

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_3^{(1,-1;a)}(E^{(b)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{18} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{18} \\ 0 & 0 & \frac{\sqrt{15}}{30} & 0 & 0 & 0 & 0 & \frac{2\sqrt{10}}{15} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{15}}{30} & 0 & 0 & 0 & 0 & \frac{2\sqrt{15}}{45} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{3}}{9} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 178 | symmetry | $z$ |
| | $\mathbb{Q}_1^{(1,0;a)}(A)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{6}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{6}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{3}}{5} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{10} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{3}}{15} & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}}{20} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{3}}{15} & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}}{20} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{3}}{5} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{10} & 0 \end{bmatrix}$ |
| 179 | symmetry | $\frac{\sqrt{2}x}{2} + \frac{\sqrt{2}iy}{2}$ |
| | $\mathbb{Q}_1^{(1,0;a)}(E^{(a)})$ | $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{3}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{1}{4} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{2}}{5} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{20} & 0 & 0 & 0 \\ 0 & 0 & \frac{2\sqrt{6}}{15} & 0 & 0 & 0 & 0 & \frac{3}{20} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{2}}{5} & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}}{20} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{20} \end{bmatrix}$ |
| 180 | symmetry | $\frac{\sqrt{2}x}{2} - \frac{\sqrt{2}iy}{2}$ |
| | $\mathbb{Q}_1^{(1,0;a)}(E^{(b)})$ | $\begin{bmatrix} \frac{1}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{3}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{20} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{2}}{5} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}}{20} & 0 & 0 & 0 & 0 \\ 0 & \frac{2\sqrt{6}}{15} & 0 & 0 & 0 & 0 & -\frac{3}{20} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{2}}{5} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{20} & 0 & 0 \end{bmatrix}$ |
| 181 | symmetry | $-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$ |

*continued ...*

Table 6

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_3^{(1,0;a)}(A,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{3} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{3} & 0 & 0 \\ -\frac{\sqrt{3}}{15} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{30} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{3}}{5} & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{30} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{3}}{5} & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{30} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{3}}{15} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{30} & 0 \end{bmatrix}$ |
| 182 | symmetry | $\frac{3\sqrt{10}x^2y}{4} - \frac{\sqrt{10}y^3}{4}$ |
| | $\mathbb{Q}_3^{(1,0;a)}(A,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{3} \\ 0 & 0 & 0 & 0 & -\frac{i}{3} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{30}i}{15} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{120} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{24} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{24} & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{30}i}{15} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{120} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 183 | symmetry | $\frac{\sqrt{10}x^3}{4} - \frac{3\sqrt{10}xy^2}{4}$ |
| | $\mathbb{Q}_3^{(1,0;a)}(A,3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{3} \\ 0 & 0 & 0 & 0 & -\frac{1}{3} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{30}}{15} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{120} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{24} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{24} & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{30}}{15} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{120} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 184 | symmetry | $-\frac{\sqrt{3}x^3}{4} - \frac{\sqrt{3}ix^2y}{4} - \frac{\sqrt{3}xy^2}{4} + \sqrt{3}xz^2 - \frac{\sqrt{3}iy^3}{4} + \sqrt{3}iyz^2$ |
| | $\mathbb{Q}_3^{(1,0;a)}(E^{(a)},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{3}}{9} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{9} & 0 \\ 0 & -\frac{2\sqrt{3}}{15} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{20} & 0 & 0 & 0 \\ 0 & 0 & \frac{2}{5} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{180} & 0 & 0 \\ 0 & 0 & 0 & -\frac{2\sqrt{3}}{15} & 0 & 0 & 0 & 0 & \frac{7\sqrt{3}}{180} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{60} \end{bmatrix}$ |
| 185 | symmetry | $\frac{\sqrt{30}x^2z}{4} - \frac{\sqrt{30}ixyz}{2} - \frac{\sqrt{30}y^2z}{4}$ |

*continued ...*

Table 6

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_3^{(1,0;a)}(E^{(a)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{9} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{9} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{36} & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{30}}{15} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{180} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{30}}{15} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{30} & 0 & 0 & 0 \end{bmatrix}$ |
| 186 | symmetry | $-\frac{\sqrt{3}x^3}{4}+\frac{\sqrt{3}ix^2y}{4}-\frac{\sqrt{3}xy^2}{4}+\sqrt{3}xz^2+\frac{\sqrt{3}iy^3}{4}-\sqrt{3}iyz^2$ |
| | $\mathbb{Q}_3^{(1,0;a)}(E^{(b)},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{9} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{3}}{9} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{60} & 0 & 0 & 0 & 0 & 0 \\ -\frac{2\sqrt{3}}{15} & 0 & 0 & 0 & 0 & -\frac{7\sqrt{3}}{180} & 0 & 0 & 0 & 0 \\ 0 & \frac{2}{5} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{180} & 0 & 0 & 0 \\ 0 & 0 & -\frac{2\sqrt{3}}{15} & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{20} & 0 & 0 \end{bmatrix}$ |
| 187 | symmetry | $\frac{\sqrt{30}x^2z}{4}+\frac{\sqrt{30}ixyz}{2}-\frac{\sqrt{30}y^2z}{4}$ |
| | $\mathbb{Q}_3^{(1,0;a)}(E^{(b)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{9} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{9} \\ 0 & 0 & -\frac{\sqrt{30}}{15} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{30} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{30}}{15} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{180} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{36} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 188 | symmetry | $z$ |
| | $\mathbb{Q}_1^{(1,1;a)}(A)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{3}{10} & 0 & 0 & 0 & 0 & \frac{1}{10} & 0 & 0 & 0 & 0 \\ 0 & -\frac{1}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{20} & 0 & 0 & 0 \\ 0 & 0 & \frac{1}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{20} & 0 & 0 \\ 0 & 0 & 0 & \frac{3}{10} & 0 & 0 & 0 & 0 & \frac{1}{10} & 0 \end{bmatrix}$ |
| 189 | symmetry | $\frac{\sqrt{2}x}{2}+\frac{\sqrt{2}iy}{2}$ |

*continued ...*

Table 6

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_1^{(1,1;a)}(E^{(a)})$ | $\begin{bmatrix} 0 & 0 & -\frac{1}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{3}}{4} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{6}}{10} & 0 & 0 & 0 & 0 & \frac{1}{20} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{2}}{5} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{20} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{6}}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{20} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{20} \end{bmatrix}$ |
| 190 | symmetry | $\frac{\sqrt{2}x}{2} - \frac{\sqrt{2}iy}{2}$ |
| | $\mathbb{Q}_1^{(1,1;a)}(E^{(b)})$ | $\begin{bmatrix} \frac{\sqrt{3}}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{1}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{20} & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{6}}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{20} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{2}}{5} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{20} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{6}}{10} & 0 & 0 & 0 & 0 & -\frac{1}{20} & 0 & 0 \end{bmatrix}$ |
| 191 | symmetry | $-\frac{x^2}{2} - \frac{y^2}{2} + z^2$ |
| | $\mathbb{G}_2^{(a)}(A)$ | $\begin{bmatrix} 0 & \frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{30} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{30} & 0 & 0 \\ -\frac{\sqrt{10}i}{20} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{10} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{10}i}{20} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{30} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{10}i}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{30} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{10}i}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{10} & 0 \end{bmatrix}$ |
| 192 | symmetry | $\frac{\sqrt{6}xz}{2} + \frac{\sqrt{6}iyz}{2}$ |
| | $\mathbb{G}_2^{(a)}(E^{(a)}, 1)$ | $\begin{bmatrix} 0 & 0 & \frac{\sqrt{30}i}{20} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{15} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{10}i}{20} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{15} & 0 \\ 0 & -\frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{20} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{12} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{30} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} \end{bmatrix}$ |
| 193 | symmetry | $\frac{\sqrt{6}x^2}{4} - \frac{\sqrt{6}ixy}{2} - \frac{\sqrt{6}y^2}{4}$ |

*continued ...*

Table 6

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_2^{(a)}(E^{(a)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{6} & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{10}i}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{30} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{i}{3} & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{5}i}{15} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{30} & 0 & 0 & 0 \end{bmatrix}$ |
| 194 | symmetry | $\frac{\sqrt{6}xz}{2} - \frac{\sqrt{6}iyz}{2}$ |
| | $\mathbb{G}_2^{(a)}(E^{(b)},1)$ | $\begin{bmatrix} -\frac{\sqrt{10}i}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{15} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{30}i}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{15} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{30} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{12} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{20} & 0 & 0 \end{bmatrix}$ |
| 195 | symmetry | $\frac{\sqrt{6}x^2}{4} + \frac{\sqrt{6}ixy}{2} - \frac{\sqrt{6}y^2}{4}$ |
| | $\mathbb{G}_2^{(a)}(E^{(b)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & \frac{\sqrt{10}i}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{30} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{6} \\ 0 & 0 & -\frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{30} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 & \frac{2\sqrt{5}i}{15} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{3} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 196 | symmetry | $-\frac{x^2}{2} - \frac{y^2}{2} + z^2$ |
| | $\mathbb{G}_2^{(1,-1;a)}(A)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{2}i}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{5} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{2}i}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{5} & 0 & 0 \\ \frac{i}{10} & 0 & 0 & 0 & 0 & \frac{3i}{10} & 0 & 0 & 0 & 0 \\ 0 & -\frac{i}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{20} & 0 & 0 & 0 \\ 0 & 0 & -\frac{i}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{20} & 0 & 0 \\ 0 & 0 & 0 & \frac{i}{10} & 0 & 0 & 0 & 0 & -\frac{3i}{10} & 0 \end{bmatrix}$ |
| 197 | symmetry | $\frac{\sqrt{6}xz}{2} + \frac{\sqrt{6}iyz}{2}$ |

*continued ...*

Table 6

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_2^{(1,-1;a)}(E^{(a)},1)$ | $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{3}i}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{5} & 0 & 0 \\ 0 & 0 & 0 & \frac{i}{20} & 0 & 0 & 0 & 0 & -\frac{2i}{5} & 0 \\ 0 & \frac{\sqrt{2}i}{10} & 0 & 0 & 0 & 0 & \frac{3\sqrt{3}i}{20} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{4} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{2}i}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{20} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{20} \end{bmatrix}$ |
| 198 | symmetry | $\frac{\sqrt{6}x^2}{4} - \frac{\sqrt{6}ixy}{2} - \frac{\sqrt{6}y^2}{4}$ |
| | $\mathbb{G}_2^{(1,-1;a)}(E^{(a)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{5} & 0 & 0 & 0 & 0 & 0 \\ \frac{i}{10} & 0 & 0 & 0 & 0 & -\frac{i}{5} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{10} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{2}i}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{5} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{2}i}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{10} & 0 & 0 & 0 \end{bmatrix}$ |
| 199 | symmetry | $\frac{\sqrt{6}xz}{2} - \frac{\sqrt{6}iyz}{2}$ |
| | $\mathbb{G}_2^{(1,-1;a)}(E^{(b)},1)$ | $\begin{bmatrix} \frac{i}{20} & 0 & 0 & 0 & 0 & \frac{2i}{5} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{3}i}{20} & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{5} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{20} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{2}i}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{20} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{4} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{2}i}{10} & 0 & 0 & 0 & 0 & \frac{3\sqrt{3}i}{20} & 0 & 0 \end{bmatrix}$ |
| 200 | symmetry | $\frac{\sqrt{6}x^2}{4} + \frac{\sqrt{6}ixy}{2} - \frac{\sqrt{6}y^2}{4}$ |
| | $\mathbb{G}_2^{(1,-1;a)}(E^{(b)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & -\frac{i}{10} & 0 & 0 & 0 & 0 & -\frac{i}{5} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{5} \\ 0 & 0 & \frac{\sqrt{2}i}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{10} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{2}i}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{5} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{10} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 201 | symmetry | $\frac{3x^4}{8} + \frac{3x^2y^2}{4} - 3x^2z^2 + \frac{3y^4}{8} - 3y^2z^2 + z^4$ |

*continued ...*

Table 6

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_4^{(1,-1;a)}(A,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{14} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 \end{bmatrix}$ |
| 202 | symmetry | $\frac{\sqrt{70}x^3z}{4} - \frac{3\sqrt{70}xy^2z}{4}$ |
| | $\mathbb{G}_4^{(1,-1;a)}(A,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{8} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{8} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{8} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{8} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 203 | symmetry | $\frac{3\sqrt{70}x^2yz}{4} - \frac{\sqrt{70}y^3z}{4}$ |
| | $\mathbb{G}_4^{(1,-1;a)}(A,3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{8} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{8} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{8} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{8} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 204 | symmetry | $-\frac{3\sqrt{5}x^3z}{4} - \frac{3\sqrt{5}ix^2yz}{4} - \frac{3\sqrt{5}xy^2z}{4} + \sqrt{5}xz^3 - \frac{3\sqrt{5}iy^3z}{4} + \sqrt{5}iyz^3$ |
| | $\mathbb{G}_4^{(1,-1;a)}(E^{(a)},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{28} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{28} \end{bmatrix}$ |
| 205 | symmetry | $\frac{\sqrt{70}x^4}{16} + \frac{\sqrt{70}ix^3y}{4} - \frac{3\sqrt{70}x^2y^2}{8} - \frac{\sqrt{70}ixy^3}{4} + \frac{\sqrt{70}y^4}{16}$ |

*continued ...*

Table 6

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_4^{(1,-1;a)}(E^{(a)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{2} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 206 | symmetry | $-\frac{\sqrt{10}x^4}{8}+\frac{\sqrt{10}ix^3y}{4}+\frac{3\sqrt{10}x^2z^2}{4}+\frac{\sqrt{10}ixy^3}{4}-\frac{3\sqrt{10}ixyz^2}{2}+\frac{\sqrt{10}y^4}{8}-\frac{3\sqrt{10}y^2z^2}{4}$ |
| | $\mathbb{G}_4^{(1,-1;a)}(E^{(a)},3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{14} & 0 & 0 & 0 \end{bmatrix}$ |
| 207 | symmetry | $-\frac{3\sqrt{5}x^3z}{4}+\frac{3\sqrt{5}ix^2yz}{4}-\frac{3\sqrt{5}xy^2z}{4}+\sqrt{5}xz^3+\frac{3\sqrt{5}iy^3z}{4}-\sqrt{5}iyz^3$ |
| | $\mathbb{G}_4^{(1,-1;a)}(E^{(b)},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{28} & 0 & 0 \end{bmatrix}$ |
| 208 | symmetry | $\frac{\sqrt{70}x^4}{16}-\frac{\sqrt{70}ix^3y}{4}-\frac{3\sqrt{70}x^2y^2}{8}+\frac{\sqrt{70}ixy^3}{4}+\frac{\sqrt{70}y^4}{16}$ |
| | $\mathbb{G}_4^{(1,-1;a)}(E^{(b)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{2} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 209 | symmetry | $-\frac{\sqrt{10}x^4}{8}-\frac{\sqrt{10}ix^3y}{4}+\frac{3\sqrt{10}x^2z^2}{4}-\frac{\sqrt{10}ixy^3}{4}+\frac{3\sqrt{10}ixyz^2}{2}+\frac{\sqrt{10}y^4}{8}-\frac{3\sqrt{10}y^2z^2}{4}$ |

*continued ...*

Table 6

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_4^{(1,-1;a)}(E^{(b)},3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{14} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{28} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{28} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 210 | symmetry | $-\frac{x^2}{2}-\frac{y^2}{2}+z^2$ |
| | $\mathbb{G}_2^{(1,0;a)}(A)$ | $\begin{bmatrix} 0 & \frac{\sqrt{30}i}{60} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{5}i}{15} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{30}i}{60} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{5}i}{15} & 0 & 0 \\ -\frac{\sqrt{15}i}{15} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{30} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{15}i}{15} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{60} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{15}i}{15} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{60} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{15}i}{15} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{30} & 0 \end{bmatrix}$ |
| 211 | symmetry | $\frac{\sqrt{6}xz}{2}+\frac{\sqrt{6}iyz}{2}$ |
| | $\mathbb{G}_2^{(1,0;a)}(E^{(a)},1)$ | $\begin{bmatrix} 0 & 0 & \frac{\sqrt{5}i}{20} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{30}i}{45} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{15}i}{60} & 0 & 0 & 0 & 0 & -\frac{4\sqrt{15}i}{45} & 0 \\ 0 & -\frac{\sqrt{30}i}{15} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{20} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{36} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{30}i}{15} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{180} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{12} \end{bmatrix}$ |
| 212 | symmetry | $\frac{\sqrt{6}x^2}{4}-\frac{\sqrt{6}ixy}{2}-\frac{\sqrt{6}y^2}{4}$ |
| | $\mathbb{G}_2^{(1,0;a)}(E^{(a)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{2\sqrt{3}i}{9} & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{15}i}{30} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{15}i}{45} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{18} & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{30}i}{15} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{45} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{30}i}{15} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{30} & 0 & 0 & 0 \end{bmatrix}$ |
| 213 | symmetry | $\frac{\sqrt{6}xz}{2}-\frac{\sqrt{6}iyz}{2}$ |

*continued ...*

Table 6

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_2^{(1,0;a)}(E^{(b)},1)$ | $\begin{bmatrix} -\frac{\sqrt{15}i}{60} & 0 & 0 & 0 & 0 & \frac{4\sqrt{15}i}{45} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{5}i}{20} & 0 & 0 & 0 & 0 & \frac{2\sqrt{30}i}{45} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{12} & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{30}i}{15} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{180} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{36} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{30}i}{15} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{20} & 0 & 0 \end{bmatrix}$ |
| 214 | symmetry | $\frac{\sqrt{6}x^2}{4}+\frac{\sqrt{6}ixy}{2}-\frac{\sqrt{6}y^2}{4}$ |
| | $\mathbb{G}_2^{(1,0;a)}(E^{(b)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & \frac{\sqrt{15}i}{30} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{15}i}{45} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{3}i}{9} \\ 0 & 0 & -\frac{\sqrt{30}i}{15} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{30} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{30}i}{15} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{45} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{18} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 215 | symmetry | 1 |
| | $\mathbb{G}_0^{(1,1;a)}(A)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 216 | symmetry | $-\frac{x^2}{2}-\frac{y^2}{2}+z^2$ |
| | $\mathbb{G}_2^{(1,1;a)}(A)$ | $\begin{bmatrix} 0 & \frac{\sqrt{42}i}{15} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{30} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{42}i}{15} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{30} & 0 & 0 \\ \frac{\sqrt{21}i}{30} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{21}i}{105} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{21}i}{30} & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{105} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{21}i}{30} & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{105} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{21}i}{30} & 0 & 0 & 0 & 0 & \frac{2\sqrt{21}i}{105} & 0 \end{bmatrix}$ |
| 217 | symmetry | $\frac{\sqrt{6}xz}{2}+\frac{\sqrt{6}iyz}{2}$ |

*continued ...*

Table 6

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_2^{(1,1;a)}(E^{(a)},1)$ | $\begin{bmatrix} 0 & 0 & \frac{\sqrt{7}i}{5} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{90} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{21}i}{15} & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{45} & 0 \\ 0 & \frac{\sqrt{42}i}{30} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{35} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{63} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{42}i}{30} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{315} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{105} \end{bmatrix}$ |
| 218 | symmetry | $\frac{\sqrt{6}x^2}{4} - \frac{\sqrt{6}ixy}{2} - \frac{\sqrt{6}y^2}{4}$ |
| | $\mathbb{G}_2^{(1,1;a)}(E^{(a)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{90} & 0 & 0 & 0 & 0 & 0 \\ -\frac{2\sqrt{21}i}{15} & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{90} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{2\sqrt{210}i}{315} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{42}i}{30} & 0 & 0 & 0 & 0 & \frac{4\sqrt{42}i}{315} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{42}i}{30} & 0 & 0 & 0 & 0 & \frac{2\sqrt{7}i}{105} & 0 & 0 & 0 \end{bmatrix}$ |
| 219 | symmetry | $\frac{\sqrt{6}xz}{2} - \frac{\sqrt{6}iyz}{2}$ |
| | $\mathbb{G}_2^{(1,1;a)}(E^{(b)},1)$ | $\begin{bmatrix} -\frac{\sqrt{21}i}{15} & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{45} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{7}i}{5} & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{90} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{105} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{42}i}{30} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{315} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{63} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{42}i}{30} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{35} & 0 & 0 \end{bmatrix}$ |
| 220 | symmetry | $\frac{\sqrt{6}x^2}{4} + \frac{\sqrt{6}ixy}{2} - \frac{\sqrt{6}y^2}{4}$ |
| | $\mathbb{G}_2^{(1,1;a)}(E^{(b)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & \frac{2\sqrt{21}i}{15} & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{90} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{90} \\ 0 & 0 & \frac{\sqrt{42}i}{30} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{7}i}{105} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{42}i}{30} & 0 & 0 & 0 & 0 & -\frac{4\sqrt{42}i}{315} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{210}i}{315} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 221 | symmetry | $z$ |

*continued ...*

Table 6

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_1^{(a)}(A)$ | $\begin{bmatrix} 0 & \frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{6}i}{20} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{10} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{6}i}{60} & 0 & 0 & 0 & 0 & \frac{3i}{10} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{6}i}{60} & 0 & 0 & 0 & 0 & \frac{3i}{10} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{6}i}{20} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{10} & 0 \end{bmatrix}$ |
| 222 | symmetry | $\frac{\sqrt{2}x}{2} + \frac{\sqrt{2}iy}{2}$ |
| | $\mathbb{T}_1^{(a)}(E^{(a)})$ | $\begin{bmatrix} 0 & 0 & \frac{\sqrt{6}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{i}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{20} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{3}i}{15} & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}i}{20} & 0 & 0 \\ 0 & 0 & 0 & -\frac{i}{10} & 0 & 0 & 0 & 0 & \frac{3i}{10} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{10} \end{bmatrix}$ |
| 223 | symmetry | $\frac{\sqrt{2}x}{2} - \frac{\sqrt{2}iy}{2}$ |
| | $\mathbb{T}_1^{(a)}(E^{(b)})$ | $\begin{bmatrix} -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{6}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{10} & 0 & 0 & 0 & 0 & 0 \\ -\frac{i}{10} & 0 & 0 & 0 & 0 & -\frac{3i}{10} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{3}i}{15} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}i}{20} & 0 & 0 & 0 \\ 0 & 0 & -\frac{i}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{20} & 0 & 0 \end{bmatrix}$ |
| 224 | symmetry | $-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$ |
| | $\mathbb{T}_3^{(a)}(A,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 \\ \frac{i}{10} & 0 & 0 & 0 & 0 & -\frac{i}{5} & 0 & 0 & 0 & 0 \\ 0 & -\frac{3i}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{15} & 0 & 0 & 0 \\ 0 & 0 & \frac{3i}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{15} & 0 & 0 \\ 0 & 0 & 0 & -\frac{i}{10} & 0 & 0 & 0 & 0 & -\frac{i}{5} & 0 \end{bmatrix}$ |
| 225 | symmetry | $\frac{3\sqrt{10}x^2y}{4} - \frac{\sqrt{10}y^3}{4}$ |

*continued ...*

Table 6

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_3^{(a)}(A,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{6} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{10}}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{20} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{12} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{12} & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{10}}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{20} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 226 | symmetry | $\frac{\sqrt{10}x^3}{4} - \frac{3\sqrt{10}xy^2}{4}$ |
| | $\mathbb{T}_3^{(a)}(A,3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{10}i}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{20} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{12} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{12} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{10}i}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{20} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 227 | symmetry | $-\frac{\sqrt{3}x^3}{4} - \frac{\sqrt{3}ix^2y}{4} - \frac{\sqrt{3}xy^2}{4} + \sqrt{3}xz^2 - \frac{\sqrt{3}iy^3}{4} + \sqrt{3}iyz^2$ |
| | $\mathbb{T}_3^{(a)}(E^{(a)},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{3} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{6} & 0 \\ 0 & \frac{i}{5} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{10} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{3}i}{5} & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{30} & 0 & 0 \\ 0 & 0 & 0 & \frac{i}{5} & 0 & 0 & 0 & 0 & \frac{7i}{30} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{30} \end{bmatrix}$ |
| 228 | symmetry | $\frac{\sqrt{30}x^2z}{4} - \frac{\sqrt{30}ixyz}{2} - \frac{\sqrt{30}y^2z}{4}$ |
| | $\mathbb{T}_3^{(a)}(E^{(a)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{i}{6} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{6} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{6} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{10}i}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{30} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{10}i}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{15} & 0 & 0 & 0 \end{bmatrix}$ |
| 229 | symmetry | $-\frac{\sqrt{3}x^3}{4} + \frac{\sqrt{3}ix^2y}{4} - \frac{\sqrt{3}xy^2}{4} + \sqrt{3}xz^2 + \frac{\sqrt{3}iy^3}{4} - \sqrt{3}iyz^2$ |

*continued ...*

Table 6

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_3^{(a)}(E^{(b)},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{6} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{3} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{30} & 0 & 0 & 0 & 0 & 0 \\ \frac{i}{5} & 0 & 0 & 0 & 0 & -\frac{7i}{30} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{3}i}{5} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{30} & 0 & 0 & 0 \\ 0 & 0 & \frac{i}{5} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{10} & 0 & 0 \end{bmatrix}$ |
| 230 | symmetry | $\frac{\sqrt{30}x^2z}{4} + \frac{\sqrt{30}ixyz}{2} - \frac{\sqrt{30}y^2z}{4}$ |
| | $\mathbb{T}_3^{(a)}(E^{(b)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{6} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{6} \\ 0 & 0 & \frac{\sqrt{10}i}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{15} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{10}i}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{30} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{6} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 231 | symmetry | $-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$ |
| | $\mathbb{T}_3^{(1,-1;a)}(A,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{6} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{6} & 0 & 0 \\ \frac{\sqrt{6}i}{60} & 0 & 0 & 0 & 0 & \frac{2\sqrt{6}i}{15} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{6}i}{20} & 0 & 0 & 0 & 0 & -\frac{4i}{15} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{6}i}{20} & 0 & 0 & 0 & 0 & -\frac{4i}{15} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{6}i}{60} & 0 & 0 & 0 & 0 & \frac{2\sqrt{6}i}{15} & 0 \end{bmatrix}$ |
| 232 | symmetry | $\frac{3\sqrt{10}x^2y}{4} - \frac{\sqrt{10}y^3}{4}$ |
| | $\mathbb{T}_3^{(1,-1;a)}(A,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{6} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{6} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{15}}{30} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{15} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{3} \\ 0 & 0 & 0 & 0 & \frac{1}{3} & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{15}}{30} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{15} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 233 | symmetry | $\frac{\sqrt{10}x^3}{4} - \frac{3\sqrt{10}xy^2}{4}$ |

*continued ...*

Table 6

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_3^{(1,-1;a)}(A,3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{6} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{6} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{15}i}{30} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{15} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{3} \\ 0 & 0 & 0 & 0 & -\frac{i}{3} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{15}i}{30} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{15} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 234 | symmetry | $-\frac{\sqrt{3}x^3}{4} - \frac{\sqrt{3}ix^2y}{4} - \frac{\sqrt{3}xy^2}{4} + \sqrt{3}xz^2 - \frac{\sqrt{3}iy^3}{4} + \sqrt{3}iyz^2$ |
| | $\mathbb{T}_3^{(1,-1;a)}(E^{(a)},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{9} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{9} & 0 \\ 0 & \frac{\sqrt{6}i}{30} & 0 & 0 & 0 & 0 & \frac{2i}{5} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{2}i}{10} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{3}i}{45} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{6}i}{30} & 0 & 0 & 0 & 0 & -\frac{7\sqrt{6}i}{45} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{15} \end{bmatrix}$ |
| 235 | symmetry | $\frac{\sqrt{30}x^2z}{4} - \frac{\sqrt{30}ixyz}{2} - \frac{\sqrt{30}y^2z}{4}$ |
| | $\mathbb{T}_3^{(1,-1;a)}(E^{(a)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{18} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{18} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{2\sqrt{3}i}{9} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{15}i}{30} & 0 & 0 & 0 & 0 & \frac{2\sqrt{15}i}{45} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{15}i}{30} & 0 & 0 & 0 & 0 & \frac{2\sqrt{10}i}{15} & 0 & 0 & 0 \end{bmatrix}$ |
| 236 | symmetry | $-\frac{\sqrt{3}x^3}{4} + \frac{\sqrt{3}ix^2y}{4} - \frac{\sqrt{3}xy^2}{4} + \sqrt{3}xz^2 + \frac{\sqrt{3}iy^3}{4} - \sqrt{3}iyz^2$ |
| | $\mathbb{T}_3^{(1,-1;a)}(E^{(b)},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{9} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{9} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{15} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{6}i}{30} & 0 & 0 & 0 & 0 & \frac{7\sqrt{6}i}{45} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{2}i}{10} & 0 & 0 & 0 & 0 & \frac{2\sqrt{3}i}{45} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{6}i}{30} & 0 & 0 & 0 & 0 & -\frac{2i}{5} & 0 & 0 \end{bmatrix}$ |
| 237 | symmetry | $\frac{\sqrt{30}x^2z}{4} + \frac{\sqrt{30}ixyz}{2} - \frac{\sqrt{30}y^2z}{4}$ |

*continued ...*

Table 6

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_3^{(1,-1;a)}(E^{(b)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{18} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{18} \\ 0 & 0 & \frac{\sqrt{15}i}{30} & 0 & 0 & 0 & 0 & \frac{2\sqrt{10}i}{15} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{15}i}{30} & 0 & 0 & 0 & 0 & \frac{2\sqrt{15}i}{45} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{3}i}{9} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 238 | symmetry | $z$ |
| | $\mathbb{T}_1^{(1,0;a)}(A)$ | $\begin{bmatrix} 0 & \frac{\sqrt{6}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{6}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{3}i}{5} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{10} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{3}i}{15} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}i}{20} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{3}i}{15} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}i}{20} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{3}i}{5} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{10} & 0 \end{bmatrix}$ |
| 239 | symmetry | $\frac{\sqrt{2}x}{2}+\frac{\sqrt{2}iy}{2}$ |
| | $\mathbb{T}_1^{(1,0;a)}(E^{(a)})$ | $\begin{bmatrix} 0 & 0 & \frac{\sqrt{3}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{i}{4} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{2}i}{5} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{20} & 0 & 0 & 0 \\ 0 & 0 & -\frac{2\sqrt{6}i}{15} & 0 & 0 & 0 & 0 & -\frac{3i}{20} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{2}i}{5} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}i}{20} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{20} \end{bmatrix}$ |
| 240 | symmetry | $\frac{\sqrt{2}x}{2}-\frac{\sqrt{2}iy}{2}$ |
| | $\mathbb{T}_1^{(1,0;a)}(E^{(b)})$ | $\begin{bmatrix} -\frac{i}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{3}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{20} & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{2}i}{5} & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}i}{20} & 0 & 0 & 0 & 0 \\ 0 & -\frac{2\sqrt{6}i}{15} & 0 & 0 & 0 & 0 & \frac{3i}{20} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{2}i}{5} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{20} & 0 & 0 \end{bmatrix}$ |
| 241 | symmetry | $-\frac{3x^2z}{2}-\frac{3y^2z}{2}+z^3$ |

*continued ...*

Table 6

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_3^{(1,0;a)}(A,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{3} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{3} & 0 & 0 \\ \frac{\sqrt{3}i}{15} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{30} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{3}i}{5} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{30} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{3}i}{5} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{30} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{3}i}{15} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{30} & 0 \end{bmatrix}$ |
| 242 | symmetry | $\frac{3\sqrt{10}x^2y}{4} - \frac{\sqrt{10}y^3}{4}$ |
| | $\mathbb{T}_3^{(1,0;a)}(A,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{3} \\ 0 & 0 & 0 & 0 & -\frac{1}{3} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{30}}{15} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{120} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{24} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{24} & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{30}}{15} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{120} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 243 | symmetry | $\frac{\sqrt{10}x^3}{4} - \frac{3\sqrt{10}xy^2}{4}$ |
| | $\mathbb{T}_3^{(1,0;a)}(A,3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{3} \\ 0 & 0 & 0 & 0 & \frac{i}{3} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{30}i}{15} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{120} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{24} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{24} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{30}i}{15} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{120} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 244 | symmetry | $-\frac{\sqrt{3}x^3}{4} - \frac{\sqrt{3}ix^2y}{4} - \frac{\sqrt{3}xy^2}{4} + \sqrt{3}xz^2 - \frac{\sqrt{3}iy^3}{4} + \sqrt{3}iyz^2$ |
| | $\mathbb{T}_3^{(1,0;a)}(E^{(a)},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{3}i}{9} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{9} & 0 \\ 0 & \frac{2\sqrt{3}i}{15} & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{20} & 0 & 0 & 0 \\ 0 & 0 & -\frac{2i}{5} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{180} & 0 & 0 \\ 0 & 0 & 0 & \frac{2\sqrt{3}i}{15} & 0 & 0 & 0 & 0 & -\frac{7\sqrt{3}i}{180} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{60} \end{bmatrix}$ |
| 245 | symmetry | $\frac{\sqrt{30}x^2z}{4} - \frac{\sqrt{30}ixyz}{2} - \frac{\sqrt{30}y^2z}{4}$ |

*continued ...*

Table 6

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_3^{(1,0;a)}(E^{(a)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{9} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{9} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{36} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{30}i}{15} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{180} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{30}i}{15} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{30} & 0 & 0 & 0 \end{bmatrix}$ |
| 246 | symmetry | $-\frac{\sqrt{3}x^3}{4}+\frac{\sqrt{3}ix^2y}{4}-\frac{\sqrt{3}xy^2}{4}+\sqrt{3}xz^2+\frac{\sqrt{3}iy^3}{4}-\sqrt{3}iyz^2$ |
| | $\mathbb{T}_3^{(1,0;a)}(E^{(b)},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{9} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{3}i}{9} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{60} & 0 & 0 & 0 & 0 & 0 \\ \frac{2\sqrt{3}i}{15} & 0 & 0 & 0 & 0 & \frac{7\sqrt{3}i}{180} & 0 & 0 & 0 & 0 \\ 0 & -\frac{2i}{5} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{180} & 0 & 0 & 0 \\ 0 & 0 & \frac{2\sqrt{3}i}{15} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{20} & 0 & 0 \end{bmatrix}$ |
| 247 | symmetry | $\frac{\sqrt{30}x^2z}{4}+\frac{\sqrt{30}ixyz}{2}-\frac{\sqrt{30}y^2z}{4}$ |
| | $\mathbb{T}_3^{(1,0;a)}(E^{(b)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{9} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{9} \\ 0 & 0 & \frac{\sqrt{30}i}{15} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{30} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{30}i}{15} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{180} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{36} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 248 | symmetry | $z$ |
| | $\mathbb{T}_1^{(1,1;a)}(A)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{3i}{10} & 0 & 0 & 0 & 0 & \frac{i}{10} & 0 & 0 & 0 & 0 \\ 0 & -\frac{i}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{20} & 0 & 0 & 0 \\ 0 & 0 & \frac{i}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{20} & 0 & 0 \\ 0 & 0 & 0 & \frac{3i}{10} & 0 & 0 & 0 & 0 & \frac{i}{10} & 0 \end{bmatrix}$ |
| 249 | symmetry | $\frac{\sqrt{2}x}{2}+\frac{\sqrt{2}iy}{2}$ |

*continued ...*

Table 6

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_1^{(1,1;a)}(E^{(a)})$ | $\begin{bmatrix} 0 & 0 & -\frac{i}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{3}i}{4} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{6}i}{10} & 0 & 0 & 0 & 0 & \frac{i}{20} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{2}i}{5} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{20} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{6}i}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{20} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{20} \end{bmatrix}$ |
| 250 | symmetry | $\frac{\sqrt{2}x}{2} - \frac{\sqrt{2}iy}{2}$ |
| | $\mathbb{T}_1^{(1,1;a)}(E^{(b)})$ | $\begin{bmatrix} \frac{\sqrt{3}i}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{i}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{20} & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{6}i}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{20} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{2}i}{5} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{20} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{6}i}{10} & 0 & 0 & 0 & 0 & -\frac{i}{20} & 0 & 0 \end{bmatrix}$ |
| 251 | symmetry | $-\frac{x^2}{2} - \frac{y^2}{2} + z^2$ |
| | $\mathbb{M}_2^{(a)}(A)$ | $\begin{bmatrix} 0 & \frac{\sqrt{5}}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{30} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{30} & 0 & 0 \\ -\frac{\sqrt{10}}{20} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{10} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{10}}{20} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{30} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{10}}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{30} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{10}}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{10} & 0 \end{bmatrix}$ |
| 252 | symmetry | $\frac{\sqrt{6}xz}{2} + \frac{\sqrt{6}iyz}{2}$ |
| | $\mathbb{M}_2^{(a)}(E^{(a)}, 1)$ | $\begin{bmatrix} 0 & 0 & \frac{\sqrt{30}}{20} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{15} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{10}}{20} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{15} & 0 \\ 0 & -\frac{\sqrt{5}}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{20} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{12} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{5}}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{30} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{6} \end{bmatrix}$ |
| 253 | symmetry | $\frac{\sqrt{6}x^2}{4} - \frac{\sqrt{6}ixy}{2} - \frac{\sqrt{6}y^2}{4}$ |

*continued ...*

Table 6

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_2^{(a)}(E^{(a)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{6} & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{10}}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{30} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{1}{3} & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{5}}{10} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{5}}{15} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{5}}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{30} & 0 & 0 & 0 \end{bmatrix}$ |
| 254 | symmetry | $\frac{\sqrt{6}xz}{2} - \frac{\sqrt{6}iyz}{2}$ |
| | $\mathbb{M}_2^{(a)}(E^{(b)},1)$ | $\begin{bmatrix} -\frac{\sqrt{10}}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{15} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{30}}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{15} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{5}}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{30} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{12} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{5}}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{20} & 0 & 0 \end{bmatrix}$ |
| 255 | symmetry | $\frac{\sqrt{6}x^2}{4} + \frac{\sqrt{6}ixy}{2} - \frac{\sqrt{6}y^2}{4}$ |
| | $\mathbb{M}_2^{(a)}(E^{(b)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & \frac{\sqrt{10}}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{30} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{6} \\ 0 & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{30} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 & 0 & 0 & \frac{2\sqrt{5}}{15} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{3} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 256 | symmetry | $-\frac{x^2}{2} - \frac{y^2}{2} + z^2$ |
| | $\mathbb{M}_2^{(1,-1;a)}(A)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{2}}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{5} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{2}}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{5} & 0 & 0 \\ \frac{1}{10} & 0 & 0 & 0 & 0 & \frac{3}{10} & 0 & 0 & 0 & 0 \\ 0 & -\frac{1}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{20} & 0 & 0 & 0 \\ 0 & 0 & -\frac{1}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{20} & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{10} & 0 & 0 & 0 & 0 & -\frac{3}{10} & 0 \end{bmatrix}$ |
| 257 | symmetry | $\frac{\sqrt{6}xz}{2} + \frac{\sqrt{6}iyz}{2}$ |

*continued ...*

Table 6

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_2^{(1,-1;a)}(E^{(a)},1)$ | $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{3}}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{5} & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{20} & 0 & 0 & 0 & 0 & -\frac{2}{5} & 0 \\ 0 & \frac{\sqrt{2}}{10} & 0 & 0 & 0 & 0 & \frac{3\sqrt{3}}{20} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{4} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{2}}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{20} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{20} \end{bmatrix}$ |
| 258 | symmetry | $\frac{\sqrt{6}x^2}{4} - \frac{\sqrt{6}ixy}{2} - \frac{\sqrt{6}y^2}{4}$ |
| | $\mathbb{M}_2^{(1,-1;a)}(E^{(a)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{5} & 0 & 0 & 0 & 0 & 0 \\ \frac{1}{10} & 0 & 0 & 0 & 0 & -\frac{1}{5} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{10} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{2}}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{5} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{2}}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{10} & 0 & 0 & 0 \end{bmatrix}$ |
| 259 | symmetry | $\frac{\sqrt{6}xz}{2} - \frac{\sqrt{6}iyz}{2}$ |
| | $\mathbb{M}_2^{(1,-1;a)}(E^{(b)},1)$ | $\begin{bmatrix} \frac{1}{20} & 0 & 0 & 0 & 0 & \frac{2}{5} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{3}}{20} & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{5} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{20} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{2}}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{20} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{4} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{2}}{10} & 0 & 0 & 0 & 0 & \frac{3\sqrt{3}}{20} & 0 & 0 \end{bmatrix}$ |
| 260 | symmetry | $\frac{\sqrt{6}x^2}{4} + \frac{\sqrt{6}ixy}{2} - \frac{\sqrt{6}y^2}{4}$ |
| | $\mathbb{M}_2^{(1,-1;a)}(E^{(b)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & -\frac{1}{10} & 0 & 0 & 0 & 0 & -\frac{1}{5} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{5} \\ 0 & 0 & \frac{\sqrt{2}}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{10} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{2}}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{5} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{10} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 261 | symmetry | $\frac{3x^4}{8} + \frac{3x^2y^2}{4} - 3x^2z^2 + \frac{3y^4}{8} - 3y^2z^2 + z^4$ |

*continued ...*

Table 6

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_4^{(1,-1;a)}(A,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{14} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{14} & 0 \end{bmatrix}$ |
| 262 | symmetry | $\frac{\sqrt{70}x^3z}{4} - \frac{3\sqrt{70}xy^2z}{4}$ |
| | $\mathbb{M}_4^{(1,-1;a)}(A,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{8} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{8} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{8} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{8} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 263 | symmetry | $\frac{3\sqrt{70}x^2yz}{4} - \frac{\sqrt{70}y^3z}{4}$ |
| | $\mathbb{M}_4^{(1,-1;a)}(A,3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{8} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{8} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{8} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{8} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 264 | symmetry | $-\frac{3\sqrt{5}x^3z}{4} - \frac{3\sqrt{5}ix^2yz}{4} - \frac{3\sqrt{5}xy^2z}{4} + \sqrt{5}xz^3 - \frac{3\sqrt{5}iy^3z}{4} + \sqrt{5}iyz^3$ |
| | $\mathbb{M}_4^{(1,-1;a)}(E^{(a)},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{28} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{28} \end{bmatrix}$ |
| 265 | symmetry | $\frac{\sqrt{70}x^4}{16} + \frac{\sqrt{70}ix^3y}{4} - \frac{3\sqrt{70}x^2y^2}{8} - \frac{\sqrt{70}ixy^3}{4} + \frac{\sqrt{70}y^4}{16}$ |

*continued ...*

Table 6

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_4^{(1,-1;a)}(E^{(a)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{2} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 266 | symmetry | $-\frac{\sqrt{10}x^4}{8}+\frac{\sqrt{10}ix^3y}{4}+\frac{3\sqrt{10}x^2z^2}{4}+\frac{\sqrt{10}ixy^3}{4}-\frac{3\sqrt{10}ixyz^2}{2}+\frac{\sqrt{10}y^4}{8}-\frac{3\sqrt{10}y^2z^2}{4}$ |
| | $\mathbb{M}_4^{(1,-1;a)}(E^{(a)},3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{14} & 0 & 0 & 0 \end{bmatrix}$ |
| 267 | symmetry | $-\frac{3\sqrt{5}x^3z}{4}+\frac{3\sqrt{5}ix^2yz}{4}-\frac{3\sqrt{5}xy^2z}{4}+\sqrt{5}xz^3+\frac{3\sqrt{5}iy^3z}{4}-\sqrt{5}iyz^3$ |
| | $\mathbb{M}_4^{(1,-1;a)}(E^{(b)},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{28} & 0 & 0 \end{bmatrix}$ |
| 268 | symmetry | $\frac{\sqrt{70}x^4}{16}-\frac{\sqrt{70}ix^3y}{4}-\frac{3\sqrt{70}x^2y^2}{8}+\frac{\sqrt{70}ixy^3}{4}+\frac{\sqrt{70}y^4}{16}$ |
| | $\mathbb{M}_4^{(1,-1;a)}(E^{(b)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{2} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 269 | symmetry | $-\frac{\sqrt{10}x^4}{8}-\frac{\sqrt{10}ix^3y}{4}+\frac{3\sqrt{10}x^2z^2}{4}-\frac{\sqrt{10}ixy^3}{4}+\frac{3\sqrt{10}ixyz^2}{2}+\frac{\sqrt{10}y^4}{8}-\frac{3\sqrt{10}y^2z^2}{4}$ |

*continued ...*

Table 6

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_4^{(1,-1;a)}(E^{(b)},3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{14} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{28} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{28} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 270 | symmetry | $-\frac{x^2}{2}-\frac{y^2}{2}+z^2$ |
| | $\mathbb{M}_2^{(1,0;a)}(A)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{30}}{60} & 0 & 0 & 0 & 0 & \frac{2\sqrt{5}}{15} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{30}}{60} & 0 & 0 & 0 & 0 & \frac{2\sqrt{5}}{15} & 0 & 0 \\ \frac{\sqrt{15}}{15} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{30} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{15}}{15} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{60} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{15}}{15} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{60} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{15}}{15} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{30} & 0 \end{bmatrix}$ |
| 271 | symmetry | $\frac{\sqrt{6}xz}{2}+\frac{\sqrt{6}iyz}{2}$ |
| | $\mathbb{M}_2^{(1,0;a)}(E^{(a)},1)$ | $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{5}}{20} & 0 & 0 & 0 & 0 & \frac{2\sqrt{30}}{45} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{15}}{60} & 0 & 0 & 0 & 0 & \frac{4\sqrt{15}}{45} & 0 \\ 0 & \frac{\sqrt{30}}{15} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{20} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{36} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{30}}{15} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{180} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{12} \end{bmatrix}$ |
| 272 | symmetry | $\frac{\sqrt{6}x^2}{4}-\frac{\sqrt{6}ixy}{2}-\frac{\sqrt{6}y^2}{4}$ |
| | $\mathbb{M}_2^{(1,0;a)}(E^{(a)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & \frac{2\sqrt{3}}{9} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{15}}{30} & 0 & 0 & 0 & 0 & \frac{2\sqrt{15}}{45} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{18} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{30}}{15} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{45} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{30}}{15} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{30} & 0 & 0 & 0 \end{bmatrix}$ |
| 273 | symmetry | $\frac{\sqrt{6}xz}{2}-\frac{\sqrt{6}iyz}{2}$ |

*continued ...*

Table 6

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_2^{(1,0;a)}(E^{(b)},1)$ | $\begin{bmatrix} \frac{\sqrt{15}}{60} & 0 & 0 & 0 & 0 & -\frac{4\sqrt{15}}{45} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{5}}{20} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{30}}{45} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{12} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{30}}{15} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{180} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{36} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{30}}{15} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{20} & 0 & 0 \end{bmatrix}$ |
| 274 | symmetry | $\frac{\sqrt{6}x^2}{4} + \frac{\sqrt{6}ixy}{2} - \frac{\sqrt{6}y^2}{4}$ |
| | $\mathbb{M}_2^{(1,0;a)}(E^{(b)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{15}}{30} & 0 & 0 & 0 & 0 & \frac{2\sqrt{15}}{45} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{3}}{9} \\ 0 & 0 & \frac{\sqrt{30}}{15} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{30} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{30}}{15} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{45} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{18} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 275 | symmetry | 1 |
| | $\mathbb{M}_0^{(1,1;a)}(A)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 276 | symmetry | $-\frac{x^2}{2} - \frac{y^2}{2} + z^2$ |
| | $\mathbb{M}_2^{(1,1;a)}(A)$ | $\begin{bmatrix} 0 & \frac{\sqrt{42}}{15} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{30} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{42}}{15} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{30} & 0 & 0 \\ \frac{\sqrt{21}}{30} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{21}}{105} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{21}}{30} & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{105} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{21}}{30} & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{105} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{21}}{30} & 0 & 0 & 0 & 0 & \frac{2\sqrt{21}}{105} & 0 \end{bmatrix}$ |
| 277 | symmetry | $\frac{\sqrt{6}xz}{2} + \frac{\sqrt{6}iyz}{2}$ |

*continued ...*

Table 6

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_2^{(1,1;a)}(E^{(a)},1)$ | $\begin{bmatrix} 0 & 0 & \frac{\sqrt{7}}{5} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{90} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{21}}{15} & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{45} & 0 \\ 0 & \frac{\sqrt{42}}{30} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{35} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{63} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{42}}{30} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{315} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{105} \end{bmatrix}$ |
| 278 | symmetry | $\frac{\sqrt{6}x^2}{4} - \frac{\sqrt{6}ixy}{2} - \frac{\sqrt{6}y^2}{4}$ |
| | $\mathbb{M}_2^{(1,1;a)}(E^{(a)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{90} & 0 & 0 & 0 & 0 & 0 \\ -\frac{2\sqrt{21}}{15} & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{90} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{2\sqrt{210}}{315} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{42}}{30} & 0 & 0 & 0 & 0 & \frac{4\sqrt{42}}{315} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{42}}{30} & 0 & 0 & 0 & 0 & \frac{2\sqrt{7}}{105} & 0 & 0 & 0 \end{bmatrix}$ |
| 279 | symmetry | $\frac{\sqrt{6}xz}{2} - \frac{\sqrt{6}iyz}{2}$ |
| | $\mathbb{M}_2^{(1,1;a)}(E^{(b)},1)$ | $\begin{bmatrix} -\frac{\sqrt{21}}{15} & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{45} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{7}}{5} & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{90} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{105} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{42}}{30} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{315} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{63} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{42}}{30} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{35} & 0 & 0 \end{bmatrix}$ |
| 280 | symmetry | $\frac{\sqrt{6}x^2}{4} + \frac{\sqrt{6}ixy}{2} - \frac{\sqrt{6}y^2}{4}$ |
| | $\mathbb{M}_2^{(1,1;a)}(E^{(b)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & \frac{2\sqrt{21}}{15} & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{90} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{90} \\ 0 & 0 & \frac{\sqrt{42}}{30} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{7}}{105} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{42}}{30} & 0 & 0 & 0 & 0 & -\frac{4\sqrt{42}}{315} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{210}}{315} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |

bra: $= \langle \frac{1}{2}, \frac{1}{2}; p|\,, \langle \frac{1}{2}, -\frac{1}{2}; p|\,, \langle \frac{3}{2}, \frac{3}{2}; p|\,, \langle \frac{3}{2}, \frac{1}{2}; p|\,, \langle \frac{3}{2}, -\frac{1}{2}; p|\,, \langle \frac{3}{2}, -\frac{3}{2}; p|$

ket: $= |\frac{5}{2}, \frac{5}{2}; f\rangle\,, |\frac{5}{2}, \frac{3}{2}; f\rangle\,, |\frac{5}{2}, \frac{1}{2}; f\rangle\,, |\frac{5}{2}, -\frac{1}{2}; f\rangle\,, |\frac{5}{2}, -\frac{3}{2}; f\rangle\,, |\frac{5}{2}, -\frac{5}{2}; f\rangle\,, |\frac{7}{2}, \frac{7}{2}; f\rangle\,, |\frac{7}{2}, \frac{5}{2}; f\rangle\,, |\frac{7}{2}, \frac{3}{2}; f\rangle\,, |\frac{7}{2}, \frac{1}{2}; f\rangle\,, |\frac{7}{2}, -\frac{1}{2}; f\rangle\,, |\frac{7}{2}, -\frac{3}{2}; f\rangle\,, |\frac{7}{2}, -\frac{5}{2}; f\rangle\,, |\frac{7}{2}, -\frac{7}{2}; f\rangle$

Table 7: (p,f) block.

| No. | multipole | matrix |
|---|---|---|
| 281 | symmetry | $-\frac{x^2}{2}-\frac{y^2}{2}+z^2$ |
| | $\mathbb{Q}_2^{(a)}(A)$ | $\begin{bmatrix} 0 & 0 & \frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{1}{7} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{14} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{6}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{6}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{7} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{14} & 0 & 0 \end{bmatrix}$ |
| 282 | symmetry | $\frac{\sqrt{6}xz}{2}+\frac{\sqrt{6}iyz}{2}$ |
| | $\mathbb{Q}_2^{(a)}(E^{(a)},1)$ | $\begin{bmatrix} 0 & 0 & 0 & \frac{\sqrt{2}}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{3} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{3}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{7} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{5}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{7} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{7} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{7} & 0 \end{bmatrix}$ |
| 283 | symmetry | $\frac{\sqrt{6}x^2}{4}-\frac{\sqrt{6}ixy}{2}-\frac{\sqrt{6}y^2}{4}$ |
| | $\mathbb{Q}_2^{(a)}(E^{(a)},2)$ | $\begin{bmatrix} \frac{\sqrt{5}}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{1}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{10}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{2\sqrt{2}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{14} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{3}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{14} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 284 | symmetry | $\frac{\sqrt{6}xz}{2}-\frac{\sqrt{6}iyz}{2}$ |
| | $\mathbb{Q}_2^{(a)}(E^{(b)},1)$ | $\begin{bmatrix} 0 & -\frac{1}{3} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{2}}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{30}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{7} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{2}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{7} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{5}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{7} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{3}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{7} & 0 & 0 & 0 \end{bmatrix}$ |

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|
| 285 | symmetry | $\frac{\sqrt{6}x^2}{4}+\frac{\sqrt{6}ixy}{2}-\frac{\sqrt{6}y^2}{4}$ |
| | $\mathbb{Q}_2^{(a)}(E^{(b)},2)$ | $\begin{bmatrix} 0&0&0&0&\frac{1}{6}&0&0&0&0&0&0&0&0&0\\ 0&0&0&0&0&\frac{\sqrt{5}}{6}&0&0&0&0&0&0&0&0\\ 0&0&0&-\frac{\sqrt{3}}{21}&0&0&0&0&0&0&\frac{1}{14}&0&0&0\\ 0&0&0&0&-\frac{2\sqrt{2}}{21}&0&0&0&0&0&0&\frac{\sqrt{5}}{14}&0&0\\ 0&0&0&0&0&-\frac{\sqrt{10}}{21}&0&0&0&0&0&0&\frac{\sqrt{15}}{14}&0\\ 0&0&0&0&0&0&0&0&0&0&0&0&0&\frac{\sqrt{35}}{14} \end{bmatrix}$ |
| 286 | symmetry | $\frac{3x^4}{8}+\frac{3x^2y^2}{4}-3x^2z^2+\frac{3y^4}{8}-3y^2z^2+z^4$ |
| | $\mathbb{Q}_4^{(a)}(A,1)$ | $\begin{bmatrix} 0&0&0&0&0&0&0&0&0&-\frac{\sqrt{3}}{6}&0&0&0&0\\ 0&0&0&0&0&0&0&0&0&0&\frac{\sqrt{3}}{6}&0&0&0\\ 0&\frac{\sqrt{3}}{14}&0&0&0&0&0&0&-\frac{\sqrt{30}}{28}&0&0&0&0&0\\ 0&0&-\frac{3\sqrt{2}}{14}&0&0&0&0&0&0&\frac{5\sqrt{6}}{84}&0&0&0&0\\ 0&0&0&\frac{3\sqrt{2}}{14}&0&0&0&0&0&0&\frac{5\sqrt{6}}{84}&0&0&0\\ 0&0&0&0&-\frac{\sqrt{3}}{14}&0&0&0&0&0&0&-\frac{\sqrt{30}}{28}&0&0 \end{bmatrix}$ |
| 287 | symmetry | $\frac{\sqrt{70}x^3z}{4}-\frac{3\sqrt{70}xy^2z}{4}$ |
| | $\mathbb{Q}_4^{(a)}(A,2)$ | $\begin{bmatrix} 0&0&0&0&0&0&\frac{\sqrt{6}}{24}&0&0&0&0&0&-\frac{\sqrt{42}}{24}&0\\ 0&0&0&0&0&0&0&-\frac{\sqrt{42}}{24}&0&0&0&0&0&\frac{\sqrt{6}}{24}\\ 0&0&0&0&\frac{\sqrt{210}}{56}&0&0&0&0&0&0&-\frac{\sqrt{21}}{28}&0&0\\ 0&0&0&0&0&-\frac{3\sqrt{14}}{56}&-\frac{\sqrt{3}}{12}&0&0&0&0&0&-\frac{\sqrt{21}}{42}&0\\ -\frac{3\sqrt{14}}{56}&0&0&0&0&0&0&\frac{\sqrt{21}}{42}&0&0&0&0&0&\frac{\sqrt{3}}{12}\\ 0&\frac{\sqrt{210}}{56}&0&0&0&0&0&0&\frac{\sqrt{21}}{28}&0&0&0&0&0 \end{bmatrix}$ |
| 288 | symmetry | $\frac{3\sqrt{70}x^2yz}{4}-\frac{\sqrt{70}y^3z}{4}$ |
| | $\mathbb{Q}_4^{(a)}(A,3)$ | $\begin{bmatrix} 0&0&0&0&0&0&\frac{\sqrt{6}i}{24}&0&0&0&0&0&\frac{\sqrt{42}i}{24}&0\\ 0&0&0&0&0&0&0&-\frac{\sqrt{42}i}{24}&0&0&0&0&0&-\frac{\sqrt{6}i}{24}\\ 0&0&0&0&-\frac{\sqrt{210}i}{56}&0&0&0&0&0&0&\frac{\sqrt{21}i}{28}&0&0\\ 0&0&0&0&0&\frac{3\sqrt{14}i}{56}&-\frac{\sqrt{3}i}{12}&0&0&0&0&0&\frac{\sqrt{21}i}{42}&0\\ -\frac{3\sqrt{14}i}{56}&0&0&0&0&0&0&\frac{\sqrt{21}i}{42}&0&0&0&0&0&-\frac{\sqrt{3}i}{12}\\ 0&\frac{\sqrt{210}i}{56}&0&0&0&0&0&0&\frac{\sqrt{21}i}{28}&0&0&0&0&0 \end{bmatrix}$ |
| 289 | symmetry | $-\frac{3\sqrt{5}x^3z}{4}-\frac{3\sqrt{5}ix^2yz}{4}-\frac{3\sqrt{5}xy^2z}{4}+\sqrt{5}xz^3-\frac{3\sqrt{5}iy^3z}{4}+\sqrt{5}iyz^3$ |

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_4^{(a)}(E^{(a)},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{12} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{4} & 0 & 0 \\ 0 & 0 & \frac{\sqrt{30}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{10}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{84} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{3\sqrt{5}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{7} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}}{28} & 0 \end{bmatrix}$ |
| 290 | symmetry | $\frac{\sqrt{70}x^4}{16}+\frac{\sqrt{70}ix^3y}{4}-\frac{3\sqrt{70}x^2y^2}{8}-\frac{\sqrt{70}ixy^3}{4}+\frac{\sqrt{70}y^4}{16}$ |
| | $\mathbb{Q}_4^{(a)}(E^{(a)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{6} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{14} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{6} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 291 | symmetry | $-\frac{\sqrt{10}x^4}{8}+\frac{\sqrt{10}ix^3y}{4}+\frac{3\sqrt{10}x^2z^2}{4}+\frac{\sqrt{10}ixy^3}{4}-\frac{3\sqrt{10}ixyz^2}{2}+\frac{\sqrt{10}y^4}{8}-\frac{3\sqrt{10}y^2z^2}{4}$ |
| | $\mathbb{Q}_4^{(a)}(E^{(a)},3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{12} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{3\sqrt{2}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{11\sqrt{3}}{84} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{3\sqrt{10}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{15}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{5}}{28} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 292 | symmetry | $-\frac{3\sqrt{5}x^3z}{4}+\frac{3\sqrt{5}ix^2yz}{4}-\frac{3\sqrt{5}xy^2z}{4}+\sqrt{5}xz^3+\frac{3\sqrt{5}iy^3z}{4}-\sqrt{5}iyz^3$ |
| | $\mathbb{Q}_4^{(a)}(E^{(b)},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{4} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{12} & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{3}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{3\sqrt{5}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{7} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{3\sqrt{10}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{84} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{30}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{14} & 0 & 0 & 0 \end{bmatrix}$ |
| 293 | symmetry | $\frac{\sqrt{70}x^4}{16}-\frac{\sqrt{70}ix^3y}{4}-\frac{3\sqrt{70}x^2y^2}{8}+\frac{\sqrt{70}ixy^3}{4}+\frac{\sqrt{70}y^4}{16}$ |

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_4^{(a)}(E^{(b)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{42}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 294 | symmetry | $-\frac{\sqrt{10}x^4}{8} - \frac{\sqrt{10}ix^3y}{4} + \frac{3\sqrt{10}x^2z^2}{4} - \frac{\sqrt{10}ixy^3}{4} + \frac{3\sqrt{10}ixyz^2}{2} + \frac{\sqrt{10}y^4}{8} - \frac{3\sqrt{10}y^2z^2}{4}$ |
| | $\mathbb{Q}_4^{(a)}(E^{(b)},3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{4} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{12} & 0 \\ 0 & 0 & 0 & \frac{\sqrt{15}}{14} & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{5}}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}}{28} & 0 & 0 & 0 & 0 & 0 & -\frac{1}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}}{28} & 0 & 0 & 0 & 0 & 0 & \frac{11\sqrt{3}}{84} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{28} \end{bmatrix}$ |
| 295 | symmetry | $\frac{3x^4}{8} + \frac{3x^2y^2}{4} - 3x^2z^2 + \frac{3y^4}{8} - 3y^2z^2 + z^4$ |
| | $\mathbb{Q}_4^{(1,-1;a)}(A,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{4} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{4} & 0 & 0 & 0 \\ 0 & \frac{1}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{10}}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{6}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{2}}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{6}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{2}}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{1}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{10}}{28} & 0 & 0 \end{bmatrix}$ |
| 296 | symmetry | $\frac{\sqrt{70}x^3z}{4} - \frac{3\sqrt{70}xy^2z}{4}$ |
| | $\mathbb{Q}_4^{(1,-1;a)}(A,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{16} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{16} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{16} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{16} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{112} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{7}}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{112} & \frac{1}{4} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{14} & 0 \\ -\frac{\sqrt{42}}{112} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & -\frac{1}{4} \\ 0 & \frac{\sqrt{70}}{112} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{7}}{28} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 297 | symmetry | $\frac{3\sqrt{70}x^2yz}{4} - \frac{\sqrt{70}y^3z}{4}$ |

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_4^{(1,-1;a)}(A,3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{16} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{16} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{16} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{16} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{112} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{7}i}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{112} & \frac{i}{4} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 \\ -\frac{\sqrt{42}i}{112} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & \frac{i}{4} \\ 0 & \frac{\sqrt{70}i}{112} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{7}i}{28} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 298 | symmetry | $-\frac{3\sqrt{5}x^3z}{4} - \frac{3\sqrt{5}ix^2yz}{4} - \frac{3\sqrt{5}xy^2z}{4} + \sqrt{5}xz^3 - \frac{3\sqrt{5}iy^3z}{4} + \sqrt{5}iyz^3$ |
| | $\mathbb{Q}_4^{(1,-1;a)}(E^{(a)},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{8} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{8} & 0 & 0 \\ 0 & 0 & \frac{\sqrt{10}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{30}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{7} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{1}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{6}}{28} & 0 \end{bmatrix}$ |
| 299 | symmetry | $\frac{\sqrt{70}x^4}{16} + \frac{\sqrt{70}ix^3y}{4} - \frac{3\sqrt{70}x^2y^2}{8} - \frac{\sqrt{70}ixy^3}{4} + \frac{\sqrt{70}y^4}{16}$ |
| | $\mathbb{Q}_4^{(1,-1;a)}(E^{(a)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{4} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{14} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 300 | symmetry | $-\frac{\sqrt{10}x^4}{8} + \frac{\sqrt{10}ix^3y}{4} + \frac{3\sqrt{10}x^2z^2}{4} + \frac{\sqrt{10}ixy^3}{4} - \frac{3\sqrt{10}ixyz^2}{2} + \frac{\sqrt{10}y^4}{8} - \frac{3\sqrt{10}y^2z^2}{4}$ |
| | $\mathbb{Q}_4^{(1,-1;a)}(E^{(a)},3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{8} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{8} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{6}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{11}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{30}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{5}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{15}}{28} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 301 | symmetry | $-\frac{3\sqrt{5}x^3z}{4} + \frac{3\sqrt{5}ix^2yz}{4} - \frac{3\sqrt{5}xy^2z}{4} + \sqrt{5}xz^3 + \frac{3\sqrt{5}iy^3z}{4} - \sqrt{5}iyz^3$ |

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_4^{(1,-1;a)}(E^{(b)},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{8} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{8} & 0 & 0 & 0 & 0 \\ -\frac{1}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{6}}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{15}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{7} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{30}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{10}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{14} & 0 & 0 & 0 \end{bmatrix}$ |
| 302 | symmetry | $\frac{\sqrt{70}x^4}{16} - \frac{\sqrt{70}ix^3y}{4} - \frac{3\sqrt{70}x^2y^2}{8} + \frac{\sqrt{70}ixy^3}{4} + \frac{\sqrt{70}y^4}{16}$ |
| | $\mathbb{Q}_4^{(1,-1;a)}(E^{(b)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{14}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{14} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 303 | symmetry | $-\frac{\sqrt{10}x^4}{8} - \frac{\sqrt{10}ix^3y}{4} + \frac{3\sqrt{10}x^2z^2}{4} - \frac{\sqrt{10}ixy^3}{4} + \frac{3\sqrt{10}ixyz^2}{2} + \frac{\sqrt{10}y^4}{8} - \frac{3\sqrt{10}y^2z^2}{4}$ |
| | $\mathbb{Q}_4^{(1,-1;a)}(E^{(b)},3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{8} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{8} & 0 \\ 0 & 0 & 0 & \frac{\sqrt{5}}{28} & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{15}}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{56} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{56} & 0 & 0 & 0 & 0 & 0 & -\frac{11}{28} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{28} \end{bmatrix}$ |
| 304 | symmetry | $-\frac{x^2}{2} - \frac{y^2}{2} + z^2$ |
| | $\mathbb{Q}_2^{(1,0;a)}(A)$ | $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{2}}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{2}}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{5\sqrt{6}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{21} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{5}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{7} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{5}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{7} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{5\sqrt{6}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{21} & 0 & 0 \end{bmatrix}$ |
| 305 | symmetry | $\frac{\sqrt{6}xz}{2} + \frac{\sqrt{6}iyz}{2}$ |

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_2^{(1,0;a)}(E^{(a)},1)$ | $\begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{3}}{9} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{9} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{5\sqrt{2}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{21} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{25\sqrt{6}}{252} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{7} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{5\sqrt{3}}{126} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{21} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{5}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{21} & 0 \end{bmatrix}$ |
| 306 | symmetry | $\frac{\sqrt{6}x^2}{4} - \frac{\sqrt{6}ixy}{2} - \frac{\sqrt{6}y^2}{4}$ |
| | $\mathbb{Q}_2^{(1,0;a)}(E^{(a)},2)$ | $\begin{bmatrix} -\frac{\sqrt{30}}{18} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{6}}{18} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{5\sqrt{15}}{63} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{10\sqrt{3}}{63} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{42} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{5\sqrt{2}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{42} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 307 | symmetry | $\frac{\sqrt{6}xz}{2} - \frac{\sqrt{6}iyz}{2}$ |
| | $\mathbb{Q}_2^{(1,0;a)}(E^{(b)},1)$ | $\begin{bmatrix} 0 & \frac{\sqrt{6}}{9} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{3}}{9} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{5\sqrt{5}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{21} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{5\sqrt{3}}{126} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{21} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{25\sqrt{6}}{252} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{7} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{5\sqrt{2}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{21} & 0 & 0 & 0 \end{bmatrix}$ |
| 308 | symmetry | $\frac{\sqrt{6}x^2}{4} + \frac{\sqrt{6}ixy}{2} - \frac{\sqrt{6}y^2}{4}$ |
| | $\mathbb{Q}_2^{(1,0;a)}(E^{(b)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{18} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{18} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{5\sqrt{2}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{42} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{10\sqrt{3}}{63} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{42} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{15}}{63} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{14} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{42} \end{bmatrix}$ |
| 309 | symmetry | $\frac{3x^4}{8} + \frac{3x^2y^2}{4} - 3x^2z^2 + \frac{3y^4}{8} - 3y^2z^2 + z^4$ |

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_4^{(1,0;a)}(A,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{12} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{12} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{15}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{3\sqrt{10}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{84} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{10}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{84} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{28} & 0 & 0 \end{bmatrix}$ |
| 310 | symmetry | $\frac{\sqrt{70}x^3z}{4} - \frac{3\sqrt{70}xy^2z}{4}$ |
| | $\mathbb{Q}_4^{(1,0;a)}(A,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{48} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{48} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{48} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{48} \\ 0 & 0 & 0 & 0 & -\frac{5\sqrt{42}}{112} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{140} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{70}}{112} & -\frac{\sqrt{15}}{60} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{210} & 0 \\ \frac{3\sqrt{70}}{112} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{210} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{60} \\ 0 & -\frac{5\sqrt{42}}{112} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{140} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 311 | symmetry | $\frac{3\sqrt{70}x^2yz}{4} - \frac{\sqrt{70}y^3z}{4}$ |
| | $\mathbb{Q}_4^{(1,0;a)}(A,3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{48} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{48} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{48} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{48} \\ 0 & 0 & 0 & 0 & \frac{5\sqrt{42}i}{112} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{140} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{70}i}{112} & -\frac{\sqrt{15}i}{60} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{210} & 0 \\ \frac{3\sqrt{70}i}{112} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{210} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{60} \\ 0 & -\frac{5\sqrt{42}i}{112} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{140} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 312 | symmetry | $-\frac{3\sqrt{5}x^3z}{4} - \frac{3\sqrt{5}ix^2yz}{4} - \frac{3\sqrt{5}xy^2z}{4} + \sqrt{5}xz^3 - \frac{3\sqrt{5}iy^3z}{4} + \sqrt{5}iyz^3$ |
| | $\mathbb{Q}_4^{(1,0;a)}(E^{(a)},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{3}}{24} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{8} & 0 & 0 \\ 0 & 0 & -\frac{5\sqrt{6}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{15\sqrt{2}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{84} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{15}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{35} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}}{140} & 0 \end{bmatrix}$ |
| 313 | symmetry | $\frac{\sqrt{70}x^4}{16} + \frac{\sqrt{70}ix^3y}{4} - \frac{3\sqrt{70}x^2y^2}{8} - \frac{\sqrt{70}ixy^3}{4} + \frac{\sqrt{70}y^4}{16}$ |

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_4^{(1,0;a)}(E^{(a)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{12} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{70} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{30} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 314 | symmetry | $-\frac{\sqrt{10}x^4}{8}+\frac{\sqrt{10}ix^3y}{4}+\frac{3\sqrt{10}x^2z^2}{4}+\frac{\sqrt{10}ixy^3}{4}-\frac{3\sqrt{10}ixyz^2}{2}+\frac{\sqrt{10}y^4}{8}-\frac{3\sqrt{10}y^2z^2}{4}$ |
| | $\mathbb{Q}_4^{(1,0;a)}(E^{(a)},3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{24} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{8} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{3\sqrt{10}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{11\sqrt{15}}{420} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{15\sqrt{2}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{140} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{5\sqrt{3}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3}{28} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 315 | symmetry | $-\frac{3\sqrt{5}x^3z}{4}+\frac{3\sqrt{5}ix^2yz}{4}-\frac{3\sqrt{5}xy^2z}{4}+\sqrt{5}xz^3+\frac{3\sqrt{5}iy^3z}{4}-\sqrt{5}iyz^3$ |
| | $\mathbb{Q}_4^{(1,0;a)}(E^{(b)},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{8} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{3}}{24} & 0 & 0 & 0 & 0 \\ \frac{\sqrt{15}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{10}}{140} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{15}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{35} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{15\sqrt{2}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{84} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{5\sqrt{6}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{14} & 0 & 0 & 0 \end{bmatrix}$ |
| 316 | symmetry | $\frac{\sqrt{70}x^4}{16}-\frac{\sqrt{70}ix^3y}{4}-\frac{3\sqrt{70}x^2y^2}{8}+\frac{\sqrt{70}ixy^3}{4}+\frac{\sqrt{70}y^4}{16}$ |
| | $\mathbb{Q}_4^{(1,0;a)}(E^{(b)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{30} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{210}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{70} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 317 | symmetry | $-\frac{\sqrt{10}x^4}{8}-\frac{\sqrt{10}ix^3y}{4}+\frac{3\sqrt{10}x^2z^2}{4}-\frac{\sqrt{10}ixy^3}{4}+\frac{3\sqrt{10}ixyz^2}{2}+\frac{\sqrt{10}y^4}{8}-\frac{3\sqrt{10}y^2z^2}{4}$ |

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_4^{(1,0;a)}(E^{(b)},3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{8} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{24} & 0 \\ 0 & 0 & 0 & -\frac{5\sqrt{3}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{15\sqrt{2}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{140} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{11\sqrt{15}}{420} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{140} \end{bmatrix}$ |
| 318 | symmetry | $-\frac{x^2}{2}-\frac{y^2}{2}+z^2$ |
| | $\mathbb{Q}_2^{(1,1;a)}(A)$ | $\begin{bmatrix} 0 & 0 & -\frac{1}{3} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{1}{3} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{4\sqrt{3}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{84} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{2\sqrt{2}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{2\sqrt{2}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{4\sqrt{3}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{84} & 0 & 0 \end{bmatrix}$ |
| 319 | symmetry | $\frac{\sqrt{6}xz}{2}+\frac{\sqrt{6}iyz}{2}$ |
| | $\mathbb{Q}_2^{(1,1;a)}(E^{(a)},1)$ | $\begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{6}}{9} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{2\sqrt{3}}{9} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{2}{7} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{42} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{10\sqrt{3}}{63} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{2\sqrt{6}}{63} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{42} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{10}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{42} & 0 \end{bmatrix}$ |
| 320 | symmetry | $\frac{\sqrt{6}x^2}{4}-\frac{\sqrt{6}ixy}{2}-\frac{\sqrt{6}y^2}{4}$ |
| | $\mathbb{Q}_2^{(1,1;a)}(E^{(a)},2)$ | $\begin{bmatrix} -\frac{\sqrt{15}}{9} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{3}}{9} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{4\sqrt{30}}{63} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{8\sqrt{6}}{63} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{84} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{4}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{84} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 321 | symmetry | $\frac{\sqrt{6}xz}{2}-\frac{\sqrt{6}iyz}{2}$ |

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_2^{(1,1;a)}(E^{(b)},1)$ | $\begin{bmatrix} 0 & \frac{2\sqrt{3}}{9} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{6}}{9} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{2\sqrt{10}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{42} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{2\sqrt{6}}{63} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{42} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{10\sqrt{3}}{63} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{2}{7} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{42} & 0 & 0 & 0 \end{bmatrix}$ |
| 322 | symmetry | $\frac{\sqrt{6}x^2}{4}+\frac{\sqrt{6}ixy}{2}-\frac{\sqrt{6}y^2}{4}$ |
| | $\mathbb{Q}_2^{(1,1;a)}(E^{(b)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{9} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{9} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{4}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{84} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{8\sqrt{6}}{63} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{84} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{4\sqrt{30}}{63} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{28} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{84} \end{bmatrix}$ |
| 323 | symmetry | $-\frac{3x^2z}{2}-\frac{3y^2z}{2}+z^3$ |
| | $\mathbb{G}_3^{(a)}(A,1)$ | $\begin{bmatrix} 0 & 0 & \frac{\sqrt{21}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{21}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{42}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{42}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{28} & 0 & 0 \end{bmatrix}$ |
| 324 | symmetry | $\frac{3\sqrt{10}x^2y}{4}-\frac{\sqrt{10}y^3}{4}$ |
| | $\mathbb{G}_3^{(a)}(A,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{21} & \frac{\sqrt{2}}{8} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{56} & 0 \\ -\frac{\sqrt{21}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{56} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{8} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{42}}{168} & -\frac{1}{4} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{14} & 0 \\ -\frac{5\sqrt{42}}{168} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & \frac{1}{4} \\ 0 & -\frac{\sqrt{70}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{28} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 325 | symmetry | $\frac{\sqrt{10}x^3}{4}-\frac{3\sqrt{10}xy^2}{4}$ |

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_3^{(a)}(A,3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{21} & -\frac{\sqrt{2}i}{8} & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{56} & 0 \\ \frac{\sqrt{21}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{56} & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{8} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{42}i}{168} & \frac{i}{4} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 \\ \frac{5\sqrt{42}i}{168} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & \frac{i}{4} \\ 0 & \frac{\sqrt{70}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{28} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 326 | symmetry | $-\frac{\sqrt{3}x^3}{4} - \frac{\sqrt{3}ix^2y}{4} - \frac{\sqrt{3}xy^2}{4} + \sqrt{3}xz^2 - \frac{\sqrt{3}iy^3}{4} + \sqrt{3}iyz^2$ |
| | $\mathbb{G}_3^{(a)}(E^{(a)},1)$ | $\begin{bmatrix} 0 & 0 & 0 & \frac{2\sqrt{7}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{28} & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{42}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{14}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{28} & 0 \end{bmatrix}$ |
| 327 | symmetry | $\frac{\sqrt{30}x^2z}{4} - \frac{\sqrt{30}ixyz}{2} - \frac{\sqrt{30}y^2z}{4}$ |
| | $\mathbb{G}_3^{(a)}(E^{(a)},2)$ | $\begin{bmatrix} \frac{\sqrt{7}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{35}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{5\sqrt{14}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{70}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{7}i}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{105}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{28} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 328 | symmetry | $-\frac{\sqrt{3}x^3}{4} + \frac{\sqrt{3}ix^2y}{4} - \frac{\sqrt{3}xy^2}{4} + \sqrt{3}xz^2 + \frac{\sqrt{3}iy^3}{4} - \sqrt{3}iyz^2$ |
| | $\mathbb{G}_3^{(a)}(E^{(b)},1)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{14}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{2\sqrt{7}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{28} & 0 & 0 & 0 & 0 \\ \frac{\sqrt{105}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{7}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{14}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{42}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{14} & 0 & 0 & 0 \end{bmatrix}$ |
| 329 | symmetry | $\frac{\sqrt{30}x^2z}{4} + \frac{\sqrt{30}ixyz}{2} - \frac{\sqrt{30}y^2z}{4}$ |

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_3^{(a)}(E^{(b)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{21} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{21} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{28} & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{105}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{7}i}{28} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{14}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{28} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{4} \end{bmatrix}$ |
| 330 | symmetry | $-\frac{3x^2z}{2}-\frac{3y^2z}{2}+z^3$ |
| | $\mathbb{G}_3^{(1,-1;a)}(A,1)$ | $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{5}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{15}i}{21} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{5}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{15}i}{21} & 0 & 0 & 0 \\ 0 & \frac{2\sqrt{15}i}{105} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{6}i}{42} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{2\sqrt{10}i}{105} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{42} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{2\sqrt{10}i}{105} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{42} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{2\sqrt{15}i}{105} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{6}i}{42} & 0 & 0 \end{bmatrix}$ |
| 331 | symmetry | $\frac{3\sqrt{10}x^2y}{4}-\frac{\sqrt{10}y^3}{4}$ |
| | $\mathbb{G}_3^{(1,-1;a)}(A,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{42} & -\frac{\sqrt{210}}{42} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{42} & 0 \\ \frac{\sqrt{5}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{42} & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{42} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{42} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{42} & -\frac{\sqrt{105}}{42} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{21} & 0 \\ \frac{\sqrt{10}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{21} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{42} \\ 0 & \frac{\sqrt{6}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{42} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 332 | symmetry | $\frac{\sqrt{10}x^3}{4}-\frac{3\sqrt{10}xy^2}{4}$ |
| | $\mathbb{G}_3^{(1,-1;a)}(A,3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{42} & \frac{\sqrt{210}i}{42} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{42} & 0 \\ -\frac{\sqrt{5}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{42} & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{42} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{42} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{42} & \frac{\sqrt{105}i}{42} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{21} & 0 \\ -\frac{\sqrt{10}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{21} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{42} \\ 0 & -\frac{\sqrt{6}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{42} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 333 | symmetry | $-\frac{\sqrt{3}x^3}{4}-\frac{\sqrt{3}ix^2y}{4}-\frac{\sqrt{3}xy^2}{4}+\sqrt{3}xz^2-\frac{\sqrt{3}iy^3}{4}+\sqrt{3}iyz^2$ |

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_3^{(1,-1;a)}(E^{(a)},1)$ | $\begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{15}i}{63} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{7} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{126} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{3}i}{21} & 0 & 0 \\ 0 & 0 & \frac{\sqrt{10}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{21} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{30}i}{315} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{45} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{6}i}{42} & 0 \end{bmatrix}$ |
| 334 | symmetry | $\frac{\sqrt{30}x^2z}{4} - \frac{\sqrt{30}ixyz}{2} - \frac{\sqrt{30}y^2z}{4}$ |
| | $\mathbb{G}_3^{(1,-1;a)}(E^{(a)},2)$ | $\begin{bmatrix} -\frac{\sqrt{15}i}{126} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{7} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{5\sqrt{3}i}{126} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{21} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{30}i}{63} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{6}i}{63} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{14} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{2i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{3}i}{42} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 335 | symmetry | $-\frac{\sqrt{3}x^3}{4} + \frac{\sqrt{3}ix^2y}{4} - \frac{\sqrt{3}xy^2}{4} + \sqrt{3}xz^2 + \frac{\sqrt{3}iy^3}{4} - \sqrt{3}iyz^2$ |
| | $\mathbb{G}_3^{(1,-1;a)}(E^{(b)},1)$ | $\begin{bmatrix} 0 & \frac{\sqrt{30}i}{126} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{3}i}{21} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{15}i}{63} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{7} & 0 & 0 & 0 & 0 \\ -\frac{i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{6}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{15}i}{45} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{30}i}{315} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{10}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{21} & 0 & 0 & 0 \end{bmatrix}$ |
| 336 | symmetry | $\frac{\sqrt{30}x^2z}{4} + \frac{\sqrt{30}ixyz}{2} - \frac{\sqrt{30}y^2z}{4}$ |
| | $\mathbb{G}_3^{(1,-1;a)}(E^{(b)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{5\sqrt{3}i}{126} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{21} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{126} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{7} & 0 \\ 0 & 0 & 0 & \frac{2i}{21} & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{3}i}{42} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{63} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{14} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{63} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{14} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{42} \end{bmatrix}$ |
| 337 | symmetry | $\frac{15x^4z}{8} + \frac{15x^2y^2z}{4} - 5x^2z^3 + \frac{15y^4z}{8} - 5y^2z^3 + z^5$ |

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_5^{(1,-1;a)}(A,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{12} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{12} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{12} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{12} & 0 & 0 \end{bmatrix}$ |
| 338 | symmetry | $-\frac{3\sqrt{70}x^4y}{16} - \frac{\sqrt{70}x^2y^3}{8} + \frac{3\sqrt{70}x^2yz^2}{2} + \frac{\sqrt{70}y^5}{16} - \frac{\sqrt{70}y^3z^2}{2}$ |
| | $\mathbb{G}_5^{(1,-1;a)}(A,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{30} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{30} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{30} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{30} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{30} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{30} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 339 | symmetry | $-\frac{\sqrt{70}x^5}{16} + \frac{\sqrt{70}x^3y^2}{8} + \frac{\sqrt{70}x^3z^2}{2} + \frac{3\sqrt{70}xy^4}{16} - \frac{3\sqrt{70}xy^2z^2}{2}$ |
| | $\mathbb{G}_5^{(1,-1;a)}(A,3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{30} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{30} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{30} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{30} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{30} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{30} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 340 | symmetry | $\frac{3\sqrt{7}x^5}{16} - \frac{15\sqrt{7}ix^4y}{16} - \frac{15\sqrt{7}x^3y^2}{8} + \frac{15\sqrt{7}ix^2y^3}{8} + \frac{15\sqrt{7}xy^4}{16} - \frac{3\sqrt{7}iy^5}{16}$ |
| | $\mathbb{G}_5^{(1,-1;a)}(E^{(a)},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 341 | symmetry | $\frac{\sqrt{30}x^5}{16} + \frac{\sqrt{30}ix^4y}{16} + \frac{\sqrt{30}x^3y^2}{8} - \frac{3\sqrt{30}x^3z^2}{4} + \frac{\sqrt{30}ix^2y^3}{8} - \frac{3\sqrt{30}ix^2yz^2}{4} + \frac{\sqrt{30}xy^4}{16} - \frac{3\sqrt{30}xy^2z^2}{4} + \frac{\sqrt{30}xz^4}{2} + \frac{\sqrt{30}iy^5}{16} - \frac{3\sqrt{30}iy^3z^2}{4} + \frac{\sqrt{30}iyz^4}{2}$ |

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_5^{(1,-1;a)}(E^{(a)},2)$ | $\begin{bmatrix} 0&0&0&0&0&0&0&0&0&0&0&0&0&0 \\ 0&0&0&0&0&0&0&0&0&0&0&0&0&0 \\ 0&0&0&0&0&0&0&0&0&-\frac{\sqrt{3}i}{6}&0&0&0&0 \\ 0&0&0&0&0&0&0&0&0&0&\frac{i}{2}&0&0&0 \\ 0&0&0&0&0&0&0&0&0&0&0&-\frac{\sqrt{15}i}{10}&0&0 \\ 0&0&0&0&0&0&0&0&0&0&0&0&\frac{\sqrt{15}i}{30}&0 \end{bmatrix}$ |
| 342 | symmetry | $\frac{3\sqrt{70}x^4z}{16}+\frac{3\sqrt{70}ix^3yz}{4}-\frac{9\sqrt{70}x^2y^2z}{8}-\frac{3\sqrt{70}ixy^3z}{4}+\frac{3\sqrt{70}y^4z}{16}$ |
| | $\mathbb{G}_5^{(1,-1;a)}(E^{(a)},3)$ | $\begin{bmatrix} 0&0&0&0&0&0&0&0&0&0&0&0&0&0 \\ 0&0&0&0&0&0&0&0&0&0&0&0&0&0 \\ 0&0&0&0&0&0&0&0&0&0&0&0&-\frac{\sqrt{35}i}{10}&0 \\ 0&0&0&0&0&0&0&0&0&0&0&0&0&\frac{\sqrt{15}i}{10} \\ 0&0&0&0&0&0&0&0&0&0&0&0&0&0 \\ 0&0&0&0&0&0&0&0&0&0&0&0&0&0 \end{bmatrix}$ |
| 343 | symmetry | $-\frac{\sqrt{210}x^4z}{8}+\frac{\sqrt{210}ix^3yz}{4}+\frac{\sqrt{210}x^2z^3}{4}+\frac{\sqrt{210}ixy^3z}{4}-\frac{\sqrt{210}ixyz^3}{2}+\frac{\sqrt{210}y^4z}{8}-\frac{\sqrt{210}y^2z^3}{4}$ |
| | $\mathbb{G}_5^{(1,-1;a)}(E^{(a)},4)$ | $\begin{bmatrix} 0&0&0&0&0&0&0&0&0&0&0&0&0&0 \\ 0&0&0&0&0&0&0&0&0&0&0&0&0&0 \\ 0&0&0&0&0&0&-\frac{\sqrt{15}i}{60}&0&0&0&0&0&0&0 \\ 0&0&0&0&0&0&0&\frac{\sqrt{35}i}{20}&0&0&0&0&0&0 \\ 0&0&0&0&0&0&0&0&-\frac{\sqrt{105}i}{20}&0&0&0&0&0 \\ 0&0&0&0&0&0&0&0&0&\frac{\sqrt{21}i}{12}&0&0&0&0 \end{bmatrix}$ |
| 344 | symmetry | $\frac{3\sqrt{7}x^5}{16}+\frac{15\sqrt{7}ix^4y}{16}-\frac{15\sqrt{7}x^3y^2}{8}-\frac{15\sqrt{7}ix^2y^3}{8}+\frac{15\sqrt{7}xy^4}{16}+\frac{3\sqrt{7}iy^5}{16}$ |
| | $\mathbb{G}_5^{(1,-1;a)}(E^{(b)},1)$ | $\begin{bmatrix} 0&0&0&0&0&0&0&0&0&0&0&0&0&0 \\ 0&0&0&0&0&0&0&0&0&0&0&0&0&0 \\ 0&0&0&0&0&0&0&0&0&0&0&0&0&-\frac{\sqrt{2}i}{2} \\ 0&0&0&0&0&0&0&0&0&0&0&0&0&0 \\ 0&0&0&0&0&0&0&0&0&0&0&0&0&0 \\ 0&0&0&0&0&0&0&0&0&0&0&0&0&0 \end{bmatrix}$ |
| 345 | symmetry | $\frac{\sqrt{30}x^5}{16}-\frac{\sqrt{30}ix^4y}{16}+\frac{\sqrt{30}x^3y^2}{8}-\frac{3\sqrt{30}x^3z^2}{4}-\frac{\sqrt{30}ix^2y^3}{8}+\frac{3\sqrt{30}ix^2yz^2}{4}+\frac{\sqrt{30}xy^4}{16}-\frac{3\sqrt{30}xy^2z^2}{4}+\frac{\sqrt{30}xz^4}{2}-\frac{\sqrt{30}iy^5}{16}+\frac{3\sqrt{30}iy^3z^2}{4}-\frac{\sqrt{30}iyz^4}{2}$ |

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_5^{(1,-1;a)}(E^{(b)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{30} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{10} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 \end{bmatrix}$ |
| 346 | symmetry | $\frac{3\sqrt{70}x^4z}{16} - \frac{3\sqrt{70}ix^3yz}{4} - \frac{9\sqrt{70}x^2y^2z}{8} + \frac{3\sqrt{70}ixy^3z}{4} + \frac{3\sqrt{70}y^4z}{16}$ |
| | $\mathbb{G}_5^{(1,-1;a)}(E^{(b)},3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{10} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 347 | symmetry | $-\frac{\sqrt{210}x^4z}{8} - \frac{\sqrt{210}ix^3yz}{4} + \frac{\sqrt{210}x^2z^3}{4} - \frac{\sqrt{210}ixy^3z}{4} + \frac{\sqrt{210}ixyz^3}{2} + \frac{\sqrt{210}y^4z}{8} - \frac{\sqrt{210}y^2z^3}{4}$ |
| | $\mathbb{G}_5^{(1,-1;a)}(E^{(b)},4)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{12} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{20} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{20} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{60} \end{bmatrix}$ |
| 348 | symmetry | $-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$ |
| | $\mathbb{G}_3^{(1,0;a)}(A,1)$ | $\begin{bmatrix} 0 & 0 & \frac{\sqrt{7}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{21}i}{84} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{7}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{21}i}{84} & 0 & 0 & 0 \\ 0 & -\frac{5\sqrt{21}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{84} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{5\sqrt{14}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{84} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{5\sqrt{14}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{84} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{5\sqrt{21}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{84} & 0 & 0 \end{bmatrix}$ |
| 349 | symmetry | $\frac{3\sqrt{10}x^2y}{4} - \frac{\sqrt{10}y^3}{4}$ |

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_3^{(1,0;a)}(A,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{21} & -\frac{5\sqrt{6}}{48} & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{42}}{336} & 0 \\ -\frac{\sqrt{7}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{42}}{336} & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{6}}{48} \\ 0 & 0 & 0 & 0 & -\frac{5\sqrt{210}}{336} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{84} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{25\sqrt{14}}{336} & \frac{\sqrt{3}}{12} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{42} & 0 \\ -\frac{25\sqrt{14}}{336} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{12} \\ 0 & -\frac{5\sqrt{210}}{336} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{84} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 350 | symmetry | $\frac{\sqrt{10}x^3}{4} - \frac{3\sqrt{10}xy^2}{4}$ |
| | $\mathbb{G}_3^{(1,0;a)}(A,3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{21} & \frac{5\sqrt{6}i}{48} & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{42}i}{336} & 0 \\ \frac{\sqrt{7}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{42}i}{336} & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{6}i}{48} \\ 0 & 0 & 0 & 0 & -\frac{5\sqrt{210}i}{336} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{84} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{25\sqrt{14}i}{336} & -\frac{\sqrt{3}i}{12} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{42} & 0 \\ \frac{25\sqrt{14}i}{336} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{12} \\ 0 & \frac{5\sqrt{210}i}{336} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{84} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 351 | symmetry | $-\frac{\sqrt{3}x^3}{4} - \frac{\sqrt{3}ix^2y}{4} - \frac{\sqrt{3}xy^2}{4} + \sqrt{3}xz^2 - \frac{\sqrt{3}iy^3}{4} + \sqrt{3}iyz^2$ |
| | $\mathbb{G}_3^{(1,0;a)}(E^{(a)},1)$ | $\begin{bmatrix} 0 & 0 & 0 & \frac{2\sqrt{21}i}{63} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{7}i}{56} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{63} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{105}i}{168} & 0 & 0 \\ 0 & 0 & -\frac{5\sqrt{14}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{42} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{5\sqrt{42}i}{504} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{5\sqrt{21}i}{72} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{35}i}{168} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{84} & 0 \end{bmatrix}$ |
| 352 | symmetry | $\frac{\sqrt{30}x^2z}{4} - \frac{\sqrt{30}ixyz}{2} - \frac{\sqrt{30}y^2z}{4}$ |
| | $\mathbb{G}_3^{(1,0;a)}(E^{(a)},2)$ | $\begin{bmatrix} \frac{\sqrt{21}i}{63} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{14}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{105}i}{63} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{42}i}{168} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{25\sqrt{42}i}{504} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{5\sqrt{210}i}{504} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{5\sqrt{35}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{84} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 353 | symmetry | $-\frac{\sqrt{3}x^3}{4} + \frac{\sqrt{3}ix^2y}{4} - \frac{\sqrt{3}xy^2}{4} + \sqrt{3}xz^2 + \frac{\sqrt{3}iy^3}{4} - \sqrt{3}iyz^2$ |

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_3^{(1,0;a)}(E^{(b)},1)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{42}i}{63} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{105}i}{168} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{2\sqrt{21}i}{63} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{7}i}{56} & 0 & 0 & 0 & 0 \\ \frac{5\sqrt{35}i}{168} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{5\sqrt{21}i}{72} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{5\sqrt{42}i}{504} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{5\sqrt{14}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{42} & 0 & 0 & 0 \end{bmatrix}$ |
| 354 | symmetry | $\frac{\sqrt{30}x^2z}{4}+\frac{\sqrt{30}ixyz}{2}-\frac{\sqrt{30}y^2z}{4}$ |
| | $\mathbb{G}_3^{(1,0;a)}(E^{(b)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{63} & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{42}i}{168} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{63} & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{14}i}{56} & 0 \\ 0 & 0 & 0 & -\frac{5\sqrt{35}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{84} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{5\sqrt{210}i}{504} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{25\sqrt{42}i}{504} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{28} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{12} \end{bmatrix}$ |
| 355 | symmetry | $z$ |
| | $\mathbb{G}_1^{(1,1;a)}(A)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{10}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{15}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{15}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 356 | symmetry | $\frac{\sqrt{2}x}{2}+\frac{\sqrt{2}iy}{2}$ |
| | $\mathbb{G}_1^{(1,1;a)}(E^{(a)})$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{10}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{30}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{i}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 357 | symmetry | $\frac{\sqrt{2}x}{2}-\frac{\sqrt{2}iy}{2}$ |

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_1^{(1,1;a)}(E^{(b)})$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{i}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{15}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{30}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{10}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 358 | symmetry | $-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$ |
| | $\mathbb{G}_3^{(1,1;a)}(A,1)$ | $\begin{bmatrix} 0 & 0 & \frac{3i}{7} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{3i}{7} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{28} & 0 & 0 & 0 \\ 0 & \frac{3\sqrt{3}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{84} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{3\sqrt{2}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{84} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{2}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{84} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{3\sqrt{3}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{84} & 0 & 0 \end{bmatrix}$ |
| 359 | symmetry | $\frac{3\sqrt{10}x^2y}{4} - \frac{\sqrt{10}y^3}{4}$ |
| | $\mathbb{G}_3^{(1,1;a)}(A,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{3}{7} & -\frac{\sqrt{42}}{112} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{112} & 0 \\ -\frac{3}{7} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{112} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{112} \\ 0 & 0 & 0 & 0 & \frac{3\sqrt{30}}{112} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{84} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{15\sqrt{2}}{112} & \frac{\sqrt{21}}{84} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{42} & 0 \\ \frac{15\sqrt{2}}{112} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{42} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{84} \\ 0 & \frac{3\sqrt{30}}{112} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{84} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 360 | symmetry | $\frac{\sqrt{10}x^3}{4} - \frac{3\sqrt{10}xy^2}{4}$ |
| | $\mathbb{G}_3^{(1,1;a)}(A,3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{3i}{7} & \frac{\sqrt{42}i}{112} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{112} & 0 \\ \frac{3i}{7} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{112} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{112} \\ 0 & 0 & 0 & 0 & \frac{3\sqrt{30}i}{112} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{84} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{15\sqrt{2}i}{112} & -\frac{\sqrt{21}i}{84} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{42} & 0 \\ -\frac{15\sqrt{2}i}{112} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{42} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{84} \\ 0 & -\frac{3\sqrt{30}i}{112} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{84} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 361 | symmetry | $-\frac{\sqrt{3}x^3}{4} - \frac{\sqrt{3}ix^2y}{4} - \frac{\sqrt{3}xy^2}{4} + \sqrt{3}xz^2 - \frac{\sqrt{3}iy^3}{4} + \sqrt{3}iyz^2$ |

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_3^{(1,1;a)}(E^{(a)},1)$ | $\begin{bmatrix} 0 & 0 & 0 & \frac{2\sqrt{3}i}{7} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3i}{56} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{7} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{56} & 0 & 0 \\ 0 & 0 & \frac{9\sqrt{2}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{42} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{6}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{5}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{84} & 0 \end{bmatrix}$ |
| 362 | symmetry | $\frac{\sqrt{30}x^2z}{4} - \frac{\sqrt{30}ixyz}{2} - \frac{\sqrt{30}y^2z}{4}$ |
| | $\mathbb{G}_3^{(1,1;a)}(E^{(a)},2)$ | $\begin{bmatrix} \frac{\sqrt{3}i}{7} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{15}i}{7} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{56} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{5\sqrt{6}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{30}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{3\sqrt{5}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{84} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 363 | symmetry | $-\frac{\sqrt{3}x^3}{4} + \frac{\sqrt{3}ix^2y}{4} - \frac{\sqrt{3}xy^2}{4} + \sqrt{3}xz^2 + \frac{\sqrt{3}iy^3}{4} - \sqrt{3}iyz^2$ |
| | $\mathbb{G}_3^{(1,1;a)}(E^{(b)},1)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{6}i}{7} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{56} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{2\sqrt{3}i}{7} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3i}{56} & 0 & 0 & 0 & 0 \\ -\frac{3\sqrt{5}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{3}i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{6}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{9\sqrt{2}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{42} & 0 & 0 & 0 \end{bmatrix}$ |
| 364 | symmetry | $\frac{\sqrt{30}x^2z}{4} + \frac{\sqrt{30}ixyz}{2} - \frac{\sqrt{30}y^2z}{4}$ |
| | $\mathbb{G}_3^{(1,1;a)}(E^{(b)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{7} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{56} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{7} & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}i}{56} & 0 \\ 0 & 0 & 0 & \frac{3\sqrt{5}i}{28} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{84} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{56} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{6}i}{56} & 0 & 0 & 0 & 0 & 0 & -\frac{i}{28} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{84} \end{bmatrix}$ |
| 365 | symmetry | $-\frac{x^2}{2} - \frac{y^2}{2} + z^2$ |

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_2^{(a)}(A)$ | $\begin{bmatrix} 0 & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{i}{7} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{14} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{6}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}i}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{6}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}i}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{i}{7} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{14} & 0 & 0 \end{bmatrix}$ |
| 366 | symmetry | $\frac{\sqrt{6}xz}{2} + \frac{\sqrt{6}iyz}{2}$ |
| | $\mathbb{T}_2^{(a)}(E^{(a)}, 1)$ | $\begin{bmatrix} 0 & 0 & 0 & \frac{\sqrt{2}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{i}{3} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{3}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{7} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{5i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{7} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{7} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{7} & 0 \end{bmatrix}$ |
| 367 | symmetry | $\frac{\sqrt{6}x^2}{4} - \frac{\sqrt{6}ixy}{2} - \frac{\sqrt{6}y^2}{4}$ |
| | $\mathbb{T}_2^{(a)}(E^{(a)}, 2)$ | $\begin{bmatrix} \frac{\sqrt{5}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{10}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{2\sqrt{2}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{14} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{3}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{14} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 368 | symmetry | $\frac{\sqrt{6}xz}{2} - \frac{\sqrt{6}iyz}{2}$ |
| | $\mathbb{T}_2^{(a)}(E^{(b)}, 1)$ | $\begin{bmatrix} 0 & -\frac{i}{3} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{2}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{30}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{7} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{2}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{7} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{5i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{7} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{3}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{7} & 0 & 0 & 0 \end{bmatrix}$ |
| 369 | symmetry | $\frac{\sqrt{6}x^2}{4} + \frac{\sqrt{6}ixy}{2} - \frac{\sqrt{6}y^2}{4}$ |

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_2^{(a)}(E^{(b)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & \frac{i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{3}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{14} & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{2\sqrt{2}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{14} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{14} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{14} \end{bmatrix}$ |
| 370 | symmetry | $\frac{3x^4}{8}+\frac{3x^2y^2}{4}-3x^2z^2+\frac{3y^4}{8}-3y^2z^2+z^4$ |
| | $\mathbb{T}_4^{(a)}(A,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{3}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{3\sqrt{2}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{6}i}{84} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{3\sqrt{2}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{6}i}{84} & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{28} & 0 \end{bmatrix}$ |
| 371 | symmetry | $\frac{\sqrt{70}x^3z}{4}-\frac{3\sqrt{70}xy^2z}{4}$ |
| | $\mathbb{T}_4^{(a)}(A,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{24} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{24} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{24} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{24} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{14}i}{56} & -\frac{\sqrt{3}i}{12} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{42} & 0 \\ -\frac{3\sqrt{14}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{12} \\ 0 & \frac{\sqrt{210}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{28} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 372 | symmetry | $\frac{3\sqrt{70}x^2yz}{4}-\frac{\sqrt{70}y^3z}{4}$ |
| | $\mathbb{T}_4^{(a)}(A,3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{24} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{24} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{24} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{24} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{14}}{56} & \frac{\sqrt{3}}{12} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{42} & 0 \\ \frac{3\sqrt{14}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{12} \\ 0 & -\frac{\sqrt{210}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{28} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 373 | symmetry | $-\frac{3\sqrt{5}x^3z}{4}-\frac{3\sqrt{5}ix^2yz}{4}-\frac{3\sqrt{5}xy^2z}{4}+\sqrt{5}xz^3-\frac{3\sqrt{5}iy^3z}{4}+\sqrt{5}iyz^3$ |

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_4^{(a)}(E^{(a)},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{12} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{4} & 0 & 0 \\ 0 & 0 & \frac{\sqrt{30}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{10}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{84} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{3\sqrt{5}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{7} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}i}{28} & 0 \end{bmatrix}$ |
| 374 | symmetry | $\frac{\sqrt{70}x^4}{16}+\frac{\sqrt{70}ix^3y}{4}-\frac{3\sqrt{70}x^2y^2}{8}-\frac{\sqrt{70}ixy^3}{4}+\frac{\sqrt{70}y^4}{16}$ |
| | $\mathbb{T}_4^{(a)}(E^{(a)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{6} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 375 | symmetry | $-\frac{\sqrt{10}x^4}{8}+\frac{\sqrt{10}ix^3y}{4}+\frac{3\sqrt{10}x^2z^2}{4}+\frac{\sqrt{10}ixy^3}{4}-\frac{3\sqrt{10}ixyz^2}{2}+\frac{\sqrt{10}y^4}{8}-\frac{3\sqrt{10}y^2z^2}{4}$ |
| | $\mathbb{T}_4^{(a)}(E^{(a)},3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{3\sqrt{2}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{11\sqrt{3}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{3\sqrt{10}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{15}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{5}i}{28} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 376 | symmetry | $-\frac{3\sqrt{5}x^3z}{4}+\frac{3\sqrt{5}ix^2yz}{4}-\frac{3\sqrt{5}xy^2z}{4}+\sqrt{5}xz^3+\frac{3\sqrt{5}iy^3z}{4}-\sqrt{5}iyz^3$ |
| | $\mathbb{T}_4^{(a)}(E^{(b)},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{4} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{12} & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{3}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{3\sqrt{5}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{7} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{3\sqrt{10}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{84} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{30}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{14} & 0 & 0 & 0 \end{bmatrix}$ |
| 377 | symmetry | $\frac{\sqrt{70}x^4}{16}-\frac{\sqrt{70}ix^3y}{4}-\frac{3\sqrt{70}x^2y^2}{8}+\frac{\sqrt{70}ixy^3}{4}+\frac{\sqrt{70}y^4}{16}$ |

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_4^{(a)}(E^{(b)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{42}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 378 | symmetry | $-\frac{\sqrt{10}x^4}{8} - \frac{\sqrt{10}ix^3y}{4} + \frac{3\sqrt{10}x^2z^2}{4} - \frac{\sqrt{10}ixy^3}{4} + \frac{3\sqrt{10}ixyz^2}{2} + \frac{\sqrt{10}y^4}{8} - \frac{3\sqrt{10}y^2z^2}{4}$ |
| | $\mathbb{T}_4^{(a)}(E^{(b)},3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{12} & 0 \\ 0 & 0 & 0 & \frac{\sqrt{15}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{5}i}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{11\sqrt{3}i}{84} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{28} \end{bmatrix}$ |
| 379 | symmetry | $\frac{3x^4}{8} + \frac{3x^2y^2}{4} - 3x^2z^2 + \frac{3y^4}{8} - 3y^2z^2 + z^4$ |
| | $\mathbb{T}_4^{(1,-1;a)}(A,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{4} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{4} & 0 & 0 & 0 \\ 0 & \frac{i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{10}i}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{6}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{2}i}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{6}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{2}i}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{10}i}{28} & 0 & 0 \end{bmatrix}$ |
| 380 | symmetry | $\frac{\sqrt{70}x^3z}{4} - \frac{3\sqrt{70}xy^2z}{4}$ |
| | $\mathbb{T}_4^{(1,-1;a)}(A,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{16} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{16} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{16} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{16} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{112} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{7}i}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{112} & \frac{i}{4} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 \\ -\frac{\sqrt{42}i}{112} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & -\frac{i}{4} \\ 0 & \frac{\sqrt{70}i}{112} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{7}i}{28} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 381 | symmetry | $\frac{3\sqrt{70}x^2yz}{4} - \frac{\sqrt{70}y^3z}{4}$ |

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_4^{(1,-1;a)}(A,3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{16} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{16} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{16} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{16} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{112} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{7}}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{112} & -\frac{1}{4} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{14} & 0 \\ \frac{\sqrt{42}}{112} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & -\frac{1}{4} \\ 0 & -\frac{\sqrt{70}}{112} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{7}}{28} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 382 | symmetry | $-\frac{3\sqrt{5}x^3z}{4} - \frac{3\sqrt{5}ix^2yz}{4} - \frac{3\sqrt{5}xy^2z}{4} + \sqrt{5}xz^3 - \frac{3\sqrt{5}iy^3z}{4} + \sqrt{5}iyz^3$ |
| | $\mathbb{T}_4^{(1,-1;a)}(E^{(a)},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{8} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{8} & 0 & 0 \\ 0 & 0 & \frac{\sqrt{10}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{30}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{7} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{6}i}{28} & 0 \end{bmatrix}$ |
| 383 | symmetry | $\frac{\sqrt{70}x^4}{16} + \frac{\sqrt{70}ix^3y}{4} - \frac{3\sqrt{70}x^2y^2}{8} - \frac{\sqrt{70}ixy^3}{4} + \frac{\sqrt{70}y^4}{16}$ |
| | $\mathbb{T}_4^{(1,-1;a)}(E^{(a)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{14} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{2} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 384 | symmetry | $-\frac{\sqrt{10}x^4}{8} + \frac{\sqrt{10}ix^3y}{4} + \frac{3\sqrt{10}x^2z^2}{4} + \frac{\sqrt{10}ixy^3}{4} - \frac{3\sqrt{10}ixyz^2}{2} + \frac{\sqrt{10}y^4}{8} - \frac{3\sqrt{10}y^2z^2}{4}$ |
| | $\mathbb{T}_4^{(1,-1;a)}(E^{(a)},3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{8} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{8} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{6}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{11i}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{30}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{5}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{15}i}{28} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 385 | symmetry | $-\frac{3\sqrt{5}x^3z}{4} + \frac{3\sqrt{5}ix^2yz}{4} - \frac{3\sqrt{5}xy^2z}{4} + \sqrt{5}xz^3 + \frac{3\sqrt{5}iy^3z}{4} - \sqrt{5}iyz^3$ |

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_4^{(1,-1;a)}(E^{(b)},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{8} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{8} & 0 & 0 & 0 & 0 \\ -\frac{i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{6}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{15}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{7} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{30}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{10}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{14} & 0 & 0 & 0 \end{bmatrix}$ |
| 386 | symmetry | $\frac{\sqrt{70}x^4}{16} - \frac{\sqrt{70}ix^3y}{4} - \frac{3\sqrt{70}x^2y^2}{8} + \frac{\sqrt{70}ixy^3}{4} + \frac{\sqrt{70}y^4}{16}$ |
| | $\mathbb{T}_4^{(1,-1;a)}(E^{(b)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{14}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 387 | symmetry | $-\frac{\sqrt{10}x^4}{8} - \frac{\sqrt{10}ix^3y}{4} + \frac{3\sqrt{10}x^2z^2}{4} - \frac{\sqrt{10}ixy^3}{4} + \frac{3\sqrt{10}ixyz^2}{2} + \frac{\sqrt{10}y^4}{8} - \frac{3\sqrt{10}y^2z^2}{4}$ |
| | $\mathbb{T}_4^{(1,-1;a)}(E^{(b)},3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{8} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{8} & 0 \\ 0 & 0 & 0 & \frac{\sqrt{5}i}{28} & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{15}i}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{56} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{56} & 0 & 0 & 0 & 0 & 0 & -\frac{11i}{28} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{28} \end{bmatrix}$ |
| 388 | symmetry | $-\frac{x^2}{2} - \frac{y^2}{2} + z^2$ |
| | $\mathbb{T}_2^{(1,0;a)}(A)$ | $\begin{bmatrix} 0 & 0 & \frac{\sqrt{2}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{2}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{5\sqrt{6}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{21} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{5i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{7} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{5i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{7} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{5\sqrt{6}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{21} & 0 & 0 \end{bmatrix}$ |
| 389 | symmetry | $\frac{\sqrt{6}xz}{2} + \frac{\sqrt{6}iyz}{2}$ |

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_2^{(1,0;a)}(E^{(a)},1)$ | $\begin{bmatrix} 0 & 0 & 0 & \frac{\sqrt{3}i}{9} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{9} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{5\sqrt{2}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{21} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{25\sqrt{6}i}{252} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{7} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{5\sqrt{3}i}{126} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{21} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{5}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{21} & 0 \end{bmatrix}$ |
| 390 | symmetry | $\frac{\sqrt{6}x^2}{4} - \frac{\sqrt{6}ixy}{2} - \frac{\sqrt{6}y^2}{4}$ |
| | $\mathbb{T}_2^{(1,0;a)}(E^{(a)},2)$ | $\begin{bmatrix} \frac{\sqrt{30}i}{18} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{6}i}{18} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{5\sqrt{15}i}{63} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{10\sqrt{3}i}{63} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{42} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{5\sqrt{2}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{42} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 391 | symmetry | $\frac{\sqrt{6}xz}{2} - \frac{\sqrt{6}iyz}{2}$ |
| | $\mathbb{T}_2^{(1,0;a)}(E^{(b)},1)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{6}i}{9} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{3}i}{9} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{5\sqrt{5}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{5\sqrt{3}i}{126} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{21} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{25\sqrt{6}i}{252} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{7} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{5\sqrt{2}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{21} & 0 & 0 & 0 \end{bmatrix}$ |
| 392 | symmetry | $\frac{\sqrt{6}x^2}{4} + \frac{\sqrt{6}ixy}{2} - \frac{\sqrt{6}y^2}{4}$ |
| | $\mathbb{T}_2^{(1,0;a)}(E^{(b)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{18} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{18} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{5\sqrt{2}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{42} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{10\sqrt{3}i}{63} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{42} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{15}i}{63} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{14} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{42} \end{bmatrix}$ |
| 393 | symmetry | $\frac{3x^4}{8} + \frac{3x^2y^2}{4} - 3x^2z^2 + \frac{3y^4}{8} - 3y^2z^2 + z^4$ |

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_4^{(1,0;a)}(A,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{12} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{12} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{15}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{3\sqrt{10}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{84} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{3\sqrt{10}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{84} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{28} & 0 & 0 \end{bmatrix}$ |
| 394 | symmetry | $\frac{\sqrt{70}x^3z}{4} - \frac{3\sqrt{70}xy^2z}{4}$ |
| | $\mathbb{T}_4^{(1,0;a)}(A,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{48} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{48} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{48} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{48} \\ 0 & 0 & 0 & 0 & \frac{5\sqrt{42}i}{112} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{140} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{70}i}{112} & \frac{\sqrt{15}i}{60} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{210} & 0 \\ -\frac{3\sqrt{70}i}{112} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{210} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{60} \\ 0 & \frac{5\sqrt{42}i}{112} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{140} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 395 | symmetry | $\frac{3\sqrt{70}x^2yz}{4} - \frac{\sqrt{70}y^3z}{4}$ |
| | $\mathbb{T}_4^{(1,0;a)}(A,3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{48} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{48} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{48} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{48} \\ 0 & 0 & 0 & 0 & \frac{5\sqrt{42}}{112} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{140} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{70}}{112} & -\frac{\sqrt{15}}{60} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{210} & 0 \\ \frac{3\sqrt{70}}{112} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{210} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{60} \\ 0 & -\frac{5\sqrt{42}}{112} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{140} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 396 | symmetry | $-\frac{3\sqrt{5}x^3z}{4} - \frac{3\sqrt{5}ix^2yz}{4} - \frac{3\sqrt{5}xy^2z}{4} + \sqrt{5}xz^3 - \frac{3\sqrt{5}iy^3z}{4} + \sqrt{5}iyz^3$ |
| | $\mathbb{T}_4^{(1,0;a)}(E^{(a)},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{3}i}{24} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{8} & 0 & 0 \\ 0 & 0 & \frac{5\sqrt{6}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{15\sqrt{2}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{84} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{15i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{35} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{10}i}{140} & 0 \end{bmatrix}$ |
| 397 | symmetry | $\frac{\sqrt{70}x^4}{16} + \frac{\sqrt{70}ix^3y}{4} - \frac{3\sqrt{70}x^2y^2}{8} - \frac{\sqrt{70}ixy^3}{4} + \frac{\sqrt{70}y^4}{16}$ |

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_4^{(1,0;a)}(E^{(a)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{12} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{28} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{70} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{30} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 398 | symmetry | $-\frac{\sqrt{10}x^4}{8}+\frac{\sqrt{10}ix^3y}{4}+\frac{3\sqrt{10}x^2z^2}{4}+\frac{\sqrt{10}ixy^3}{4}-\frac{3\sqrt{10}ixyz^2}{2}+\frac{\sqrt{10}y^4}{8}-\frac{3\sqrt{10}y^2z^2}{4}$ |
| | $\mathbb{T}_4^{(1,0;a)}(E^{(a)},3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{24} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{8} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{3\sqrt{10}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{11\sqrt{15}i}{420} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{15\sqrt{2}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{140} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{5\sqrt{3}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3i}{28} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 399 | symmetry | $-\frac{3\sqrt{5}x^3z}{4}+\frac{3\sqrt{5}ix^2yz}{4}-\frac{3\sqrt{5}xy^2z}{4}+\sqrt{5}xz^3+\frac{3\sqrt{5}iy^3z}{4}-\sqrt{5}iyz^3$ |
| | $\mathbb{T}_4^{(1,0;a)}(E^{(b)},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{8} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{3}i}{24} & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{15}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{15i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{35} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{15\sqrt{2}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{84} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{5\sqrt{6}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{14} & 0 & 0 & 0 \end{bmatrix}$ |
| 400 | symmetry | $\frac{\sqrt{70}x^4}{16}-\frac{\sqrt{70}ix^3y}{4}-\frac{3\sqrt{70}x^2y^2}{8}+\frac{\sqrt{70}ixy^3}{4}+\frac{\sqrt{70}y^4}{16}$ |
| | $\mathbb{T}_4^{(1,0;a)}(E^{(b)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{30} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{210}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 401 | symmetry | $-\frac{\sqrt{10}x^4}{8}-\frac{\sqrt{10}ix^3y}{4}+\frac{3\sqrt{10}x^2z^2}{4}-\frac{\sqrt{10}ixy^3}{4}+\frac{3\sqrt{10}ixyz^2}{2}+\frac{\sqrt{10}y^4}{8}-\frac{3\sqrt{10}y^2z^2}{4}$ |

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_4^{(1,0;a)}(E^{(b)},3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{8} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{24} & 0 \\ 0 & 0 & 0 & \frac{5\sqrt{3}i}{28} & 0 & 0 & 0 & 0 & 0 & \frac{3i}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{15\sqrt{2}i}{56} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{140} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{10}i}{56} & 0 & 0 & 0 & 0 & 0 & -\frac{11\sqrt{15}i}{420} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{140} \end{bmatrix}$ |
| 402 | symmetry | $-\frac{x^2}{2}-\frac{y^2}{2}+z^2$ |
| | $\mathbb{T}_2^{(1,1;a)}(A)$ | $\begin{bmatrix} 0 & 0 & -\frac{i}{3} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{i}{3} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{4\sqrt{3}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{84} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{2\sqrt{2}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{2\sqrt{2}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{4\sqrt{3}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{84} & 0 & 0 \end{bmatrix}$ |
| 403 | symmetry | $\frac{\sqrt{6}xz}{2}+\frac{\sqrt{6}iyz}{2}$ |
| | $\mathbb{T}_2^{(1,1;a)}(E^{(a)},1)$ | $\begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{6}i}{9} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{2\sqrt{3}i}{9} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{2i}{7} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{42} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{10\sqrt{3}i}{63} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{2\sqrt{6}i}{63} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{42} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{10}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{42} & 0 \end{bmatrix}$ |
| 404 | symmetry | $\frac{\sqrt{6}x^2}{4}-\frac{\sqrt{6}ixy}{2}-\frac{\sqrt{6}y^2}{4}$ |
| | $\mathbb{T}_2^{(1,1;a)}(E^{(a)},2)$ | $\begin{bmatrix} -\frac{\sqrt{15}i}{9} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{3}i}{9} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{4\sqrt{30}i}{63} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{8\sqrt{6}i}{63} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{84} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{4i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{84} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 405 | symmetry | $\frac{\sqrt{6}xz}{2}-\frac{\sqrt{6}iyz}{2}$ |

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_2^{(1,1;a)}(E^{(b)},1)$ | $\begin{bmatrix} 0 & \frac{2\sqrt{3}i}{9} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{6}i}{9} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{2\sqrt{10}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{2\sqrt{6}i}{63} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{42} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{10\sqrt{3}i}{63} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{2i}{7} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{42} & 0 & 0 & 0 \end{bmatrix}$ |
| 406 | symmetry | $\frac{\sqrt{6}x^2}{4}+\frac{\sqrt{6}ixy}{2}-\frac{\sqrt{6}y^2}{4}$ |
| | $\mathbb{T}_2^{(1,1;a)}(E^{(b)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{9} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{9} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{4i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{84} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{8\sqrt{6}i}{63} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{84} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{4\sqrt{30}i}{63} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{28} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{84} \end{bmatrix}$ |
| 407 | symmetry | $-\frac{3x^2z}{2}-\frac{3y^2z}{2}+z^3$ |
| | $\mathbb{M}_3^{(a)}(A,1)$ | $\begin{bmatrix} 0 & 0 & \frac{\sqrt{21}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{21}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{42}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{42}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{28} & 0 & 0 \end{bmatrix}$ |
| 408 | symmetry | $\frac{3\sqrt{10}x^2y}{4}-\frac{\sqrt{10}y^3}{4}$ |
| | $\mathbb{M}_3^{(a)}(A,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{21} & -\frac{\sqrt{2}i}{8} & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{56} & 0 \\ \frac{\sqrt{21}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{56} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{8} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{42}i}{168} & \frac{i}{4} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 \\ \frac{5\sqrt{42}i}{168} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & -\frac{i}{4} \\ 0 & \frac{\sqrt{70}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{28} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 409 | symmetry | $\frac{\sqrt{10}x^3}{4}-\frac{3\sqrt{10}xy^2}{4}$ |

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_3^{(a)}(A,3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{21} & -\frac{\sqrt{2}}{8} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{56} & 0 \\ \frac{\sqrt{21}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{56} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{8} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{42}}{168} & \frac{1}{4} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{14} & 0 \\ \frac{5\sqrt{42}}{168} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & \frac{1}{4} \\ 0 & \frac{\sqrt{70}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{28} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 410 | symmetry | $-\frac{\sqrt{3}x^3}{4} - \frac{\sqrt{3}ix^2y}{4} - \frac{\sqrt{3}xy^2}{4} + \sqrt{3}xz^2 - \frac{\sqrt{3}iy^3}{4} + \sqrt{3}iyz^2$ |
| | $\mathbb{M}_3^{(a)}(E^{(a)},1)$ | $\begin{bmatrix} 0 & 0 & 0 & \frac{2\sqrt{7}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{28} & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{42}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{14}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{28} & 0 \end{bmatrix}$ |
| 411 | symmetry | $\frac{\sqrt{30}x^2z}{4} - \frac{\sqrt{30}ixyz}{2} - \frac{\sqrt{30}y^2z}{4}$ |
| | $\mathbb{M}_3^{(a)}(E^{(a)},2)$ | $\begin{bmatrix} \frac{\sqrt{7}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{35}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{5\sqrt{14}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{70}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{7}}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{105}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{28} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 412 | symmetry | $-\frac{\sqrt{3}x^3}{4} + \frac{\sqrt{3}ix^2y}{4} - \frac{\sqrt{3}xy^2}{4} + \sqrt{3}xz^2 + \frac{\sqrt{3}iy^3}{4} - \sqrt{3}iyz^2$ |
| | $\mathbb{M}_3^{(a)}(E^{(b)},1)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{14}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{2\sqrt{7}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{28} & 0 & 0 & 0 & 0 \\ \frac{\sqrt{105}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{7}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{14}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{42}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 & 0 \end{bmatrix}$ |
| 413 | symmetry | $\frac{\sqrt{30}x^2z}{4} + \frac{\sqrt{30}ixyz}{2} - \frac{\sqrt{30}y^2z}{4}$ |

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_3^{(a)}(E^{(b)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{28} & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{105}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{7}}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{14}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{28} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{4} \end{bmatrix}$ |
| 414 | symmetry | $-\frac{3x^2z}{2}-\frac{3y^2z}{2}+z^3$ |
| | $\mathbb{M}_3^{(1,-1;a)}(A,1)$ | $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{5}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{15}}{21} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{5}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{15}}{21} & 0 & 0 & 0 \\ 0 & \frac{2\sqrt{15}}{105} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{6}}{42} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{2\sqrt{10}}{105} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{42} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{2\sqrt{10}}{105} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{42} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{2\sqrt{15}}{105} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{6}}{42} & 0 & 0 \end{bmatrix}$ |
| 415 | symmetry | $\frac{3\sqrt{10}x^2y}{4}-\frac{\sqrt{10}y^3}{4}$ |
| | $\mathbb{M}_3^{(1,-1;a)}(A,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{42} & \frac{\sqrt{210}i}{42} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{42} & 0 \\ -\frac{\sqrt{5}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{42} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{42} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{42} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{42} & \frac{\sqrt{105}i}{42} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{21} & 0 \\ -\frac{\sqrt{10}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{21} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{42} \\ 0 & -\frac{\sqrt{6}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{42} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 416 | symmetry | $\frac{\sqrt{10}x^3}{4}-\frac{3\sqrt{10}xy^2}{4}$ |
| | $\mathbb{M}_3^{(1,-1;a)}(A,3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{42} & \frac{\sqrt{210}}{42} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{42} & 0 \\ -\frac{\sqrt{5}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{42} & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{42} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{42} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{42} & \frac{\sqrt{105}}{42} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{21} & 0 \\ -\frac{\sqrt{10}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{21} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{42} \\ 0 & -\frac{\sqrt{6}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{42} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 417 | symmetry | $-\frac{\sqrt{3}x^3}{4}-\frac{\sqrt{3}ix^2y}{4}-\frac{\sqrt{3}xy^2}{4}+\sqrt{3}xz^2-\frac{\sqrt{3}iy^3}{4}+\sqrt{3}iyz^2$ |

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_3^{(1,-1;a)}(E^{(a)},1)$ | $\begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{15}}{63} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{7} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{126} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{3}}{21} & 0 & 0 \\ 0 & 0 & \frac{\sqrt{10}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{21} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{30}}{315} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{45} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{6}}{42} & 0 \end{bmatrix}$ |
| 418 | symmetry | $\frac{\sqrt{30}x^2z}{4} - \frac{\sqrt{30}ixyz}{2} - \frac{\sqrt{30}y^2z}{4}$ |
| | $\mathbb{M}_3^{(1,-1;a)}(E^{(a)},2)$ | $\begin{bmatrix} -\frac{\sqrt{15}}{126} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{7} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{5\sqrt{3}}{126} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{21} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{30}}{63} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{6}}{63} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{14} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{2}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{3}}{42} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 419 | symmetry | $-\frac{\sqrt{3}x^3}{4} + \frac{\sqrt{3}ix^2y}{4} - \frac{\sqrt{3}xy^2}{4} + \sqrt{3}xz^2 + \frac{\sqrt{3}iy^3}{4} - \sqrt{3}iyz^2$ |
| | $\mathbb{M}_3^{(1,-1;a)}(E^{(b)},1)$ | $\begin{bmatrix} 0 & \frac{\sqrt{30}}{126} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{3}}{21} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{15}}{63} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{7} & 0 & 0 & 0 & 0 \\ -\frac{1}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{6}}{42} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{15}}{45} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{30}}{315} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{10}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{21} & 0 & 0 & 0 \end{bmatrix}$ |
| 420 | symmetry | $\frac{\sqrt{30}x^2z}{4} + \frac{\sqrt{30}ixyz}{2} - \frac{\sqrt{30}y^2z}{4}$ |
| | $\mathbb{M}_3^{(1,-1;a)}(E^{(b)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{5\sqrt{3}}{126} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{21} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{126} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{7} & 0 \\ 0 & 0 & 0 & \frac{2}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{3}}{42} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{63} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{14} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{63} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{14} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{42} \end{bmatrix}$ |
| 421 | symmetry | $\frac{15x^4z}{8} + \frac{15x^2y^2z}{4} - 5x^2z^3 + \frac{15y^4z}{8} - 5y^2z^3 + z^5$ |

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_5^{(1,-1;a)}(A,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{12} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{12} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{12} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{12} & 0 & 0 \end{bmatrix}$ |
| 422 | symmetry | $-\frac{3\sqrt{70}x^4y}{16} - \frac{\sqrt{70}x^2y^3}{8} + \frac{3\sqrt{70}x^2yz^2}{2} + \frac{\sqrt{70}y^5}{16} - \frac{\sqrt{70}y^3z^2}{2}$ |
| | $\mathbb{M}_5^{(1,-1;a)}(A,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{30} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{30} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{30} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{30} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{30} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{30} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 423 | symmetry | $-\frac{\sqrt{70}x^5}{16} + \frac{\sqrt{70}x^3y^2}{8} + \frac{\sqrt{70}x^3z^2}{2} + \frac{3\sqrt{70}xy^4}{16} - \frac{3\sqrt{70}xy^2z^2}{2}$ |
| | $\mathbb{M}_5^{(1,-1;a)}(A,3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{30} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{30} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{30} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{30} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{30} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{30} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 424 | symmetry | $\frac{3\sqrt{7}x^5}{16} - \frac{15\sqrt{7}ix^4y}{16} - \frac{15\sqrt{7}x^3y^2}{8} + \frac{15\sqrt{7}ix^2y^3}{8} + \frac{15\sqrt{7}xy^4}{16} - \frac{3\sqrt{7}iy^5}{16}$ |
| | $\mathbb{M}_5^{(1,-1;a)}(E^{(a)},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 425 | symmetry | $\frac{\sqrt{30}x^5}{16} + \frac{\sqrt{30}ix^4y}{16} + \frac{\sqrt{30}x^3y^2}{8} - \frac{3\sqrt{30}x^3z^2}{4} + \frac{\sqrt{30}ix^2y^3}{8} - \frac{3\sqrt{30}ix^2yz^2}{4} + \frac{\sqrt{30}xy^4}{16} - \frac{3\sqrt{30}xy^2z^2}{4} + \frac{\sqrt{30}xz^4}{2} + \frac{\sqrt{30}iy^5}{16} - \frac{3\sqrt{30}iy^3z^2}{4} + \frac{\sqrt{30}iyz^4}{2}$ |

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_5^{(1,-1;a)}(E^{(a)},2)$ | $\begin{bmatrix} 0&0&0&0&0&0&0&0&0&0&0&0&0&0 \\ 0&0&0&0&0&0&0&0&0&0&0&0&0&0 \\ 0&0&0&0&0&0&0&0&0&-\frac{\sqrt{3}}{6}&0&0&0&0 \\ 0&0&0&0&0&0&0&0&0&0&\frac{1}{2}&0&0&0 \\ 0&0&0&0&0&0&0&0&0&0&0&-\frac{\sqrt{15}}{10}&0&0 \\ 0&0&0&0&0&0&0&0&0&0&0&0&\frac{\sqrt{15}}{30}&0 \end{bmatrix}$ |
| 426 | symmetry | $\frac{3\sqrt{70}x^4z}{16}+\frac{3\sqrt{70}ix^3yz}{4}-\frac{9\sqrt{70}x^2y^2z}{8}-\frac{3\sqrt{70}ixy^3z}{4}+\frac{3\sqrt{70}y^4z}{16}$ |
| | $\mathbb{M}_5^{(1,-1;a)}(E^{(a)},3)$ | $\begin{bmatrix} 0&0&0&0&0&0&0&0&0&0&0&0&0&0&0 \\ 0&0&0&0&0&0&0&0&0&0&0&0&0&0&0 \\ 0&0&0&0&0&0&0&0&0&0&0&0&0&-\frac{\sqrt{35}}{10}&0 \\ 0&0&0&0&0&0&0&0&0&0&0&0&0&0&\frac{\sqrt{15}}{10} \\ 0&0&0&0&0&0&0&0&0&0&0&0&0&0&0 \\ 0&0&0&0&0&0&0&0&0&0&0&0&0&0&0 \end{bmatrix}$ |
| 427 | symmetry | $-\frac{\sqrt{210}x^4z}{8}+\frac{\sqrt{210}ix^3yz}{4}+\frac{\sqrt{210}x^2z^3}{4}+\frac{\sqrt{210}ixy^3z}{4}-\frac{\sqrt{210}ixyz^3}{2}+\frac{\sqrt{210}y^4z}{8}-\frac{\sqrt{210}y^2z^3}{4}$ |
| | $\mathbb{M}_5^{(1,-1;a)}(E^{(a)},4)$ | $\begin{bmatrix} 0&0&0&0&0&0&0&0&0&0&0&0&0&0 \\ 0&0&0&0&0&0&0&0&0&0&0&0&0&0 \\ 0&0&0&0&0&0&-\frac{\sqrt{15}}{60}&0&0&0&0&0&0&0 \\ 0&0&0&0&0&0&0&\frac{\sqrt{35}}{20}&0&0&0&0&0&0 \\ 0&0&0&0&0&0&0&0&-\frac{\sqrt{105}}{20}&0&0&0&0&0 \\ 0&0&0&0&0&0&0&0&0&\frac{\sqrt{21}}{12}&0&0&0&0 \end{bmatrix}$ |
| 428 | symmetry | $\frac{3\sqrt{7}x^5}{16}+\frac{15\sqrt{7}ix^4y}{16}-\frac{15\sqrt{7}x^3y^2}{8}-\frac{15\sqrt{7}ix^2y^3}{8}+\frac{15\sqrt{7}xy^4}{16}+\frac{3\sqrt{7}iy^5}{16}$ |
| | $\mathbb{M}_5^{(1,-1;a)}(E^{(b)},1)$ | $\begin{bmatrix} 0&0&0&0&0&0&0&0&0&0&0&0&0&0&0 \\ 0&0&0&0&0&0&0&0&0&0&0&0&0&0&0 \\ 0&0&0&0&0&0&0&0&0&0&0&0&0&0&-\frac{\sqrt{2}}{2} \\ 0&0&0&0&0&0&0&0&0&0&0&0&0&0&0 \\ 0&0&0&0&0&0&0&0&0&0&0&0&0&0&0 \\ 0&0&0&0&0&0&0&0&0&0&0&0&0&0&0 \end{bmatrix}$ |
| 429 | symmetry | $\frac{\sqrt{30}x^5}{16}-\frac{\sqrt{30}ix^4y}{16}+\frac{\sqrt{30}x^3y^2}{8}-\frac{3\sqrt{30}x^3z^2}{4}-\frac{\sqrt{30}ix^2y^3}{8}+\frac{3\sqrt{30}ix^2yz^2}{4}+\frac{\sqrt{30}xy^4}{16}-\frac{3\sqrt{30}xy^2z^2}{4}+\frac{\sqrt{30}xz^4}{2}-\frac{\sqrt{30}iy^5}{16}+\frac{3\sqrt{30}iy^3z^2}{4}-\frac{\sqrt{30}iyz^4}{2}$ |

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_5^{(1,-1;a)}(E^{(b)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{30} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{10} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & 0 \end{bmatrix}$ |
| 430 | symmetry | $\frac{3\sqrt{70}x^4z}{16} - \frac{3\sqrt{70}ix^3yz}{4} - \frac{9\sqrt{70}x^2y^2z}{8} + \frac{3\sqrt{70}ixy^3z}{4} + \frac{3\sqrt{70}y^4z}{16}$ |
| | $\mathbb{M}_5^{(1,-1;a)}(E^{(b)},3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{10} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 431 | symmetry | $-\frac{\sqrt{210}x^4z}{8} - \frac{\sqrt{210}ix^3yz}{4} + \frac{\sqrt{210}x^2z^3}{4} - \frac{\sqrt{210}ixy^3z}{4} + \frac{\sqrt{210}ixyz^3}{2} + \frac{\sqrt{210}y^4z}{8} - \frac{\sqrt{210}y^2z^3}{4}$ |
| | $\mathbb{M}_5^{(1,-1;a)}(E^{(b)},4)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{12} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{20} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{20} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{60} \end{bmatrix}$ |
| 432 | symmetry | $-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$ |
| | $\mathbb{M}_3^{(1,0;a)}(A,1)$ | $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{7}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{21}}{84} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{7}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{21}}{84} & 0 & 0 & 0 \\ 0 & \frac{5\sqrt{21}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{84} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{5\sqrt{14}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{84} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{5\sqrt{14}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{84} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{5\sqrt{21}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{84} & 0 & 0 \end{bmatrix}$ |
| 433 | symmetry | $\frac{3\sqrt{10}x^2y}{4} - \frac{\sqrt{10}y^3}{4}$ |

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_3^{(1,0;a)}(A,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{21} & -\frac{5\sqrt{6}i}{48} & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{42}i}{336} & 0 \\ -\frac{\sqrt{7}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{42}i}{336} & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{6}i}{48} \\ 0 & 0 & 0 & 0 & -\frac{5\sqrt{210}i}{336} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{84} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{25\sqrt{14}i}{336} & \frac{\sqrt{3}i}{12} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{42} & 0 \\ -\frac{25\sqrt{14}i}{336} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{12} \\ 0 & -\frac{5\sqrt{210}i}{336} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{84} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 434 | symmetry | $\frac{\sqrt{10}x^3}{4} - \frac{3\sqrt{10}xy^2}{4}$ |
| | $\mathbb{M}_3^{(1,0;a)}(A,3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{21} & -\frac{5\sqrt{6}}{48} & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{42}}{336} & 0 \\ -\frac{\sqrt{7}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{42}}{336} & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{6}}{48} \\ 0 & 0 & 0 & 0 & \frac{5\sqrt{210}}{336} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{84} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{25\sqrt{14}}{336} & \frac{\sqrt{3}}{12} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{42} & 0 \\ -\frac{25\sqrt{14}}{336} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{12} \\ 0 & -\frac{5\sqrt{210}}{336} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{84} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 435 | symmetry | $-\frac{\sqrt{3}x^3}{4} - \frac{\sqrt{3}ix^2y}{4} - \frac{\sqrt{3}xy^2}{4} + \sqrt{3}xz^2 - \frac{\sqrt{3}iy^3}{4} + \sqrt{3}iyz^2$ |
| | $\mathbb{M}_3^{(1,0;a)}(E^{(a)},1)$ | $\begin{bmatrix} 0 & 0 & 0 & -\frac{2\sqrt{21}}{63} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{7}}{56} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{63} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{105}}{168} & 0 & 0 \\ 0 & 0 & \frac{5\sqrt{14}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{42} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{5\sqrt{42}}{504} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{5\sqrt{21}}{72} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{35}}{168} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{84} & 0 \end{bmatrix}$ |
| 436 | symmetry | $\frac{\sqrt{30}x^2z}{4} - \frac{\sqrt{30}ixyz}{2} - \frac{\sqrt{30}y^2z}{4}$ |
| | $\mathbb{M}_3^{(1,0;a)}(E^{(a)},2)$ | $\begin{bmatrix} -\frac{\sqrt{21}}{63} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{14}}{56} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{105}}{63} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{42}}{168} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{25\sqrt{42}}{504} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{5\sqrt{210}}{504} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{5\sqrt{35}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{84} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 437 | symmetry | $-\frac{\sqrt{3}x^3}{4} + \frac{\sqrt{3}ix^2y}{4} - \frac{\sqrt{3}xy^2}{4} + \sqrt{3}xz^2 + \frac{\sqrt{3}iy^3}{4} - \sqrt{3}iyz^2$ |

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_3^{(1,0;a)}(E^{(b)},1)$ | $\begin{bmatrix} 0 & \frac{\sqrt{42}}{63} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{105}}{168} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{2\sqrt{21}}{63} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{7}}{56} & 0 & 0 & 0 & 0 \\ -\frac{5\sqrt{35}}{168} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{84} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{5\sqrt{21}}{72} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{5\sqrt{42}}{504} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{5\sqrt{14}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{42} & 0 & 0 & 0 \end{bmatrix}$ |
| 438 | symmetry | $\frac{\sqrt{30}x^2z}{4} + \frac{\sqrt{30}ixyz}{2} - \frac{\sqrt{30}y^2z}{4}$ |
| | $\mathbb{M}_3^{(1,0;a)}(E^{(b)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{63} & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{42}}{168} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{63} & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{14}}{56} & 0 \\ 0 & 0 & 0 & \frac{5\sqrt{35}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{84} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{5\sqrt{210}}{504} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{28} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{25\sqrt{42}}{504} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{28} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{12} \end{bmatrix}$ |
| 439 | symmetry | $z$ |
| | $\mathbb{M}_1^{(1,1;a)}(A)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{10}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{15}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{15}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 440 | symmetry | $\frac{\sqrt{2}x}{2} + \frac{\sqrt{2}iy}{2}$ |
| | $\mathbb{M}_1^{(1,1;a)}(E^{(a)})$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{10}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{30}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 441 | symmetry | $\frac{\sqrt{2}x}{2} - \frac{\sqrt{2}iy}{2}$ |

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_1^{(1,1;a)}(E^{(b)})$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{15}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{30}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{10}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 442 | symmetry | $-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$ |
| | $\mathbb{M}_3^{(1,1;a)}(A,1)$ | $\begin{bmatrix} 0 & 0 & \frac{3}{7} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{3}{7} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{28} & 0 & 0 & 0 \\ 0 & \frac{3\sqrt{3}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{84} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{3\sqrt{2}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{84} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{2}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{84} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{3\sqrt{3}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{84} & 0 & 0 \end{bmatrix}$ |
| 443 | symmetry | $\frac{3\sqrt{10}x^2y}{4} - \frac{\sqrt{10}y^3}{4}$ |
| | $\mathbb{M}_3^{(1,1;a)}(A,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{3i}{7} & \frac{\sqrt{42}i}{112} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{112} & 0 \\ \frac{3i}{7} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{112} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{112} \\ 0 & 0 & 0 & 0 & -\frac{3\sqrt{30}i}{112} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{84} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{15\sqrt{2}i}{112} & -\frac{\sqrt{21}i}{84} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{42} & 0 \\ -\frac{15\sqrt{2}i}{112} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{42} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{84} \\ 0 & -\frac{3\sqrt{30}i}{112} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{84} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 444 | symmetry | $\frac{\sqrt{10}x^3}{4} - \frac{3\sqrt{10}xy^2}{4}$ |
| | $\mathbb{M}_3^{(1,1;a)}(A,3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{3}{7} & \frac{\sqrt{42}}{112} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{112} & 0 \\ \frac{3}{7} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{112} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{112} \\ 0 & 0 & 0 & 0 & \frac{3\sqrt{30}}{112} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{84} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{15\sqrt{2}}{112} & -\frac{\sqrt{21}}{84} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{42} & 0 \\ -\frac{15\sqrt{2}}{112} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{42} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{84} \\ 0 & -\frac{3\sqrt{30}}{112} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{84} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 445 | symmetry | $-\frac{\sqrt{3}x^3}{4} - \frac{\sqrt{3}ix^2y}{4} - \frac{\sqrt{3}xy^2}{4} + \sqrt{3}xz^2 - \frac{\sqrt{3}iy^3}{4} + \sqrt{3}iyz^2$ |

*continued ...*

Table 7

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_3^{(1,1;a)}(E^{(a)},1)$ | $\begin{bmatrix} 0 & 0 & 0 & \frac{2\sqrt{3}}{7} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3}{56} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{7} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{56} & 0 & 0 \\ 0 & 0 & \frac{9\sqrt{2}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{42} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{6}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{5}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{84} & 0 \end{bmatrix}$ |
| 446 | symmetry | $\frac{\sqrt{30}x^2z}{4} - \frac{\sqrt{30}ixyz}{2} - \frac{\sqrt{30}y^2z}{4}$ |
| | $\mathbb{M}_3^{(1,1;a)}(E^{(a)},2)$ | $\begin{bmatrix} \frac{\sqrt{3}}{7} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}}{56} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{15}}{7} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{56} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{5\sqrt{6}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{30}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{3\sqrt{5}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{84} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 447 | symmetry | $-\frac{\sqrt{3}x^3}{4} + \frac{\sqrt{3}ix^2y}{4} - \frac{\sqrt{3}xy^2}{4} + \sqrt{3}xz^2 + \frac{\sqrt{3}iy^3}{4} - \sqrt{3}iyz^2$ |
| | $\mathbb{M}_3^{(1,1;a)}(E^{(b)},1)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{6}}{7} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{56} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{2\sqrt{3}}{7} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3}{56} & 0 & 0 & 0 & 0 \\ -\frac{3\sqrt{5}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{84} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{3}}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{6}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{9\sqrt{2}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{42} & 0 & 0 & 0 \end{bmatrix}$ |
| 448 | symmetry | $\frac{\sqrt{30}x^2z}{4} + \frac{\sqrt{30}ixyz}{2} - \frac{\sqrt{30}y^2z}{4}$ |
| | $\mathbb{M}_3^{(1,1;a)}(E^{(b)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{7} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{56} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{7} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}}{56} & 0 \\ 0 & 0 & 0 & \frac{3\sqrt{5}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{84} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{6}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{28} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{84} \end{bmatrix}$ |

bra: $= \langle \frac{3}{2}, \frac{3}{2}; d|\,, \langle \frac{3}{2}, \frac{1}{2}; d|\,, \langle \frac{3}{2}, -\frac{1}{2}; d|\,, \langle \frac{3}{2}, -\frac{3}{2}; d|\,, \langle \frac{5}{2}, \frac{5}{2}; d|\,, \langle \frac{5}{2}, \frac{3}{2}; d|\,, \langle \frac{5}{2}, \frac{1}{2}; d|\,, \langle \frac{5}{2}, -\frac{1}{2}; d|\,, \langle \frac{5}{2}, -\frac{3}{2}; d|\,, \langle \frac{5}{2}, -\frac{5}{2}; d|$

ket: $= |\frac{3}{2}, \frac{3}{2}; d\rangle\,, |\frac{3}{2}, \frac{1}{2}; d\rangle\,, |\frac{3}{2}, -\frac{1}{2}; d\rangle\,, |\frac{3}{2}, -\frac{3}{2}; d\rangle\,, |\frac{5}{2}, \frac{5}{2}; d\rangle\,, |\frac{5}{2}, \frac{3}{2}; d\rangle\,, |\frac{5}{2}, \frac{1}{2}; d\rangle\,, |\frac{5}{2}, -\frac{1}{2}; d\rangle\,, |\frac{5}{2}, -\frac{3}{2}; d\rangle\,, |\frac{5}{2}, -\frac{5}{2}; d\rangle$

Table 8: (d,d) block.

| No. | multipole | matrix |
|---|---|---|
| 449 | symmetry | 1 |
| | $\mathbb{Q}_0^{(a)}(A)$ | $\begin{bmatrix} \frac{\sqrt{10}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{10}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{10}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{10}}{10} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{10} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{10} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{10} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{10} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{10} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{10} \end{bmatrix}$ |
| 450 | symmetry | $-\frac{x^2}{2} - \frac{y^2}{2} + z^2$ |
| | $\mathbb{Q}_2^{(a)}(A)$ | $\begin{bmatrix} -\frac{\sqrt{7}}{10} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{7}}{35} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{7}}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{70} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{7}}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{70} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{7}}{10} & 0 & 0 & 0 & 0 & \frac{3\sqrt{7}}{35} & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{7} & 0 & 0 & 0 & 0 & 0 \\ -\frac{3\sqrt{7}}{35} & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{35} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{42}}{70} & 0 & 0 & 0 & 0 & \frac{4\sqrt{7}}{35} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{42}}{70} & 0 & 0 & 0 & 0 & \frac{4\sqrt{7}}{35} & 0 & 0 \\ 0 & 0 & 0 & \frac{3\sqrt{7}}{35} & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{35} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{7} \end{bmatrix}$ |
| 451 | symmetry | $\frac{\sqrt{6}xz}{2} + \frac{\sqrt{6}iyz}{2}$ |

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_2^{(a)}(E^{(a)},1)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{14}}{10} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{21}}{70} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{14} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{14}}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{70} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{70} \\ \frac{\sqrt{210}}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{35} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{14}}{70} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{21}}{35} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{21}}{70} & 0 & 0 & 0 & 0 & \frac{2\sqrt{21}}{35} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{35} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 452 | symmetry | $\frac{\sqrt{6}x^2}{4} - \frac{\sqrt{6}ixy}{2} - \frac{\sqrt{6}y^2}{4}$ |
| | $\mathbb{Q}_2^{(a)}(E^{(a)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{35} & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{14}}{10} & 0 & 0 & 0 & 0 & \frac{2\sqrt{14}}{35} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{14}}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{35} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{35} & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{21}}{35} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{21}}{35} & 0 & 0 & 0 & 0 \\ 0 & -\frac{2\sqrt{14}}{35} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{21}}{35} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{70}}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{35} & 0 & 0 \end{bmatrix}$ |
| 453 | symmetry | $\frac{\sqrt{6}xz}{2} - \frac{\sqrt{6}iyz}{2}$ |

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_2^{(a)}(E^{(b)},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{70} & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{14}}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{70} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{14} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{14}}{10} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{21}}{70} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{35} & 0 & 0 & 0 & 0 & 0 \\ -\frac{3\sqrt{21}}{70} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{21}}{35} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{14}}{70} & 0 & 0 & 0 & 0 & \frac{2\sqrt{21}}{35} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{210}}{70} & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{35} & 0 \end{bmatrix}$ |
| 454 | symmetry | $\frac{\sqrt{6}x^2}{4}+\frac{\sqrt{6}ixy}{2}-\frac{\sqrt{6}y^2}{4}$ |
| | $\mathbb{Q}_2^{(a)}(E^{(b)},2)$ | $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{14}}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{35} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{14}}{10} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{14}}{35} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{35} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{70}}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{35} & 0 & 0 & 0 \\ 0 & 0 & \frac{2\sqrt{14}}{35} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{21}}{35} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{21}}{35} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{21}}{35} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{35} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 455 | symmetry | $\frac{3x^4}{8}+\frac{3x^2y^2}{4}-3x^2z^2+\frac{3y^4}{8}-3y^2z^2+z^4$ |

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_4^{(a)}(A,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{35} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{35} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{35} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{35} & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{70} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{35}}{35} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{35}}{70} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{210}}{35} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{35} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{210}}{35} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{35} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{35}}{35} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{35}}{70} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{70} \end{bmatrix}$ |
| 456 | symmetry | $\frac{\sqrt{70}x^3z}{4} - \frac{3\sqrt{70}xy^2z}{4}$ |
| | $\mathbb{Q}_4^{(a)}(A,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{4} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{20} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{20} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{30}}{20} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{10} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{10} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{10} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{30}}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 & 0 \end{bmatrix}$ |
| 457 | symmetry | $\frac{3\sqrt{70}x^2yz}{4} - \frac{\sqrt{70}y^3z}{4}$ |

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_4^{(a)}(A,3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{20} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{20} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{30}i}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{10} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{10} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{30}i}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{10} & 0 & 0 & 0 \end{bmatrix}$ |
| 458 | symmetry | $-\frac{3\sqrt{5}x^3z}{4} - \frac{3\sqrt{5}ix^2yz}{4} - \frac{3\sqrt{5}xy^2z}{4} + \sqrt{5}xz^3 - \frac{3\sqrt{5}iy^3z}{4} + \sqrt{5}iyz^3$ |
| | $\mathbb{Q}_4^{(a)}(E^{(a)},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{14} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{14} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{70} \\ -\frac{\sqrt{35}}{70} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{35} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{21}}{14} & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{14} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{42}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{35} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 459 | symmetry | $\frac{\sqrt{70}x^4}{16} + \frac{\sqrt{70}ix^3y}{4} - \frac{3\sqrt{70}x^2y^2}{8} - \frac{\sqrt{70}ixy^3}{4} + \frac{\sqrt{70}y^4}{16}$ |

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_4^{(a)}(E^{(a)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{5} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{10}}{5} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{10} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{10} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 460 | symmetry | $-\frac{\sqrt{10}x^4}{8}+\frac{\sqrt{10}ix^3y}{4}+\frac{3\sqrt{10}x^2z^2}{4}+\frac{\sqrt{10}ixy^3}{4}-\frac{3\sqrt{10}ixyz^2}{2}+\frac{\sqrt{10}y^4}{8}-\frac{3\sqrt{10}y^2z^2}{4}$ |
| | $\mathbb{Q}_4^{(a)}(E^{(a)},3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{70} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{7} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{3\sqrt{35}}{70} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{7}}{7} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{42}}{14} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{14} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{210}}{70} & 0 & 0 & 0 & 0 & \frac{3\sqrt{35}}{70} & 0 & 0 \end{bmatrix}$ |
| 461 | symmetry | $-\frac{3\sqrt{5}x^3z}{4}+\frac{3\sqrt{5}ix^2yz}{4}-\frac{3\sqrt{5}xy^2z}{4}+\sqrt{5}xz^3+\frac{3\sqrt{5}iy^3z}{4}-\sqrt{5}iyz^3$ |

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_4^{(a)}(E^{(b)},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{70} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{35} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{42}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{21}}{14} & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{35}}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{35} & 0 \end{bmatrix}$ |
| 462 | symmetry | $\frac{\sqrt{70}x^4}{16} - \frac{\sqrt{70}ix^3y}{4} - \frac{3\sqrt{70}x^2y^2}{8} + \frac{\sqrt{70}ixy^3}{4} + \frac{\sqrt{70}y^4}{16}$ |
| | $\mathbb{Q}_4^{(a)}(E^{(b)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{5} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{10} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{10}}{5} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{10} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 463 | symmetry | $-\frac{\sqrt{10}x^4}{8} - \frac{\sqrt{10}ix^3y}{4} + \frac{3\sqrt{10}x^2z^2}{4} - \frac{\sqrt{10}ixy^3}{4} + \frac{3\sqrt{10}ixyz^2}{2} + \frac{\sqrt{10}y^4}{8} - \frac{3\sqrt{10}y^2z^2}{4}$ |

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_4^{(a)}(E^{(b)},3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{7} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{14} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{70} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{210}}{70} & 0 & 0 & 0 & 0 & \frac{3\sqrt{35}}{70} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{42}}{14} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{14} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{7}}{7} & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{14} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{35}}{70} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 464 | symmetry | $-\frac{x^2}{2}-\frac{y^2}{2}+z^2$ |
| | $\mathbb{Q}_2^{(1,-1;a)}(A)$ | $\begin{bmatrix} -\frac{\sqrt{30}}{50} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{30}}{50} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{30}}{50} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{5}}{50} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{30}}{50} & 0 & 0 & 0 & 0 & \frac{3\sqrt{5}}{50} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{30}}{50} & 0 & 0 & 0 & 0 & \frac{3\sqrt{30}}{50} & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{15} & 0 & 0 & 0 & 0 & 0 \\ -\frac{3\sqrt{30}}{50} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{75} & 0 & 0 & 0 & 0 \\ 0 & -\frac{3\sqrt{5}}{50} & 0 & 0 & 0 & 0 & -\frac{4\sqrt{30}}{75} & 0 & 0 & 0 \\ 0 & 0 & \frac{3\sqrt{5}}{50} & 0 & 0 & 0 & 0 & -\frac{4\sqrt{30}}{75} & 0 & 0 \\ 0 & 0 & 0 & \frac{3\sqrt{30}}{50} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{75} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{15} \end{bmatrix}$ |
| 465 | symmetry | $\frac{\sqrt{6}xz}{2}+\frac{\sqrt{6}iyz}{2}$ |

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_2^{(1,-1;a)}(E^{(a)},1)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{15}}{25} & 0 & 0 & 0 & 0 & -\frac{9\sqrt{10}}{100} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{20} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{15}}{25} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{50} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3}{10} \\ \frac{3}{10} & 0 & 0 & 0 & 0 & \frac{2}{5} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{15}}{50} & 0 & 0 & 0 & 0 & \frac{2\sqrt{10}}{25} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{30}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{9\sqrt{10}}{100} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{10}}{25} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2}{5} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 466 | symmetry | $\frac{\sqrt{6}x^2}{4} - \frac{\sqrt{6}ixy}{2} - \frac{\sqrt{6}y^2}{4}$ |
| | $\mathbb{Q}_2^{(1,-1;a)}(E^{(a)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{5} & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{15}}{25} & 0 & 0 & 0 & 0 & \frac{2\sqrt{15}}{25} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{15}}{25} & 0 & 0 & 0 & 0 & \frac{3\sqrt{10}}{50} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{5} & 0 & 0 & 0 & 0 & 0 \\ -\frac{3\sqrt{10}}{50} & 0 & 0 & 0 & 0 & \frac{3\sqrt{10}}{25} & 0 & 0 & 0 & 0 \\ 0 & -\frac{2\sqrt{15}}{25} & 0 & 0 & 0 & 0 & \frac{3\sqrt{10}}{25} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{3}}{5} & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{5} & 0 & 0 \end{bmatrix}$ |
| 467 | symmetry | $\frac{\sqrt{6}xz}{2} - \frac{\sqrt{6}iyz}{2}$ |

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_2^{(1,-1;a)}(E^{(b)},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & \frac{3}{10} & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{15}}{25} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{50} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{20} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{15}}{25} & 0 & 0 & 0 & 0 & -\frac{9\sqrt{10}}{100} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{2}{5} & 0 & 0 & 0 & 0 & 0 \\ -\frac{9\sqrt{10}}{100} & 0 & 0 & 0 & 0 & \frac{2\sqrt{10}}{25} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{30}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{15}}{50} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{10}}{25} & 0 & 0 \\ 0 & 0 & 0 & \frac{3}{10} & 0 & 0 & 0 & 0 & -\frac{2}{5} & 0 \end{bmatrix}$ |
| 468 | symmetry | $\frac{\sqrt{6}x^2}{4} + \frac{\sqrt{6}ixy}{2} - \frac{\sqrt{6}y^2}{4}$ |
| | $\mathbb{Q}_2^{(1,-1;a)}(E^{(b)},2)$ | $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{15}}{25} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}}{50} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{15}}{25} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{15}}{25} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{5} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{3}}{5} & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{5} & 0 & 0 & 0 \\ 0 & 0 & \frac{2\sqrt{15}}{25} & 0 & 0 & 0 & 0 & \frac{3\sqrt{10}}{25} & 0 & 0 \\ 0 & 0 & 0 & \frac{3\sqrt{10}}{50} & 0 & 0 & 0 & 0 & \frac{3\sqrt{10}}{25} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{5} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 469 | symmetry | $\frac{3x^4}{8} + \frac{3x^2y^2}{4} - 3x^2z^2 + \frac{3y^4}{8} - 3y^2z^2 + z^4$ |

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_4^{(1,-1;a)}(A,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{70} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{70} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{70} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{70} & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{35} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{35}}{70} & 0 & 0 & 0 & 0 & \frac{3\sqrt{35}}{35} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{210}}{70} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{35}}{35} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{210}}{70} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{35}}{35} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{35}}{70} & 0 & 0 & 0 & 0 & \frac{3\sqrt{35}}{35} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{35} \end{bmatrix}$ |
| 470 | symmetry | $\frac{\sqrt{70}x^3z}{4} - \frac{3\sqrt{70}xy^2z}{4}$ |
| | $\mathbb{Q}_4^{(1,-1;a)}(A,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{8} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{40} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{40} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{8} & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{30}}{40} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{5} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{2}}{8} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{5} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{5} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{2}}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{30}}{40} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{5} & 0 & 0 & 0 \end{bmatrix}$ |
| 471 | symmetry | $\frac{3\sqrt{70}x^2yz}{4} - \frac{\sqrt{70}y^3z}{4}$ |

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_4^{(1,-1;a)}(A,3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{8} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{40} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{40} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{8} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{30}i}{40} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{5} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{2}i}{8} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{5} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{5} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{2}i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{30}i}{40} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{5} & 0 & 0 & 0 \end{bmatrix}$ |
| 472 | symmetry | $-\frac{3\sqrt{5}x^3z}{4} - \frac{3\sqrt{5}ix^2yz}{4} - \frac{3\sqrt{5}xy^2z}{4} + \sqrt{5}xz^3 - \frac{3\sqrt{5}iy^3z}{4} + \sqrt{5}iyz^3$ |
| | $\mathbb{Q}_4^{(1,-1;a)}(E^{(a)},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{28} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{140} \\ -\frac{\sqrt{35}}{140} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{35}}{35} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{21}}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{7} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{42}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{14}}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{7} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{35}}{35} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 473 | symmetry | $\frac{\sqrt{70}x^4}{16} + \frac{\sqrt{70}ix^3y}{4} - \frac{3\sqrt{70}x^2y^2}{8} - \frac{\sqrt{70}ixy^3}{4} + \frac{\sqrt{70}y^4}{16}$ |

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_4^{(1,-1;a)}(E^{(a)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{10} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{10}}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{5} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{5} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 474 | symmetry | $-\frac{\sqrt{10}x^4}{8}+\frac{\sqrt{10}ix^3y}{4}+\frac{3\sqrt{10}x^2z^2}{4}+\frac{\sqrt{10}ixy^3}{4}-\frac{3\sqrt{10}ixyz^2}{2}+\frac{\sqrt{10}y^4}{8}-\frac{3\sqrt{10}y^2z^2}{4}$ |
| | $\mathbb{Q}_4^{(1,-1;a)}(E^{(a)},3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{140} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{3\sqrt{35}}{35} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{7} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{42}}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{7} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{210}}{140} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{35}}{35} & 0 & 0 \end{bmatrix}$ |
| 475 | symmetry | $-\frac{3\sqrt{5}x^3z}{4}+\frac{3\sqrt{5}ix^2yz}{4}-\frac{3\sqrt{5}xy^2z}{4}+\sqrt{5}xz^3+\frac{3\sqrt{5}iy^3z}{4}-\sqrt{5}iyz^3$ |

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_4^{(1,-1;a)}(E^{(b)},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{140} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{2\sqrt{35}}{35} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{14}}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{7} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{42}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{21}}{28} & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{7} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{35}}{140} & 0 & 0 & 0 & 0 & \frac{2\sqrt{35}}{35} & 0 \end{bmatrix}$ |
| 476 | symmetry | $\frac{\sqrt{70}x^4}{16} - \frac{\sqrt{70}ix^3y}{4} - \frac{3\sqrt{70}x^2y^2}{8} + \frac{\sqrt{70}ixy^3}{4} + \frac{\sqrt{70}y^4}{16}$ |
| | $\mathbb{Q}_4^{(1,-1;a)}(E^{(b)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{10} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{5} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{10}}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{5} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 477 | symmetry | $-\frac{\sqrt{10}x^4}{8} - \frac{\sqrt{10}ix^3y}{4} + \frac{3\sqrt{10}x^2z^2}{4} - \frac{\sqrt{10}ixy^3}{4} + \frac{3\sqrt{10}ixyz^2}{2} + \frac{\sqrt{10}y^4}{8} - \frac{3\sqrt{10}y^2z^2}{4}$ |

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_4^{(1,-1;a)}(E^{(b)},3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{14} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{28} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{140} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{210}}{140} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{35}}{35} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{42}}{28} & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{7} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{7} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{35}}{35} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 478 | symmetry | 1 |
| | $\mathbb{Q}_0^{(1,1;a)}(A)$ | $\begin{bmatrix} -\frac{\sqrt{15}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{15}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{15}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{15}}{10} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{15} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{15} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{15} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{15} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{15} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{15} \end{bmatrix}$ |
| 479 | symmetry | $-\frac{x^2}{2}-\frac{y^2}{2}+z^2$ |

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_2^{(1,1;a)}(A)$ | $\begin{bmatrix} \frac{\sqrt{105}}{25} & 0 & 0 & 0 & 0 & -\frac{4\sqrt{105}}{175} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{105}}{25} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{70}}{175} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{105}}{25} & 0 & 0 & 0 & 0 & \frac{2\sqrt{70}}{175} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{105}}{25} & 0 & 0 & 0 & 0 & \frac{4\sqrt{105}}{175} & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{70} & 0 & 0 & 0 & 0 & 0 \\ -\frac{4\sqrt{105}}{175} & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{350} & 0 & 0 & 0 & 0 \\ 0 & -\frac{2\sqrt{70}}{175} & 0 & 0 & 0 & 0 & \frac{2\sqrt{105}}{175} & 0 & 0 & 0 \\ 0 & 0 & \frac{2\sqrt{70}}{175} & 0 & 0 & 0 & 0 & \frac{2\sqrt{105}}{175} & 0 & 0 \\ 0 & 0 & 0 & \frac{4\sqrt{105}}{175} & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{350} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{70} \end{bmatrix}$ |
| 480 | symmetry | $\frac{\sqrt{6}xz}{2} + \frac{\sqrt{6}iyz}{2}$ |
| | $\mathbb{Q}_2^{(1,1;a)}(E^{(a)},1)$ | $\begin{bmatrix} 0 & \frac{\sqrt{210}}{25} & 0 & 0 & 0 & 0 & -\frac{6\sqrt{35}}{175} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{105}}{105} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{210}}{25} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{210}}{525} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{14}}{35} \\ \frac{2\sqrt{14}}{35} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{14}}{70} & 0 & 0 & 0 & 0 \\ 0 & -\frac{2\sqrt{210}}{525} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{35}}{175} & 0 & 0 & 0 \\ 0 & 0 & -\frac{2\sqrt{105}}{105} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{6\sqrt{35}}{175} & 0 & 0 & 0 & 0 & \frac{3\sqrt{35}}{175} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{14}}{70} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 481 | symmetry | $\frac{\sqrt{6}x^2}{4} - \frac{\sqrt{6}ixy}{2} - \frac{\sqrt{6}y^2}{4}$ |

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_2^{(1,1;a)}(E^{(a)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{4\sqrt{42}}{105} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{210}}{25} & 0 & 0 & 0 & 0 & \frac{8\sqrt{210}}{525} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{210}}{25} & 0 & 0 & 0 & 0 & \frac{4\sqrt{35}}{175} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{3\sqrt{7}}{70} & 0 & 0 & 0 & 0 & 0 \\ -\frac{4\sqrt{35}}{175} & 0 & 0 & 0 & 0 & -\frac{9\sqrt{35}}{350} & 0 & 0 & 0 & 0 \\ 0 & -\frac{8\sqrt{210}}{525} & 0 & 0 & 0 & 0 & -\frac{9\sqrt{35}}{350} & 0 & 0 & 0 \\ 0 & 0 & -\frac{4\sqrt{42}}{105} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{7}}{70} & 0 & 0 \end{bmatrix}$ |
| 482 | symmetry | $\frac{\sqrt{6}xz}{2} - \frac{\sqrt{6}iyz}{2}$ |
| | $\mathbb{Q}_2^{(1,1;a)}(E^{(b)},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & \frac{2\sqrt{14}}{35} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{210}}{25} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{210}}{525} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{105}}{105} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{210}}{25} & 0 & 0 & 0 & 0 & -\frac{6\sqrt{35}}{175} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{3\sqrt{14}}{70} & 0 & 0 & 0 & 0 & 0 \\ -\frac{6\sqrt{35}}{175} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{35}}{175} & 0 & 0 & 0 & 0 \\ 0 & -\frac{2\sqrt{105}}{105} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{2\sqrt{210}}{525} & 0 & 0 & 0 & 0 & \frac{3\sqrt{35}}{175} & 0 & 0 \\ 0 & 0 & 0 & \frac{2\sqrt{14}}{35} & 0 & 0 & 0 & 0 & \frac{3\sqrt{14}}{70} & 0 \end{bmatrix}$ |
| 483 | symmetry | $\frac{\sqrt{6}x^2}{4} + \frac{\sqrt{6}ixy}{2} - \frac{\sqrt{6}y^2}{4}$ |

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_2^{(1,1;a)}(E^{(b)},2)$ | $\begin{bmatrix} 0 & 0 & \frac{\sqrt{210}}{25} & 0 & 0 & 0 & 0 & -\frac{4\sqrt{35}}{175} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{210}}{25} & 0 & 0 & 0 & 0 & -\frac{8\sqrt{210}}{525} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{4\sqrt{42}}{105} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{4\sqrt{42}}{105} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{7}}{70} & 0 & 0 & 0 \\ 0 & 0 & \frac{8\sqrt{210}}{525} & 0 & 0 & 0 & 0 & -\frac{9\sqrt{35}}{350} & 0 & 0 \\ 0 & 0 & 0 & \frac{4\sqrt{35}}{175} & 0 & 0 & 0 & 0 & -\frac{9\sqrt{35}}{350} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{7}}{70} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 484 | symmetry | $z$ |
| | $\mathbb{G}_1^{(1,0;a)}(A)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{10} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{10} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{10} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{10} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{10}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{15}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{15}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{10}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 485 | symmetry | $\frac{\sqrt{2}x}{2} + \frac{\sqrt{2}iy}{2}$ |

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_1^{(1,0;a)}(E^{(a)})$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{20} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{20} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{10} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{2} \\ -\frac{i}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{15}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{30}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{10}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 486 | symmetry | $\frac{\sqrt{2}x}{2} - \frac{\sqrt{2}iy}{2}$ |
| | $\mathbb{G}_1^{(1,0;a)}(E^{(b)})$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & \frac{i}{2} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{10} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{20} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{20} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{10}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{30}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{15}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{i}{2} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 487 | symmetry | $-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$ |

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_3^{(1,0;a)}(A,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{10} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{10} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{10} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{10} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{15}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{10}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{10}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{15}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 488 | symmetry | $\frac{3\sqrt{10}x^2y}{4} - \frac{\sqrt{10}y^3}{4}$ |
| | $\mathbb{G}_3^{(1,0;a)}(A,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{8} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{8} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{8} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{8} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{10}}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{6}}{8} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{6}}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{10}}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 489 | symmetry | $\frac{\sqrt{10}x^3}{4} - \frac{3\sqrt{10}xy^2}{4}$ |

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_3^{(1,0;a)}(A,3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{8} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{8} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{8} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{8} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{10}i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{6}i}{8} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{6}i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{10}i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 490 | symmetry | $-\frac{\sqrt{3}x^3}{4} - \frac{\sqrt{3}ix^2y}{4} - \frac{\sqrt{3}xy^2}{4} + \sqrt{3}xz^2 - \frac{\sqrt{3}iy^3}{4} + \sqrt{3}iyz^2$ |
| | $\mathbb{G}_3^{(1,0;a)}(E^{(a)},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{10}i}{20} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{60} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{7\sqrt{15}i}{60} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{4} \\ \frac{i}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{7\sqrt{15}i}{60} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{30}i}{60} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{3\sqrt{10}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 491 | symmetry | $\frac{\sqrt{30}x^2z}{4} - \frac{\sqrt{30}ixyz}{2} - \frac{\sqrt{30}y^2z}{4}$ |

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_3^{(1,0;a)}(E^{(a)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{12} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{12} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{i}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{6}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{30}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 492 | symmetry | $-\frac{\sqrt{3}x^3}{4}+\frac{\sqrt{3}ix^2y}{4}-\frac{\sqrt{3}xy^2}{4}+\sqrt{3}xz^2+\frac{\sqrt{3}iy^3}{4}-\sqrt{3}iyz^2$ |
| | $\mathbb{G}_3^{(1,0;a)}(E^{(b)},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{i}{4} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{7\sqrt{15}i}{60} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{60} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}i}{20} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{3\sqrt{10}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{30}i}{60} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{7\sqrt{15}i}{60} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{i}{4} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 493 | symmetry | $\frac{\sqrt{30}x^2z}{4}+\frac{\sqrt{30}ixyz}{2}-\frac{\sqrt{30}y^2z}{4}$ |

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_3^{(1,0;a)}(E^{(b)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{2} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{12} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{12} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{30}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{6}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{i}{2} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 494 | symmetry | $-\frac{x^2}{2}-\frac{y^2}{2}+z^2$ |
| | $\mathbb{T}_2^{(1,0;a)}(A)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{14} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{42}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{42}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 495 | symmetry | $\frac{\sqrt{6}xz}{2}+\frac{\sqrt{6}iyz}{2}$ |

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_2^{(1,0;a)}(E^{(a)},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{14}i}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{42}i}{84} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{42} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{14} \\ \frac{\sqrt{35}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{5\sqrt{42}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{14}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 496 | symmetry | $\frac{\sqrt{6}x^2}{4} - \frac{\sqrt{6}ixy}{2} - \frac{\sqrt{6}y^2}{4}$ |
| | $\mathbb{T}_2^{(1,0;a)}(E^{(a)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{21} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{21}i}{21} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{14}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{2\sqrt{21}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{105}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 497 | symmetry | $\frac{\sqrt{6}xz}{2} - \frac{\sqrt{6}iyz}{2}$ |

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_2^{(1,0;a)}(E^{(b)},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{14} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{42}i}{84} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{14}i}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{3\sqrt{14}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{5\sqrt{42}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{35}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 498 | symmetry | $\frac{\sqrt{6}x^2}{4}+\frac{\sqrt{6}ixy}{2}-\frac{\sqrt{6}y^2}{4}$ |
| | $\mathbb{T}_2^{(1,0;a)}(E^{(b)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{14} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{21}i}{21} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{21} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{105}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{2\sqrt{21}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{14}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 499 | symmetry | $\frac{3x^4}{8}+\frac{3x^2y^2}{4}-3x^2z^2+\frac{3y^4}{8}-3y^2z^2+z^4$ |

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_4^{(1,0;a)}(A,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{14} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{42}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{42}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 500 | symmetry | $\frac{\sqrt{70}x^3z}{4} - \frac{3\sqrt{70}xy^2z}{4}$ |
| | $\mathbb{T}_4^{(1,0;a)}(A,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{8} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{8} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{8} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{8} & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{6}i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{10}i}{8} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{10}i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{6}i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 501 | symmetry | $\frac{3\sqrt{70}x^2yz}{4} - \frac{\sqrt{70}y^3z}{4}$ |

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_4^{(1,0;a)}(A,3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{8} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{8} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{8} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{8} & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{6}}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{10}}{8} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{10}}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{6}}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 502 | symmetry | $-\frac{3\sqrt{5}x^3z}{4} - \frac{3\sqrt{5}ix^2yz}{4} - \frac{3\sqrt{5}xy^2z}{4} + \sqrt{5}xz^3 - \frac{3\sqrt{5}iy^3z}{4} + \sqrt{5}iyz^3$ |
| | $\mathbb{T}_4^{(1,0;a)}(E^{(a)},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{28} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{28} \\ -\frac{\sqrt{7}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{105}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{210}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{70}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 503 | symmetry | $\frac{\sqrt{70}x^4}{16} + \frac{\sqrt{70}ix^3y}{4} - \frac{3\sqrt{70}x^2y^2}{8} - \frac{\sqrt{70}ixy^3}{4} + \frac{\sqrt{70}y^4}{16}$ |

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_4^{(1,0;a)}(E^{(a)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{2} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{2}i}{2} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 504 | symmetry | $-\frac{\sqrt{10}x^4}{8}+\frac{\sqrt{10}ix^3y}{4}+\frac{3\sqrt{10}x^2z^2}{4}+\frac{\sqrt{10}ixy^3}{4}-\frac{3\sqrt{10}ixyz^2}{2}+\frac{\sqrt{10}y^4}{8}-\frac{3\sqrt{10}y^2z^2}{4}$ |
| | $\mathbb{T}_4^{(1,0;a)}(E^{(a)},3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{35}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{210}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{42}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 505 | symmetry | $-\frac{3\sqrt{5}x^3z}{4}+\frac{3\sqrt{5}ix^2yz}{4}-\frac{3\sqrt{5}xy^2z}{4}+\sqrt{5}xz^3+\frac{3\sqrt{5}iy^3z}{4}-\sqrt{5}iyz^3$ |

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_4^{(1,0;a)}(E^{(b)},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{70}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{210}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{105}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{7}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 506 | symmetry | $\frac{\sqrt{70}x^4}{16} - \frac{\sqrt{70}ix^3y}{4} - \frac{3\sqrt{70}x^2y^2}{8} + \frac{\sqrt{70}ixy^3}{4} + \frac{\sqrt{70}y^4}{16}$ |
| | $\mathbb{T}_4^{(1,0;a)}(E^{(b)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{2} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{2}i}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 507 | symmetry | $-\frac{\sqrt{10}x^4}{8} - \frac{\sqrt{10}ix^3y}{4} + \frac{3\sqrt{10}x^2z^2}{4} - \frac{\sqrt{10}ixy^3}{4} + \frac{3\sqrt{10}ixyz^2}{2} + \frac{\sqrt{10}y^4}{8} - \frac{3\sqrt{10}y^2z^2}{4}$ |

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_4^{(1,0;a)}(E^{(b)},3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{14} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{28} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{28} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{42}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{210}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{35}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 508 | symmetry | $z$ |
| | $\mathbb{M}_1^{(a)}(A)$ | $\begin{bmatrix} \frac{9\sqrt{5}}{50} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{25} & 0 & 0 & 0 & 0 \\ 0 & \frac{3\sqrt{5}}{50} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{50} & 0 & 0 & 0 \\ 0 & 0 & -\frac{3\sqrt{5}}{50} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{50} & 0 & 0 \\ 0 & 0 & 0 & -\frac{9\sqrt{5}}{50} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{25} & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{5} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{5}}{25} & 0 & 0 & 0 & 0 & \frac{3\sqrt{5}}{25} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{30}}{50} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{25} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{30}}{50} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{25} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{5}}{25} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{5}}{25} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{5} \end{bmatrix}$ |
| 509 | symmetry | $\frac{\sqrt{2}x}{2} + \frac{\sqrt{2}iy}{2}$ |

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_1^{(a)}(E^{(a)})$ | $\begin{bmatrix} 0 & \frac{3\sqrt{30}}{50} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{50} & 0 & 0 & 0 \\ 0 & 0 & \frac{3\sqrt{10}}{25} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{50} & 0 & 0 \\ 0 & 0 & 0 & \frac{3\sqrt{30}}{50} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{50} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{10} \\ -\frac{\sqrt{2}}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{5} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{30}}{50} & 0 & 0 & 0 & 0 & \frac{4\sqrt{5}}{25} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{15}}{50} & 0 & 0 & 0 & 0 & \frac{3\sqrt{10}}{25} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{5}}{50} & 0 & 0 & 0 & 0 & \frac{4\sqrt{5}}{25} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{5} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 510 | symmetry | $\frac{\sqrt{2}x}{2} - \frac{\sqrt{2}iy}{2}$ |
| | $\mathbb{M}_1^{(a)}(E^{(b)})$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{10} & 0 & 0 & 0 & 0 & 0 \\ \frac{3\sqrt{30}}{50} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{50} & 0 & 0 & 0 & 0 \\ 0 & \frac{3\sqrt{10}}{25} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{50} & 0 & 0 & 0 \\ 0 & 0 & \frac{3\sqrt{30}}{50} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{50} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{5} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{5}}{50} & 0 & 0 & 0 & 0 & \frac{4\sqrt{5}}{25} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{15}}{50} & 0 & 0 & 0 & 0 & \frac{3\sqrt{10}}{25} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{30}}{50} & 0 & 0 & 0 & 0 & \frac{4\sqrt{5}}{25} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{2}}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{5} & 0 \end{bmatrix}$ |
| 511 | symmetry | $-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$ |

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_3^{(a)}(A,1)$ | $\begin{bmatrix} -\frac{\sqrt{5}}{25} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{5}}{25} & 0 & 0 & 0 & 0 \\ 0 & \frac{3\sqrt{5}}{25} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{25} & 0 & 0 & 0 \\ 0 & 0 & -\frac{3\sqrt{5}}{25} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{25} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{5}}{25} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{5}}{25} & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 & 0 & 0 & 0 \\ -\frac{3\sqrt{5}}{25} & 0 & 0 & 0 & 0 & \frac{7\sqrt{5}}{50} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{30}}{25} & 0 & 0 & 0 & 0 & \frac{2\sqrt{5}}{25} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{30}}{25} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{5}}{25} & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{5}}{25} & 0 & 0 & 0 & 0 & -\frac{7\sqrt{5}}{50} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{10} \end{bmatrix}$ |
| 512 | symmetry | $\frac{3\sqrt{10}x^2y}{4} - \frac{\sqrt{10}y^3}{4}$ |
| | $\mathbb{M}_3^{(a)}(A,2)$ | $\begin{bmatrix} 0 & 0 & 0 & \frac{\sqrt{2}i}{5} & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}i}{20} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{20} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{20} & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{2}i}{5} & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}i}{20} & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{30}i}{20} & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{10} & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{2}i}{20} & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{5} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{10} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 & 0 \\ -\frac{3\sqrt{2}i}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{5} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{30}i}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{10} & 0 & 0 & 0 \end{bmatrix}$ |
| 513 | symmetry | $\frac{\sqrt{10}x^3}{4} - \frac{3\sqrt{10}xy^2}{4}$ |

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_3^{(a)}(A,3)$ | $\begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{2}}{5} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}}{20} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{20} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{20} & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{2}}{5} & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}}{20} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{30}}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 \\ 0 & 0 & 0 & \frac{3\sqrt{2}}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{5} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{10} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 & 0 & 0 & 0 \\ -\frac{3\sqrt{2}}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{5} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{30}}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 & 0 \end{bmatrix}$ |
| 514 | symmetry | $-\frac{\sqrt{3}x^3}{4} - \frac{\sqrt{3}ix^2y}{4} - \frac{\sqrt{3}xy^2}{4} + \sqrt{3}xz^2 - \frac{\sqrt{3}iy^3}{4} + \sqrt{3}iyz^2$ |
| | $\mathbb{M}_3^{(a)}(E^{(a)},1)$ | $\begin{bmatrix} 0 & -\frac{2\sqrt{5}}{25} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{30}}{50} & 0 & 0 & 0 \\ 0 & 0 & \frac{2\sqrt{15}}{25} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{50} & 0 & 0 \\ 0 & 0 & 0 & -\frac{2\sqrt{5}}{25} & 0 & 0 & 0 & 0 & \frac{7\sqrt{5}}{50} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{10} \\ \frac{\sqrt{3}}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{5} & 0 & 0 & 0 & 0 \\ 0 & -\frac{7\sqrt{5}}{50} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{50} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{10}}{50} & 0 & 0 & 0 & 0 & \frac{2\sqrt{15}}{25} & 0 & 0 \\ 0 & 0 & 0 & \frac{3\sqrt{30}}{50} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{50} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{5} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 515 | symmetry | $\frac{\sqrt{30}x^2z}{4} - \frac{\sqrt{30}ixyz}{2} - \frac{\sqrt{30}y^2z}{4}$ |

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_3^{(a)}(E^{(a)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{10} & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{2}}{5} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{10} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{2}}{5} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{5} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{10} & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{3}}{5} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{10} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{2}}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{10} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{10}}{10} & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{10} & 0 & 0 \end{bmatrix}$ |
| 516 | symmetry | $-\frac{\sqrt{3}x^3}{4}+\frac{\sqrt{3}ix^2y}{4}-\frac{\sqrt{3}xy^2}{4}+\sqrt{3}xz^2+\frac{\sqrt{3}iy^3}{4}-\sqrt{3}iyz^2$ |
| | $\mathbb{M}_3^{(a)}(E^{(b)},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{10} & 0 & 0 & 0 & 0 & 0 \\ -\frac{2\sqrt{5}}{25} & 0 & 0 & 0 & 0 & -\frac{7\sqrt{5}}{50} & 0 & 0 & 0 & 0 \\ 0 & \frac{2\sqrt{15}}{25} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{50} & 0 & 0 & 0 \\ 0 & 0 & -\frac{2\sqrt{5}}{25} & 0 & 0 & 0 & 0 & \frac{3\sqrt{30}}{50} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{5} & 0 & 0 & 0 & 0 & 0 \\ -\frac{3\sqrt{30}}{50} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{50} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{10}}{50} & 0 & 0 & 0 & 0 & \frac{2\sqrt{15}}{25} & 0 & 0 & 0 \\ 0 & 0 & \frac{7\sqrt{5}}{50} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{50} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{3}}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{5} & 0 \end{bmatrix}$ |
| 517 | symmetry | $\frac{\sqrt{30}x^2z}{4}+\frac{\sqrt{30}ixyz}{2}-\frac{\sqrt{30}y^2z}{4}$ |

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_3^{(a)}(E^{(b)},2)$ | $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{2}}{5} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{5} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{2}}{5} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{10} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{10} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{10}}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{10} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{2}}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{10} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{3}}{5} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{10} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{10} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 518 | symmetry | $z$ |
| | $\mathbb{M}_1^{(1,-1;a)}(A)$ | $\begin{bmatrix} -\frac{3\sqrt{10}}{50} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{10}}{25} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{10}}{50} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{15}}{25} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{10}}{50} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{15}}{25} & 0 & 0 \\ 0 & 0 & 0 & \frac{3\sqrt{10}}{50} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{10}}{25} & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{10} & 0 & 0 & 0 & 0 & 0 \\ -\frac{2\sqrt{10}}{25} & 0 & 0 & 0 & 0 & \frac{3\sqrt{10}}{50} & 0 & 0 & 0 & 0 \\ 0 & -\frac{2\sqrt{15}}{25} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{50} & 0 & 0 & 0 \\ 0 & 0 & -\frac{2\sqrt{15}}{25} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{50} & 0 & 0 \\ 0 & 0 & 0 & -\frac{2\sqrt{10}}{25} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}}{50} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{10} \end{bmatrix}$ |
| 519 | symmetry | $\frac{\sqrt{2}x}{2}+\frac{\sqrt{2}iy}{2}$ |

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_1^{(1,-1;a)}(E^{(a)})$ | $\begin{bmatrix} 0 & -\frac{\sqrt{15}}{25} & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{25} & 0 & 0 & 0 \\ 0 & 0 & -\frac{2\sqrt{5}}{25} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{25} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{15}}{25} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{15}}{25} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2}{5} \\ \frac{2}{5} & 0 & 0 & 0 & 0 & \frac{1}{5} & 0 & 0 & 0 & 0 \\ 0 & \frac{2\sqrt{15}}{25} & 0 & 0 & 0 & 0 & \frac{2\sqrt{10}}{25} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{30}}{25} & 0 & 0 & 0 & 0 & \frac{3\sqrt{5}}{25} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{10}}{25} & 0 & 0 & 0 & 0 & \frac{2\sqrt{10}}{25} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{5} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 520 | symmetry | $\frac{\sqrt{2}x}{2} - \frac{\sqrt{2}iy}{2}$ |
| | $\mathbb{M}_1^{(1,-1;a)}(E^{(b)})$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & \frac{2}{5} & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{15}}{25} & 0 & 0 & 0 & 0 & \frac{2\sqrt{15}}{25} & 0 & 0 & 0 & 0 \\ 0 & -\frac{2\sqrt{5}}{25} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{25} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{15}}{25} & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{25} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{5} & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{10}}{25} & 0 & 0 & 0 & 0 & \frac{2\sqrt{10}}{25} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{30}}{25} & 0 & 0 & 0 & 0 & \frac{3\sqrt{5}}{25} & 0 & 0 & 0 \\ 0 & 0 & -\frac{2\sqrt{15}}{25} & 0 & 0 & 0 & 0 & \frac{2\sqrt{10}}{25} & 0 & 0 \\ 0 & 0 & 0 & -\frac{2}{5} & 0 & 0 & 0 & 0 & \frac{1}{5} & 0 \end{bmatrix}$ |
| 521 | symmetry | $-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$ |

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_3^{(1,-1;a)}(A,1)$ | $\begin{bmatrix} \frac{\sqrt{105}}{350} & 0 & 0 & 0 & 0 & \frac{4\sqrt{105}}{175} & 0 & 0 & 0 & 0 \\ 0 & -\frac{3\sqrt{105}}{350} & 0 & 0 & 0 & 0 & -\frac{4\sqrt{70}}{175} & 0 & 0 & 0 \\ 0 & 0 & \frac{3\sqrt{105}}{350} & 0 & 0 & 0 & 0 & -\frac{4\sqrt{70}}{175} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{105}}{350} & 0 & 0 & 0 & 0 & \frac{4\sqrt{105}}{175} & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{35} & 0 & 0 & 0 & 0 & 0 \\ \frac{4\sqrt{105}}{175} & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{25} & 0 & 0 & 0 & 0 \\ 0 & -\frac{4\sqrt{70}}{175} & 0 & 0 & 0 & 0 & \frac{4\sqrt{105}}{175} & 0 & 0 & 0 \\ 0 & 0 & -\frac{4\sqrt{70}}{175} & 0 & 0 & 0 & 0 & -\frac{4\sqrt{105}}{175} & 0 & 0 \\ 0 & 0 & 0 & \frac{4\sqrt{105}}{175} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{25} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{35} \end{bmatrix}$ |
| 522 | symmetry | $\frac{3\sqrt{10}x^2y}{4} - \frac{\sqrt{10}y^3}{4}$ |
| | $\mathbb{M}_3^{(1,-1;a)}(A,2)$ | $\begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{42}i}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{35} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{35} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{35} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{42}i}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{35} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{70}i}{35} & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{35} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{42}i}{35} & 0 & 0 & 0 & 0 & \frac{2\sqrt{42}i}{35} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{35} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{35} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{42}i}{35} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{42}i}{35} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{70}i}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{35} & 0 & 0 & 0 \end{bmatrix}$ |
| 523 | symmetry | $\frac{\sqrt{10}x^3}{4} - \frac{3\sqrt{10}xy^2}{4}$ |

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_3^{(1,-1;a)}(A,3)$ | $\begin{bmatrix} 0 & 0 & 0 & \frac{\sqrt{42}}{70} & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{35} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{35} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{35} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{42}}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{35} & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{70}}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{35} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{42}}{35} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{42}}{35} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{35} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{35} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{42}}{35} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{42}}{35} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{70}}{35} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{35} & 0 & 0 & 0 \end{bmatrix}$ |
| 524 | symmetry | $-\frac{\sqrt{3}x^3}{4} - \frac{\sqrt{3}ix^2y}{4} - \frac{\sqrt{3}xy^2}{4} + \sqrt{3}xz^2 - \frac{\sqrt{3}iy^3}{4} + \sqrt{3}iyz^2$ |
| | $\mathbb{M}_3^{(1,-1;a)}(E^{(a)},1)$ | $\begin{bmatrix} 0 & \frac{\sqrt{105}}{175} & 0 & 0 & 0 & 0 & \frac{6\sqrt{70}}{175} & 0 & 0 & 0 \\ 0 & 0 & -\frac{3\sqrt{35}}{175} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{210}}{525} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{105}}{175} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{105}}{75} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{7}}{35} \\ -\frac{2\sqrt{7}}{35} & 0 & 0 & 0 & 0 & -\frac{6\sqrt{7}}{35} & 0 & 0 & 0 & 0 \\ 0 & \frac{2\sqrt{105}}{75} & 0 & 0 & 0 & 0 & \frac{3\sqrt{70}}{175} & 0 & 0 & 0 \\ 0 & 0 & \frac{2\sqrt{210}}{525} & 0 & 0 & 0 & 0 & \frac{12\sqrt{35}}{175} & 0 & 0 \\ 0 & 0 & 0 & -\frac{6\sqrt{70}}{175} & 0 & 0 & 0 & 0 & \frac{3\sqrt{70}}{175} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{6\sqrt{7}}{35} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 525 | symmetry | $\frac{\sqrt{30}x^2z}{4} - \frac{\sqrt{30}ixyz}{2} - \frac{\sqrt{30}y^2z}{4}$ |

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_3^{(1,-1;a)}(E^{(a)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{2\sqrt{210}}{105} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{42}}{70} & 0 & 0 & 0 & 0 & \frac{2\sqrt{42}}{105} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{42}}{70} & 0 & 0 & 0 & 0 & \frac{4\sqrt{7}}{35} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{3\sqrt{35}}{35} & 0 & 0 & 0 & 0 & 0 \\ \frac{4\sqrt{7}}{35} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{7}}{35} & 0 & 0 & 0 & 0 \\ 0 & \frac{2\sqrt{42}}{105} & 0 & 0 & 0 & 0 & \frac{3\sqrt{7}}{35} & 0 & 0 & 0 \\ 0 & 0 & -\frac{2\sqrt{210}}{105} & 0 & 0 & 0 & 0 & \frac{3\sqrt{35}}{35} & 0 & 0 \end{bmatrix}$ |
| 526 | symmetry | $-\frac{\sqrt{3}x^3}{4}+\frac{\sqrt{3}ix^2y}{4}-\frac{\sqrt{3}xy^2}{4}+\sqrt{3}xz^2+\frac{\sqrt{3}iy^3}{4}-\sqrt{3}iyz^2$ |
| | $\mathbb{M}_3^{(1,-1;a)}(E^{(b)},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{2\sqrt{7}}{35} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{105}}{175} & 0 & 0 & 0 & 0 & \frac{2\sqrt{105}}{75} & 0 & 0 & 0 & 0 \\ 0 & -\frac{3\sqrt{35}}{175} & 0 & 0 & 0 & 0 & \frac{2\sqrt{210}}{525} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{105}}{175} & 0 & 0 & 0 & 0 & -\frac{6\sqrt{70}}{175} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{6\sqrt{7}}{35} & 0 & 0 & 0 & 0 & 0 \\ \frac{6\sqrt{70}}{175} & 0 & 0 & 0 & 0 & \frac{3\sqrt{70}}{175} & 0 & 0 & 0 & 0 \\ 0 & -\frac{2\sqrt{210}}{525} & 0 & 0 & 0 & 0 & \frac{12\sqrt{35}}{175} & 0 & 0 & 0 \\ 0 & 0 & -\frac{2\sqrt{105}}{75} & 0 & 0 & 0 & 0 & \frac{3\sqrt{70}}{175} & 0 & 0 \\ 0 & 0 & 0 & \frac{2\sqrt{7}}{35} & 0 & 0 & 0 & 0 & -\frac{6\sqrt{7}}{35} & 0 \end{bmatrix}$ |
| 527 | symmetry | $\frac{\sqrt{30}x^2z}{4}+\frac{\sqrt{30}ixyz}{2}-\frac{\sqrt{30}y^2z}{4}$ |

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_3^{(1,-1;a)}(E^{(b)},2)$ | $\begin{bmatrix} 0 & 0 & \frac{\sqrt{42}}{70} & 0 & 0 & 0 & 0 & \frac{4\sqrt{7}}{35} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{42}}{70} & 0 & 0 & 0 & 0 & \frac{2\sqrt{42}}{105} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{210}}{105} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{2\sqrt{210}}{105} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{35}}{35} & 0 & 0 & 0 \\ 0 & 0 & \frac{2\sqrt{42}}{105} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{7}}{35} & 0 & 0 \\ 0 & 0 & 0 & \frac{4\sqrt{7}}{35} & 0 & 0 & 0 & 0 & \frac{3\sqrt{7}}{35} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{35}}{35} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 528 | symmetry | $\frac{15x^4z}{8}+\frac{15x^2y^2z}{4}-5x^2z^3+\frac{15y^4z}{8}-5y^2z^3+z^5$ |
| | $\mathbb{M}_5^{(1,-1;a)}(A,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{42} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{7}}{42} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{7}}{21} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{7}}{21} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{7}}{42} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{42} \end{bmatrix}$ |
| 529 | symmetry | $-\frac{3\sqrt{70}x^4y}{16}-\frac{\sqrt{70}x^2y^3}{8}+\frac{3\sqrt{70}x^2yz^2}{2}+\frac{\sqrt{70}y^5}{16}-\frac{\sqrt{70}y^3z^2}{2}$ |

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_5^{(1,-1;a)}(A,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{3} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{6} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{3} \\ 0 & 0 & 0 & 0 & \frac{i}{3} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{6} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{3} & 0 & 0 & 0 \end{bmatrix}$ |
| 530 | symmetry | $-\frac{\sqrt{70}x^5}{16}+\frac{\sqrt{70}x^3y^2}{8}+\frac{\sqrt{70}x^3z^2}{2}+\frac{3\sqrt{70}xy^4}{16}-\frac{3\sqrt{70}xy^2z^2}{2}$ |
| | $\mathbb{M}_5^{(1,-1;a)}(A,3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{3} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{6} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{3} \\ 0 & 0 & 0 & 0 & \frac{1}{3} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{6} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{3} & 0 & 0 & 0 \end{bmatrix}$ |
| 531 | symmetry | $\frac{3\sqrt{7}x^5}{16}-\frac{15\sqrt{7}ix^4y}{16}-\frac{15\sqrt{7}x^3y^2}{8}+\frac{15\sqrt{7}ix^2y^3}{8}+\frac{15\sqrt{7}xy^4}{16}-\frac{3\sqrt{7}iy^5}{16}$ |

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_5^{(1,-1;a)}(E^{(a)},1)$ | $\begin{bmatrix} 0&0&0&0&0&0&0&0&0&0\\ 0&0&0&0&0&0&0&0&0&0\\ 0&0&0&0&0&0&0&0&0&0\\ 0&0&0&0&0&0&0&0&0&0\\ 0&0&0&0&0&0&0&0&0&0\\ 0&0&0&0&0&0&0&0&0&0\\ 0&0&0&0&0&0&0&0&0&0\\ 0&0&0&0&0&0&0&0&0&0\\ 0&0&0&0&1&0&0&0&0&0 \end{bmatrix}$ |
| 532 | symmetry | $\frac{\sqrt{30}x^5}{16}+\frac{\sqrt{30}ix^4y}{16}+\frac{\sqrt{30}x^3y^2}{8}-\frac{3\sqrt{30}x^3z^2}{4}+\frac{\sqrt{30}ix^2y^3}{8}-\frac{3\sqrt{30}ix^2yz^2}{4}+\frac{\sqrt{30}xy^4}{16}-\frac{3\sqrt{30}xy^2z^2}{4}+\frac{\sqrt{30}xz^4}{2}+\frac{\sqrt{30}iy^5}{16}-\frac{3\sqrt{30}iy^3z^2}{4}+\frac{\sqrt{30}iyz^4}{2}$ |
| | $\mathbb{M}_5^{(1,-1;a)}(E^{(a)},2)$ | $\begin{bmatrix} 0&0&0&0&0&0&0&0&0&0\\ 0&0&0&0&0&0&0&0&0&0\\ 0&0&0&0&0&0&0&0&0&0\\ 0&0&0&0&0&0&0&0&0&0\\ 0&0&0&0&0&\frac{\sqrt{42}}{42}&0&0&0&0\\ 0&0&0&0&0&0&-\frac{\sqrt{105}}{21}&0&0&0\\ 0&0&0&0&0&0&0&\frac{\sqrt{210}}{21}&0&0\\ 0&0&0&0&0&0&0&0&-\frac{\sqrt{105}}{21}&0\\ 0&0&0&0&0&0&0&0&0&\frac{\sqrt{42}}{42}\\ 0&0&0&0&0&0&0&0&0&0 \end{bmatrix}$ |
| 533 | symmetry | $\frac{3\sqrt{70}x^4z}{16}+\frac{3\sqrt{70}ix^3yz}{4}-\frac{9\sqrt{70}x^2y^2z}{8}-\frac{3\sqrt{70}ixy^3z}{4}+\frac{3\sqrt{70}y^4z}{16}$ |

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_5^{(1,-1;a)}(E^{(a)},3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{2} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 534 | symmetry | $-\frac{\sqrt{210}x^4z}{8}+\frac{\sqrt{210}ix^3yz}{4}+\frac{\sqrt{210}x^2z^3}{4}+\frac{\sqrt{210}ixy^3z}{4}-\frac{\sqrt{210}ixyz^3}{2}+\frac{\sqrt{210}y^4z}{8}-\frac{\sqrt{210}y^2z^3}{4}$ |
| | $\mathbb{M}_5^{(1,-1;a)}(E^{(a)},4)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{6} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{6} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 \end{bmatrix}$ |
| 535 | symmetry | $\frac{3\sqrt{7}x^5}{16}+\frac{15\sqrt{7}ix^4y}{16}-\frac{15\sqrt{7}x^3y^2}{8}-\frac{15\sqrt{7}ix^2y^3}{8}+\frac{15\sqrt{7}xy^4}{16}+\frac{3\sqrt{7}iy^5}{16}$ |

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_5^{(1,-1;a)}(E^{(b)},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 536 | symmetry | $\frac{\sqrt{30}x^5}{16} - \frac{\sqrt{30}ix^4y}{16} + \frac{\sqrt{30}x^3y^2}{8} - \frac{3\sqrt{30}x^3z^2}{4} - \frac{\sqrt{30}ix^2y^3}{8} + \frac{3\sqrt{30}ix^2yz^2}{4} + \frac{\sqrt{30}xy^4}{16} - \frac{3\sqrt{30}xy^2z^2}{4} + \frac{\sqrt{30}xz^4}{2} - \frac{\sqrt{30}iy^5}{16} + \frac{3\sqrt{30}iy^3z^2}{4} - \frac{\sqrt{30}iyz^4}{2}$ |
| | $\mathbb{M}_5^{(1,-1;a)}(E^{(b)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{42} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{21} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{21} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{21} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{42} & 0 \end{bmatrix}$ |
| 537 | symmetry | $\frac{3\sqrt{70}x^4z}{16} - \frac{3\sqrt{70}ix^3yz}{4} - \frac{9\sqrt{70}x^2y^2z}{8} + \frac{3\sqrt{70}ixy^3z}{4} + \frac{3\sqrt{70}y^4z}{16}$ |

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_5^{(1,-1;a)}(E^{(b)},3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{2} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{2} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 538 | symmetry | $-\frac{\sqrt{210}x^4z}{8} - \frac{\sqrt{210}ix^3yz}{4} + \frac{\sqrt{210}x^2z^3}{4} - \frac{\sqrt{210}ixy^3z}{4} + \frac{\sqrt{210}ixyz^3}{2} + \frac{\sqrt{210}y^4z}{8} - \frac{\sqrt{210}y^2z^3}{4}$ |
| | $\mathbb{M}_5^{(1,-1;a)}(E^{(b)},4)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{6} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{6} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{6} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{6} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 539 | symmetry | $z$ |

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_1^{(1,1;a)}(A)$ | $\begin{bmatrix} \frac{3\sqrt{70}}{50} & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{50} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{70}}{50} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{50} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{70}}{50} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{50} & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{70}}{50} & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{50} & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{35} & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{70}}{50} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{70}}{175} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{105}}{50} & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{175} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{105}}{50} & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{175} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{70}}{50} & 0 & 0 & 0 & 0 & \frac{3\sqrt{70}}{175} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{35} \end{bmatrix}$ |
| 540 | symmetry | $\frac{\sqrt{2}x}{2} + \frac{\sqrt{2}iy}{2}$ |
| | $\mathbb{M}_1^{(1,1;a)}(E^{(a)})$ | $\begin{bmatrix} 0 & \frac{\sqrt{105}}{25} & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{100} & 0 & 0 & 0 \\ 0 & 0 & \frac{2\sqrt{35}}{25} & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{100} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{105}}{25} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{50} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{10} \\ \frac{\sqrt{7}}{10} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{7}}{35} & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{105}}{50} & 0 & 0 & 0 & 0 & -\frac{4\sqrt{70}}{175} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{210}}{100} & 0 & 0 & 0 & 0 & -\frac{6\sqrt{35}}{175} & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{70}}{100} & 0 & 0 & 0 & 0 & -\frac{4\sqrt{70}}{175} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{7}}{35} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 541 | symmetry | $\frac{\sqrt{2}x}{2} - \frac{\sqrt{2}iy}{2}$ |

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_1^{(1,1;a)}(E^{(b)})$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{10} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{105}}{25} & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{50} & 0 & 0 & 0 & 0 \\ 0 & \frac{2\sqrt{35}}{25} & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{100} & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{105}}{25} & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{100} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{2\sqrt{7}}{35} & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{70}}{100} & 0 & 0 & 0 & 0 & -\frac{4\sqrt{70}}{175} & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{210}}{100} & 0 & 0 & 0 & 0 & -\frac{6\sqrt{35}}{175} & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{105}}{50} & 0 & 0 & 0 & 0 & -\frac{4\sqrt{70}}{175} & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{7}}{10} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{7}}{35} & 0 \end{bmatrix}$ |
| 542 | symmetry | $-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$ |
| | $\mathbb{M}_3^{(1,1;a)}(A,1)$ | $\begin{bmatrix} -\frac{6\sqrt{35}}{175} & 0 & 0 & 0 & 0 & \frac{9\sqrt{35}}{350} & 0 & 0 & 0 & 0 \\ 0 & \frac{18\sqrt{35}}{175} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{210}}{350} & 0 & 0 & 0 \\ 0 & 0 & -\frac{18\sqrt{35}}{175} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{210}}{350} & 0 & 0 \\ 0 & 0 & 0 & \frac{6\sqrt{35}}{175} & 0 & 0 & 0 & 0 & \frac{9\sqrt{35}}{350} & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{105} & 0 & 0 & 0 & 0 & 0 \\ \frac{9\sqrt{35}}{350} & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{75} & 0 & 0 & 0 & 0 \\ 0 & -\frac{3\sqrt{210}}{350} & 0 & 0 & 0 & 0 & -\frac{4\sqrt{35}}{525} & 0 & 0 & 0 \\ 0 & 0 & -\frac{3\sqrt{210}}{350} & 0 & 0 & 0 & 0 & \frac{4\sqrt{35}}{525} & 0 & 0 \\ 0 & 0 & 0 & \frac{9\sqrt{35}}{350} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{75} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{105} \end{bmatrix}$ |
| 543 | symmetry | $\frac{3\sqrt{10}x^2y}{4} - \frac{\sqrt{10}y^3}{4}$ |

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_3^{(1,1;a)}(A,2)$ | $\begin{bmatrix} 0 & 0 & 0 & \frac{6\sqrt{14}i}{35} & 0 & 0 & 0 & 0 & -\frac{9\sqrt{14}i}{280} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{210}i}{280} \\ 0 & 0 & 0 & 0 & -\frac{3\sqrt{210}i}{280} & 0 & 0 & 0 & 0 & 0 \\ -\frac{6\sqrt{14}i}{35} & 0 & 0 & 0 & 0 & -\frac{9\sqrt{14}i}{280} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{3\sqrt{210}i}{280} & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{105} & 0 & 0 \\ 0 & 0 & 0 & \frac{9\sqrt{14}i}{280} & 0 & 0 & 0 & 0 & -\frac{2\sqrt{14}i}{105} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{105} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{105} & 0 & 0 & 0 & 0 & 0 \\ \frac{9\sqrt{14}i}{280} & 0 & 0 & 0 & 0 & \frac{2\sqrt{14}i}{105} & 0 & 0 & 0 & 0 \\ 0 & \frac{3\sqrt{210}i}{280} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{105} & 0 & 0 & 0 \end{bmatrix}$ |
| 544 | symmetry | $\frac{\sqrt{10}x^3}{4} - \frac{3\sqrt{10}xy^2}{4}$ |
| | $\mathbb{M}_3^{(1,1;a)}(A,3)$ | $\begin{bmatrix} 0 & 0 & 0 & -\frac{6\sqrt{14}}{35} & 0 & 0 & 0 & 0 & \frac{9\sqrt{14}}{280} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{210}}{280} \\ 0 & 0 & 0 & 0 & -\frac{3\sqrt{210}}{280} & 0 & 0 & 0 & 0 & 0 \\ -\frac{6\sqrt{14}}{35} & 0 & 0 & 0 & 0 & -\frac{9\sqrt{14}}{280} & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{3\sqrt{210}}{280} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{105} & 0 & 0 \\ 0 & 0 & 0 & -\frac{9\sqrt{14}}{280} & 0 & 0 & 0 & 0 & \frac{2\sqrt{14}}{105} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{105} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{105} & 0 & 0 & 0 & 0 & 0 \\ \frac{9\sqrt{14}}{280} & 0 & 0 & 0 & 0 & \frac{2\sqrt{14}}{105} & 0 & 0 & 0 & 0 \\ 0 & \frac{3\sqrt{210}}{280} & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{105} & 0 & 0 & 0 \end{bmatrix}$ |
| 545 | symmetry | $-\frac{\sqrt{3}x^3}{4} - \frac{\sqrt{3}ix^2y}{4} - \frac{\sqrt{3}xy^2}{4} + \sqrt{3}xz^2 - \frac{\sqrt{3}iy^3}{4} + \sqrt{3}iyz^2$ |

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_3^{(1,1;a)}(E^{(a)},1)$ | $\begin{bmatrix} 0 & -\frac{12\sqrt{35}}{175} & 0 & 0 & 0 & 0 & \frac{9\sqrt{210}}{700} & 0 & 0 & 0 \\ 0 & 0 & \frac{12\sqrt{105}}{175} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{70}}{700} & 0 & 0 \\ 0 & 0 & 0 & -\frac{12\sqrt{35}}{175} & 0 & 0 & 0 & 0 & -\frac{3\sqrt{35}}{100} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{21}}{140} \\ -\frac{3\sqrt{21}}{140} & 0 & 0 & 0 & 0 & \frac{2\sqrt{21}}{105} & 0 & 0 & 0 & 0 \\ 0 & \frac{3\sqrt{35}}{100} & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{525} & 0 & 0 & 0 \\ 0 & 0 & \frac{3\sqrt{70}}{700} & 0 & 0 & 0 & 0 & -\frac{4\sqrt{105}}{525} & 0 & 0 \\ 0 & 0 & 0 & -\frac{9\sqrt{210}}{700} & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{525} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{21}}{105} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 546 | $\mathbb{M}_3^{(1,1;a)}(E^{(a)},2)$ | $\frac{\sqrt{30}x^2z}{4} - \frac{\sqrt{30}ixyz}{2} - \frac{\sqrt{30}y^2z}{4}$ $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{3\sqrt{70}}{140} & 0 & 0 & 0 & 0 & 0 \\ -\frac{6\sqrt{14}}{35} & 0 & 0 & 0 & 0 & \frac{3\sqrt{14}}{140} & 0 & 0 & 0 & 0 \\ 0 & \frac{6\sqrt{14}}{35} & 0 & 0 & 0 & 0 & \frac{3\sqrt{21}}{70} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{105} & 0 & 0 & 0 & 0 & 0 \\ \frac{3\sqrt{21}}{70} & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{105} & 0 & 0 & 0 & 0 \\ 0 & \frac{3\sqrt{14}}{140} & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{105} & 0 & 0 & 0 \\ 0 & 0 & -\frac{3\sqrt{70}}{140} & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{105} & 0 & 0 \end{bmatrix}$ |
| 547 | symmetry | $-\frac{\sqrt{3}x^3}{4} + \frac{\sqrt{3}ix^2y}{4} - \frac{\sqrt{3}xy^2}{4} + \sqrt{3}xz^2 + \frac{\sqrt{3}iy^3}{4} - \sqrt{3}iyz^2$ |

*continued ...*

Table 8

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_3^{(1,1;a)}(E^{(b)},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{3\sqrt{21}}{140} & 0 & 0 & 0 & 0 & 0 \\ -\frac{12\sqrt{35}}{175} & 0 & 0 & 0 & 0 & \frac{3\sqrt{35}}{100} & 0 & 0 & 0 & 0 \\ 0 & \frac{12\sqrt{105}}{175} & 0 & 0 & 0 & 0 & \frac{3\sqrt{70}}{700} & 0 & 0 & 0 \\ 0 & 0 & -\frac{12\sqrt{35}}{175} & 0 & 0 & 0 & 0 & -\frac{9\sqrt{210}}{700} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{2\sqrt{21}}{105} & 0 & 0 & 0 & 0 & 0 \\ \frac{9\sqrt{210}}{700} & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{525} & 0 & 0 & 0 & 0 \\ 0 & -\frac{3\sqrt{70}}{700} & 0 & 0 & 0 & 0 & -\frac{4\sqrt{105}}{525} & 0 & 0 & 0 \\ 0 & 0 & -\frac{3\sqrt{35}}{100} & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{525} & 0 & 0 \\ 0 & 0 & 0 & \frac{3\sqrt{21}}{140} & 0 & 0 & 0 & 0 & \frac{2\sqrt{21}}{105} & 0 \end{bmatrix}$ |
| 548 | symmetry | $\frac{\sqrt{30}x^2z}{4}+\frac{\sqrt{30}ixyz}{2}-\frac{\sqrt{30}y^2z}{4}$ |
| | $\mathbb{M}_3^{(1,1;a)}(E^{(b)},2)$ | $\begin{bmatrix} 0 & 0 & -\frac{6\sqrt{14}}{35} & 0 & 0 & 0 & 0 & \frac{3\sqrt{21}}{70} & 0 & 0 \\ 0 & 0 & 0 & \frac{6\sqrt{14}}{35} & 0 & 0 & 0 & 0 & \frac{3\sqrt{14}}{140} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{70}}{140} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{3\sqrt{70}}{140} & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{105} & 0 & 0 & 0 \\ 0 & 0 & \frac{3\sqrt{14}}{140} & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{105} & 0 & 0 \\ 0 & 0 & 0 & \frac{3\sqrt{21}}{70} & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{105} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{105} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |

bra: $=\langle\frac{3}{2},\frac{3}{2};d|\,,\langle\frac{3}{2},\frac{1}{2};d|\,,\langle\frac{3}{2},-\frac{1}{2};d|\,,\langle\frac{3}{2},-\frac{3}{2};d|\,,\langle\frac{5}{2},\frac{5}{2};d|\,,\langle\frac{5}{2},\frac{3}{2};d|\,,\langle\frac{5}{2},\frac{1}{2};d|\,,\langle\frac{5}{2},-\frac{1}{2};d|\,,\langle\frac{5}{2},-\frac{3}{2};d|\,,\langle\frac{5}{2},-\frac{5}{2};d|$

ket: $=|\frac{5}{2},\frac{5}{2};f\rangle\,,|\frac{5}{2},\frac{3}{2};f\rangle\,,|\frac{5}{2},\frac{1}{2};f\rangle\,,|\frac{5}{2},-\frac{1}{2};f\rangle\,,|\frac{5}{2},-\frac{3}{2};f\rangle\,,|\frac{5}{2},-\frac{5}{2};f\rangle\,,|\frac{7}{2},\frac{7}{2};f\rangle\,,|\frac{7}{2},\frac{5}{2};f\rangle\,,|\frac{7}{2},\frac{3}{2};f\rangle\,,|\frac{7}{2},\frac{1}{2};f\rangle\,,|\frac{7}{2},-\frac{1}{2};f\rangle\,,|\frac{7}{2},-\frac{3}{2};f\rangle\,,|\frac{7}{2},-\frac{5}{2};f\rangle\,,|\frac{7}{2},-\frac{7}{2};f\rangle$

Table 9: (d,f) block.

| No. | multipole | matrix |
|---|---|---|
| 549 | symmetry | $z$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_1^{(a)}(A)$ | $\begin{bmatrix} 0 & \frac{1}{5} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{6}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{6}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{5} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{1}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{3}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{14} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{1}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{7} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{7} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{3}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{14} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{14} & 0 \end{bmatrix}$ |
| 550 | symmetry | $\frac{\sqrt{2}x}{2} + \frac{\sqrt{2}iy}{2}$ |
| | $\mathbb{Q}_1^{(a)}(E^{(a)})$ | $\begin{bmatrix} 0 & 0 & \frac{1}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{3}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{10}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{14} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{2}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{2}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{2}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{14} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{14} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{14} \end{bmatrix}$ |
| 551 | symmetry | $\frac{\sqrt{2}x}{2} - \frac{\sqrt{2}iy}{2}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_1^{(a)}(E^{(b)})$ | $\begin{bmatrix} -\frac{\sqrt{10}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{6}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{3}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{1}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{10}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{2}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{14} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{3\sqrt{2}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{2}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{14} & 0 & 0 \end{bmatrix}$ |
| 552 | symmetry | $-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$ |
| | $\mathbb{Q}_3^{(a)}(A,1)$ | $\begin{bmatrix} 0 & -\frac{3\sqrt{21}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{84} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{3\sqrt{14}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{84} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{3\sqrt{14}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{84} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{3\sqrt{21}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{84} & 0 & 0 \\ \frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{21}}{30} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{2\sqrt{21}}{105} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{2\sqrt{21}}{105} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{30} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{14} & 0 \end{bmatrix}$ |
| 553 | symmetry | $\frac{3\sqrt{10}x^2y}{4} - \frac{\sqrt{10}y^3}{4}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_3^{(a)}(A,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & \frac{3\sqrt{210}i}{280} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{84} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{14}i}{56} & -\frac{\sqrt{3}i}{12} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{42} & 0 \\ \frac{3\sqrt{14}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{12} \\ 0 & \frac{3\sqrt{210}i}{280} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{84} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{105} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{42} & \frac{\sqrt{2}i}{8} & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{14}i}{56} & 0 \\ \frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{14}i}{56} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{8} \\ 0 & \frac{\sqrt{210}i}{105} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{28} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 554 | symmetry | $\frac{\sqrt{10}x^3}{4} - \frac{3\sqrt{10}xy^2}{4}$ |
| | $\mathbb{Q}_3^{(a)}(A,3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{3\sqrt{210}}{280} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{84} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{14}}{56} & -\frac{\sqrt{3}}{12} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{42} & 0 \\ \frac{3\sqrt{14}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{12} \\ 0 & \frac{3\sqrt{210}}{280} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{84} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{105} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{42} & \frac{\sqrt{2}}{8} & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{14}}{56} & 0 \\ \frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{14}}{56} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{8} \\ 0 & \frac{\sqrt{210}}{105} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{28} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 555 | symmetry | $-\frac{\sqrt{3}x^3}{4} - \frac{\sqrt{3}ix^2y}{4} - \frac{\sqrt{3}xy^2}{4} + \sqrt{3}xz^2 - \frac{\sqrt{3}iy^3}{4} + \sqrt{3}iyz^2$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_3^{(a)}(E^{(a)},1)$ | $\begin{bmatrix} 0 & 0 & -\frac{9\sqrt{14}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{42} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{42}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{35}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{84} & 0 \\ 0 & \frac{\sqrt{35}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{14}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{42} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{2\sqrt{7}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{84} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{84} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{12} \end{bmatrix}$ |
| 556 | symmetry | $\frac{\sqrt{30}x^2z}{4} - \frac{\sqrt{30}ixyz}{2} - \frac{\sqrt{30}y^2z}{4}$ |
| | $\mathbb{Q}_3^{(a)}(E^{(a)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{42}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{210}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{3\sqrt{35}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{84} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{84} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{35}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{35}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{42} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{21} & 0 & 0 & 0 \end{bmatrix}$ |
| 557 | symmetry | $-\frac{\sqrt{3}x^3}{4} + \frac{\sqrt{3}ix^2y}{4} - \frac{\sqrt{3}xy^2}{4} + \sqrt{3}xz^2 + \frac{\sqrt{3}iy^3}{4} - \sqrt{3}iyz^2$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_3^{(a)}(E^{(b)},1)$ | $\begin{bmatrix} \frac{3\sqrt{35}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{84} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{21}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{42}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{9\sqrt{14}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{42} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{35}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{84} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{14}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{2\sqrt{7}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{84} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{14}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{42} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 \end{bmatrix}$ |
| 558 | symmetry | $\frac{\sqrt{30}x^2z}{4}+\frac{\sqrt{30}ixyz}{2}-\frac{\sqrt{30}y^2z}{4}$ |
| | $\mathbb{Q}_3^{(a)}(E^{(b)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & -\frac{3\sqrt{35}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{84} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{28} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{12} \\ 0 & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{21} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{35}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{42} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{84} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{6} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 559 | symmetry | $\frac{15x^4z}{8}+\frac{15x^2y^2z}{4}-5x^2z^3+\frac{15y^4z}{8}-5y^2z^3+z^5$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_5^{(a)}(A,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{30} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{6} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{30} & 0 & 0 \\ -\frac{\sqrt{6}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{5\sqrt{6}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{15}}{70} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{5\sqrt{6}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{5\sqrt{6}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{5\sqrt{6}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{15}}{70} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{14} & 0 \end{bmatrix}$ |
| 560 | symmetry | $-\frac{3\sqrt{70}x^4y}{16} - \frac{\sqrt{70}x^2y^3}{8} + \frac{3\sqrt{70}x^2yz^2}{2} + \frac{\sqrt{70}y^5}{16} - \frac{\sqrt{70}y^3z^2}{2}$ |
| | $\mathbb{Q}_5^{(a)}(A,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{30} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{30} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{30} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{30} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{30} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{30} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{42}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{140} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{42} & -\frac{i}{10} & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{7}i}{70} & 0 \\ -\frac{\sqrt{42}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{7}i}{70} & 0 & 0 & 0 & 0 & 0 & -\frac{i}{10} \\ 0 & \frac{\sqrt{105}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{140} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{42}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{28} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 561 | symmetry | $-\frac{\sqrt{70}x^5}{16} + \frac{\sqrt{70}x^3y^2}{8} + \frac{\sqrt{70}x^3z^2}{2} + \frac{3\sqrt{70}xy^4}{16} - \frac{3\sqrt{70}xy^2z^2}{2}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_5^{(a)}(A,3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{30} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{30} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{30} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{30} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{30} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{30} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{42}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{140} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{42} & -\frac{1}{10} & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{7}}{70} & 0 \\ -\frac{\sqrt{42}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{7}}{70} & 0 & 0 & 0 & 0 & 0 & \frac{1}{10} \\ 0 & \frac{\sqrt{105}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{140} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{42}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{28} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 562 | symmetry | $\frac{3\sqrt{7}x^5}{16} - \frac{15\sqrt{7}ix^4y}{16} - \frac{15\sqrt{7}x^3y^2}{8} + \frac{15\sqrt{7}ix^2y^3}{8} + \frac{15\sqrt{7}xy^4}{16} - \frac{3\sqrt{7}iy^5}{16}$ |
| | $\mathbb{Q}_5^{(a)}(E^{(a)},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{5} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{42}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 563 | symmetry | $\frac{\sqrt{30}x^5}{16} + \frac{\sqrt{30}ix^4y}{16} + \frac{\sqrt{30}x^3y^2}{8} - \frac{3\sqrt{30}x^3z^2}{4} + \frac{\sqrt{30}ix^2y^3}{8} - \frac{3\sqrt{30}ix^2yz^2}{4} + \frac{\sqrt{30}xy^4}{16} - \frac{3\sqrt{30}xy^2z^2}{4} + \frac{\sqrt{30}xz^4}{2} + \frac{\sqrt{30}iy^5}{16} - \frac{3\sqrt{30}iy^3z^2}{4} + \frac{\sqrt{30}iyz^4}{2}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_5^{(a)}(E^{(a)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{30} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{10} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{10} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{30} & 0 \\ 0 & -\frac{1}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{10}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{13\sqrt{30}}{420} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{5}}{7} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{210} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{11}{70} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{1}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{23\sqrt{6}}{420} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{420} \end{bmatrix}$ |
| 564 | symmetry | $\frac{3\sqrt{70}x^4z}{16}+\frac{3\sqrt{70}ix^3yz}{4}-\frac{9\sqrt{70}x^2y^2z}{8}-\frac{3\sqrt{70}ixy^3z}{4}+\frac{3\sqrt{70}y^4z}{16}$ |
| | $\mathbb{Q}_5^{(a)}(E^{(a)},3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{10} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{10} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{14} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{70} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{14} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{70} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{5} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 565 | symmetry | $-\frac{\sqrt{210}x^4z}{8}+\frac{\sqrt{210}ix^3yz}{4}+\frac{\sqrt{210}x^2z^3}{4}+\frac{\sqrt{210}ixy^3z}{4}-\frac{\sqrt{210}ixyz^3}{2}+\frac{\sqrt{210}y^4z}{8}-\frac{\sqrt{210}y^2z^3}{4}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_5^{(a)}(E^{(a)},4)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{60} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{20} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{20} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{60} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{30} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{14}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{4\sqrt{21}}{105} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{70}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{35} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{70}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{105} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{14}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{42} & 0 & 0 & 0 \end{bmatrix}$ |
| 566 | symmetry | $\frac{3\sqrt{7}x^5}{16}+\frac{15\sqrt{7}ix^4y}{16}-\frac{15\sqrt{7}x^3y^2}{8}-\frac{15\sqrt{7}ix^2y^3}{8}+\frac{15\sqrt{7}xy^4}{16}+\frac{3\sqrt{7}iy^5}{16}$ |
| | $\mathbb{Q}_5^{(a)}(E^{(b)},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{5} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{14} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{14} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{10} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 567 | symmetry | $\frac{\sqrt{30}x^5}{16}-\frac{\sqrt{30}ix^4y}{16}+\frac{\sqrt{30}x^3y^2}{8}-\frac{3\sqrt{30}x^3z^2}{4}-\frac{\sqrt{30}ix^2y^3}{8}+\frac{3\sqrt{30}ix^2yz^2}{4}+\frac{\sqrt{30}xy^4}{16}-\frac{3\sqrt{30}xy^2z^2}{4}+\frac{\sqrt{30}xz^4}{2}-\frac{\sqrt{30}iy^5}{16}+\frac{3\sqrt{30}iy^3z^2}{4}-\frac{\sqrt{30}iyz^4}{2}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_5^{(a)}(E^{(b)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{30} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{10} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{10} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{30} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{420} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{1}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{23\sqrt{6}}{420} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{10}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{11}{70} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{5}}{7} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{210} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{10}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{13\sqrt{30}}{420} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{1}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{28} & 0 & 0 \end{bmatrix}$ |
| 568 | symmetry | $\frac{3\sqrt{70}x^4z}{16} - \frac{3\sqrt{70}ix^3yz}{4} - \frac{9\sqrt{70}x^2y^2z}{8} + \frac{3\sqrt{70}ixy^3z}{4} + \frac{3\sqrt{70}y^4z}{16}$ |
| | $\mathbb{Q}_5^{(a)}(E^{(b)},3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{10} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{5} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{21}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{70} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{21}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{70} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 569 | symmetry | $-\frac{\sqrt{210}x^4z}{8} - \frac{\sqrt{210}ix^3yz}{4} + \frac{\sqrt{210}x^2z^3}{4} - \frac{\sqrt{210}ixy^3z}{4} + \frac{\sqrt{210}ixyz^3}{2} + \frac{\sqrt{210}y^4z}{8} - \frac{\sqrt{210}y^2z^3}{4}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_5^{(a)}(E^{(b)},4)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{60} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{20} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{20} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{60} \\ 0 & 0 & -\frac{\sqrt{14}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{42} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{70}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{105} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{35} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{4\sqrt{21}}{105} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{30} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 570 | symmetry | $-\frac{3x^2z}{2}-\frac{3y^2z}{2}+z^3$ |
| | $\mathbb{Q}_3^{(1,-1;a)}(A,1)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{6}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{14} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{2}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{2}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{14} & 0 & 0 \\ \frac{\sqrt{6}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2}{7} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{6}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{6}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{7} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{6}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{7} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2}{7} & 0 \end{bmatrix}$ |
| 571 | symmetry | $\frac{3\sqrt{10}x^2y}{4}-\frac{\sqrt{10}y^3}{4}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_3^{(1,-1;a)}(A,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{i}{14} & -\frac{\sqrt{42}i}{28} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{14} & 0 \\ \frac{i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{14} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{28} \\ 0 & \frac{\sqrt{15}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{6}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{14} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{28} & -\frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & -\frac{3i}{14} & 0 \\ \frac{\sqrt{6}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3i}{14} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{14} \\ 0 & \frac{\sqrt{15}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{14} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{6}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{14} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 572 | symmetry | $\frac{\sqrt{10}x^3}{4} - \frac{3\sqrt{10}xy^2}{4}$ |
| | $\mathbb{Q}_3^{(1,-1;a)}(A,3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{1}{14} & -\frac{\sqrt{42}}{28} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{14} & 0 \\ \frac{1}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{14} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{28} \\ 0 & \frac{\sqrt{15}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{6}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{14} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{28} & -\frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & \frac{3}{14} & 0 \\ \frac{\sqrt{6}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3}{14} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{14} \\ 0 & \frac{\sqrt{15}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{14} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{6}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{14} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 573 | symmetry | $-\frac{\sqrt{3}x^3}{4} - \frac{\sqrt{3}ix^2y}{4} - \frac{\sqrt{3}xy^2}{4} + \sqrt{3}xz^2 - \frac{\sqrt{3}iy^3}{4} + \sqrt{3}iyz^2$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_3^{(1,-1;a)}(E^{(a)},1)$ | $\begin{bmatrix} 0 & 0 & -\frac{3}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{7} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{3}}{105} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{30} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{14} & 0 \\ 0 & \frac{3\sqrt{10}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2}{7} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{3}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{3}}{21} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{2}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{42} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{3}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{14} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{10}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{21} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{21} \end{bmatrix}$ |
| 574 | symmetry | $\frac{\sqrt{30}x^2z}{4} - \frac{\sqrt{30}ixyz}{2} - \frac{\sqrt{30}y^2z}{4}$ |
| | $\mathbb{Q}_3^{(1,-1;a)}(E^{(a)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{3}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{15}}{105} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{6}}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{10}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{3\sqrt{2}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{21} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{3\sqrt{10}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{7} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{3\sqrt{10}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{21} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{2}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{6}}{21} & 0 & 0 & 0 \end{bmatrix}$ |
| 575 | symmetry | $-\frac{\sqrt{3}x^3}{4} + \frac{\sqrt{3}ix^2y}{4} - \frac{\sqrt{3}xy^2}{4} + \sqrt{3}xz^2 + \frac{\sqrt{3}iy^3}{4} - \sqrt{3}iyz^2$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_3^{(1,-1;a)}(E^{(b)},1)$ | $\begin{bmatrix} \frac{\sqrt{10}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{6}}{30} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{3}}{105} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{3}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{7} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{3\sqrt{10}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{21} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{3}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{14} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{3\sqrt{2}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{42} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{3}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{3}}{21} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{3\sqrt{10}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2}{7} & 0 & 0 \end{bmatrix}$ |
| 576 | symmetry | $\frac{\sqrt{30}x^2z}{4}+\frac{\sqrt{30}ixyz}{2}-\frac{\sqrt{30}y^2z}{4}$ |
| | $\mathbb{Q}_3^{(1,-1;a)}(E^{(b)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{10}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{105} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{6}}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}}{28} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{28} \\ 0 & 0 & \frac{3\sqrt{2}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{6}}{21} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{3\sqrt{10}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{21} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{7} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{21} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{21} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 577 | symmetry | $\frac{15x^4z}{8}+\frac{15x^2y^2z}{4}-5x^2z^3+\frac{15y^4z}{8}-5y^2z^3+z^5$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_5^{(1,-1;a)}(A,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{20} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{20} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{20} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{20} & 0 & 0 \\ -\frac{\sqrt{5}}{210} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{35} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{5}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{2}}{35} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{5}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{15}}{35} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{5}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{15}}{35} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{2}}{35} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{210} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{35} & 0 \end{bmatrix}$ |
| 578 | symmetry | $-\frac{3\sqrt{70}x^4y}{16} - \frac{\sqrt{70}x^2y^3}{8} + \frac{3\sqrt{70}x^2yz^2}{2} + \frac{\sqrt{70}y^5}{16} - \frac{\sqrt{70}y^3z^2}{2}$ |
| | $\mathbb{Q}_5^{(1,-1;a)}(A,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{50} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{50} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{50} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{50} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{50} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{50} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{35}i}{105} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{35} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{35}i}{175} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{105} & \frac{\sqrt{30}i}{25} & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{210}i}{175} & 0 \\ -\frac{\sqrt{35}i}{105} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{210}i}{175} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{25} \\ 0 & \frac{\sqrt{14}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{35}i}{175} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{35}i}{105} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{35} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 579 | symmetry | $-\frac{\sqrt{70}x^5}{16} + \frac{\sqrt{70}x^3y^2}{8} + \frac{\sqrt{70}x^3z^2}{2} + \frac{3\sqrt{70}xy^4}{16} - \frac{3\sqrt{70}xy^2z^2}{2}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_5^{(1,-1;a)}(A,3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{50} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{50} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{50} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{50} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{50} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{50} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{35}}{105} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{35} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{35}}{175} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{105} & \frac{\sqrt{30}}{25} & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{210}}{175} & 0 \\ -\frac{\sqrt{35}}{105} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{210}}{175} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{25} \\ 0 & \frac{\sqrt{14}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{35}}{175} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{35}}{105} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{35} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 580 | symmetry | $\frac{3\sqrt{7}x^5}{16} - \frac{15\sqrt{7}ix^4y}{16} - \frac{15\sqrt{7}x^3y^2}{8} + \frac{15\sqrt{7}ix^2y^3}{8} + \frac{15\sqrt{7}xy^4}{16} - \frac{3\sqrt{7}iy^5}{16}$ |
| | $\mathbb{Q}_5^{(1,-1;a)}(E^{(a)},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{5} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{35}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{35} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 581 | symmetry | $\frac{\sqrt{30}x^5}{16} + \frac{\sqrt{30}ix^4y}{16} + \frac{\sqrt{30}x^3y^2}{8} - \frac{3\sqrt{30}x^3z^2}{4} + \frac{\sqrt{30}ix^2y^3}{8} - \frac{3\sqrt{30}ix^2yz^2}{4} + \frac{\sqrt{30}xy^4}{16} - \frac{3\sqrt{30}xy^2z^2}{4} + \frac{\sqrt{30}xz^4}{2} + \frac{\sqrt{30}iy^5}{16} - \frac{3\sqrt{30}iy^3z^2}{4} + \frac{\sqrt{30}iyz^4}{2}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_5^{(1,-1;a)}(E^{(a)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{10} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{10} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{5}}{50} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{50} & 0 \\ 0 & -\frac{\sqrt{30}}{210} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{7} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{3}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{13}{35} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{6}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{35} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{11\sqrt{30}}{175} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{210} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{23\sqrt{5}}{175} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{35} \end{bmatrix}$ |
| 582 | symmetry | $\frac{3\sqrt{70}x^4z}{16}+\frac{3\sqrt{70}ix^3yz}{4}-\frac{9\sqrt{70}x^2y^2z}{8}-\frac{3\sqrt{70}ixy^3z}{4}+\frac{3\sqrt{70}y^4z}{16}$ |
| | $\mathbb{Q}_5^{(1,-1;a)}(E^{(a)},3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{50} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{5}}{50} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{6\sqrt{7}}{35} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{105}}{175} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{30}}{25} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 583 | symmetry | $-\frac{\sqrt{210}x^4z}{8}+\frac{\sqrt{210}ix^3yz}{4}+\frac{\sqrt{210}x^2z^3}{4}+\frac{\sqrt{210}ixy^3z}{4}-\frac{\sqrt{210}ixyz^3}{2}+\frac{\sqrt{210}y^4z}{8}-\frac{\sqrt{210}y^2z^3}{4}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_5^{(1,-1;a)}(E^{(a)},4)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{100} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{100} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{35}}{100} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{20} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{5}}{25} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{105}}{210} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{8\sqrt{70}}{175} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{210}}{175} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{4\sqrt{7}}{35} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{105}}{210} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{35}}{35} & 0 & 0 & 0 \end{bmatrix}$ |
| 584 | $\mathbb{Q}_5^{(1,-1;a)}(E^{(b)},1)$ | $\frac{3\sqrt{7}x^5}{16}+\frac{15\sqrt{7}ix^4y}{16}-\frac{15\sqrt{7}x^3y^2}{8}-\frac{15\sqrt{7}ix^2y^3}{8}+\frac{15\sqrt{7}xy^4}{16}+\frac{3\sqrt{7}iy^5}{16}$ $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{10} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{35} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{5} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 585 | symmetry | $\frac{\sqrt{30}x^5}{16}-\frac{\sqrt{30}ix^4y}{16}+\frac{\sqrt{30}x^3y^2}{8}-\frac{3\sqrt{30}x^3z^2}{4}-\frac{\sqrt{30}ix^2y^3}{8}+\frac{3\sqrt{30}ix^2yz^2}{4}+\frac{\sqrt{30}xy^4}{16}-\frac{3\sqrt{30}xy^2z^2}{4}+\frac{\sqrt{30}xz^4}{2}-\frac{\sqrt{30}iy^5}{16}+\frac{3\sqrt{30}iy^3z^2}{4}-\frac{\sqrt{30}iyz^4}{2}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_5^{(1,-1;a)}(E^{(b)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{50} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{5}}{50} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{10} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{10} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{30}}{210} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{23\sqrt{5}}{175} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{3}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{11\sqrt{30}}{175} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{6}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{35} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{3}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{13}{35} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{210} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{7} & 0 & 0 \end{bmatrix}$ |
| 586 | symmetry | $\frac{3\sqrt{70}x^4z}{16} - \frac{3\sqrt{70}ix^3yz}{4} - \frac{9\sqrt{70}x^2y^2z}{8} + \frac{3\sqrt{70}ixy^3z}{4} + \frac{3\sqrt{70}y^4z}{16}$ |
| | $\mathbb{Q}_5^{(1,-1;a)}(E^{(b)},3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{5}}{50} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{50} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{30}}{25} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{70}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{105}}{175} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{70}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{6\sqrt{7}}{35} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 587 | symmetry | $-\frac{\sqrt{210}x^4z}{8} - \frac{\sqrt{210}ix^3yz}{4} + \frac{\sqrt{210}x^2z^3}{4} - \frac{\sqrt{210}ixy^3z}{4} + \frac{\sqrt{210}ixyz^3}{2} + \frac{\sqrt{210}y^4z}{8} - \frac{\sqrt{210}y^2z^3}{4}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_5^{(1,-1;a)}(E^{(b)},4)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{20} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{35}}{100} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{100} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{100} \\ 0 & 0 & -\frac{\sqrt{105}}{210} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{35}}{35} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{4\sqrt{7}}{35} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{210}}{175} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{210} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{8\sqrt{70}}{175} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{5}}{25} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 588 | symmetry | $z$ |
| | $\mathbb{Q}_1^{(1,0;a)}(A)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{2}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{3}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{3}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{3\sqrt{2}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{9\sqrt{2}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{14} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{3\sqrt{2}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{2}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{9\sqrt{2}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{14} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{14} & 0 \end{bmatrix}$ |
| 589 | symmetry | $\frac{\sqrt{2}x}{2} + \frac{\sqrt{2}iy}{2}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_1^{(1,0;a)}(E^{(a)})$ | $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{2}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{6}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{3\sqrt{5}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{6\sqrt{2}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{9}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{6\sqrt{2}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{14} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{5}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{28} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{28} \end{bmatrix}$ |
| 590 | symmetry | $\frac{\sqrt{2}x}{2} - \frac{\sqrt{2}iy}{2}$ |
| | $\mathbb{Q}_1^{(1,0;a)}(E^{(b)})$ | $\begin{bmatrix} \frac{\sqrt{5}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{3}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{6}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{2}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{3\sqrt{5}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{6\sqrt{2}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{14} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{9}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{6\sqrt{2}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{3\sqrt{5}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{28} & 0 & 0 \end{bmatrix}$ |
| 591 | symmetry | $-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_3^{(1,0;a)}(A,1)$ | $\begin{bmatrix} 0 & \frac{3\sqrt{7}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{42}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{42}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{3\sqrt{7}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{28} & 0 & 0 \\ -\frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{84} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{7}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{2\sqrt{7}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{84} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{2\sqrt{7}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{84} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{84} & 0 \end{bmatrix}$ |
| 592 | symmetry | $\frac{3\sqrt{10}x^2y}{4} - \frac{\sqrt{10}y^3}{4}$ |
| | $\mathbb{Q}_3^{(1,0;a)}(A,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{3\sqrt{70}i}{560} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{112} & -\frac{i}{4} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 \\ -\frac{\sqrt{42}i}{112} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & \frac{i}{4} \\ 0 & -\frac{3\sqrt{70}i}{560} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{168} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{56} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & \frac{\sqrt{6}i}{48} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{112} & 0 \\ -\frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{112} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{48} \\ 0 & -\frac{\sqrt{70}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{56} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{168} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 593 | symmetry | $\frac{\sqrt{10}x^3}{4} - \frac{3\sqrt{10}xy^2}{4}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_3^{(1,0;a)}(A,3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & \frac{3\sqrt{70}}{560} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{112} & -\frac{1}{4} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{14} & 0 \\ -\frac{\sqrt{42}}{112} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & -\frac{1}{4} \\ 0 & -\frac{3\sqrt{70}}{560} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{168} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{56} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{14} & \frac{\sqrt{6}}{48} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{112} & 0 \\ -\frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{112} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{48} \\ 0 & -\frac{\sqrt{70}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{56} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{168} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 594 | symmetry | $-\frac{\sqrt{3}x^3}{4} - \frac{\sqrt{3}ix^2y}{4} - \frac{\sqrt{3}xy^2}{4} + \sqrt{3}xz^2 - \frac{\sqrt{3}iy^3}{4} + \sqrt{3}iyz^2$ |
| | $\mathbb{Q}_3^{(1,0;a)}(E^{(a)},1)$ | $\begin{bmatrix} 0 & 0 & \frac{3\sqrt{42}}{280} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{14}}{280} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{40} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{280} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{28} & 0 \\ 0 & -\frac{\sqrt{105}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{84} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{42}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{84} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{2\sqrt{21}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{168} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{168} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{168} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{24} \end{bmatrix}$ |
| 595 | symmetry | $\frac{\sqrt{30}x^2z}{4} - \frac{\sqrt{30}ixyz}{2} - \frac{\sqrt{30}y^2z}{4}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_3^{(1,0;a)}(E^{(a)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{14}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{70}}{280} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{7}}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{105}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{21}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{168} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{105}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{168} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{105}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{84} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{21}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{42} & 0 & 0 & 0 \end{bmatrix}$ |
| 596 | symmetry | $-\frac{\sqrt{3}x^3}{4}+\frac{\sqrt{3}ix^2y}{4}-\frac{\sqrt{3}xy^2}{4}+\sqrt{3}xz^2+\frac{\sqrt{3}iy^3}{4}-\sqrt{3}iyz^2$ |
| | $\mathbb{Q}_3^{(1,0;a)}(E^{(b)},1)$ | $\begin{bmatrix} -\frac{\sqrt{105}}{280} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{7}}{40} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{14}}{280} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{42}}{280} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{105}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{168} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{42}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{168} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{2\sqrt{21}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{168} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{42}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{84} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{84} & 0 & 0 \end{bmatrix}$ |
| 597 | symmetry | $\frac{\sqrt{30}x^2z}{4}+\frac{\sqrt{30}ixyz}{2}-\frac{\sqrt{30}y^2z}{4}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_3^{(1,0;a)}(E^{(b)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & \frac{\sqrt{105}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{280} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{7}}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{28} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{4} \\ 0 & 0 & -\frac{\sqrt{21}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{42} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{105}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{84} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{168} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{168} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{12} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 598 | symmetry | $\frac{15x^4z}{8}+\frac{15x^2y^2z}{4}-5x^2z^3+\frac{15y^4z}{8}-5y^2z^3+z^5$ |
| | $\mathbb{Q}_5^{(1,0;a)}(A,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{10} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{10} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{10} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{10} & 0 & 0 \\ \frac{\sqrt{5}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{420} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{5}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}}{140} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{5}}{7} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{210} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{5}}{7} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{210} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}}{140} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{420} & 0 \end{bmatrix}$ |
| 599 | symmetry | $-\frac{3\sqrt{70}x^4y}{16}-\frac{\sqrt{70}x^2y^3}{8}+\frac{3\sqrt{70}x^2yz^2}{2}+\frac{\sqrt{70}y^5}{16}-\frac{\sqrt{70}y^3z^2}{2}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_5^{(1,0;a)}(A,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{25} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{25} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{25} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{25} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{25} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{25} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{35}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{420} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{700} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{35} & -\frac{\sqrt{30}i}{300} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{700} & 0 \\ \frac{\sqrt{35}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{700} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{300} \\ 0 & -\frac{\sqrt{14}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{700} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{35}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{420} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 600 | symmetry | $-\frac{\sqrt{70}x^5}{16} + \frac{\sqrt{70}x^3y^2}{8} + \frac{\sqrt{70}x^3z^2}{2} + \frac{3\sqrt{70}xy^4}{16} - \frac{3\sqrt{70}xy^2z^2}{2}$ |
| | $\mathbb{Q}_5^{(1,0;a)}(A,3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{25} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{25} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{25} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{25} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{25} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{25} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{35}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{420} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{700} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{35} & -\frac{\sqrt{30}}{300} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{700} & 0 \\ \frac{\sqrt{35}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{700} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{300} \\ 0 & -\frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{700} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{35}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{420} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 601 | symmetry | $\frac{3\sqrt{7}x^5}{16} - \frac{15\sqrt{7}ix^4y}{16} - \frac{15\sqrt{7}x^3y^2}{8} + \frac{15\sqrt{7}ix^2y^3}{8} + \frac{15\sqrt{7}xy^4}{16} - \frac{3\sqrt{7}iy^5}{16}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_5^{(1,0;a)}(E^{(a)},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{5} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{60} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{3\sqrt{35}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{420} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 602 | symmetry | $\frac{\sqrt{30}x^5}{16}+\frac{\sqrt{30}ix^4y}{16}+\frac{\sqrt{30}x^3y^2}{8}-\frac{3\sqrt{30}x^3z^2}{4}+\frac{\sqrt{30}ix^2y^3}{8}-\frac{3\sqrt{30}ix^2yz^2}{4}+\frac{\sqrt{30}xy^4}{16}-\frac{3\sqrt{30}xy^2z^2}{4}+\frac{\sqrt{30}xz^4}{2}+\frac{\sqrt{30}iy^5}{16}-\frac{3\sqrt{30}iy^3z^2}{4}+\frac{\sqrt{30}iyz^4}{2}$ |
| | $\mathbb{Q}_5^{(1,0;a)}(E^{(a)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{5} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{5} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{5}}{25} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{25} & 0 \\ 0 & \frac{\sqrt{30}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{84} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{3}}{7} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{13}{420} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{6}}{7} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{420} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{7} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{11\sqrt{30}}{2100} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{23\sqrt{5}}{2100} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{420} \end{bmatrix}$ |
| 603 | symmetry | $\frac{3\sqrt{70}x^4z}{16}+\frac{3\sqrt{70}ix^3yz}{4}-\frac{9\sqrt{70}x^2y^2z}{8}-\frac{3\sqrt{70}ixy^3z}{4}+\frac{3\sqrt{70}y^4z}{16}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_5^{(1,0;a)}(E^{(a)},3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{25} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{5}}{25} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{3\sqrt{70}}{70} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{70} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{70}}{70} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{1050} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{150} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 604 | symmetry | $-\frac{\sqrt{210}x^4z}{8}+\frac{\sqrt{210}ix^3yz}{4}+\frac{\sqrt{210}x^2z^3}{4}+\frac{\sqrt{210}ixy^3z}{4}-\frac{\sqrt{210}ixyz^3}{2}+\frac{\sqrt{210}y^4z}{8}-\frac{\sqrt{210}y^2z^3}{4}$ |
| | $\mathbb{Q}_5^{(1,0;a)}(E^{(a)},4)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{50} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{50} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{35}}{50} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{10} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{150} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{105}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{70}}{525} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{21}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{1050} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{21}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{105} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{105}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{210} & 0 & 0 & 0 \end{bmatrix}$ |
| 605 | symmetry | $\frac{3\sqrt{7}x^5}{16}+\frac{15\sqrt{7}ix^4y}{16}-\frac{15\sqrt{7}x^3y^2}{8}-\frac{15\sqrt{7}ix^2y^3}{8}+\frac{15\sqrt{7}xy^4}{16}+\frac{3\sqrt{7}iy^5}{16}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_5^{(1,0;a)}(E^{(b)},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{5} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{35}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{420} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{60} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 606 | symmetry | $\frac{\sqrt{30}x^5}{16} - \frac{\sqrt{30}ix^4y}{16} + \frac{\sqrt{30}x^3y^2}{8} - \frac{3\sqrt{30}x^3z^2}{4} - \frac{\sqrt{30}ix^2y^3}{8} + \frac{3\sqrt{30}ix^2yz^2}{4} + \frac{\sqrt{30}xy^4}{16} - \frac{3\sqrt{30}xy^2z^2}{4} + \frac{\sqrt{30}xz^4}{2} - \frac{\sqrt{30}iy^5}{16} + \frac{3\sqrt{30}iy^3z^2}{4} - \frac{\sqrt{30}iyz^4}{2}$ |
| | $\mathbb{Q}_5^{(1,0;a)}(E^{(b)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{25} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{5}}{25} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{5} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{5} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{420} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{30}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{23\sqrt{5}}{2100} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{3}}{7} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{11\sqrt{30}}{2100} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{6}}{7} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{420} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{3}}{7} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{13}{420} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{84} & 0 & 0 \end{bmatrix}$ |
| 607 | symmetry | $\frac{3\sqrt{70}x^4z}{16} - \frac{3\sqrt{70}ix^3yz}{4} - \frac{9\sqrt{70}x^2y^2z}{8} + \frac{3\sqrt{70}ixy^3z}{4} + \frac{3\sqrt{70}y^4z}{16}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_5^{(1,0;a)}(E^{(b)},3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{5}}{25} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{25} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{150} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{3\sqrt{70}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{1050} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{3\sqrt{70}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{70} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 608 | symmetry | $-\frac{\sqrt{210}x^4z}{8} - \frac{\sqrt{210}ix^3yz}{4} + \frac{\sqrt{210}x^2z^3}{4} - \frac{\sqrt{210}ixy^3z}{4} + \frac{\sqrt{210}ixyz^3}{2} + \frac{\sqrt{210}y^4z}{8} - \frac{\sqrt{210}y^2z^3}{4}$ |
| | $\mathbb{Q}_5^{(1,0;a)}(E^{(b)},4)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{10} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{35}}{50} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{50} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{50} \\ 0 & 0 & \frac{\sqrt{105}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{210} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{21}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{105} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{1050} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{70}}{525} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{150} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 609 | symmetry | $z$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_1^{(1,1;a)}(A)$ | $\begin{bmatrix} 0 & -\frac{1}{5} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{6}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{6}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{1}{5} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{2}{7} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{6}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{2}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{2}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{6}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{2}{7} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{28} & 0 \end{bmatrix}$ |
| 610 | symmetry | $\frac{\sqrt{2}x}{2} + \frac{\sqrt{2}iy}{2}$ |
| | $\mathbb{Q}_1^{(1,1;a)}(E^{(a)})$ | $\begin{bmatrix} 0 & 0 & -\frac{1}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{3}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{2\sqrt{10}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{8}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{6\sqrt{2}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{8}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{10}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{28} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{28} \end{bmatrix}$ |
| 611 | symmetry | $\frac{\sqrt{2}x}{2} - \frac{\sqrt{2}iy}{2}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_1^{(1,1;a)}(E^{(b)})$ | $\begin{bmatrix} \frac{\sqrt{10}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{6}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{3}}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{2\sqrt{10}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{8}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{6\sqrt{2}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{8}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{2\sqrt{10}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{28} & 0 & 0 \end{bmatrix}$ |
| 612 | symmetry | $-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$ |
| | $\mathbb{Q}_3^{(1,1;a)}(A,1)$ | $\begin{bmatrix} 0 & \frac{9}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{3\sqrt{6}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{6}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{9}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{28} & 0 & 0 \\ \frac{5}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{1}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{2}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{2}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{5}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{28} & 0 \end{bmatrix}$ |
| 613 | symmetry | $\frac{3\sqrt{10}x^2y}{4} - \frac{\sqrt{10}y^3}{4}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_3^{(1,1;a)}(A,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{9\sqrt{10}i}{112} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{15\sqrt{6}i}{112} & -\frac{\sqrt{7}i}{28} & 0 & 0 & 0 & 0 & 0 & \frac{i}{14} & 0 \\ -\frac{15\sqrt{6}i}{112} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{14} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{28} \\ 0 & -\frac{9\sqrt{10}i}{112} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{5i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{56} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3i}{56} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{5i}{42} & \frac{\sqrt{42}i}{112} & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{6}i}{112} & 0 \\ \frac{5i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{6}i}{112} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{112} \\ 0 & \frac{\sqrt{10}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3i}{56} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{5i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{56} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 614 | symmetry | $\frac{\sqrt{10}x^3}{4} - \frac{3\sqrt{10}xy^2}{4}$ |
| | $\mathbb{Q}_3^{(1,1;a)}(A,3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & \frac{9\sqrt{10}}{112} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{15\sqrt{6}}{112} & -\frac{\sqrt{7}}{28} & 0 & 0 & 0 & 0 & 0 & -\frac{1}{14} & 0 \\ -\frac{15\sqrt{6}}{112} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{14} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{28} \\ 0 & -\frac{9\sqrt{10}}{112} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{5}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{56} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3}{56} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{5}{42} & \frac{\sqrt{42}}{112} & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{6}}{112} & 0 \\ \frac{5}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{6}}{112} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{112} \\ 0 & \frac{\sqrt{10}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3}{56} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{5}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{56} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 615 | symmetry | $-\frac{\sqrt{3}x^3}{4} - \frac{\sqrt{3}ix^2y}{4} - \frac{\sqrt{3}xy^2}{4} + \sqrt{3}xz^2 - \frac{\sqrt{3}iy^3}{4} + \sqrt{3}iyz^2$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_3^{(1,1;a)}(E^{(a)},1)$ | $\begin{bmatrix} 0 & 0 & \frac{9\sqrt{6}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{2}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{3}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{15}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{28} & 0 \\ 0 & \frac{\sqrt{15}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{6}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{2\sqrt{3}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{56} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{56} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{56} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{56} \end{bmatrix}$ |
| 616 | symmetry | $\frac{\sqrt{30}x^2z}{4} - \frac{\sqrt{30}ixyz}{2} - \frac{\sqrt{30}y^2z}{4}$ |
| | $\mathbb{Q}_3^{(1,1;a)}(E^{(a)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{15\sqrt{2}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{3\sqrt{10}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{3\sqrt{15}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{5\sqrt{3}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{56} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{15}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{56} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{15}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{5\sqrt{3}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{14} & 0 & 0 & 0 \end{bmatrix}$ |
| 617 | symmetry | $-\frac{\sqrt{3}x^3}{4} + \frac{\sqrt{3}ix^2y}{4} - \frac{\sqrt{3}xy^2}{4} + \sqrt{3}xz^2 + \frac{\sqrt{3}iy^3}{4} - \sqrt{3}iyz^2$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_3^{(1,1;a)}(E^{(b)},1)$ | $\begin{bmatrix} -\frac{3\sqrt{15}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{3}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{3\sqrt{2}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{9\sqrt{6}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{15}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{56} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{6}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{56} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{2\sqrt{3}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{56} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{6}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{28} & 0 & 0 \end{bmatrix}$ |
| 618 | symmetry | $\frac{\sqrt{30}x^2z}{4} + \frac{\sqrt{30}ixyz}{2} - \frac{\sqrt{30}y^2z}{4}$ |
| | $\mathbb{Q}_3^{(1,1;a)}(E^{(b)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & \frac{3\sqrt{15}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{3\sqrt{10}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{15\sqrt{2}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{28} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{28} \\ 0 & 0 & \frac{5\sqrt{3}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{15}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{56} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{3}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{56} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{28} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 619 | symmetry | $-\frac{x^2}{2} - \frac{y^2}{2} + z^2$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_2^{(a)}(A)$ | $\begin{bmatrix} 0 & \frac{3\sqrt{10}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{14} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{15}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{5}i}{70} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{15}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{5}i}{70} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{14} & 0 & 0 \\ -\frac{\sqrt{10}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{10}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3i}{14} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{10}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{70} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{10}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{70} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3i}{14} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{14} & 0 \end{bmatrix}$ |
| 620 | symmetry | $\frac{\sqrt{6}xz}{2} + \frac{\sqrt{6}iyz}{2}$ |
| | $\mathbb{G}_2^{(a)}(E^{(a)}, 1)$ | $\begin{bmatrix} 0 & 0 & \frac{3\sqrt{30}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{70} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{10}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{70} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{14} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{7} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{14} & 0 \\ 0 & -\frac{\sqrt{3}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{30}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{11\sqrt{10}i}{140} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{10} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{14} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{28} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{28} \end{bmatrix}$ |
| 621 | symmetry | $\frac{\sqrt{6}x^2}{4} - \frac{\sqrt{6}ixy}{2} - \frac{\sqrt{6}y^2}{4}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_2^{(a)}(E^{(a)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{2i}{7} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{4\sqrt{5}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{30}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{140} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{6}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2i}{7} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{3\sqrt{30}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{7} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{3\sqrt{30}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{10}i}{35} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{6}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{14} & 0 & 0 & 0 \end{bmatrix}$ |
| 622 | symmetry | $\frac{\sqrt{6}xz}{2} - \frac{\sqrt{6}iyz}{2}$ |
| | $\mathbb{G}_2^{(a)}(E^{(b)},1)$ | $\begin{bmatrix} -\frac{\sqrt{3}i}{7} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{5}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{14} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{10}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{70} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{3\sqrt{30}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{70} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{3}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{30}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{14} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{30}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{11\sqrt{10}i}{140} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{28} & 0 & 0 \end{bmatrix}$ |
| 623 | symmetry | $\frac{\sqrt{6}x^2}{4} + \frac{\sqrt{6}ixy}{2} - \frac{\sqrt{6}y^2}{4}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_2^{(a)}(E^{(b)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & \frac{\sqrt{30}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{140} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{4\sqrt{5}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{2i}{7} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{28} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{28} \\ 0 & 0 & -\frac{\sqrt{6}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{30}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{10}i}{35} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{3\sqrt{30}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{7} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2i}{7} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{14} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 624 | symmetry | $\frac{3x^4}{8}+\frac{3x^2y^2}{4}-3x^2z^2+\frac{3y^4}{8}-3y^2z^2+z^4$ |
| | $\mathbb{G}_4^{(a)}(A,1)$ | $\begin{bmatrix} 0 & -\frac{i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{10}i}{140} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{6}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}i}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{6}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}i}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{6}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{10}i}{140} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{3i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{10}i}{35} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{i}{7} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{i}{7} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{3i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{10}i}{35} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{14} & 0 \end{bmatrix}$ |
| 625 | symmetry | $\frac{\sqrt{70}x^3z}{4}-\frac{3\sqrt{70}xy^2z}{4}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_4^{(a)}(A,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{7}i}{140} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{56} & -\frac{3i}{20} & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{7}i}{70} & 0 \\ \frac{\sqrt{42}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{7}i}{70} & 0 & 0 & 0 & 0 & 0 & \frac{3i}{20} \\ 0 & -\frac{\sqrt{70}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{7}i}{140} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{20} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{14} & \frac{3\sqrt{6}i}{40} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{280} & 0 \\ \frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{280} & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{6}i}{40} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{20} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{28} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 626 | symmetry | $\frac{3\sqrt{70}x^2yz}{4} - \frac{\sqrt{70}y^3z}{4}$ |
| | $\mathbb{G}_4^{(a)}(A,3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{7}}{140} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{56} & \frac{3}{20} & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{7}}{70} & 0 \\ -\frac{\sqrt{42}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{7}}{70} & 0 & 0 & 0 & 0 & 0 & \frac{3}{20} \\ 0 & \frac{\sqrt{70}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{7}}{140} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{20} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{14} & -\frac{3\sqrt{6}}{40} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{280} & 0 \\ -\frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{280} & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{6}}{40} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{20} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{28} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 627 | symmetry | $-\frac{3\sqrt{5}x^3z}{4} - \frac{3\sqrt{5}ix^2yz}{4} - \frac{3\sqrt{5}xy^2z}{4} + \sqrt{5}xz^3 - \frac{3\sqrt{5}iy^3z}{4} + \sqrt{5}iyz^3$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_4^{(a)}(E^{(a)},1)$ | $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{10}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{30}i}{70} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{30}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{10}i}{140} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{6}i}{35} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{6}i}{140} & 0 \\ 0 & \frac{i}{7} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{14} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{10}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{70} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{20} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{140} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{i}{7} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{13\sqrt{6}i}{140} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{140} \end{bmatrix}$ |
| 628 | symmetry | $\frac{\sqrt{70}x^4}{16} + \frac{\sqrt{70}ix^3y}{4} - \frac{3\sqrt{70}x^2y^2}{8} - \frac{\sqrt{70}ixy^3}{4} + \frac{\sqrt{70}y^4}{16}$ |
| | $\mathbb{G}_4^{(a)}(E^{(a)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{21}i}{70} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3i}{10} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{35} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{21}i}{35} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{10} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 629 | symmetry | $-\frac{\sqrt{10}x^4}{8} + \frac{\sqrt{10}ix^3y}{4} + \frac{3\sqrt{10}x^2z^2}{4} + \frac{\sqrt{10}ixy^3}{4} - \frac{3\sqrt{10}ixyz^2}{2} + \frac{\sqrt{10}y^4}{8} - \frac{3\sqrt{10}y^2z^2}{4}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_4^{(a)}(E^{(a)},3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{21}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{6}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{33i}{140} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{30}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{3}i}{140} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{5}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{15}i}{140} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{21}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{3i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{6}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{5}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{17\sqrt{2}i}{140} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{5}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{70} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{3i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{7} & 0 & 0 & 0 \end{bmatrix}$ |
| 630 | symmetry | $-\frac{3\sqrt{5}x^3z}{4}+\frac{3\sqrt{5}ix^2yz}{4}-\frac{3\sqrt{5}xy^2z}{4}+\sqrt{5}xz^3+\frac{3\sqrt{5}iy^3z}{4}-\sqrt{5}iyz^3$ |
| | $\mathbb{G}_4^{(a)}(E^{(b)},1)$ | $\begin{bmatrix} \frac{i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{6}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{15}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{6}i}{35} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{30}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}i}{140} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{10}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{30}i}{70} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{i}{7} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{13\sqrt{6}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{10}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{140} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{20} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{10}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{70} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{i}{7} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{14} & 0 & 0 \end{bmatrix}$ |
| 631 | symmetry | $\frac{\sqrt{70}x^4}{16}-\frac{\sqrt{70}ix^3y}{4}-\frac{3\sqrt{70}x^2y^2}{8}+\frac{\sqrt{70}ixy^3}{4}+\frac{\sqrt{70}y^4}{16}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_4^{(a)}(E^{(b)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3i}{10} & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{14}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{21}i}{70} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{14}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{21}i}{35} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{14}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{35} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 632 | symmetry | $-\frac{\sqrt{10}x^4}{8} - \frac{\sqrt{10}ix^3y}{4} + \frac{3\sqrt{10}x^2z^2}{4} - \frac{\sqrt{10}ixy^3}{4} + \frac{3\sqrt{10}ixyz^2}{2} + \frac{\sqrt{10}y^4}{8} - \frac{3\sqrt{10}y^2z^2}{4}$ |
| | $\mathbb{G}_4^{(a)}(E^{(b)},3)$ | $\begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{5}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{15}i}{140} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{3}i}{140} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{33i}{140} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{21}i}{140} \\ 0 & 0 & \frac{3i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{7} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{5}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{70} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{17\sqrt{2}i}{140} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{3i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{6}i}{140} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{21}i}{70} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 633 | symmetry | $-\frac{x^2}{2} - \frac{y^2}{2} + z^2$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_2^{(1,-1;a)}(A)$ | $\begin{bmatrix} 0 & -\frac{3\sqrt{6}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{15}i}{35} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{3i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{6\sqrt{3}i}{35} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{3i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{6\sqrt{3}i}{35} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{3\sqrt{6}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{15}i}{35} & 0 & 0 \\ \frac{\sqrt{6}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3i}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{6}i}{105} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{15}i}{70} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{4\sqrt{6}i}{105} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}i}{70} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{4\sqrt{6}i}{105} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}i}{70} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{105} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{15}i}{70} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3i}{14} & 0 \end{bmatrix}$ |
| 634 | symmetry | $\frac{\sqrt{6}xz}{2} + \frac{\sqrt{6}iyz}{2}$ |
| | $\mathbb{G}_2^{(1,-1;a)}(E^{(a)}, 1)$ | $\begin{bmatrix} 0 & 0 & -\frac{9\sqrt{2}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{6}i}{35} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{6}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{6\sqrt{2}i}{35} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{30}i}{35} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{5}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{30}i}{35} & 0 \\ 0 & \frac{2\sqrt{5}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}i}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{2\sqrt{2}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{11\sqrt{6}i}{140} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{10} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{2\sqrt{2}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{5}i}{70} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{5}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{140} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{28} \end{bmatrix}$ |
| 635 | symmetry | $\frac{\sqrt{6}x^2}{4} - \frac{\sqrt{6}ixy}{2} - \frac{\sqrt{6}y^2}{4}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_2^{(1,-1;a)}(E^{(a)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{15}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{2\sqrt{3}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{35} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{3\sqrt{2}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{35} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{10}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{15}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{3\sqrt{2}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{5}i}{35} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{3\sqrt{2}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{6}i}{35} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{10}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{70} & 0 & 0 & 0 \end{bmatrix}$ |
| 636 | symmetry | $\frac{\sqrt{6}xz}{2} - \frac{\sqrt{6}iyz}{2}$ |
| | $\mathbb{G}_2^{(1,-1;a)}(E^{(b)},1)$ | $\begin{bmatrix} \frac{3\sqrt{5}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{30}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{3}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{30}i}{35} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{6}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{6\sqrt{2}i}{35} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{9\sqrt{2}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{6}i}{35} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{2\sqrt{5}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{2\sqrt{2}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{5}i}{70} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{10} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{2\sqrt{2}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{11\sqrt{6}i}{140} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{2\sqrt{5}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}i}{28} & 0 & 0 \end{bmatrix}$ |
| 637 | symmetry | $\frac{\sqrt{6}x^2}{4} + \frac{\sqrt{6}ixy}{2} - \frac{\sqrt{6}y^2}{4}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_2^{(1,-1;a)}(E^{(b)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & -\frac{3\sqrt{2}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{35} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{2\sqrt{3}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{35} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}i}{35} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{35} \\ 0 & 0 & \frac{\sqrt{10}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{70} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{3\sqrt{2}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{6}i}{35} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{3\sqrt{2}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{5}i}{35} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{15}i}{35} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{70} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 638 | symmetry | $\frac{3x^4}{8}+\frac{3x^2y^2}{4}-3x^2z^2+\frac{3y^4}{8}-3y^2z^2+z^4$ |
| | $\mathbb{G}_4^{(1,-1;a)}(A,1)$ | $\begin{bmatrix} 0 & \frac{\sqrt{3}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{2}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{6}i}{84} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{2}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{6}i}{84} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{28} & 0 & 0 \\ -\frac{\sqrt{3}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{2}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{3}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{21} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{3}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5i}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{3}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5i}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{21} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{2}i}{28} & 0 \end{bmatrix}$ |
| 639 | symmetry | $\frac{\sqrt{70}x^3z}{4}-\frac{3\sqrt{70}xy^2z}{4}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_4^{(1,-1;a)}(A,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{336} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{112} & \frac{\sqrt{3}i}{12} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{42} & 0 \\ -\frac{\sqrt{14}i}{112} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{12} \\ 0 & \frac{\sqrt{210}i}{336} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{7}i}{56} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{24} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{42} & \frac{3\sqrt{2}i}{16} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{112} & 0 \\ -\frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{112} & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}i}{16} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{24} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{7}i}{56} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 640 | symmetry | $\frac{3\sqrt{70}x^2yz}{4} - \frac{\sqrt{70}y^3z}{4}$ |
| | $\mathbb{G}_4^{(1,-1;a)}(A,3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{336} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{112} & -\frac{\sqrt{3}}{12} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{42} & 0 \\ \frac{\sqrt{14}}{112} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{12} \\ 0 & -\frac{\sqrt{210}}{336} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{7}}{56} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{24} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{42} & -\frac{3\sqrt{2}}{16} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{112} & 0 \\ \frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{112} & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}}{16} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{24} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{7}}{56} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 641 | symmetry | $-\frac{3\sqrt{5}x^3z}{4} - \frac{3\sqrt{5}ix^2yz}{4} - \frac{3\sqrt{5}xy^2z}{4} + \sqrt{5}xz^3 - \frac{3\sqrt{5}iy^3z}{4} + \sqrt{5}iyz^3$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_4^{(1,-1;a)}(E^{(a)},1)$ | $\begin{bmatrix} 0 & 0 & \frac{\sqrt{30}i}{168} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{10}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{84} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{7} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{168} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}i}{28} & 0 \\ 0 & -\frac{\sqrt{3}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{30}i}{84} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{30}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{8} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{168} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{13\sqrt{2}i}{56} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{56} \end{bmatrix}$ |
| 642 | symmetry | $\frac{\sqrt{70}x^4}{16} + \frac{\sqrt{70}ix^3y}{4} - \frac{3\sqrt{70}x^2y^2}{8} - \frac{\sqrt{70}ixy^3}{4} + \frac{\sqrt{70}y^4}{16}$ |
| | $\mathbb{G}_4^{(1,-1;a)}(E^{(a)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{6} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{42} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{7} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 643 | symmetry | $-\frac{\sqrt{10}x^4}{8} + \frac{\sqrt{10}ix^3y}{4} + \frac{3\sqrt{10}x^2z^2}{4} + \frac{\sqrt{10}ixy^3}{4} - \frac{3\sqrt{10}ixyz^2}{2} + \frac{\sqrt{10}y^4}{8} - \frac{3\sqrt{10}y^2z^2}{4}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_4^{(1,-1;a)}(E^{(a)},3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{2}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{11\sqrt{3}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{10}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{15}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{5}i}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{7}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{3}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{2}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{15}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{17\sqrt{6}i}{168} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{15}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{3}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5i}{14} & 0 & 0 & 0 \end{bmatrix}$ |
| 644 | symmetry | $-\frac{3\sqrt{5}x^3z}{4}+\frac{3\sqrt{5}ix^2yz}{4}-\frac{3\sqrt{5}xy^2z}{4}+\sqrt{5}xz^3+\frac{3\sqrt{5}iy^3z}{4}-\sqrt{5}iyz^3$ |
| | $\mathbb{G}_4^{(1,-1;a)}(E^{(b)},1)$ | $\begin{bmatrix} -\frac{\sqrt{3}i}{168} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{5}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{7} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{10}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{84} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{30}i}{168} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{3}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{13\sqrt{2}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{30}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{168} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{8} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{30}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{30}i}{84} & 0 & 0 \end{bmatrix}$ |
| 645 | symmetry | $\frac{\sqrt{70}x^4}{16}-\frac{\sqrt{70}ix^3y}{4}-\frac{3\sqrt{70}x^2y^2}{8}+\frac{\sqrt{70}ixy^3}{4}+\frac{\sqrt{70}y^4}{16}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_4^{(1,-1;a)}(E^{(b)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{42}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{42}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{7} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{42}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{42} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 646 | symmetry | $-\frac{\sqrt{10}x^4}{8} - \frac{\sqrt{10}ix^3y}{4} + \frac{3\sqrt{10}x^2z^2}{4} - \frac{\sqrt{10}ixy^3}{4} + \frac{3\sqrt{10}ixyz^2}{2} + \frac{\sqrt{10}y^4}{8} - \frac{3\sqrt{10}y^2z^2}{4}$ |
| | $\mathbb{G}_4^{(1,-1;a)}(E^{(b)},3)$ | $\begin{bmatrix} 0 & 0 & 0 & \frac{\sqrt{15}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{5}i}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{11\sqrt{3}i}{84} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{28} \\ 0 & 0 & -\frac{\sqrt{3}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5i}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{15}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{17\sqrt{6}i}{168} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{2}i}{56} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{7}i}{28} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 647 | symmetry | $-\frac{5x^6}{16} - \frac{15x^4y^2}{16} + \frac{45x^4z^2}{8} - \frac{15x^2y^4}{16} + \frac{45x^2y^2z^2}{4} - \frac{15x^2z^4}{2} - \frac{5y^6}{16} + \frac{45y^4z^2}{8} - \frac{15y^2z^4}{2} + z^6$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_6^{(1,-1;a)}(A,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{66}i}{132} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{110}i}{44} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{33}i}{66} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{33}i}{66} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{110}i}{44} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{66}i}{132} & 0 \end{bmatrix}$ |
| 648 | symmetry | $\frac{\sqrt{462}x^6}{32} - \frac{15\sqrt{462}x^4y^2}{32} + \frac{15\sqrt{462}x^2y^4}{32} - \frac{\sqrt{462}y^6}{32}$ |
| | $\mathbb{G}_6^{(1,-1;a)}(A,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 649 | symmetry | $\frac{3\sqrt{462}x^5y}{16} - \frac{5\sqrt{462}x^3y^3}{8} + \frac{3\sqrt{462}xy^5}{16}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_6^{(1,-1;a)}(A,3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 650 | symmetry | $-\frac{3\sqrt{210}x^5z}{16}+\frac{3\sqrt{210}x^3y^2z}{8}+\frac{\sqrt{210}x^3z^3}{2}+\frac{9\sqrt{210}xy^4z}{16}-\frac{3\sqrt{210}xy^2z^3}{2}$ |
| | $\mathbb{G}_6^{(1,-1;a)}(A,4)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_6^{(1,-1;a)}(A,5)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{77}}{44} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{231}}{44} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{22}}{44} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{154}}{44} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{154}}{44} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{22}}{44} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{231}}{44} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{77}}{44} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 652 | symmetry | $\frac{3\sqrt{77}x^5z}{16} - \frac{15\sqrt{77}ix^4yz}{16} - \frac{15\sqrt{77}x^3y^2z}{8} + \frac{15\sqrt{77}ix^2y^3z}{8} + \frac{15\sqrt{77}xy^4z}{16} - \frac{3\sqrt{77}iy^5z}{16}$ |
| | $\mathbb{G}_6^{(1,-1;a)}(E^{(a)},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 653 | symmetry | $\frac{5\sqrt{42}x^5z}{16} + \frac{5\sqrt{42}ix^4yz}{16} + \frac{5\sqrt{42}x^3y^2z}{8} - \frac{5\sqrt{42}x^3z^3}{4} + \frac{5\sqrt{42}ix^2y^3z}{8} - \frac{5\sqrt{42}ix^2yz^3}{4} + \frac{5\sqrt{42}xy^4z}{16} - \frac{5\sqrt{42}xy^2z^3}{4} + \frac{\sqrt{42}xz^5}{2} + \frac{5\sqrt{42}iy^5z}{16} - \frac{5\sqrt{42}iy^3z^3}{4} + \frac{\sqrt{42}iyz^5}{2}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_6^{(1,-1;a)}(E^{(a)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{231}i}{132} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{77}i}{132} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{154}i}{132} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}i}{132} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{385}i}{132} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{11}i}{132} \end{bmatrix}$ |
| 654 | symmetry | $-\frac{3\sqrt{14}x^6}{32} - \frac{3\sqrt{14}ix^5y}{8} + \frac{15\sqrt{14}x^4y^2}{32} + \frac{15\sqrt{14}x^4z^2}{16} + \frac{15\sqrt{14}ix^3yz^2}{4} + \frac{15\sqrt{14}x^2y^4}{32} - \frac{45\sqrt{14}x^2y^2z^2}{8} + \frac{3\sqrt{14}ixy^5}{8} - \frac{15\sqrt{14}ixy^3z^2}{4} - \frac{3\sqrt{14}y^6}{32} + \frac{15\sqrt{14}y^4z^2}{16}$ |
| | $\mathbb{G}_6^{(1,-1;a)}(E^{(a)},3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 655 | symmetry | $\frac{\sqrt{105}x^6}{32} - \frac{\sqrt{105}ix^5y}{16} + \frac{\sqrt{105}x^4y^2}{32} - \frac{\sqrt{105}x^4z^2}{2} - \frac{\sqrt{105}ix^3y^3}{8} + \sqrt{105}ix^3yz^2 - \frac{\sqrt{105}x^2y^4}{32} + \frac{\sqrt{105}x^2z^4}{2} - \frac{\sqrt{105}ixy^5}{16} + \sqrt{105}ixy^3z^2 - \sqrt{105}ixyz^4 - \frac{\sqrt{105}y^6}{32} + \frac{\sqrt{105}y^4z^2}{2} - \frac{\sqrt{105}y^2z^4}{2}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_6^{(1,-1;a)}(E^{(a)},4)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{22}i}{66} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{77}i}{33} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{231}i}{33} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{770}i}{66} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{154}i}{66} & 0 & 0 \end{bmatrix}$ |
| 656 | symmetry | $\frac{3\sqrt{77}x^5z}{16}+\frac{15\sqrt{77}ix^4yz}{16}-\frac{15\sqrt{77}x^3y^2z}{8}-\frac{15\sqrt{77}ix^2y^3z}{8}+\frac{15\sqrt{77}xy^4z}{16}+\frac{3\sqrt{77}iy^5z}{16}$ |
| | $\mathbb{G}_6^{(1,-1;a)}(E^{(b)},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 657 | symmetry | $\frac{5\sqrt{42}x^5z}{16}-\frac{5\sqrt{42}ix^4yz}{16}+\frac{5\sqrt{42}x^3y^2z}{8}-\frac{5\sqrt{42}x^3z^3}{4}-\frac{5\sqrt{42}ix^2y^3z}{8}+\frac{5\sqrt{42}ix^2yz^3}{4}+\frac{5\sqrt{42}xy^4z}{16}-\frac{5\sqrt{42}xy^2z^3}{4}+\frac{\sqrt{42}xz^5}{2}-\frac{5\sqrt{42}iy^5z}{16}+\frac{5\sqrt{42}iy^3z^3}{4}-\frac{\sqrt{42}iyz^5}{2}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_6^{(1,-1;a)}(E^{(b)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{11}i}{132} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{385}i}{132} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}i}{132} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{154}i}{132} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{77}i}{132} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{231}i}{132} & 0 & 0 \end{bmatrix}$ |
| 658 | symmetry | $-\frac{3\sqrt{14}x^6}{32}+\frac{3\sqrt{14}ix^5y}{8}+\frac{15\sqrt{14}x^4y^2}{32}+\frac{15\sqrt{14}x^4z^2}{16}-\frac{15\sqrt{14}ix^3yz^2}{4}+\frac{15\sqrt{14}x^2y^4}{32}-\frac{45\sqrt{14}x^2y^2z^2}{8}-\frac{3\sqrt{14}ixy^5}{8}+\frac{15\sqrt{14}ixy^3z^2}{4}-\frac{3\sqrt{14}y^6}{32}+\frac{15\sqrt{14}y^4z^2}{16}$ |
| | $\mathbb{G}_6^{(1,-1;a)}(E^{(b)},3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}i}{66} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{1155}i}{66} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{77}i}{22} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 659 | symmetry | $\frac{\sqrt{105}x^6}{32}+\frac{\sqrt{105}ix^5y}{16}+\frac{\sqrt{105}x^4y^2}{32}-\frac{\sqrt{105}x^4z^2}{2}+\frac{\sqrt{105}ix^3y^3}{8}-\sqrt{105}ix^3yz^2-\frac{\sqrt{105}x^2y^4}{32}+\frac{\sqrt{105}x^2z^4}{2}+\frac{\sqrt{105}ixy^5}{16}-\sqrt{105}ixy^3z^2+\sqrt{105}ixyz^4-\frac{\sqrt{105}y^6}{32}+\frac{\sqrt{105}y^4z^2}{2}-\frac{\sqrt{105}y^2z^4}{2}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_6^{(1,-1;a)}(E^{(b)},4)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{154}i}{66} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{770}i}{66} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{231}i}{33} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{77}i}{33} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{22}i}{66} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 660 | symmetry | $-\frac{x^2}{2}-\frac{y^2}{2}+z^2$ |
| | $\mathbb{G}_2^{(1,0;a)}(A)$ | $\begin{bmatrix} 0 & \frac{\sqrt{15}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{14} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{10}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{30}i}{70} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{10}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{30}i}{70} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{14} & 0 & 0 \\ -\frac{\sqrt{15}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{15}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{2\sqrt{15}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{70} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{2\sqrt{15}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{70} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{28} & 0 \end{bmatrix}$ |
| 661 | symmetry | $\frac{\sqrt{6}xz}{2}+\frac{\sqrt{6}iyz}{2}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_2^{(1,0;a)}(E^{(a)},1)$ | $\begin{bmatrix} 0 & 0 & \frac{3\sqrt{5}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{35} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{15}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{5}i}{35} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{210} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{7} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{7} & 0 \\ 0 & -\frac{3\sqrt{2}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{3\sqrt{5}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{11\sqrt{15}i}{420} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{60} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{3\sqrt{5}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{84} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{84} \end{bmatrix}$ |
| 662 | symmetry | $\frac{\sqrt{6}x^2}{4} - \frac{\sqrt{6}ixy}{2} - \frac{\sqrt{6}y^2}{4}$ |
| | $\mathbb{G}_2^{(1,0;a)}(E^{(a)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{6}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3i}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{2\sqrt{30}i}{105} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{14} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{5}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{70} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{3i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{9\sqrt{5}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{14} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{9\sqrt{5}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{15}i}{105} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{3i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{42} & 0 & 0 & 0 \end{bmatrix}$ |
| 663 | symmetry | $\frac{\sqrt{6}xz}{2} - \frac{\sqrt{6}iyz}{2}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_2^{(1,0;a)}(E^{(b)},1)$ | $\begin{bmatrix} -\frac{\sqrt{2}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{7} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{30}i}{210} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{7} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{15}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{5}i}{35} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{3\sqrt{5}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{35} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{3\sqrt{2}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{3\sqrt{5}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{60} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{3\sqrt{5}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{11\sqrt{15}i}{420} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{3\sqrt{2}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{28} & 0 & 0 \end{bmatrix}$ |
| 664 | symmetry | $\frac{\sqrt{6}x^2}{4}+\frac{\sqrt{6}ixy}{2}-\frac{\sqrt{6}y^2}{4}$ |
| | $\mathbb{G}_2^{(1,0;a)}(E^{(b)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & \frac{\sqrt{5}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{70} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{2\sqrt{30}i}{105} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{14} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3i}{14} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{14} \\ 0 & 0 & -\frac{3i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{42} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{9\sqrt{5}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{15}i}{105} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{9\sqrt{5}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{14} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{3i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{21} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{42} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 665 | symmetry | $\frac{3x^4}{8}+\frac{3x^2y^2}{4}-3x^2z^2+\frac{3y^4}{8}-3y^2z^2+z^4$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_4^{(1,0;a)}(A,1)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{5}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{27\sqrt{2}i}{140} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{30}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{10}i}{140} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{30}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{10}i}{140} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{27\sqrt{2}i}{140} & 0 & 0 \\ \frac{3\sqrt{5}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{9\sqrt{5}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{35} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{3\sqrt{5}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{140} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{3\sqrt{5}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{140} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{9\sqrt{5}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{35} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{5}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{140} & 0 \end{bmatrix}$ |
| 666 | symmetry | $\frac{\sqrt{70}x^3z}{4} - \frac{3\sqrt{70}xy^2z}{4}$ |
| | $\mathbb{G}_4^{(1,0;a)}(A,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{112} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{27\sqrt{35}i}{700} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{560} & \frac{9\sqrt{5}i}{100} & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{35}i}{350} & 0 \\ \frac{\sqrt{210}i}{560} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{35}i}{350} & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{5}i}{100} \\ 0 & -\frac{\sqrt{14}i}{112} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{27\sqrt{35}i}{700} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{3\sqrt{35}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{280} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{200} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{35}i}{70} & -\frac{3\sqrt{30}i}{400} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{2800} & 0 \\ \frac{3\sqrt{35}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{2800} & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{30}i}{400} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{200} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{3\sqrt{35}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{280} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 667 | symmetry | $\frac{3\sqrt{70}x^2yz}{4} - \frac{\sqrt{70}y^3z}{4}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_4^{(1,0;a)}(A,3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{112} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{27\sqrt{35}}{700} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{560} & -\frac{9\sqrt{5}}{100} & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{35}}{350} & 0 \\ -\frac{\sqrt{210}}{560} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{35}}{350} & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{5}}{100} \\ 0 & \frac{\sqrt{14}}{112} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{27\sqrt{35}}{700} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{3\sqrt{35}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{280} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{200} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{35}}{70} & \frac{3\sqrt{30}}{400} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{2800} & 0 \\ -\frac{3\sqrt{35}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{2800} & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{30}}{400} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{200} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{3\sqrt{35}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{280} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 668 | symmetry | $-\frac{3\sqrt{5}x^3z}{4} - \frac{3\sqrt{5}ix^2yz}{4} - \frac{3\sqrt{5}xy^2z}{4} + \sqrt{5}xz^3 - \frac{3\sqrt{5}iy^3z}{4} + \sqrt{5}iyz^3$ |
| | $\mathbb{G}_4^{(1,0;a)}(E^{(a)},1)$ | $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{2}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{6}i}{70} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{6}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{2}i}{140} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{30}i}{175} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{280} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{27\sqrt{30}i}{700} & 0 \\ 0 & \frac{3\sqrt{5}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{3\sqrt{2}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{140} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{40} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{3\sqrt{2}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{1400} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{5}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{13\sqrt{30}i}{1400} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{280} \end{bmatrix}$ |
| 669 | symmetry | $\frac{\sqrt{70}x^4}{16} + \frac{\sqrt{70}ix^3y}{4} - \frac{3\sqrt{70}x^2y^2}{8} - \frac{\sqrt{70}ixy^3}{4} + \frac{\sqrt{70}y^4}{16}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_4^{(1,0;a)}(E^{(a)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{105}i}{350} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{5}i}{50} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{3\sqrt{70}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{70} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{70}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{175} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{100} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 670 | symmetry | $-\frac{\sqrt{10}x^4}{8}+\frac{\sqrt{10}ix^3y}{4}+\frac{3\sqrt{10}x^2z^2}{4}+\frac{\sqrt{10}ixy^3}{4}-\frac{3\sqrt{10}ixyz^2}{2}+\frac{\sqrt{10}y^4}{8}-\frac{3\sqrt{10}y^2z^2}{4}$ |
| | $\mathbb{G}_4^{(1,0;a)}(E^{(a)},3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{105}i}{700} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{30}i}{280} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{99\sqrt{5}i}{700} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{6}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{15}i}{700} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{27\sqrt{3}i}{140} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{105}i}{700} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{9\sqrt{5}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{30}i}{1400} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{3i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{17\sqrt{10}i}{1400} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{3i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{140} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{9\sqrt{5}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{70} & 0 & 0 & 0 \end{bmatrix}$ |
| 671 | symmetry | $-\frac{3\sqrt{5}x^3z}{4}+\frac{3\sqrt{5}ix^2yz}{4}-\frac{3\sqrt{5}xy^2z}{4}+\sqrt{5}xz^3+\frac{3\sqrt{5}iy^3z}{4}-\sqrt{5}iyz^3$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_4^{(1,0;a)}(E^{(b)},1)$ | $\begin{bmatrix} \frac{\sqrt{5}i}{280} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{27\sqrt{30}i}{700} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{3}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{30}i}{175} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{6}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{2}i}{140} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{2}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{6}i}{70} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{280} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{3\sqrt{5}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{13\sqrt{30}i}{1400} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{3\sqrt{2}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{1400} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{40} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{3\sqrt{2}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{140} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{3\sqrt{5}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{28} & 0 & 0 \end{bmatrix}$ |
| 672 | symmetry | $\frac{\sqrt{70}x^4}{16} - \frac{\sqrt{70}ix^3y}{4} - \frac{3\sqrt{70}x^2y^2}{8} + \frac{\sqrt{70}ixy^3}{4} + \frac{\sqrt{70}y^4}{16}$ |
| | $\mathbb{G}_4^{(1,0;a)}(E^{(b)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{5}i}{50} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{70}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{105}i}{350} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{100} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{3\sqrt{70}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{175} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{3\sqrt{70}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{70} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 673 | symmetry | $-\frac{\sqrt{10}x^4}{8} - \frac{\sqrt{10}ix^3y}{4} + \frac{3\sqrt{10}x^2z^2}{4} - \frac{\sqrt{10}ixy^3}{4} + \frac{3\sqrt{10}ixyz^2}{2} + \frac{\sqrt{10}y^4}{8} - \frac{3\sqrt{10}y^2z^2}{4}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_4^{(1,0;a)}(E^{(b)},3)$ | $\begin{bmatrix} 0 & 0 & 0 & -\frac{i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{27\sqrt{3}i}{140} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{15}i}{700} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{280} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{99\sqrt{5}i}{700} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{105}i}{700} \\ 0 & 0 & \frac{9\sqrt{5}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{70} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{3i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{140} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{3i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{17\sqrt{10}i}{1400} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{5}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{30}i}{1400} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{105}i}{700} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 674 | symmetry | 1 |
| | $\mathbb{G}_0^{(1,1;a)}(A)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 675 | symmetry | $-\frac{x^2}{2}-\frac{y^2}{2}+z^2$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_2^{(1,1;a)}(A)$ | $\begin{bmatrix} 0 & \frac{12i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}i}{140} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{2\sqrt{6}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{2}i}{140} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{2\sqrt{6}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{2}i}{140} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{12i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}i}{140} & 0 & 0 \\ \frac{3i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{3i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{35} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{6i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{3}i}{105} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{6i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{3}i}{105} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{3i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{35} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{3i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{21} & 0 \end{bmatrix}$ |
| 676 | symmetry | $\frac{\sqrt{6}xz}{2} + \frac{\sqrt{6}iyz}{2}$ |
| | $\mathbb{G}_2^{(1,1;a)}(E^{(a)}, 1)$ | $\begin{bmatrix} 0 & 0 & \frac{6\sqrt{3}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3i}{70} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{2i}{7} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{3}i}{70} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{2\sqrt{2}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{5}i}{70} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{30}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{5}i}{70} & 0 \\ 0 & \frac{3\sqrt{30}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{21} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{3\sqrt{3}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{11i}{105} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{15} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{3\sqrt{3}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{105} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{30}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{105} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{21} \end{bmatrix}$ |
| 677 | symmetry | $\frac{\sqrt{6}x^2}{4} - \frac{\sqrt{6}ixy}{2} - \frac{\sqrt{6}y^2}{4}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_2^{(1,1;a)}(E^{(a)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{35}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{4\sqrt{10}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{15}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{8\sqrt{2}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{5}i}{140} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{4\sqrt{3}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3i}{140} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{35}i}{105} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{3\sqrt{15}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{4\sqrt{10}i}{105} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{9\sqrt{3}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{30}i}{105} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{9\sqrt{3}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{8i}{105} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{3\sqrt{15}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{5}i}{105} & 0 & 0 & 0 \end{bmatrix}$ |
| 678 | symmetry | $\frac{\sqrt{6}xz}{2} - \frac{\sqrt{6}iyz}{2}$ |
| | $\mathbb{G}_2^{(1,1;a)}(E^{(b)},1)$ | $\begin{bmatrix} -\frac{2\sqrt{30}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{5}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{2\sqrt{2}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{5}i}{70} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{2i}{7} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{3}i}{70} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{6\sqrt{3}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3i}{70} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{3\sqrt{30}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{105} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{3\sqrt{3}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{105} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{15} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{3}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{11i}{105} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{3\sqrt{30}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{21} & 0 & 0 \end{bmatrix}$ |
| 679 | symmetry | $\frac{\sqrt{6}x^2}{4} + \frac{\sqrt{6}ixy}{2} - \frac{\sqrt{6}y^2}{4}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_2^{(1,1;a)}(E^{(b)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & \frac{4\sqrt{3}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3i}{140} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{8\sqrt{2}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{5}i}{140} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{4\sqrt{10}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{15}i}{140} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{35}i}{140} \\ 0 & 0 & \frac{3\sqrt{15}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{5}i}{105} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{9\sqrt{3}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{8i}{105} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{9\sqrt{3}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{30}i}{105} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{15}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{4\sqrt{10}i}{105} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{35}i}{105} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 680 | symmetry | $\frac{3x^4}{8}+\frac{3x^2y^2}{4}-3x^2z^2+\frac{3y^4}{8}-3y^2z^2+z^4$ |
| | $\mathbb{G}_4^{(1,1;a)}(A,1)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{330}i}{105} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{33}i}{70} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{2\sqrt{55}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}i}{210} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{2\sqrt{55}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}i}{210} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{330}i}{105} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{33}i}{70} & 0 & 0 \\ -\frac{\sqrt{330}i}{420} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{55}i}{385} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{330}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{8\sqrt{33}i}{1155} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{330}i}{210} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{110}i}{385} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{330}i}{210} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{110}i}{385} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{330}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{8\sqrt{33}i}{1155} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}i}{420} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{55}i}{385} & 0 \end{bmatrix}$ |
| 681 | symmetry | $\frac{\sqrt{70}x^3z}{4}-\frac{3\sqrt{70}xy^2z}{4}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_4^{(1,1;a)}(A,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{\sqrt{231}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}i}{700} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{385}i}{70} & \frac{\sqrt{330}i}{300} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}i}{1050} & 0 \\ \frac{\sqrt{385}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}i}{1050} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}i}{300} \\ 0 & -\frac{\sqrt{231}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}i}{700} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{2310}i}{420} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{770}i}{770} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}i}{1650} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}i}{420} & -\frac{3\sqrt{55}i}{550} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{385}i}{3850} & 0 \\ -\frac{\sqrt{2310}i}{420} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{385}i}{3850} & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{55}i}{550} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}i}{1650} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{2310}i}{420} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{770}i}{770} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 682 | symmetry | $\frac{3\sqrt{70}x^2yz}{4} - \frac{\sqrt{70}y^3z}{4}$ |
| | $\mathbb{G}_4^{(1,1;a)}(A,3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{\sqrt{231}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}}{700} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{385}}{70} & -\frac{\sqrt{330}}{300} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}}{1050} & 0 \\ -\frac{\sqrt{385}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}}{1050} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}}{300} \\ 0 & \frac{\sqrt{231}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}}{700} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{2310}}{420} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{770}}{770} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}}{1650} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}}{420} & \frac{3\sqrt{55}}{550} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{385}}{3850} & 0 \\ \frac{\sqrt{2310}}{420} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{385}}{3850} & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{55}}{550} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}}{1650} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{2310}}{420} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{770}}{770} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 683 | symmetry | $-\frac{3\sqrt{5}x^3z}{4} - \frac{3\sqrt{5}ix^2yz}{4} - \frac{3\sqrt{5}xy^2z}{4} + \sqrt{5}xz^3 - \frac{3\sqrt{5}iy^3z}{4} + \sqrt{5}iyz^3$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_4^{(1,1;a)}(E^{(a)},1)$ | $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{33}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{11}i}{35} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{11}i}{7} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{33}i}{210} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{22}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{55}i}{175} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{330}i}{210} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{55}i}{350} & 0 \\ 0 & -\frac{\sqrt{330}i}{210} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{33}i}{231} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{33}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{11}i}{385} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{22}i}{110} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{33}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}i}{11550} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{330}i}{210} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{13\sqrt{55}i}{1925} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{77}i}{385} \end{bmatrix}$ |
| 684 | symmetry | $\frac{\sqrt{70}x^4}{16}+\frac{\sqrt{70}ix^3y}{4}-\frac{3\sqrt{70}x^2y^2}{8}-\frac{\sqrt{70}ixy^3}{4}+\frac{\sqrt{70}y^4}{16}$ |
| | $\mathbb{G}_4^{(1,1;a)}(E^{(a)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{1155}i}{105} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{770}i}{350} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{330}i}{150} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{1155}i}{210} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{462}i}{1155} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{1155}i}{210} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{4\sqrt{770}i}{1925} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{55}i}{275} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 685 | symmetry | $-\frac{\sqrt{10}x^4}{8}+\frac{\sqrt{10}ix^3y}{4}+\frac{3\sqrt{10}x^2z^2}{4}+\frac{\sqrt{10}ixy^3}{4}-\frac{3\sqrt{10}ixyz^2}{2}+\frac{\sqrt{10}y^4}{8}-\frac{3\sqrt{10}y^2z^2}{4}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_4^{(1,1;a)}(E^{(a)},3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{770}i}{700} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{55}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{11\sqrt{330}i}{2100} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{11}i}{7} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{110}i}{700} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{66}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{22}i}{140} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{770}i}{1925} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{330}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{55}i}{1925} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{66}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{17\sqrt{165}i}{5775} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{66}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{22}i}{385} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{330}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{110}i}{385} & 0 & 0 & 0 \end{bmatrix}$ |
| 686 | symmetry | $-\frac{3\sqrt{5}x^3z}{4}+\frac{3\sqrt{5}ix^2yz}{4}-\frac{3\sqrt{5}xy^2z}{4}+\sqrt{5}xz^3+\frac{3\sqrt{5}iy^3z}{4}-\sqrt{5}iyz^3$ |
| | $\mathbb{G}_4^{(1,1;a)}(E^{(b)},1)$ | $\begin{bmatrix} \frac{\sqrt{330}i}{210} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{55}i}{350} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{22}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{55}i}{175} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{11}i}{7} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{33}i}{210} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{33}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{11}i}{35} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{77}i}{385} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{330}i}{210} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{13\sqrt{55}i}{1925} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{33}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}i}{11550} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{22}i}{110} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{33}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{11}i}{385} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{330}i}{210} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{33}i}{231} & 0 & 0 \end{bmatrix}$ |
| 687 | symmetry | $\frac{\sqrt{70}x^4}{16}-\frac{\sqrt{70}ix^3y}{4}-\frac{3\sqrt{70}x^2y^2}{8}+\frac{\sqrt{70}ixy^3}{4}+\frac{\sqrt{70}y^4}{16}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_4^{(1,1;a)}(E^{(b)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{330}i}{150} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{2\sqrt{1155}i}{105} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{770}i}{350} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{55}i}{275} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{1155}i}{210} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{4\sqrt{770}i}{1925} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{1155}i}{210} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{462}i}{1155} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 688 | symmetry | $-\frac{\sqrt{10}x^4}{8} - \frac{\sqrt{10}ix^3y}{4} + \frac{3\sqrt{10}x^2z^2}{4} - \frac{\sqrt{10}ixy^3}{4} + \frac{3\sqrt{10}ixyz^2}{2} + \frac{\sqrt{10}y^4}{8} - \frac{3\sqrt{10}y^2z^2}{4}$ |
| | $\mathbb{G}_4^{(1,1;a)}(E^{(b)},3)$ | $\begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{66}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{22}i}{140} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{11}i}{7} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{110}i}{700} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{55}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{11\sqrt{330}i}{2100} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{770}i}{700} \\ 0 & 0 & -\frac{\sqrt{330}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{110}i}{385} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{66}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{22}i}{385} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{66}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{17\sqrt{165}i}{5775} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{55}i}{1925} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{770}i}{1925} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 689 | symmetry | $z$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_1^{(a)}(A)$ | $\begin{bmatrix} 0 & \frac{i}{5} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{6}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{6}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{i}{5} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{3i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{14} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{7} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{7} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{3i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{14} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{14} & 0 \end{bmatrix}$ |
| 690 | symmetry | $\frac{\sqrt{2}x}{2} + \frac{\sqrt{2}iy}{2}$ |
| | $\mathbb{T}_1^{(a)}(E^{(a)})$ | $\begin{bmatrix} 0 & 0 & \frac{i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{3}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{10}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{14} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{2i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{2}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{2i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{14} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{14} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{14} \end{bmatrix}$ |
| 691 | symmetry | $\frac{\sqrt{2}x}{2} - \frac{\sqrt{2}iy}{2}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_1^{(a)}(E^{(b)})$ | $\begin{bmatrix} -\frac{\sqrt{10}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{6}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{3}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{10}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{2i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{14} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{3\sqrt{2}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{2i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{14} & 0 & 0 \end{bmatrix}$ |
| 692 | symmetry | $-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$ |
| | $\mathbb{T}_3^{(a)}(A,1)$ | $\begin{bmatrix} 0 & -\frac{3\sqrt{21}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{84} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{3\sqrt{14}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{84} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{3\sqrt{14}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{84} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{3\sqrt{21}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{84} & 0 & 0 \\ \frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{21}i}{30} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{2\sqrt{21}i}{105} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{2\sqrt{21}i}{105} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{30} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{14} & 0 \end{bmatrix}$ |
| 693 | symmetry | $\frac{3\sqrt{10}x^2y}{4} - \frac{\sqrt{10}y^3}{4}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_3^{(a)}(A,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{3\sqrt{210}}{280} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{84} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{14}}{56} & \frac{\sqrt{3}}{12} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{42} & 0 \\ -\frac{3\sqrt{14}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{12} \\ 0 & -\frac{3\sqrt{210}}{280} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{84} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{105} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{42} & -\frac{\sqrt{2}}{8} & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{14}}{56} & 0 \\ -\frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{14}}{56} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{8} \\ 0 & -\frac{\sqrt{210}}{105} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{28} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 694 | symmetry | $\frac{\sqrt{10}x^3}{4} - \frac{3\sqrt{10}xy^2}{4}$ |
| | $\mathbb{T}_3^{(a)}(A,3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{3\sqrt{210}i}{280} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{84} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{14}i}{56} & -\frac{\sqrt{3}i}{12} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{42} & 0 \\ \frac{3\sqrt{14}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{12} \\ 0 & \frac{3\sqrt{210}i}{280} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{84} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{105} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{42} & \frac{\sqrt{2}i}{8} & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{14}i}{56} & 0 \\ \frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{14}i}{56} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{8} \\ 0 & \frac{\sqrt{210}i}{105} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{28} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 695 | symmetry | $-\frac{\sqrt{3}x^3}{4} - \frac{\sqrt{3}ix^2y}{4} - \frac{\sqrt{3}xy^2}{4} + \sqrt{3}xz^2 - \frac{\sqrt{3}iy^3}{4} + \sqrt{3}iyz^2$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_3^{(a)}(E^{(a)},1)$ | $\begin{bmatrix} 0 & 0 & -\frac{9\sqrt{14}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{42} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{42}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{35}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{84} & 0 \\ 0 & \frac{\sqrt{35}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{14} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{14}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{42} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{2\sqrt{7}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{84} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{84} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{12} \end{bmatrix}$ |
| 696 | symmetry | $\frac{\sqrt{30}x^2z}{4} - \frac{\sqrt{30}ixyz}{2} - \frac{\sqrt{30}y^2z}{4}$ |
| | $\mathbb{T}_3^{(a)}(E^{(a)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{42}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{210}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{3\sqrt{35}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{84} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{35}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{35}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{42} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{21} & 0 & 0 & 0 \end{bmatrix}$ |
| 697 | symmetry | $-\frac{\sqrt{3}x^3}{4} + \frac{\sqrt{3}ix^2y}{4} - \frac{\sqrt{3}xy^2}{4} + \sqrt{3}xz^2 + \frac{\sqrt{3}iy^3}{4} - \sqrt{3}iyz^2$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_3^{(a)}(E^{(b)},1)$ | $\begin{bmatrix} \frac{3\sqrt{35}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{21}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{42}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{9\sqrt{14}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{42} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{35}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{14}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{2\sqrt{7}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{84} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{14}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{42} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{14} & 0 & 0 \end{bmatrix}$ |
| 698 | symmetry | $\frac{\sqrt{30}x^2z}{4}+\frac{\sqrt{30}ixyz}{2}-\frac{\sqrt{30}y^2z}{4}$ |
| | $\mathbb{T}_3^{(a)}(E^{(b)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & -\frac{3\sqrt{35}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{84} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{28} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{12} \\ 0 & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{21} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{35}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{42} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{84} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{6} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 699 | symmetry | $\frac{15x^4z}{8}+\frac{15x^2y^2z}{4}-5x^2z^3+\frac{15y^4z}{8}-5y^2z^3+z^5$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_5^{(a)}(A,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{30} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{30} & 0 & 0 \\ -\frac{\sqrt{6}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{5\sqrt{6}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{15}i}{70} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{5\sqrt{6}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{5\sqrt{6}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{5\sqrt{6}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{15}i}{70} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{14} & 0 \end{bmatrix}$ |
| 700 | symmetry | $-\frac{3\sqrt{70}x^4y}{16} - \frac{\sqrt{70}x^2y^3}{8} + \frac{3\sqrt{70}x^2yz^2}{2} + \frac{\sqrt{70}y^5}{16} - \frac{\sqrt{70}y^3z^2}{2}$ |
| | $\mathbb{T}_5^{(a)}(A,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{30} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{30} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{30} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{30} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{30} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{30} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{42}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{140} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{42} & \frac{1}{10} & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{7}}{70} & 0 \\ \frac{\sqrt{42}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{7}}{70} & 0 & 0 & 0 & 0 & 0 & \frac{1}{10} \\ 0 & -\frac{\sqrt{105}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{140} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{42}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{28} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 701 | symmetry | $-\frac{\sqrt{70}x^5}{16} + \frac{\sqrt{70}x^3y^2}{8} + \frac{\sqrt{70}x^3z^2}{2} + \frac{3\sqrt{70}xy^4}{16} - \frac{3\sqrt{70}xy^2z^2}{2}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_5^{(a)}(A,3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{30} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{30} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{30} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{30} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{30} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{30} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{42}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{140} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{42} & -\frac{i}{10} & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{7}i}{70} & 0 \\ -\frac{\sqrt{42}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{7}i}{70} & 0 & 0 & 0 & 0 & 0 & \frac{i}{10} \\ 0 & \frac{\sqrt{105}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{140} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{42}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{28} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 702 | symmetry | $\frac{3\sqrt{7}x^5}{16} - \frac{15\sqrt{7}ix^4y}{16} - \frac{15\sqrt{7}x^3y^2}{8} + \frac{15\sqrt{7}ix^2y^3}{8} + \frac{15\sqrt{7}xy^4}{16} - \frac{3\sqrt{7}iy^5}{16}$ |
| | $\mathbb{T}_5^{(a)}(E^{(a)},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{5} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{42}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 703 | symmetry | $\frac{\sqrt{30}x^5}{16} + \frac{\sqrt{30}ix^4y}{16} + \frac{\sqrt{30}x^3y^2}{8} - \frac{3\sqrt{30}x^3z^2}{4} + \frac{\sqrt{30}ix^2y^3}{8} - \frac{3\sqrt{30}ix^2yz^2}{4} + \frac{\sqrt{30}xy^4}{16} - \frac{3\sqrt{30}xy^2z^2}{4} + \frac{\sqrt{30}xz^4}{2} + \frac{\sqrt{30}iy^5}{16} - \frac{3\sqrt{30}iy^3z^2}{4} + \frac{\sqrt{30}iyz^4}{2}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_5^{(a)}(E^{(a)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{30} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{10} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{10} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{30} & 0 \\ 0 & -\frac{i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{10}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{13\sqrt{30}i}{420} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{5}i}{7} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{210} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{11i}{70} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{23\sqrt{6}i}{420} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{420} \end{bmatrix}$ |
| 704 | $\mathbb{T}_5^{(a)}(E^{(a)},3)$ | $\frac{3\sqrt{70}x^4z}{16}+\frac{3\sqrt{70}ix^3yz}{4}-\frac{9\sqrt{70}x^2y^2z}{8}-\frac{3\sqrt{70}ixy^3z}{4}+\frac{3\sqrt{70}y^4z}{16}$ $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{10} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{10} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{14} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{70} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{14} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{70} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{5} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 705 | symmetry | $-\frac{\sqrt{210}x^4z}{8}+\frac{\sqrt{210}ix^3yz}{4}+\frac{\sqrt{210}x^2z^3}{4}+\frac{\sqrt{210}ixy^3z}{4}-\frac{\sqrt{210}ixyz^3}{2}+\frac{\sqrt{210}y^4z}{8}-\frac{\sqrt{210}y^2z^3}{4}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_5^{(a)}(E^{(a)},4)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{60} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{20} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{60} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{30} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{14}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{4\sqrt{21}i}{105} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{70}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{35} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{70}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{105} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{14}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{42} & 0 & 0 & 0 \end{bmatrix}$ |
| 706 | symmetry | $\frac{3\sqrt{7}x^5}{16}+\frac{15\sqrt{7}ix^4y}{16}-\frac{15\sqrt{7}x^3y^2}{8}-\frac{15\sqrt{7}ix^2y^3}{8}+\frac{15\sqrt{7}xy^4}{16}+\frac{3\sqrt{7}iy^5}{16}$ |
| | $\mathbb{T}_5^{(a)}(E^{(b)},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{5} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{14} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{10} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 707 | symmetry | $\frac{\sqrt{30}x^5}{16}-\frac{\sqrt{30}ix^4y}{16}+\frac{\sqrt{30}x^3y^2}{8}-\frac{3\sqrt{30}x^3z^2}{4}-\frac{\sqrt{30}ix^2y^3}{8}+\frac{3\sqrt{30}ix^2yz^2}{4}+\frac{\sqrt{30}xy^4}{16}-\frac{3\sqrt{30}xy^2z^2}{4}+\frac{\sqrt{30}xz^4}{2}-\frac{\sqrt{30}iy^5}{16}+\frac{3\sqrt{30}iy^3z^2}{4}-\frac{\sqrt{30}iyz^4}{2}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_5^{(a)}(E^{(b)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{30} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{10} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{10} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{30} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{420} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{23\sqrt{6}i}{420} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{10}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{11i}{70} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{5}i}{7} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{210} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{10}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{13\sqrt{30}i}{420} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{28} & 0 & 0 \end{bmatrix}$ |
| 708 | symmetry | $\frac{3\sqrt{70}x^4z}{16} - \frac{3\sqrt{70}ix^3yz}{4} - \frac{9\sqrt{70}x^2y^2z}{8} + \frac{3\sqrt{70}ixy^3z}{4} + \frac{3\sqrt{70}y^4z}{16}$ |
| | $\mathbb{T}_5^{(a)}(E^{(b)},3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{5} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{21}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{21}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{70} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 709 | symmetry | $-\frac{\sqrt{210}x^4z}{8} - \frac{\sqrt{210}ix^3yz}{4} + \frac{\sqrt{210}x^2z^3}{4} - \frac{\sqrt{210}ixy^3z}{4} + \frac{\sqrt{210}ixyz^3}{2} + \frac{\sqrt{210}y^4z}{8} - \frac{\sqrt{210}y^2z^3}{4}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_5^{(a)}(E^{(b)},4)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{60} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{20} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{20} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{60} \\ 0 & 0 & -\frac{\sqrt{14}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{42} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{70}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{105} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{35} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{4\sqrt{21}i}{105} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{30} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 710 | symmetry | $-\frac{3x^2z}{2}-\frac{3y^2z}{2}+z^3$ |
| | $\mathbb{T}_3^{(1,-1;a)}(A,1)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{6}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{14} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{2i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{2i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{14} & 0 & 0 \\ \frac{\sqrt{6}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2i}{7} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{6}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{6}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{7} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{6}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{7} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2i}{7} & 0 \end{bmatrix}$ |
| 711 | symmetry | $\frac{3\sqrt{10}x^2y}{4}-\frac{\sqrt{10}y^3}{4}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_3^{(1,-1;a)}(A,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{1}{14} & \frac{\sqrt{42}}{28} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{14} & 0 \\ -\frac{1}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{14} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{28} \\ 0 & -\frac{\sqrt{15}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{6}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{14} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{28} & \frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & \frac{3}{14} & 0 \\ -\frac{\sqrt{6}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3}{14} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{14} \\ 0 & -\frac{\sqrt{15}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{14} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{6}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{14} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 712 | symmetry | $\frac{\sqrt{10}x^3}{4} - \frac{3\sqrt{10}xy^2}{4}$ |
| | $\mathbb{T}_3^{(1,-1;a)}(A,3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{i}{14} & -\frac{\sqrt{42}i}{28} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{14} & 0 \\ \frac{i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{14} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{28} \\ 0 & \frac{\sqrt{15}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{6}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{14} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{28} & -\frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & \frac{3i}{14} & 0 \\ \frac{\sqrt{6}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3i}{14} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} \\ 0 & \frac{\sqrt{15}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{14} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{6}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{14} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 713 | symmetry | $-\frac{\sqrt{3}x^3}{4} - \frac{\sqrt{3}ix^2y}{4} - \frac{\sqrt{3}xy^2}{4} + \sqrt{3}xz^2 - \frac{\sqrt{3}iy^3}{4} + \sqrt{3}iyz^2$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_3^{(1,-1;a)}(E^{(a)},1)$ | $\begin{bmatrix} 0 & 0 & -\frac{3i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{7} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{3}i}{105} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3i}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{30} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{14} & 0 \\ 0 & \frac{3\sqrt{10}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2i}{7} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{3i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{3}i}{21} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{2}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{42} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{3i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{14} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{10}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{21} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{21} \end{bmatrix}$ |
| 714 | symmetry | $\frac{\sqrt{30}x^2z}{4} - \frac{\sqrt{30}ixyz}{2} - \frac{\sqrt{30}y^2z}{4}$ |
| | $\mathbb{T}_3^{(1,-1;a)}(E^{(a)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{3}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{15}i}{105} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{6}i}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{10}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{3\sqrt{2}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{3\sqrt{10}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{7} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{3\sqrt{10}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{21} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{2}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{6}i}{21} & 0 & 0 & 0 \end{bmatrix}$ |
| 715 | symmetry | $-\frac{\sqrt{3}x^3}{4} + \frac{\sqrt{3}ix^2y}{4} - \frac{\sqrt{3}xy^2}{4} + \sqrt{3}xz^2 + \frac{\sqrt{3}iy^3}{4} - \sqrt{3}iyz^2$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_3^{(1,-1;a)}(E^{(b)},1)$ | $\begin{bmatrix} \frac{\sqrt{10}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{6}i}{30} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{3}i}{105} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3i}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{3i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{7} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{3\sqrt{10}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{3i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{14} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{3\sqrt{2}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{42} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{3i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{3}i}{21} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{3\sqrt{10}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2i}{7} & 0 & 0 \end{bmatrix}$ |
| 716 | symmetry | $\frac{\sqrt{30}x^2z}{4}+\frac{\sqrt{30}ixyz}{2}-\frac{\sqrt{30}y^2z}{4}$ |
| | $\mathbb{T}_3^{(1,-1;a)}(E^{(b)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{10}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{105} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{6}i}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}i}{28} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{28} \\ 0 & 0 & \frac{3\sqrt{2}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{6}i}{21} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{3\sqrt{10}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{21} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{7} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{21} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{21} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 717 | symmetry | $\frac{15x^4z}{8}+\frac{15x^2y^2z}{4}-5x^2z^3+\frac{15y^4z}{8}-5y^2z^3+z^5$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_5^{(1,-1;a)}(A,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{20} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{20} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{20} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{20} & 0 & 0 \\ -\frac{\sqrt{5}i}{210} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{5}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{2}i}{35} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{5}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{15}i}{35} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{5}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{15}i}{35} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{2}i}{35} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{210} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{35} & 0 \end{bmatrix}$ |
| 718 | symmetry | $-\frac{3\sqrt{70}x^4y}{16} - \frac{\sqrt{70}x^2y^3}{8} + \frac{3\sqrt{70}x^2yz^2}{2} + \frac{\sqrt{70}y^5}{16} - \frac{\sqrt{70}y^3z^2}{2}$ |
| | $\mathbb{T}_5^{(1,-1;a)}(A,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{50} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{50} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{50} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{50} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{50} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{50} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{35}}{105} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{35} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{35}}{175} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{105} & -\frac{\sqrt{30}}{25} & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{210}}{175} & 0 \\ \frac{\sqrt{35}}{105} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{210}}{175} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{25} \\ 0 & -\frac{\sqrt{14}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{35}}{175} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{35}}{105} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{35} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 719 | symmetry | $-\frac{\sqrt{70}x^5}{16} + \frac{\sqrt{70}x^3y^2}{8} + \frac{\sqrt{70}x^3z^2}{2} + \frac{3\sqrt{70}xy^4}{16} - \frac{3\sqrt{70}xy^2z^2}{2}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_5^{(1,-1;a)}(A,3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{50} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{50} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{50} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{50} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{50} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{50} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{35}i}{105} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{35} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{35}i}{175} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{105} & \frac{\sqrt{30}i}{25} & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{210}i}{175} & 0 \\ -\frac{\sqrt{35}i}{105} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{210}i}{175} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{25} \\ 0 & \frac{\sqrt{14}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{35}i}{175} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{35}i}{105} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{35} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 720 | symmetry | $\frac{3\sqrt{7}x^5}{16} - \frac{15\sqrt{7}ix^4y}{16} - \frac{15\sqrt{7}x^3y^2}{8} + \frac{15\sqrt{7}ix^2y^3}{8} + \frac{15\sqrt{7}xy^4}{16} - \frac{3\sqrt{7}iy^5}{16}$ |
| | $\mathbb{T}_5^{(1,-1;a)}(E^{(a)},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{5} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{35}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 721 | symmetry | $\frac{\sqrt{30}x^5}{16} + \frac{\sqrt{30}ix^4y}{16} + \frac{\sqrt{30}x^3y^2}{8} - \frac{3\sqrt{30}x^3z^2}{4} + \frac{\sqrt{30}ix^2y^3}{8} - \frac{3\sqrt{30}ix^2yz^2}{4} + \frac{\sqrt{30}xy^4}{16} - \frac{3\sqrt{30}xy^2z^2}{4} + \frac{\sqrt{30}xz^4}{2} + \frac{\sqrt{30}iy^5}{16} - \frac{3\sqrt{30}iy^3z^2}{4} + \frac{\sqrt{30}iyz^4}{2}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_5^{(1,-1;a)}(E^{(a)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{10} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{10} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{5}i}{50} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{50} & 0 \\ 0 & -\frac{\sqrt{30}i}{210} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{7} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{3}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{13i}{35} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{6}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{35} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{11\sqrt{30}i}{175} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{210} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{23\sqrt{5}i}{175} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{35} \end{bmatrix}$ |
| 722 | symmetry | $\frac{3\sqrt{70}x^4z}{16}+\frac{3\sqrt{70}ix^3yz}{4}-\frac{9\sqrt{70}x^2y^2z}{8}-\frac{3\sqrt{70}ixy^3z}{4}+\frac{3\sqrt{70}y^4z}{16}$ |
| | $\mathbb{T}_5^{(1,-1;a)}(E^{(a)},3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{50} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{5}i}{50} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{70} & 0 & 0 & 0 & 0 & 0 & -\frac{6\sqrt{7}i}{35} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{70} & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{105}i}{175} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{30}i}{25} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 723 | symmetry | $-\frac{\sqrt{210}x^4z}{8}+\frac{\sqrt{210}ix^3yz}{4}+\frac{\sqrt{210}x^2z^3}{4}+\frac{\sqrt{210}ixy^3z}{4}-\frac{\sqrt{210}ixyz^3}{2}+\frac{\sqrt{210}y^4z}{8}-\frac{\sqrt{210}y^2z^3}{4}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_5^{(1,-1;a)}(E^{(a)},4)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{100} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{100} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{35}i}{100} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{20} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{5}i}{25} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{105}i}{210} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{8\sqrt{70}i}{175} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{210}i}{175} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{4\sqrt{7}i}{35} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{105}i}{210} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{35}i}{35} & 0 & 0 & 0 \end{bmatrix}$ |
| 724 | symmetry | $\frac{3\sqrt{7}x^5}{16}+\frac{15\sqrt{7}ix^4y}{16}-\frac{15\sqrt{7}x^3y^2}{8}-\frac{15\sqrt{7}ix^2y^3}{8}+\frac{15\sqrt{7}xy^4}{16}+\frac{3\sqrt{7}iy^5}{16}$ |
| | $\mathbb{T}_5^{(1,-1;a)}(E^{(b)},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{10} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{35} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{5} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 725 | symmetry | $\frac{\sqrt{30}x^5}{16}-\frac{\sqrt{30}ix^4y}{16}+\frac{\sqrt{30}x^3y^2}{8}-\frac{3\sqrt{30}x^3z^2}{4}-\frac{\sqrt{30}ix^2y^3}{8}+\frac{3\sqrt{30}ix^2yz^2}{4}+\frac{\sqrt{30}xy^4}{16}-\frac{3\sqrt{30}xy^2z^2}{4}+\frac{\sqrt{30}xz^4}{2}-\frac{\sqrt{30}iy^5}{16}+\frac{3\sqrt{30}iy^3z^2}{4}-\frac{\sqrt{30}iyz^4}{2}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_5^{(1,-1;a)}(E^{(b)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{50} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{5}i}{50} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{10} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{10} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{30}i}{210} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{23\sqrt{5}i}{175} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{3}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{11\sqrt{30}i}{175} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{6}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{35} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{3}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{13i}{35} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{210} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{7} & 0 & 0 \end{bmatrix}$ |
| 726 | symmetry | $\frac{3\sqrt{70}x^4z}{16} - \frac{3\sqrt{70}ix^3yz}{4} - \frac{9\sqrt{70}x^2y^2z}{8} + \frac{3\sqrt{70}ixy^3z}{4} + \frac{3\sqrt{70}y^4z}{16}$ |
| | $\mathbb{T}_5^{(1,-1;a)}(E^{(b)},3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{5}i}{50} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{50} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{30}i}{25} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{70}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{105}i}{175} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{70}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{6\sqrt{7}i}{35} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 727 | symmetry | $-\frac{\sqrt{210}x^4z}{8} - \frac{\sqrt{210}ix^3yz}{4} + \frac{\sqrt{210}x^2z^3}{4} - \frac{\sqrt{210}ixy^3z}{4} + \frac{\sqrt{210}ixyz^3}{2} + \frac{\sqrt{210}y^4z}{8} - \frac{\sqrt{210}y^2z^3}{4}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_5^{(1,-1;a)}(E^{(b)},4)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{20} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{35}i}{100} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{100} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{100} \\ 0 & 0 & -\frac{\sqrt{105}i}{210} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{35}i}{35} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{4\sqrt{7}i}{35} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{210}i}{175} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{210} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{8\sqrt{70}i}{175} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{5}i}{25} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 728 | symmetry | $z$ |
| | $\mathbb{T}_1^{(1,0;a)}(A)$ | $\begin{bmatrix} 0 & \frac{\sqrt{2}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{3}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{3}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{3\sqrt{2}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{9\sqrt{2}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{14} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{3\sqrt{2}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{3\sqrt{2}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{9\sqrt{2}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{14} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{14} & 0 \end{bmatrix}$ |
| 729 | symmetry | $\frac{\sqrt{2}x}{2} + \frac{\sqrt{2}iy}{2}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_1^{(1,0;a)}(E^{(a)})$ | $\begin{bmatrix} 0 & 0 & \frac{\sqrt{2}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{6}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{3\sqrt{5}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{6\sqrt{2}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{9i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{6\sqrt{2}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{14} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{5}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{28} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{28} \end{bmatrix}$ |
| 730 | symmetry | $\frac{\sqrt{2}x}{2} - \frac{\sqrt{2}iy}{2}$ |
| | $\mathbb{T}_1^{(1,0;a)}(E^{(b)})$ | $\begin{bmatrix} -\frac{\sqrt{5}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{3}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{6}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{2}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{3\sqrt{5}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{6\sqrt{2}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{14} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{9i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{6\sqrt{2}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{3\sqrt{5}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{28} & 0 & 0 \end{bmatrix}$ |
| 731 | symmetry | $-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_3^{(1,0;a)}(A,1)$ | $\begin{bmatrix} 0 & -\frac{3\sqrt{7}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{42}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{42}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{3\sqrt{7}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{28} & 0 & 0 \\ \frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{7}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{2\sqrt{7}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{84} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{2\sqrt{7}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{84} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{84} & 0 \end{bmatrix}$ |
| 732 | symmetry | $\frac{3\sqrt{10}x^2y}{4} - \frac{\sqrt{10}y^3}{4}$ |
| | $\mathbb{T}_3^{(1,0;a)}(A,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{3\sqrt{70}}{560} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{112} & -\frac{1}{4} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{14} & 0 \\ -\frac{\sqrt{42}}{112} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & \frac{1}{4} \\ 0 & -\frac{3\sqrt{70}}{560} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{168} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{56} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{14} & \frac{\sqrt{6}}{48} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{112} & 0 \\ -\frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{112} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{48} \\ 0 & -\frac{\sqrt{70}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{56} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{168} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 733 | symmetry | $\frac{\sqrt{10}x^3}{4} - \frac{3\sqrt{10}xy^2}{4}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_3^{(1,0;a)}(A,3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{3\sqrt{70}i}{560} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{112} & \frac{i}{4} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 \\ \frac{\sqrt{42}i}{112} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & \frac{i}{4} \\ 0 & \frac{3\sqrt{70}i}{560} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{168} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{56} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & -\frac{\sqrt{6}i}{48} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{112} & 0 \\ \frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{112} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{48} \\ 0 & \frac{\sqrt{70}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{56} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{168} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 734 | symmetry | $-\frac{\sqrt{3}x^3}{4} - \frac{\sqrt{3}ix^2y}{4} - \frac{\sqrt{3}xy^2}{4} + \sqrt{3}xz^2 - \frac{\sqrt{3}iy^3}{4} + \sqrt{3}iyz^2$ |
| | $\mathbb{T}_3^{(1,0;a)}(E^{(a)},1)$ | $\begin{bmatrix} 0 & 0 & -\frac{3\sqrt{42}i}{280} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{14}i}{280} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{40} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{280} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{28} & 0 \\ 0 & \frac{\sqrt{105}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{84} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{42}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{84} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{2\sqrt{21}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{168} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{168} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{168} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{24} \end{bmatrix}$ |
| 735 | symmetry | $\frac{\sqrt{30}x^2z}{4} - \frac{\sqrt{30}ixyz}{2} - \frac{\sqrt{30}y^2z}{4}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_3^{(1,0;a)}(E^{(a)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{14}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{70}i}{280} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{7}i}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{105}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{21}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{168} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{105}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{168} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{105}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{84} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{21}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{42} & 0 & 0 & 0 \end{bmatrix}$ |
| 736 | symmetry | $-\frac{\sqrt{3}x^3}{4}+\frac{\sqrt{3}ix^2y}{4}-\frac{\sqrt{3}xy^2}{4}+\sqrt{3}xz^2+\frac{\sqrt{3}iy^3}{4}-\sqrt{3}iyz^2$ |
| | $\mathbb{T}_3^{(1,0;a)}(E^{(b)},1)$ | $\begin{bmatrix} \frac{\sqrt{105}i}{280} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{7}i}{40} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{14}i}{280} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{3\sqrt{42}i}{280} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{105}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{168} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{42}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{168} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{2\sqrt{21}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{168} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{42}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{84} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{84} & 0 & 0 \end{bmatrix}$ |
| 737 | symmetry | $\frac{\sqrt{30}x^2z}{4}+\frac{\sqrt{30}ixyz}{2}-\frac{\sqrt{30}y^2z}{4}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_3^{(1,0;a)}(E^{(b)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{105}i}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{280} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{7}i}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{28} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{4} \\ 0 & 0 & \frac{\sqrt{21}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{42} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{105}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{84} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{168} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{168} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{12} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 738 | symmetry | $\frac{15x^4z}{8}+\frac{15x^2y^2z}{4}-5x^2z^3+\frac{15y^4z}{8}-5y^2z^3+z^5$ |
| | $\mathbb{T}_5^{(1,0;a)}(A,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{10} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{10} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{10} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{10} & 0 & 0 \\ -\frac{\sqrt{5}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{420} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{5}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}i}{140} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{5}i}{7} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{210} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{5}i}{7} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{210} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}i}{140} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{420} & 0 \end{bmatrix}$ |
| 739 | symmetry | $-\frac{3\sqrt{70}x^4y}{16}-\frac{\sqrt{70}x^2y^3}{8}+\frac{3\sqrt{70}x^2yz^2}{2}+\frac{\sqrt{70}y^5}{16}-\frac{\sqrt{70}y^3z^2}{2}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_5^{(1,0;a)}(A,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{25} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{25} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{25} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{25} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{25} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{25} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{35}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{420} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{700} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{35} & -\frac{\sqrt{30}}{300} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{700} & 0 \\ \frac{\sqrt{35}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{700} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{300} \\ 0 & -\frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{700} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{35}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{420} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 740 | symmetry | $-\frac{\sqrt{70}x^5}{16} + \frac{\sqrt{70}x^3y^2}{8} + \frac{\sqrt{70}x^3z^2}{2} + \frac{3\sqrt{70}xy^4}{16} - \frac{3\sqrt{70}xy^2z^2}{2}$ |
| | $\mathbb{T}_5^{(1,0;a)}(A,3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{25} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{25} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{25} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{25} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{25} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{25} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{35}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{420} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{700} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{35} & \frac{\sqrt{30}i}{300} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{700} & 0 \\ -\frac{\sqrt{35}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{700} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{300} \\ 0 & \frac{\sqrt{14}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{700} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{35}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{420} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 741 | symmetry | $\frac{3\sqrt{7}x^5}{16} - \frac{15\sqrt{7}ix^4y}{16} - \frac{15\sqrt{7}x^3y^2}{8} + \frac{15\sqrt{7}ix^2y^3}{8} + \frac{15\sqrt{7}xy^4}{16} - \frac{3\sqrt{7}iy^5}{16}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_5^{(1,0;a)}(E^{(a)},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{5} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{60} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{3\sqrt{35}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{420} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 742 | symmetry | $\frac{\sqrt{30}x^5}{16}+\frac{\sqrt{30}ix^4y}{16}+\frac{\sqrt{30}x^3y^2}{8}-\frac{3\sqrt{30}x^3z^2}{4}+\frac{\sqrt{30}ix^2y^3}{8}-\frac{3\sqrt{30}ix^2yz^2}{4}+\frac{\sqrt{30}xy^4}{16}-\frac{3\sqrt{30}xy^2z^2}{4}+\frac{\sqrt{30}xz^4}{2}+\frac{\sqrt{30}iy^5}{16}-\frac{3\sqrt{30}iy^3z^2}{4}+\frac{\sqrt{30}iyz^4}{2}$ |
| | $\mathbb{T}_5^{(1,0;a)}(E^{(a)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{5} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{5} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{5}i}{25} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{25} & 0 \\ 0 & -\frac{\sqrt{30}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{84} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{3}i}{7} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{13i}{420} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{6}i}{7} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{420} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{7} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{11\sqrt{30}i}{2100} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{23\sqrt{5}i}{2100} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{420} \end{bmatrix}$ |
| 743 | symmetry | $\frac{3\sqrt{70}x^4z}{16}+\frac{3\sqrt{70}ix^3yz}{4}-\frac{9\sqrt{70}x^2y^2z}{8}-\frac{3\sqrt{70}ixy^3z}{4}+\frac{3\sqrt{70}y^4z}{16}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_5^{(1,0;a)}(E^{(a)},3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{25} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{5}i}{25} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{3\sqrt{70}i}{70} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{70} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{70}i}{70} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{1050} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{150} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 744 | symmetry | $-\frac{\sqrt{210}x^4z}{8}+\frac{\sqrt{210}ix^3yz}{4}+\frac{\sqrt{210}x^2z^3}{4}+\frac{\sqrt{210}ixy^3z}{4}-\frac{\sqrt{210}ixyz^3}{2}+\frac{\sqrt{210}y^4z}{8}-\frac{\sqrt{210}y^2z^3}{4}$ |
| | $\mathbb{T}_5^{(1,0;a)}(E^{(a)},4)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{50} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{50} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{35}i}{50} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{10} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{150} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{105}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{70}i}{525} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{21}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{1050} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{21}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{105} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{105}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{210} & 0 & 0 & 0 \end{bmatrix}$ |
| 745 | symmetry | $\frac{3\sqrt{7}x^5}{16}+\frac{15\sqrt{7}ix^4y}{16}-\frac{15\sqrt{7}x^3y^2}{8}-\frac{15\sqrt{7}ix^2y^3}{8}+\frac{15\sqrt{7}xy^4}{16}+\frac{3\sqrt{7}iy^5}{16}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_5^{(1,0;a)}(E^{(b)},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{5} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{35}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{420} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{60} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 746 | symmetry | $\frac{\sqrt{30}x^5}{16} - \frac{\sqrt{30}ix^4y}{16} + \frac{\sqrt{30}x^3y^2}{8} - \frac{3\sqrt{30}x^3z^2}{4} - \frac{\sqrt{30}ix^2y^3}{8} + \frac{3\sqrt{30}ix^2yz^2}{4} + \frac{\sqrt{30}xy^4}{16} - \frac{3\sqrt{30}xy^2z^2}{4} + \frac{\sqrt{30}xz^4}{2} - \frac{\sqrt{30}iy^5}{16} + \frac{3\sqrt{30}iy^3z^2}{4} - \frac{\sqrt{30}iyz^4}{2}$ |
| | $\mathbb{T}_5^{(1,0;a)}(E^{(b)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{25} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{5}i}{25} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{5} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{5} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{420} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{30}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{23\sqrt{5}i}{2100} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{3}i}{7} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{11\sqrt{30}i}{2100} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{6}i}{7} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{420} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{3}i}{7} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{13i}{420} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{84} & 0 & 0 \end{bmatrix}$ |
| 747 | symmetry | $\frac{3\sqrt{70}x^4z}{16} - \frac{3\sqrt{70}ix^3yz}{4} - \frac{9\sqrt{70}x^2y^2z}{8} + \frac{3\sqrt{70}ixy^3z}{4} + \frac{3\sqrt{70}y^4z}{16}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_5^{(1,0;a)}(E^{(b)},3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{5}i}{25} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{25} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{150} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{3\sqrt{70}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{1050} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{3\sqrt{70}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{70} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 748 | symmetry | $-\frac{\sqrt{210}x^4z}{8} - \frac{\sqrt{210}ix^3yz}{4} + \frac{\sqrt{210}x^2z^3}{4} - \frac{\sqrt{210}ixy^3z}{4} + \frac{\sqrt{210}ixyz^3}{2} + \frac{\sqrt{210}y^4z}{8} - \frac{\sqrt{210}y^2z^3}{4}$ |
| | $\mathbb{T}_5^{(1,0;a)}(E^{(b)},4)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{10} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{35}i}{50} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{50} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{50} \\ 0 & 0 & -\frac{\sqrt{105}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{210} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{21}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{105} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{1050} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{70}i}{525} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{150} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 749 | symmetry | $z$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_1^{(1,1;a)}(A)$ | $\begin{bmatrix} 0 & -\frac{i}{5} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{6}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{6}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{i}{5} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{2i}{7} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{6i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{2i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{2i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{6i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{2i}{7} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{28} & 0 \end{bmatrix}$ |
| 750 | symmetry | $\frac{\sqrt{2}x}{2} + \frac{\sqrt{2}iy}{2}$ |
| | $\mathbb{T}_1^{(1,1;a)}(E^{(a)})$ | $\begin{bmatrix} 0 & 0 & -\frac{i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{3}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{2\sqrt{10}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{8i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{6\sqrt{2}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{8i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{10}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{28} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{28} \end{bmatrix}$ |
| 751 | symmetry | $\frac{\sqrt{2}x}{2} - \frac{\sqrt{2}iy}{2}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_1^{(1,1;a)}(E^{(b)})$ | $\begin{bmatrix} \frac{\sqrt{10}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{6}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{3}i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{i}{10} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{2\sqrt{10}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{8i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{6\sqrt{2}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{8i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{2\sqrt{10}i}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{28} & 0 & 0 \end{bmatrix}$ |
| 752 | symmetry | $-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$ |
| | $\mathbb{T}_3^{(1,1;a)}(A,1)$ | $\begin{bmatrix} 0 & \frac{9i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{3\sqrt{6}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{6}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{9i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{28} & 0 & 0 \\ \frac{5i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{2i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{2i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{5i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{28} & 0 \end{bmatrix}$ |
| 753 | symmetry | $\frac{3\sqrt{10}x^2y}{4} - \frac{\sqrt{10}y^3}{4}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_3^{(1,1;a)}(A,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & \frac{9\sqrt{10}}{112} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{15\sqrt{6}}{112} & \frac{\sqrt{7}}{28} & 0 & 0 & 0 & 0 & 0 & -\frac{1}{14} & 0 \\ \frac{15\sqrt{6}}{112} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{14} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{28} \\ 0 & \frac{9\sqrt{10}}{112} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{5}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{56} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3}{56} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{5}{42} & -\frac{\sqrt{42}}{112} & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{6}}{112} & 0 \\ -\frac{5}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{6}}{112} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{112} \\ 0 & -\frac{\sqrt{10}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3}{56} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{5}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{56} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 754 | symmetry | $\frac{\sqrt{10}x^3}{4} - \frac{3\sqrt{10}xy^2}{4}$ |
| | $\mathbb{T}_3^{(1,1;a)}(A,3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & \frac{9\sqrt{10}i}{112} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{15\sqrt{6}i}{112} & -\frac{\sqrt{7}i}{28} & 0 & 0 & 0 & 0 & 0 & -\frac{i}{14} & 0 \\ -\frac{15\sqrt{6}i}{112} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{14} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{28} \\ 0 & -\frac{9\sqrt{10}i}{112} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{5i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{56} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3i}{56} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{5i}{42} & \frac{\sqrt{42}i}{112} & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{6}i}{112} & 0 \\ \frac{5i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{6}i}{112} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{112} \\ 0 & \frac{\sqrt{10}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3i}{56} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{5i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{56} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 755 | symmetry | $-\frac{\sqrt{3}x^3}{4} - \frac{\sqrt{3}ix^2y}{4} - \frac{\sqrt{3}xy^2}{4} + \sqrt{3}xz^2 - \frac{\sqrt{3}iy^3}{4} + \sqrt{3}iyz^2$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_3^{(1,1;a)}(E^{(a)},1)$ | $\begin{bmatrix} 0 & 0 & \frac{9\sqrt{6}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{2}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{3i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{15}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{28} & 0 \\ 0 & \frac{\sqrt{15}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{6}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{2\sqrt{3}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{56} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{56} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{56} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{56} \end{bmatrix}$ |
| 756 | symmetry | $\frac{\sqrt{30}x^2z}{4} - \frac{\sqrt{30}ixyz}{2} - \frac{\sqrt{30}y^2z}{4}$ |
| | $\mathbb{T}_3^{(1,1;a)}(E^{(a)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{15\sqrt{2}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{3\sqrt{10}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3i}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{3\sqrt{15}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{5\sqrt{3}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{15}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{56} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{15}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{5\sqrt{3}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{14} & 0 & 0 & 0 \end{bmatrix}$ |
| 757 | symmetry | $-\frac{\sqrt{3}x^3}{4} + \frac{\sqrt{3}ix^2y}{4} - \frac{\sqrt{3}xy^2}{4} + \sqrt{3}xz^2 + \frac{\sqrt{3}iy^3}{4} - \sqrt{3}iyz^2$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_3^{(1,1;a)}(E^{(b)},1)$ | $\begin{bmatrix} -\frac{3\sqrt{15}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{3i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{3\sqrt{2}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{9\sqrt{6}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{15}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{6}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{56} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{2\sqrt{3}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{56} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{6}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{28} & 0 & 0 \end{bmatrix}$ |
| 758 | symmetry | $\frac{\sqrt{30}x^2z}{4} + \frac{\sqrt{30}ixyz}{2} - \frac{\sqrt{30}y^2z}{4}$ |
| | $\mathbb{T}_3^{(1,1;a)}(E^{(b)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & \frac{3\sqrt{15}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{3\sqrt{10}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3i}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{15\sqrt{2}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{28} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{28} \\ 0 & 0 & \frac{5\sqrt{3}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{15}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{56} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{3}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{56} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{28} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 759 | symmetry | $-\frac{x^2}{2} - \frac{y^2}{2} + z^2$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_2^{(a)}(A)$ | $\begin{bmatrix} 0 & \frac{3\sqrt{10}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{14} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{15}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{5}}{70} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{15}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{5}}{70} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{14} & 0 & 0 \\ -\frac{\sqrt{10}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{10}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3}{14} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{10}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{70} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{10}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{70} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3}{14} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{14} & 0 \end{bmatrix}$ |
| 760 | symmetry | $\frac{\sqrt{6}xz}{2} + \frac{\sqrt{6}iyz}{2}$ |
| | $\mathbb{M}_2^{(a)}(E^{(a)}, 1)$ | $\begin{bmatrix} 0 & 0 & \frac{3\sqrt{30}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{70} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{10}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{70} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{14} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{7} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{14} & 0 \\ 0 & -\frac{\sqrt{3}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{30}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{11\sqrt{10}}{140} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{10} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{14} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{28} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{28} \end{bmatrix}$ |
| 761 | symmetry | $\frac{\sqrt{6}x^2}{4} - \frac{\sqrt{6}ixy}{2} - \frac{\sqrt{6}y^2}{4}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_2^{(a)}(E^{(a)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{2}{7} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{4\sqrt{5}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{30}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{140} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{6}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2}{7} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{3\sqrt{30}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{7} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{3\sqrt{30}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{10}}{35} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{6}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{14} & 0 & 0 & 0 \end{bmatrix}$ |
| 762 | symmetry | $\frac{\sqrt{6}xz}{2} - \frac{\sqrt{6}iyz}{2}$ |
| | $\mathbb{M}_2^{(a)}(E^{(b)},1)$ | $\begin{bmatrix} -\frac{\sqrt{3}}{7} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{5}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{14} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{10}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{70} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{3\sqrt{30}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{70} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{3}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{30}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{14} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{10} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{30}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{11\sqrt{10}}{140} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{28} & 0 & 0 \end{bmatrix}$ |
| 763 | symmetry | $\frac{\sqrt{6}x^2}{4} + \frac{\sqrt{6}ixy}{2} - \frac{\sqrt{6}y^2}{4}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_2^{(a)}(E^{(b)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & \frac{\sqrt{30}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{140} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{4\sqrt{5}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{2}{7} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{28} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{28} \\ 0 & 0 & -\frac{\sqrt{6}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{30}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{10}}{35} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{3\sqrt{30}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{7} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2}{7} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 764 | symmetry | $\frac{3x^4}{8}+\frac{3x^2y^2}{4}-3x^2z^2+\frac{3y^4}{8}-3y^2z^2+z^4$ |
| | $\mathbb{M}_4^{(a)}(A,1)$ | $\begin{bmatrix} 0 & -\frac{1}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{10}}{140} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{6}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{6}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{10}}{140} & 0 & 0 \\ \frac{1}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{3}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{10}}{35} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{1}{7} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{7} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{3}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{10}}{35} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{14} & 0 \end{bmatrix}$ |
| 765 | symmetry | $\frac{\sqrt{70}x^3z}{4}-\frac{3\sqrt{70}xy^2z}{4}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_4^{(a)}(A,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{7}}{140} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{56} & -\frac{3}{20} & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{7}}{70} & 0 \\ \frac{\sqrt{42}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{7}}{70} & 0 & 0 & 0 & 0 & 0 & \frac{3}{20} \\ 0 & -\frac{\sqrt{70}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{7}}{140} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{20} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{14} & \frac{3\sqrt{6}}{40} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{280} & 0 \\ \frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{280} & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{6}}{40} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{20} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{28} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 766 | symmetry | $\frac{3\sqrt{70}x^2yz}{4} - \frac{\sqrt{70}y^3z}{4}$ |
| | $\mathbb{M}_4^{(a)}(A,3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{7}i}{140} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{56} & -\frac{3i}{20} & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{7}i}{70} & 0 \\ \frac{\sqrt{42}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{7}i}{70} & 0 & 0 & 0 & 0 & 0 & -\frac{3i}{20} \\ 0 & -\frac{\sqrt{70}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{7}i}{140} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{20} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & \frac{3\sqrt{6}i}{40} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{280} & 0 \\ \frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{280} & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{6}i}{40} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{20} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{28} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 767 | symmetry | $-\frac{3\sqrt{5}x^3z}{4} - \frac{3\sqrt{5}ix^2yz}{4} - \frac{3\sqrt{5}xy^2z}{4} + \sqrt{5}xz^3 - \frac{3\sqrt{5}iy^3z}{4} + \sqrt{5}iyz^3$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_4^{(a)}(E^{(a)},1)$ | $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{10}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{30}}{70} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{30}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{10}}{140} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{6}}{35} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{6}}{140} & 0 \\ 0 & \frac{1}{7} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{14} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{10}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{70} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{20} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{140} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{1}{7} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{13\sqrt{6}}{140} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{140} \end{bmatrix}$ |
| 768 | symmetry | $\frac{\sqrt{70}x^4}{16} + \frac{\sqrt{70}ix^3y}{4} - \frac{3\sqrt{70}x^2y^2}{8} - \frac{\sqrt{70}ixy^3}{4} + \frac{\sqrt{70}y^4}{16}$ |
| | $\mathbb{M}_4^{(a)}(E^{(a)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{21}}{70} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3}{10} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{35} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{21}}{35} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{10} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 769 | symmetry | $-\frac{\sqrt{10}x^4}{8} + \frac{\sqrt{10}ix^3y}{4} + \frac{3\sqrt{10}x^2z^2}{4} + \frac{\sqrt{10}ixy^3}{4} - \frac{3\sqrt{10}ixyz^2}{2} + \frac{\sqrt{10}y^4}{8} - \frac{3\sqrt{10}y^2z^2}{4}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_4^{(a)}(E^{(a)},3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{21}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{6}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{33}{140} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{30}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{3}}{140} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{5}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{15}}{140} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{21}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{3}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{6}}{140} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{5}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{17\sqrt{2}}{140} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{5}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{70} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{3}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{7} & 0 & 0 & 0 \end{bmatrix}$ |
| 770 | symmetry | $-\frac{3\sqrt{5}x^3z}{4}+\frac{3\sqrt{5}ix^2yz}{4}-\frac{3\sqrt{5}xy^2z}{4}+\sqrt{5}xz^3+\frac{3\sqrt{5}iy^3z}{4}-\sqrt{5}iyz^3$ |
| | $\mathbb{M}_4^{(a)}(E^{(b)},1)$ | $\begin{bmatrix} \frac{1}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{6}}{140} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{15}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{6}}{35} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{30}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}}{140} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{10}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{30}}{70} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{1}{7} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{13\sqrt{6}}{140} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{10}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{140} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{20} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{10}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{70} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{1}{7} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{14} & 0 & 0 \end{bmatrix}$ |
| 771 | symmetry | $\frac{\sqrt{70}x^4}{16}-\frac{\sqrt{70}ix^3y}{4}-\frac{3\sqrt{70}x^2y^2}{8}+\frac{\sqrt{70}ixy^3}{4}+\frac{\sqrt{70}y^4}{16}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_4^{(a)}(E^{(b)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3}{10} & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{21}}{70} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{10} & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{21}}{35} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{35} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 772 | symmetry | $-\frac{\sqrt{10}x^4}{8} - \frac{\sqrt{10}ix^3y}{4} + \frac{3\sqrt{10}x^2z^2}{4} - \frac{\sqrt{10}ixy^3}{4} + \frac{3\sqrt{10}ixyz^2}{2} + \frac{\sqrt{10}y^4}{8} - \frac{3\sqrt{10}y^2z^2}{4}$ |
| | $\mathbb{M}_4^{(a)}(E^{(b)},3)$ | $\begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{5}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{15}}{140} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{3}}{140} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{33}{140} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{21}}{140} \\ 0 & 0 & \frac{3}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{7} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{5}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{70} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{17\sqrt{2}}{140} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{3}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{6}}{140} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{21}}{70} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 773 | symmetry | $-\frac{x^2}{2} - \frac{y^2}{2} + z^2$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_2^{(1,-1;a)}(A)$ | $\begin{bmatrix} 0 & -\frac{3\sqrt{6}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{15}}{35} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{3}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{6\sqrt{3}}{35} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{3}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{6\sqrt{3}}{35} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{3\sqrt{6}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{15}}{35} & 0 & 0 \\ \frac{\sqrt{6}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{6}}{105} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{15}}{70} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{4\sqrt{6}}{105} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}}{70} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{4\sqrt{6}}{105} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}}{70} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{105} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{15}}{70} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3}{14} & 0 \end{bmatrix}$ |
| 774 | symmetry | $\frac{\sqrt{6}xz}{2} + \frac{\sqrt{6}iyz}{2}$ |
| | $\mathbb{M}_2^{(1,-1;a)}(E^{(a)},1)$ | $\begin{bmatrix} 0 & 0 & -\frac{9\sqrt{2}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{6}}{35} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{6}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{6\sqrt{2}}{35} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{30}}{35} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{5}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{30}}{35} & 0 \\ 0 & \frac{2\sqrt{5}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{2\sqrt{2}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{11\sqrt{6}}{140} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{10} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{2\sqrt{2}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{5}}{70} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{5}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{140} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{28} \end{bmatrix}$ |
| 775 | symmetry | $\frac{\sqrt{6}x^2}{4} - \frac{\sqrt{6}ixy}{2} - \frac{\sqrt{6}y^2}{4}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_2^{(1,-1;a)}(E^{(a)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{15}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}}{35} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{2\sqrt{3}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{35} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{3\sqrt{2}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{35} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{10}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{15}}{35} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{3\sqrt{2}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{5}}{35} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{3\sqrt{2}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{6}}{35} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{10}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{70} & 0 & 0 & 0 \end{bmatrix}$ |
| 776 | symmetry | $\frac{\sqrt{6}xz}{2} - \frac{\sqrt{6}iyz}{2}$ |
| | $\mathbb{M}_2^{(1,-1;a)}(E^{(b)},1)$ | $\begin{bmatrix} \frac{3\sqrt{5}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{30}}{35} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{3}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{30}}{35} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{6}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{6\sqrt{2}}{35} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{9\sqrt{2}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{6}}{35} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{140} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{2\sqrt{2}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{5}}{70} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{10} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{2\sqrt{2}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{11\sqrt{6}}{140} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{2\sqrt{5}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}}{28} & 0 & 0 \end{bmatrix}$ |
| 777 | symmetry | $\frac{\sqrt{6}x^2}{4} + \frac{\sqrt{6}ixy}{2} - \frac{\sqrt{6}y^2}{4}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_2^{(1,-1;a)}(E^{(b)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & -\frac{3\sqrt{2}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{35} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{2\sqrt{3}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{35} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}}{35} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{35} \\ 0 & 0 & \frac{\sqrt{10}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{70} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{3\sqrt{2}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{6}}{35} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{3\sqrt{2}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{5}}{35} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{15}}{35} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{70} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 778 | symmetry | $\frac{3x^4}{8}+\frac{3x^2y^2}{4}-3x^2z^2+\frac{3y^4}{8}-3y^2z^2+z^4$ |
| | $\mathbb{M}_4^{(1,-1;a)}(A,1)$ | $\begin{bmatrix} 0 & \frac{\sqrt{3}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{2}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{6}}{84} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{2}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{6}}{84} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{28} & 0 & 0 \\ -\frac{\sqrt{3}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{2}}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{3}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{21} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{3}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{3}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{21} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{2}}{28} & 0 \end{bmatrix}$ |
| 779 | symmetry | $\frac{\sqrt{70}x^3z}{4}-\frac{3\sqrt{70}xy^2z}{4}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_4^{(1,-1;a)}(A,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{336} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{112} & \frac{\sqrt{3}}{12} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{42} & 0 \\ -\frac{\sqrt{14}}{112} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{12} \\ 0 & \frac{\sqrt{210}}{336} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{7}}{56} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{24} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{42} & \frac{3\sqrt{2}}{16} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{112} & 0 \\ -\frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{112} & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}}{16} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{24} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{7}}{56} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 780 | symmetry | $\frac{3\sqrt{70}x^2yz}{4} - \frac{\sqrt{70}y^3z}{4}$ |
| | $\mathbb{M}_4^{(1,-1;a)}(A,3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{336} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{112} & \frac{\sqrt{3}i}{12} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{42} & 0 \\ -\frac{\sqrt{14}i}{112} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{12} \\ 0 & \frac{\sqrt{210}i}{336} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{7}i}{56} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{24} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{42} & \frac{3\sqrt{2}i}{16} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{112} & 0 \\ -\frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{112} & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}i}{16} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{24} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{7}i}{56} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 781 | symmetry | $-\frac{3\sqrt{5}x^3z}{4} - \frac{3\sqrt{5}ix^2yz}{4} - \frac{3\sqrt{5}xy^2z}{4} + \sqrt{5}xz^3 - \frac{3\sqrt{5}iy^3z}{4} + \sqrt{5}iyz^3$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_4^{(1,-1;a)}(E^{(a)},1)$ | $\begin{bmatrix} 0 & 0 & \frac{\sqrt{30}}{168} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{10}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{84} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{7} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{168} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}}{28} & 0 \\ 0 & -\frac{\sqrt{3}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{30}}{84} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{30}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{8} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{168} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{13\sqrt{2}}{56} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{56} \end{bmatrix}$ |
| 782 | symmetry | $\frac{\sqrt{70}x^4}{16} + \frac{\sqrt{70}ix^3y}{4} - \frac{3\sqrt{70}x^2y^2}{8} - \frac{\sqrt{70}ixy^3}{4} + \frac{\sqrt{70}y^4}{16}$ |
| | $\mathbb{M}_4^{(1,-1;a)}(E^{(a)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{14} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{6} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{42} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{7} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{4} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 783 | symmetry | $-\frac{\sqrt{10}x^4}{8} + \frac{\sqrt{10}ix^3y}{4} + \frac{3\sqrt{10}x^2z^2}{4} + \frac{\sqrt{10}ixy^3}{4} - \frac{3\sqrt{10}ixyz^2}{2} + \frac{\sqrt{10}y^4}{8} - \frac{3\sqrt{10}y^2z^2}{4}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_4^{(1,-1;a)}(E^{(a)},3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{2}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{11\sqrt{3}}{84} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{10}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{15}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{5}}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{7}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{3}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{2}}{56} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{15}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{17\sqrt{6}}{168} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{15}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{3}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5}{14} & 0 & 0 & 0 \end{bmatrix}$ |
| 784 | symmetry | $-\frac{3\sqrt{5}x^3z}{4}+\frac{3\sqrt{5}ix^2yz}{4}-\frac{3\sqrt{5}xy^2z}{4}+\sqrt{5}xz^3+\frac{3\sqrt{5}iy^3z}{4}-\sqrt{5}iyz^3$ |
| | $\mathbb{M}_4^{(1,-1;a)}(E^{(b)},1)$ | $\begin{bmatrix} -\frac{\sqrt{3}}{168} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{5}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{7} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{10}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{84} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{30}}{168} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{3}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{13\sqrt{2}}{56} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{30}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{168} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{8} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{30}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{30}}{84} & 0 & 0 \end{bmatrix}$ |
| 785 | symmetry | $\frac{\sqrt{70}x^4}{16}-\frac{\sqrt{70}ix^3y}{4}-\frac{3\sqrt{70}x^2y^2}{8}+\frac{\sqrt{70}ixy^3}{4}+\frac{\sqrt{70}y^4}{16}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_4^{(1,-1;a)}(E^{(b)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{42}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{42}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{7} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{42}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{42} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 786 | symmetry | $-\frac{\sqrt{10}x^4}{8} - \frac{\sqrt{10}ix^3y}{4} + \frac{3\sqrt{10}x^2z^2}{4} - \frac{\sqrt{10}ixy^3}{4} + \frac{3\sqrt{10}ixyz^2}{2} + \frac{\sqrt{10}y^4}{8} - \frac{3\sqrt{10}y^2z^2}{4}$ |
| | $\mathbb{M}_4^{(1,-1;a)}(E^{(b)},3)$ | $\begin{bmatrix} 0 & 0 & 0 & \frac{\sqrt{15}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{5}}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{11\sqrt{3}}{84} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{28} \\ 0 & 0 & -\frac{\sqrt{3}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{15}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{17\sqrt{6}}{168} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{2}}{56} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{7}}{28} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 787 | symmetry | $-\frac{5x^6}{16} - \frac{15x^4y^2}{16} + \frac{45x^4z^2}{8} - \frac{15x^2y^4}{16} + \frac{45x^2y^2z^2}{4} - \frac{15x^2z^4}{2} - \frac{5y^6}{16} + \frac{45y^4z^2}{8} - \frac{15y^2z^4}{2} + z^6$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_6^{(1,-1;a)}(A,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{66}}{132} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{110}}{44} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{33}}{66} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{33}}{66} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{110}}{44} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{66}}{132} & 0 \end{bmatrix}$ |
| 788 | symmetry | $\frac{\sqrt{462}x^6}{32} - \frac{15\sqrt{462}x^4y^2}{32} + \frac{15\sqrt{462}x^2y^4}{32} - \frac{\sqrt{462}y^6}{32}$ |
| | $\mathbb{M}_6^{(1,-1;a)}(A,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 789 | symmetry | $\frac{3\sqrt{462}x^5y}{16} - \frac{5\sqrt{462}x^3y^3}{8} + \frac{3\sqrt{462}xy^5}{16}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_6^{(1,-1;a)}(A,3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{2} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 790 | symmetry | $-\frac{3\sqrt{210}x^5z}{16}+\frac{3\sqrt{210}x^3y^2z}{8}+\frac{\sqrt{210}x^3z^3}{2}+\frac{9\sqrt{210}xy^4z}{16}-\frac{3\sqrt{210}xy^2z^3}{2}$ |
| | $\mathbb{M}_6^{(1,-1;a)}(A,4)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{77}}{44} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{231}}{44} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{22}}{44} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{154}}{44} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{154}}{44} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{22}}{44} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{231}}{44} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{77}}{44} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 791 | symmetry | $-\frac{9\sqrt{210}x^4yz}{16}-\frac{3\sqrt{210}x^2y^3z}{8}+\frac{3\sqrt{210}x^2yz^3}{2}+\frac{3\sqrt{210}y^5z}{16}-\frac{\sqrt{210}y^3z^3}{2}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_6^{(1,-1;a)}(A,5)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{77}i}{44} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{231}i}{44} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{22}i}{44} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{154}i}{44} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{154}i}{44} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{22}i}{44} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{231}i}{44} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{77}i}{44} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 792 | symmetry | $\frac{3\sqrt{77}x^5z}{16} - \frac{15\sqrt{77}ix^4yz}{16} - \frac{15\sqrt{77}x^3y^2z}{8} + \frac{15\sqrt{77}ix^2y^3z}{8} + \frac{15\sqrt{77}xy^4z}{16} - \frac{3\sqrt{77}iy^5z}{16}$ |
| | $\mathbb{M}_6^{(1,-1;a)}(E^{(a)},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{12} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 793 | symmetry | $\frac{5\sqrt{42}x^5z}{16} + \frac{5\sqrt{42}ix^4yz}{16} + \frac{5\sqrt{42}x^3y^2z}{8} - \frac{5\sqrt{42}x^3z^3}{4} + \frac{5\sqrt{42}ix^2y^3z}{8} - \frac{5\sqrt{42}ix^2yz^3}{4} + \frac{5\sqrt{42}xy^4z}{16} - \frac{5\sqrt{42}xy^2z^3}{4} + \frac{\sqrt{42}xz^5}{2} + \frac{5\sqrt{42}iy^5z}{16} - \frac{5\sqrt{42}iy^3z^3}{4} + \frac{\sqrt{42}iyz^5}{2}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_6^{(1,-1;a)}(E^{(a)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{231}}{132} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{77}}{132} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{154}}{132} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}}{132} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{385}}{132} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{11}}{132} \end{bmatrix}$ |
| 794 | symmetry | $-\frac{3\sqrt{14}x^6}{32} - \frac{3\sqrt{14}ix^5y}{8} + \frac{15\sqrt{14}x^4y^2}{32} + \frac{15\sqrt{14}x^4z^2}{16} + \frac{15\sqrt{14}ix^3yz^2}{4} + \frac{15\sqrt{14}x^2y^4}{32} - \frac{45\sqrt{14}x^2y^2z^2}{8} + \frac{3\sqrt{14}ixy^5}{8} - \frac{15\sqrt{14}ixy^3z^2}{4} - \frac{3\sqrt{14}y^6}{32} + \frac{15\sqrt{14}y^4z^2}{16}$ |
| | $\mathbb{M}_6^{(1,-1;a)}(E^{(a)},3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{77}}{22} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{1155}}{66} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{330}}{66} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 795 | symmetry | $\frac{\sqrt{105}x^6}{32} - \frac{\sqrt{105}ix^5y}{16} + \frac{\sqrt{105}x^4y^2}{32} - \frac{\sqrt{105}x^4z^2}{2} - \frac{\sqrt{105}ix^3y^3}{8} + \sqrt{105}ix^3yz^2 - \frac{\sqrt{105}x^2y^4}{32} + \frac{\sqrt{105}x^2z^4}{2} - \frac{\sqrt{105}ixy^5}{16} + \sqrt{105}ixy^3z^2 - \sqrt{105}ixyz^4 - \frac{\sqrt{105}y^6}{32} + \frac{\sqrt{105}y^4z^2}{2} - \frac{\sqrt{105}y^2z^4}{2}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_6^{(1,-1;a)}(E^{(a)},4)$ | $\begin{bmatrix} 0&0&0&0&0&0&0&0&0&0&0&0&0&0 \\ 0&0&0&0&0&0&0&0&0&0&0&0&0&0 \\ 0&0&0&0&0&0&0&0&0&0&0&0&0&0 \\ 0&0&0&0&0&0&0&0&0&0&0&0&0&0 \\ 0&0&0&0&0&0&0&0&0&0&0&0&0&0 \\ 0&0&0&0&0&0&-\frac{\sqrt{22}}{66}&0&0&0&0&0&0&0 \\ 0&0&0&0&0&0&0&\frac{\sqrt{77}}{33}&0&0&0&0&0&0 \\ 0&0&0&0&0&0&0&0&-\frac{\sqrt{231}}{33}&0&0&0&0&0 \\ 0&0&0&0&0&0&0&0&0&\frac{\sqrt{770}}{66}&0&0&0&0 \\ 0&0&0&0&0&0&0&0&0&0&-\frac{\sqrt{154}}{66}&0&0&0 \end{bmatrix}$ |
| 796 | symmetry | $\frac{3\sqrt{77}x^5z}{16}+\frac{15\sqrt{77}ix^4yz}{16}-\frac{15\sqrt{77}x^3y^2z}{8}-\frac{15\sqrt{77}ix^2y^3z}{8}+\frac{15\sqrt{77}xy^4z}{16}+\frac{3\sqrt{77}iy^5z}{16}$ |
| | $\mathbb{M}_6^{(1,-1;a)}(E^{(b)},1)$ | $\begin{bmatrix} 0&0&0&0&0&0&0&0&0&0&0&0&0&0&0 \\ 0&0&0&0&0&0&0&0&0&0&0&0&0&0&0 \\ 0&0&0&0&0&0&0&0&0&0&0&0&0&0&0 \\ 0&0&0&0&0&0&0&0&0&0&0&0&0&0&0 \\ 0&0&0&0&0&0&0&0&0&0&0&0&0&0&0 \\ 0&0&0&0&0&0&0&0&0&0&0&0&0&0&0 \\ 0&0&0&0&0&0&0&0&0&0&0&0&0&0&0 \\ 0&0&0&0&0&0&0&0&0&0&0&0&0&0&0 \\ 0&0&0&0&0&0&0&0&0&0&0&0&0&0&0 \\ 0&0&0&0&0&0&0&0&0&0&0&0&0&0&0 \\ 0&0&0&0&0&0&0&0&0&0&0&0&\frac{\sqrt{42}}{12}&0&0 \\ 0&0&0&0&0&0&0&0&0&0&0&0&0&-\frac{\sqrt{30}}{12}&0 \\ 0&0&0&0&0&0&0&0&0&0&0&0&0&0&0 \\ 0&0&0&0&0&0&0&0&0&0&0&0&0&0&0 \\ 0&0&0&0&0&0&0&0&0&0&0&0&0&0&0 \end{bmatrix}$ |
| 797 | symmetry | $\frac{5\sqrt{42}x^5z}{16}-\frac{5\sqrt{42}ix^4yz}{16}+\frac{5\sqrt{42}x^3y^2z}{8}-\frac{5\sqrt{42}x^3z^3}{4}-\frac{5\sqrt{42}ix^2y^3z}{8}+\frac{5\sqrt{42}ix^2yz^3}{4}+\frac{5\sqrt{42}xy^4z}{16}-\frac{5\sqrt{42}xy^2z^3}{4}+\frac{\sqrt{42}xz^5}{2}-\frac{5\sqrt{42}iy^5z}{16}+\frac{5\sqrt{42}iy^3z^3}{4}-\frac{\sqrt{42}iyz^5}{2}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_6^{(1,-1;a)}(E^{(b)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{11}}{132} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{385}}{132} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}}{132} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{154}}{132} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{77}}{132} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{231}}{132} & 0 & 0 \end{bmatrix}$ |
| 798 | symmetry | $-\frac{3\sqrt{14}x^6}{32}+\frac{3\sqrt{14}ix^5y}{8}+\frac{15\sqrt{14}x^4y^2}{32}+\frac{15\sqrt{14}x^4z^2}{16}-\frac{15\sqrt{14}ix^3yz^2}{4}+\frac{15\sqrt{14}x^2y^4}{32}-\frac{45\sqrt{14}x^2y^2z^2}{8}-\frac{3\sqrt{14}ixy^5}{8}+\frac{15\sqrt{14}ixy^3z^2}{4}-\frac{3\sqrt{14}y^6}{32}+\frac{15\sqrt{14}y^4z^2}{16}$ |
| | $\mathbb{M}_6^{(1,-1;a)}(E^{(b)},3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}}{66} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{1155}}{66} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{77}}{22} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 799 | symmetry | $\frac{\sqrt{105}x^6}{32}+\frac{\sqrt{105}ix^5y}{16}+\frac{\sqrt{105}x^4y^2}{32}-\frac{\sqrt{105}x^4z^2}{2}+\frac{\sqrt{105}ix^3y^3}{8}-\sqrt{105}ix^3yz^2-\frac{\sqrt{105}x^2y^4}{32}+\frac{\sqrt{105}x^2z^4}{2}+\frac{\sqrt{105}ixy^5}{16}-\sqrt{105}ixy^3z^2+\sqrt{105}ixyz^4-\frac{\sqrt{105}y^6}{32}+\frac{\sqrt{105}y^4z^2}{2}-\frac{\sqrt{105}y^2z^4}{2}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_6^{(1,-1;a)}(E^{(b)},4)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{154}}{66} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{770}}{66} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{231}}{33} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{77}}{33} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{22}}{66} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 800 | symmetry | $-\frac{x^2}{2}-\frac{y^2}{2}+z^2$ |
| | $\mathbb{M}_2^{(1,0;a)}(A)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{15}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{14} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{10}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{30}}{70} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{10}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{30}}{70} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{14} & 0 & 0 \\ \frac{\sqrt{15}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{15}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{2\sqrt{15}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{70} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{2\sqrt{15}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{70} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{28} & 0 \end{bmatrix}$ |
| 801 | symmetry | $\frac{\sqrt{6}xz}{2}+\frac{\sqrt{6}iyz}{2}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_2^{(1,0;a)}(E^{(a)},1)$ | $\begin{bmatrix} 0 & 0 & -\frac{3\sqrt{5}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{35} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{15}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{5}}{35} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{210} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{7} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{7} & 0 \\ 0 & \frac{3\sqrt{2}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{3\sqrt{5}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{11\sqrt{15}}{420} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{60} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{3\sqrt{5}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{84} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{84} \end{bmatrix}$ |
| 802 | symmetry | $\frac{\sqrt{6}x^2}{4} - \frac{\sqrt{6}ixy}{2} - \frac{\sqrt{6}y^2}{4}$ |
| | $\mathbb{M}_2^{(1,0;a)}(E^{(a)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{6}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{2\sqrt{30}}{105} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{14} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{5}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{70} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{3}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{21} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{9\sqrt{5}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{14} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{9\sqrt{5}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{15}}{105} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{3}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{42} & 0 & 0 & 0 \end{bmatrix}$ |
| 803 | symmetry | $\frac{\sqrt{6}xz}{2} - \frac{\sqrt{6}iyz}{2}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_2^{(1,0;a)}(E^{(b)},1)$ | $\begin{bmatrix} \frac{\sqrt{2}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{7} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{30}}{210} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{7} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{15}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{5}}{35} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{5}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{35} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{3\sqrt{2}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{84} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{3\sqrt{5}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{60} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{5}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{11\sqrt{15}}{420} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{28} & 0 & 0 \end{bmatrix}$ |
| 804 | symmetry | $\frac{\sqrt{6}x^2}{4} + \frac{\sqrt{6}ixy}{2} - \frac{\sqrt{6}y^2}{4}$ |
| | $\mathbb{M}_2^{(1,0;a)}(E^{(b)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{5}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{70} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{2\sqrt{30}}{105} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{14} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3}{14} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{14} \\ 0 & 0 & \frac{3}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{42} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{9\sqrt{5}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{15}}{105} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{9\sqrt{5}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{14} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{3}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{21} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{42} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 805 | symmetry | $\frac{3x^4}{8} + \frac{3x^2y^2}{4} - 3x^2z^2 + \frac{3y^4}{8} - 3y^2z^2 + z^4$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_4^{(1,0;a)}(A,1)$ | $\begin{bmatrix} 0 & \frac{\sqrt{5}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{27\sqrt{2}}{140} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{30}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{10}}{140} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{30}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{10}}{140} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{27\sqrt{2}}{140} & 0 & 0 \\ -\frac{3\sqrt{5}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{140} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{9\sqrt{5}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{35} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{3\sqrt{5}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{140} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{5}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{140} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{9\sqrt{5}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{35} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{5}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{140} & 0 \end{bmatrix}$ |
| 806 | symmetry | $\frac{\sqrt{70}x^3z}{4} - \frac{3\sqrt{70}xy^2z}{4}$ |
| | $\mathbb{M}_4^{(1,0;a)}(A,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{112} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{27\sqrt{35}}{700} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{560} & -\frac{9\sqrt{5}}{100} & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{35}}{350} & 0 \\ -\frac{\sqrt{210}}{560} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{35}}{350} & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{5}}{100} \\ 0 & \frac{\sqrt{14}}{112} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{27\sqrt{35}}{700} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{35}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{280} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{200} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{35}}{70} & \frac{3\sqrt{30}}{400} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{2800} & 0 \\ -\frac{3\sqrt{35}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{2800} & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{30}}{400} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{200} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{3\sqrt{35}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{280} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 807 | symmetry | $\frac{3\sqrt{70}x^2yz}{4} - \frac{\sqrt{70}y^3z}{4}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_4^{(1,0;a)}(A,3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{112} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{27\sqrt{35}i}{700} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{560} & -\frac{9\sqrt{5}i}{100} & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{35}i}{350} & 0 \\ -\frac{\sqrt{210}i}{560} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{35}i}{350} & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{5}i}{100} \\ 0 & \frac{\sqrt{14}i}{112} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{27\sqrt{35}i}{700} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{3\sqrt{35}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{280} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{200} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{35}i}{70} & \frac{3\sqrt{30}i}{400} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{2800} & 0 \\ -\frac{3\sqrt{35}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{2800} & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{30}i}{400} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{200} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{3\sqrt{35}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{280} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 808 | $\mathbb{M}_4^{(1,0;a)}(E^{(a)},1)$ | $-\frac{3\sqrt{5}x^3z}{4} - \frac{3\sqrt{5}ix^2yz}{4} - \frac{3\sqrt{5}xy^2z}{4} + \sqrt{5}xz^3 - \frac{3\sqrt{5}iy^3z}{4} + \sqrt{5}iyz^3$ $\begin{bmatrix} 0 & 0 & \frac{\sqrt{2}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{6}}{70} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{6}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{2}}{140} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{30}}{175} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{280} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{27\sqrt{30}}{700} & 0 \\ 0 & -\frac{3\sqrt{5}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{3\sqrt{2}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{140} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{40} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{1400} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{5}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{13\sqrt{30}}{1400} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{280} \end{bmatrix}$ |
| 809 | symmetry | $\frac{\sqrt{70}x^4}{16} + \frac{\sqrt{70}ix^3y}{4} - \frac{3\sqrt{70}x^2y^2}{8} - \frac{\sqrt{70}ixy^3}{4} + \frac{\sqrt{70}y^4}{16}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_4^{(1,0;a)}(E^{(a)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{105}}{350} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{5}}{50} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{3\sqrt{70}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{70} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{70}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{175} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{100} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 810 | symmetry | $-\frac{\sqrt{10}x^4}{8}+\frac{\sqrt{10}ix^3y}{4}+\frac{3\sqrt{10}x^2z^2}{4}+\frac{\sqrt{10}ixy^3}{4}-\frac{3\sqrt{10}ixyz^2}{2}+\frac{\sqrt{10}y^4}{8}-\frac{3\sqrt{10}y^2z^2}{4}$ |
| | $\mathbb{M}_4^{(1,0;a)}(E^{(a)},3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{105}}{700} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{30}}{280} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{99\sqrt{5}}{700} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{6}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{15}}{700} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{1}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{27\sqrt{3}}{140} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{105}}{700} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{9\sqrt{5}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{30}}{1400} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{3}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{17\sqrt{10}}{1400} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{3}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{140} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{9\sqrt{5}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{70} & 0 & 0 & 0 \end{bmatrix}$ |
| 811 | symmetry | $-\frac{3\sqrt{5}x^3z}{4}+\frac{3\sqrt{5}ix^2yz}{4}-\frac{3\sqrt{5}xy^2z}{4}+\sqrt{5}xz^3+\frac{3\sqrt{5}iy^3z}{4}-\sqrt{5}iyz^3$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_4^{(1,0;a)}(E^{(b)},1)$ | $\begin{bmatrix} -\frac{\sqrt{5}}{280} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{27\sqrt{30}}{700} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{3}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{30}}{175} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{6}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{2}}{140} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{2}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{6}}{70} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{280} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{3\sqrt{5}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{13\sqrt{30}}{1400} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{3\sqrt{2}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{1400} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{40} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{2}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{140} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{3\sqrt{5}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{28} & 0 & 0 \end{bmatrix}$ |
| 812 | symmetry | $\frac{\sqrt{70}x^4}{16} - \frac{\sqrt{70}ix^3y}{4} - \frac{3\sqrt{70}x^2y^2}{8} + \frac{\sqrt{70}ixy^3}{4} + \frac{\sqrt{70}y^4}{16}$ |
| | $\mathbb{M}_4^{(1,0;a)}(E^{(b)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{5}}{50} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{70}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{105}}{350} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{100} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{3\sqrt{70}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{175} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{3\sqrt{70}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{70} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 813 | symmetry | $-\frac{\sqrt{10}x^4}{8} - \frac{\sqrt{10}ix^3y}{4} + \frac{3\sqrt{10}x^2z^2}{4} - \frac{\sqrt{10}ixy^3}{4} + \frac{3\sqrt{10}ixyz^2}{2} + \frac{\sqrt{10}y^4}{8} - \frac{3\sqrt{10}y^2z^2}{4}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_4^{(1,0;a)}(E^{(b)},3)$ | $\begin{bmatrix} 0 & 0 & 0 & \frac{1}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{27\sqrt{3}}{140} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{15}}{700} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{280} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{99\sqrt{5}}{700} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{105}}{700} \\ 0 & 0 & -\frac{9\sqrt{5}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{70} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{3}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{140} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{3}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{17\sqrt{10}}{1400} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{5}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{30}}{1400} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{105}}{700} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 814 | symmetry | 1 |
| | $\mathbb{M}_0^{(1,1;a)}(A)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 815 | symmetry | $-\frac{x^2}{2}-\frac{y^2}{2}+z^2$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_2^{(1,1;a)}(A)$ | $\begin{bmatrix} 0 & \frac{12}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}}{140} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{2\sqrt{6}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{2}}{140} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{2\sqrt{6}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{2}}{140} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{12}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{10}}{140} & 0 & 0 \\ \frac{3}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{21} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{3}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{35} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{6}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{3}}{105} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{6}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{3}}{105} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{3}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}}{35} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{3}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{21} & 0 \end{bmatrix}$ |
| 816 | symmetry | $\frac{\sqrt{6}xz}{2} + \frac{\sqrt{6}iyz}{2}$ |
| | $\mathbb{M}_2^{(1,1;a)}(E^{(a)},1)$ | $\begin{bmatrix} 0 & 0 & \frac{6\sqrt{3}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3}{70} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{2}{7} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{3}}{70} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{2\sqrt{2}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{5}}{70} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{30}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{5}}{70} & 0 \\ 0 & \frac{3\sqrt{30}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{21} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{3\sqrt{3}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{11}{105} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{15} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{3\sqrt{3}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{105} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{30}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{105} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{21} \end{bmatrix}$ |
| 817 | symmetry | $\frac{\sqrt{6}x^2}{4} - \frac{\sqrt{6}ixy}{2} - \frac{\sqrt{6}y^2}{4}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_2^{(1,1;a)}(E^{(a)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{35}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{4\sqrt{10}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{15}}{140} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{8\sqrt{2}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{5}}{140} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{4\sqrt{3}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3}{140} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{35}}{105} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{3\sqrt{15}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{4\sqrt{10}}{105} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{9\sqrt{3}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{30}}{105} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{9\sqrt{3}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{8}{105} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{3\sqrt{15}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{5}}{105} & 0 & 0 & 0 \end{bmatrix}$ |
| 818 | symmetry | $\frac{\sqrt{6}xz}{2} - \frac{\sqrt{6}iyz}{2}$ |
| | $\mathbb{M}_2^{(1,1;a)}(E^{(b)},1)$ | $\begin{bmatrix} -\frac{2\sqrt{30}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{5}}{70} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{2\sqrt{2}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{5}}{70} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{2}{7} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{3}}{70} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{6\sqrt{3}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3}{70} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{3\sqrt{30}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{105} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{3\sqrt{3}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{105} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{15} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{3}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{11}{105} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{3\sqrt{30}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{21} & 0 & 0 \end{bmatrix}$ |
| 819 | symmetry | $\frac{\sqrt{6}x^2}{4} + \frac{\sqrt{6}ixy}{2} - \frac{\sqrt{6}y^2}{4}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_2^{(1,1;a)}(E^{(b)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & \frac{4\sqrt{3}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3}{140} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{8\sqrt{2}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{5}}{140} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{4\sqrt{10}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{15}}{140} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{35}}{140} \\ 0 & 0 & \frac{3\sqrt{15}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{5}}{105} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{9\sqrt{3}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{8}{105} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{9\sqrt{3}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{30}}{105} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{15}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{4\sqrt{10}}{105} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{35}}{105} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 820 | symmetry | $\frac{3x^4}{8}+\frac{3x^2y^2}{4}-3x^2z^2+\frac{3y^4}{8}-3y^2z^2+z^4$ |
| | $\mathbb{M}_4^{(1,1;a)}(A,1)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{330}}{105} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{33}}{70} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{2\sqrt{55}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}}{210} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{2\sqrt{55}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}}{210} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{330}}{105} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{33}}{70} & 0 & 0 \\ -\frac{\sqrt{330}}{420} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{55}}{385} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{330}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{8\sqrt{33}}{1155} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{330}}{210} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{110}}{385} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{330}}{210} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{110}}{385} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{330}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{8\sqrt{33}}{1155} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}}{420} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{55}}{385} & 0 \end{bmatrix}$ |
| 821 | symmetry | $\frac{\sqrt{70}x^3z}{4}-\frac{3\sqrt{70}xy^2z}{4}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_4^{(1,1;a)}(A,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{\sqrt{231}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}}{700} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{385}}{70} & \frac{\sqrt{330}}{300} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}}{1050} & 0 \\ \frac{\sqrt{385}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}}{1050} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}}{300} \\ 0 & -\frac{\sqrt{231}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}}{700} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{2310}}{420} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{770}}{770} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}}{1650} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}}{420} & -\frac{3\sqrt{55}}{550} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{385}}{3850} & 0 \\ -\frac{\sqrt{2310}}{420} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{385}}{3850} & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{55}}{550} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}}{1650} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{2310}}{420} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{770}}{770} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 822 | symmetry | $\frac{3\sqrt{70}x^2yz}{4} - \frac{\sqrt{70}y^3z}{4}$ |
| | $\mathbb{M}_4^{(1,1;a)}(A,3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{231}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}i}{700} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{385}i}{70} & \frac{\sqrt{330}i}{300} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}i}{1050} & 0 \\ \frac{\sqrt{385}i}{70} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}i}{1050} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{330}i}{300} \\ 0 & -\frac{\sqrt{231}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}i}{700} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{2310}i}{420} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{770}i}{770} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}i}{1650} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}i}{420} & -\frac{3\sqrt{55}i}{550} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{385}i}{3850} & 0 \\ -\frac{\sqrt{2310}i}{420} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{385}i}{3850} & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{55}i}{550} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}i}{1650} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{2310}i}{420} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{770}i}{770} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 823 | symmetry | $-\frac{3\sqrt{5}x^3z}{4} - \frac{3\sqrt{5}ix^2yz}{4} - \frac{3\sqrt{5}xy^2z}{4} + \sqrt{5}xz^3 - \frac{3\sqrt{5}iy^3z}{4} + \sqrt{5}iyz^3$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_4^{(1,1;a)}(E^{(a)},1)$ | $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{33}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{11}}{35} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{11}}{7} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{33}}{210} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{22}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{55}}{175} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{330}}{210} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{55}}{350} & 0 \\ 0 & -\frac{\sqrt{330}}{210} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{33}}{231} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{33}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{11}}{385} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{22}}{110} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{33}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}}{11550} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{330}}{210} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{13\sqrt{55}}{1925} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{77}}{385} \end{bmatrix}$ |
| 824 | symmetry | $\frac{\sqrt{70}x^4}{16} + \frac{\sqrt{70}ix^3y}{4} - \frac{3\sqrt{70}x^2y^2}{8} - \frac{\sqrt{70}ixy^3}{4} + \frac{\sqrt{70}y^4}{16}$ |
| | $\mathbb{M}_4^{(1,1;a)}(E^{(a)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{1155}}{105} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{770}}{350} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{330}}{150} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{1155}}{210} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{462}}{1155} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{1155}}{210} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{4\sqrt{770}}{1925} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{55}}{275} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 825 | symmetry | $-\frac{\sqrt{10}x^4}{8} + \frac{\sqrt{10}ix^3y}{4} + \frac{3\sqrt{10}x^2z^2}{4} + \frac{\sqrt{10}ixy^3}{4} - \frac{3\sqrt{10}ixyz^2}{2} + \frac{\sqrt{10}y^4}{8} - \frac{3\sqrt{10}y^2z^2}{4}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_4^{(1,1;a)}(E^{(a)},3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{770}}{700} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{55}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{11\sqrt{330}}{2100} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{11}}{7} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{110}}{700} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{66}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{22}}{140} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{770}}{1925} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{330}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{55}}{1925} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{66}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{17\sqrt{165}}{5775} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{66}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{22}}{385} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{330}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{110}}{385} & 0 & 0 & 0 \end{bmatrix}$ |
| 826 | symmetry | $-\frac{3\sqrt{5}x^3z}{4}+\frac{3\sqrt{5}ix^2yz}{4}-\frac{3\sqrt{5}xy^2z}{4}+\sqrt{5}xz^3+\frac{3\sqrt{5}iy^3z}{4}-\sqrt{5}iyz^3$ |
| | $\mathbb{M}_4^{(1,1;a)}(E^{(b)},1)$ | $\begin{bmatrix} \frac{\sqrt{330}}{210} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{55}}{350} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{22}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{55}}{175} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{11}}{7} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{33}}{210} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{33}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{11}}{35} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{77}}{385} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{330}}{210} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{13\sqrt{55}}{1925} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{33}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}}{11550} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{22}}{110} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{33}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{11}}{385} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{330}}{210} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{33}}{231} & 0 & 0 \end{bmatrix}$ |
| 827 | symmetry | $\frac{\sqrt{70}x^4}{16}-\frac{\sqrt{70}ix^3y}{4}-\frac{3\sqrt{70}x^2y^2}{8}+\frac{\sqrt{70}ixy^3}{4}+\frac{\sqrt{70}y^4}{16}$ |

*continued ...*

Table 9

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_4^{(1,1;a)}(E^{(b)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{330}}{150} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{2\sqrt{1155}}{105} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{770}}{350} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{55}}{275} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{1155}}{210} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{4\sqrt{770}}{1925} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{1155}}{210} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{462}}{1155} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 828 | symmetry | $-\frac{\sqrt{10}x^4}{8}-\frac{\sqrt{10}ix^3y}{4}+\frac{3\sqrt{10}x^2z^2}{4}-\frac{\sqrt{10}ixy^3}{4}+\frac{3\sqrt{10}ixyz^2}{2}+\frac{\sqrt{10}y^4}{8}-\frac{3\sqrt{10}y^2z^2}{4}$ |
| | $\mathbb{M}_4^{(1,1;a)}(E^{(b)},3)$ | $\begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{66}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{22}}{140} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{11}}{7} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{110}}{700} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{55}}{35} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{11\sqrt{330}}{2100} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{770}}{700} \\ 0 & 0 & -\frac{\sqrt{330}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{110}}{385} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{66}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{22}}{385} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{66}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{17\sqrt{165}}{5775} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}}{140} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{55}}{1925} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{770}}{1925} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |

bra: $=\langle\frac{5}{2},\frac{5}{2};f|\,,\langle\frac{5}{2},\frac{3}{2};f|\,,\langle\frac{5}{2},\frac{1}{2};f|\,,\langle\frac{5}{2},-\frac{1}{2};f|\,,\langle\frac{5}{2},-\frac{3}{2};f|\,,\langle\frac{5}{2},-\frac{5}{2};f|\,,\langle\frac{7}{2},\frac{7}{2};f|\,,\langle\frac{7}{2},\frac{5}{2};f|\,,\langle\frac{7}{2},\frac{3}{2};f|\,,\langle\frac{7}{2},\frac{1}{2};f|\,,\langle\frac{7}{2},-\frac{1}{2};f|\,,\langle\frac{7}{2},-\frac{3}{2};f|\,,\langle\frac{7}{2},-\frac{5}{2};f|\,,\langle\frac{7}{2},-\frac{7}{2};f|$
ket: $=|\frac{5}{2},\frac{5}{2};f\rangle\,,|\frac{5}{2},\frac{3}{2};f\rangle\,,|\frac{5}{2},\frac{1}{2};f\rangle\,,|\frac{5}{2},-\frac{1}{2};f\rangle\,,|\frac{5}{2},-\frac{3}{2};f\rangle\,,|\frac{5}{2},-\frac{5}{2};f\rangle\,,|\frac{7}{2},\frac{7}{2};f\rangle\,,|\frac{7}{2},\frac{5}{2};f\rangle\,,|\frac{7}{2},\frac{3}{2};f\rangle\,,|\frac{7}{2},\frac{1}{2};f\rangle\,,|\frac{7}{2},-\frac{1}{2};f\rangle\,,|\frac{7}{2},-\frac{3}{2};f\rangle\,,|\frac{7}{2},-\frac{5}{2};f\rangle\,,|\frac{7}{2},-\frac{7}{2};f\rangle$

Table 10: (f,f) block.

| No. | multipole | matrix |
|---|---|---|
| 829 | symmetry | 1 |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_0^{(a)}(A)$ | $\begin{bmatrix} \frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} \end{bmatrix}$ |
| 830 | symmetry | $-\frac{x^2}{2} - \frac{y^2}{2} + z^2$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_2^{(a)}(A)$ | $\begin{pmatrix} -\frac{5\sqrt{42}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{7}}{98} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{42}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{98} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{2\sqrt{42}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{98} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{2\sqrt{42}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{98} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{98} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{42}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{7}}{98} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{42}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{5\sqrt{7}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{42}}{588} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{105}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{42}}{196} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{14}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{25\sqrt{42}}{588} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{14}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{25\sqrt{42}}{588} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{42}}{196} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{7}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{42}}{588} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{42}}{84} \end{pmatrix}$ |
| 831 | symmetry | $\frac{\sqrt{6}xz}{2} + \frac{\sqrt{6}iyz}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_2^{(a)}(E^{(a)},1)$ | $\begin{pmatrix} 0 & -\frac{3\sqrt{35}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{14}}{196} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{3\sqrt{14}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{11\sqrt{42}}{588} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{42} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{3\sqrt{14}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{98} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{35}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{588} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{6}}{84} \\ \frac{5\sqrt{6}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{210}}{588} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{10\sqrt{21}}{147} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{35}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{105}}{294} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{11\sqrt{42}}{588} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{105}}{294} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{14}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{10\sqrt{21}}{147} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5}{14} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$ |
| 832 | symmetry | $\frac{\sqrt{6}x^2}{4} - \frac{\sqrt{6}ixy}{2} - \frac{\sqrt{6}y^2}{4}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_2^{(a)}(E^{(a)},2)$ | $\begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{3\sqrt{70}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{105}}{147} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{9\sqrt{14}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{49} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{9\sqrt{14}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{42}}{147} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{70}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{294} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{3}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{210}}{294} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{35}}{98} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{2\sqrt{42}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{105}}{147} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{35}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{105}}{147} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{2\sqrt{105}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{35}}{98} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{3}}{42} & 0 & 0 \end{pmatrix}$ |
| 833 | symmetry | $\frac{\sqrt{6}xz}{2} - \frac{\sqrt{6}iyz}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_2^{(a)}(E^{(b)},1)$ | $\begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{6}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{3\sqrt{35}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{588} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{3\sqrt{14}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{98} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{3\sqrt{14}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{11\sqrt{42}}{588} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{3\sqrt{35}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{14}}{196} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{5\sqrt{14}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{10\sqrt{21}}{147} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{11\sqrt{42}}{588} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{105}}{294} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{21}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{35}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{105}}{294} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{588} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{10\sqrt{21}}{147} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{6}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5}{14} & 0 \end{pmatrix}$ |
| 834 | symmetry | $\frac{\sqrt{6}x^2}{4} + \frac{\sqrt{6}ixy}{2} - \frac{\sqrt{6}y^2}{4}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_2^{(a)}(E^{(b)},2)$ | $\begin{bmatrix} 0 & 0 & -\frac{3\sqrt{70}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{294} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{9\sqrt{14}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{42}}{147} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{9\sqrt{14}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{49} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{70}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{105}}{147} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{42} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{30}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{3}}{42} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{2\sqrt{105}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{35}}{98} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{35}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{105}}{147} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{2\sqrt{42}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{105}}{147} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{294} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{35}}{98} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{3}}{42} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 835 | symmetry | $\frac{3x^4}{8}+\frac{3x^2y^2}{4}-3x^2z^2+\frac{3y^4}{8}-3y^2z^2+z^4$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_4^{(a)}(A,1)$ | $\begin{pmatrix} \frac{\sqrt{77}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{462}}{539} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{3\sqrt{77}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{4\sqrt{770}}{539} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{77}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{231}}{539} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{77}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{231}}{539} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{3\sqrt{77}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{4\sqrt{770}}{539} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{77}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{462}}{539} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{77}}{154} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{5\sqrt{462}}{539} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{39\sqrt{77}}{1078} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{4\sqrt{770}}{539} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{77}}{1078} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{5\sqrt{231}}{539} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{27\sqrt{77}}{1078} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{5\sqrt{231}}{539} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{27\sqrt{77}}{1078} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{4\sqrt{770}}{539} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{77}}{1078} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{462}}{539} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{39\sqrt{77}}{1078} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{77}}{154} \end{pmatrix}$ |
| 836 | symmetry | $\frac{\sqrt{70}x^3z}{4} - \frac{3\sqrt{70}xy^2z}{4}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_4^{(a)}(A,2)$ | $\begin{pmatrix} 0 & 0 & 0 & \frac{\sqrt{11}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{33}}{154} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{11}}{22} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{11}}{14} & -\frac{3\sqrt{462}}{308} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{66}}{308} & 0 \\ \frac{\sqrt{11}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{66}}{308} & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{462}}{308} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{11}}{22} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{11}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{33}}{154} & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{3\sqrt{462}}{308} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{154}}{154} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{66}}{308} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{22}}{77} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{11}}{22} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{33}}{154} & \frac{3\sqrt{154}}{154} & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{22}}{77} & 0 \\ \frac{5\sqrt{33}}{154} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{22}}{77} & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{154}}{154} \\ 0 & \frac{\sqrt{11}}{22} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{66}}{308} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{22}}{77} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{462}}{308} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{154}}{154} & 0 & 0 & 0 \end{pmatrix}$ |
| 837 | symmetry | $\frac{3\sqrt{70}x^2yz}{4} - \frac{\sqrt{70}y^3z}{4}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_4^{(a)}(A,3)$ | $\begin{pmatrix} 0 & 0 & 0 & -\frac{\sqrt{11}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{33}i}{154} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{11}i}{22} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{11}i}{14} & -\frac{3\sqrt{462}i}{308} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{66}i}{308} & 0 \\ \frac{\sqrt{11}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{66}i}{308} & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{462}i}{308} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{11}i}{22} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{11}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{33}i}{154} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{3\sqrt{462}i}{308} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{154}i}{154} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{66}i}{308} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{22}i}{77} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{11}i}{22} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{33}i}{154} & \frac{3\sqrt{154}i}{154} & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{22}i}{77} & 0 \\ \frac{5\sqrt{33}i}{154} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{22}i}{77} & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{154}i}{154} \\ 0 & \frac{\sqrt{11}i}{22} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{66}i}{308} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{22}i}{77} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{462}i}{308} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{154}i}{154} & 0 & 0 & 0 \end{pmatrix}$ |
| 838 | symmetry | $-\frac{3\sqrt{5}x^3z}{4} - \frac{3\sqrt{5}ix^2yz}{4} - \frac{3\sqrt{5}xy^2z}{4} + \sqrt{5}xz^3 - \frac{3\sqrt{5}iy^3z}{4} + \sqrt{5}iyz^3$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_4^{(a)}(E^{(a)},1)$ | $\begin{pmatrix} 0 & \frac{\sqrt{77}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{770}}{539} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{770}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}}{539} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{1155}}{154} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{770}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{77}}{1078} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{77}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{13\sqrt{462}}{1078} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}}{154} \\ -\frac{\sqrt{330}}{154} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{55}}{77} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{13\sqrt{462}}{1078} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{1155}}{539} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{77}}{1078} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{231}}{539} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{1155}}{154} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}}{539} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{231}}{539} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{770}}{539} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{1155}}{539} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{55}}{77} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$ |
| 839 | symmetry | $\frac{\sqrt{70}x^4}{16} + \frac{\sqrt{70}ix^3y}{4} - \frac{3\sqrt{70}x^2y^2}{8} - \frac{\sqrt{70}ixy^3}{4} + \frac{\sqrt{70}y^4}{16}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_4^{(a)}(E^{(a)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & \frac{\sqrt{22}}{14} & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{55}}{77} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{22}}{14} & 0 & 0 & 0 & 0 & 0 & \frac{4\sqrt{33}}{77} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{462}}{77} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{462}}{77} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{154}}{154} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{4\sqrt{33}}{77} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{330}}{154} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{55}}{77} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{330}}{154} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{154}}{154} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 840 | symmetry | $-\frac{\sqrt{10}x^4}{8}+\frac{\sqrt{10}ix^3y}{4}+\frac{3\sqrt{10}x^2z^2}{4}+\frac{\sqrt{10}ixy^3}{4}-\frac{3\sqrt{10}ixyz^2}{2}+\frac{\sqrt{10}y^4}{8}-\frac{3\sqrt{10}y^2z^2}{4}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_4^{(a)}(E^{(a)},3)$ | $\begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{33}}{77} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{3\sqrt{77}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{462}}{1078} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{385}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{17\sqrt{154}}{1078} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{385}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{1155}}{539} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{3\sqrt{77}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{10\sqrt{231}}{539} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{330}}{154} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{10\sqrt{231}}{539} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{154}}{1078} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{1155}}{539} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{6\sqrt{462}}{539} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{17\sqrt{154}}{1078} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{6\sqrt{462}}{539} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{9\sqrt{462}}{1078} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{154}}{1078} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{3\sqrt{33}}{77} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{330}}{154} & 0 & 0 \end{pmatrix}$ |
| 841 | symmetry | $-\frac{3\sqrt{5}x^3z}{4}+\frac{3\sqrt{5}ix^2yz}{4}-\frac{3\sqrt{5}xy^2z}{4}+\sqrt{5}xz^3+\frac{3\sqrt{5}iy^3z}{4}-\sqrt{5}iyz^3$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_4^{(a)}(E^{(b)},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}}{154} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{77}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{13\sqrt{462}}{1078} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{770}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{77}}{1078} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{1155}}{154} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{770}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}}{539} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{77}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{770}}{539} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{55}}{77} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{5\sqrt{770}}{539} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{1155}}{539} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{2310}}{539} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{231}}{539} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{1155}}{154} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{77}}{1078} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{231}}{539} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{13\sqrt{462}}{1078} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{1155}}{539} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}}{154} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{55}}{77} & 0 \end{bmatrix}$ |
| 842 | symmetry | $\frac{\sqrt{70}x^4}{16} - \frac{\sqrt{70}ix^3y}{4} - \frac{3\sqrt{70}x^2y^2}{8} + \frac{\sqrt{70}ixy^3}{4} + \frac{\sqrt{70}y^4}{16}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_4^{(a)}(E^{(b)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{462}}{77} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{22}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{4\sqrt{33}}{77} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{22}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{55}}{77} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{154}}{154} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{2\sqrt{55}}{77} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{330}}{154} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{4\sqrt{33}}{77} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{330}}{154} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{462}}{77} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{154}}{154} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 843 | symmetry | $-\frac{\sqrt{10}x^4}{8} - \frac{\sqrt{10}ix^3y}{4} + \frac{3\sqrt{10}x^2z^2}{4} - \frac{\sqrt{10}ixy^3}{4} + \frac{3\sqrt{10}ixyz^2}{2} + \frac{\sqrt{10}y^4}{8} - \frac{3\sqrt{10}y^2z^2}{4}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_4^{(a)}(E^{(b)},3)$ | $\begin{bmatrix} 0 & 0 & \frac{3\sqrt{77}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{10\sqrt{231}}{539} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{385}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{1155}}{539} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{385}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{17\sqrt{154}}{1078} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{77}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{462}}{1078} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{33}}{77} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{3\sqrt{33}}{77} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{330}}{154} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{9\sqrt{462}}{1078} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{154}}{1078} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{17\sqrt{154}}{1078} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{6\sqrt{462}}{539} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{1155}}{539} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{6\sqrt{462}}{539} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{10\sqrt{231}}{539} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{154}}{1078} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{330}}{154} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 844 | symmetry | $-\frac{5x^6}{16}-\frac{15x^4y^2}{16}+\frac{45x^4z^2}{8}-\frac{15x^2y^4}{16}+\frac{45x^2y^2z^2}{4}-\frac{15x^2z^4}{2}-\frac{5y^6}{16}+\frac{45y^4z^2}{8}-\frac{15y^2z^4}{2}+z^6$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_6^{(a)}(A,1)$ | $\begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{77}}{154} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{1155}}{154} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{154}}{154} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{154}}{154} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{1155}}{154} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{77}}{154} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{462}}{924} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{77}}{154} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{462}}{924} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{1155}}{154} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{462}}{308} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{5\sqrt{154}}{154} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{462}}{924} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{5\sqrt{154}}{154} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{462}}{924} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{1155}}{154} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{462}}{308} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{77}}{154} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{462}}{924} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{462}}{924} \end{pmatrix}$ |
| 845 | symmetry | $\frac{\sqrt{462}x^6}{32} - \frac{15\sqrt{462}x^4y^2}{32} + \frac{15\sqrt{462}x^2y^4}{32} - \frac{\sqrt{462}y^6}{32}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_6^{(a)}(A,2)$ | $\begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{14} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{14} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{14} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{42}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$ |
| 846 | symmetry | $\frac{3\sqrt{462}x^5y}{16} - \frac{5\sqrt{462}x^3y^3}{8} + \frac{3\sqrt{462}xy^5}{16}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_6^{(a)}(A,3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{14} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{42}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 847 | symmetry | $-\frac{3\sqrt{210}x^5z}{16}+\frac{3\sqrt{210}x^3y^2z}{8}+\frac{\sqrt{210}x^3z^3}{2}+\frac{9\sqrt{210}xy^4z}{16}-\frac{3\sqrt{210}xy^2z^3}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_6^{(a)}(A,4)$ | $\begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{66}}{44} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{22}}{44} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{231}}{154} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{33}}{22} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{33}}{22} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{231}}{154} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{22}}{44} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{66}}{44} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{231}}{154} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{77}}{77} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{33}}{22} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{11}}{22} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{3\sqrt{22}}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{66}}{44} & -\frac{\sqrt{77}}{77} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{11}}{22} & 0 \\ -\frac{\sqrt{66}}{44} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{11}}{22} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{77}}{77} \\ 0 & \frac{3\sqrt{22}}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{33}}{22} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{11}}{22} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{231}}{154} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{77}}{77} & 0 & 0 & 0 \end{pmatrix}$ |
| 848 | symmetry | $-\frac{9\sqrt{210}x^4yz}{16} - \frac{3\sqrt{210}x^2y^3z}{8} + \frac{3\sqrt{210}x^2yz^3}{2} + \frac{3\sqrt{210}y^5z}{16} - \frac{\sqrt{210}y^3z^3}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_6^{(a)}(A,5)$ | $\begin{matrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{66}i}{44} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{22}i}{44} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{231}i}{154} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{33}i}{22} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{33}i}{22} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{231}i}{154} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{22}i}{44} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{66}i}{44} & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{231}i}{154} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{77}i}{77} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{33}i}{22} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{11}i}{22} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{3\sqrt{22}i}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{66}i}{44} & -\frac{\sqrt{77}i}{77} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{11}i}{22} & 0 \\ -\frac{\sqrt{66}i}{44} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{11}i}{22} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{77}i}{77} \\ 0 & \frac{3\sqrt{22}i}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{33}i}{22} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{11}i}{22} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{231}i}{154} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{77}i}{77} & 0 & 0 & 0 \end{matrix}$ |
| 849 | symmetry | $\frac{3\sqrt{77}x^5z}{16} - \frac{15\sqrt{77}ix^4yz}{16} - \frac{15\sqrt{77}x^3y^2z}{8} + \frac{15\sqrt{77}ix^2y^3z}{8} + \frac{15\sqrt{77}xy^4z}{16} - \frac{3\sqrt{77}iy^5z}{16}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_6^{(a)}(E^{(a)},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{35}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 850 | symmetry | $\frac{5\sqrt{42}x^5z}{16}+\frac{5\sqrt{42}ix^4yz}{16}+\frac{5\sqrt{42}x^3y^2z}{8}-\frac{5\sqrt{42}x^3z^3}{4}+\frac{5\sqrt{42}ix^2y^3z}{8}-\frac{5\sqrt{42}ix^2yz^3}{4}+\frac{5\sqrt{42}xy^4z}{16}-\frac{5\sqrt{42}xy^2z^3}{4}+\frac{\sqrt{42}xz^5}{2}+\frac{5\sqrt{42}iy^5z}{16}-\frac{5\sqrt{42}iy^3z^3}{4}+\frac{\sqrt{42}iyz^5}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_6^{(a)}(E^{(a)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{22}}{44} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{66}}{132} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{33}}{66} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{55}}{22} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}}{132} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{462}}{924} \\ \frac{\sqrt{462}}{924} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{77}}{154} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{330}}{132} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{33}}{33} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{55}}{22} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}}{66} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{5\sqrt{33}}{66} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{5\sqrt{66}}{132} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{165}}{66} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{22}}{44} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{33}}{33} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{77}}{154} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 851 | symmetry | $-\frac{3\sqrt{14}x^6}{32}-\frac{3\sqrt{14}ix^5y}{8}+\frac{15\sqrt{14}x^4y^2}{32}+\frac{15\sqrt{14}x^4z^2}{16}+\frac{15\sqrt{14}ix^3yz^2}{4}+\frac{15\sqrt{14}x^2y^4}{32}-\frac{45\sqrt{14}x^2y^2z^2}{8}+\frac{3\sqrt{14}ixy^5}{8}-\frac{15\sqrt{14}ixy^3z^2}{4}-\frac{3\sqrt{14}y^6}{32}+\frac{15\sqrt{14}y^4z^2}{16}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_6^{(a)}(E^{(a)},3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{66}}{22} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{110}}{22} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{385}}{77} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{385}}{77} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{1155}}{154} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{110}}{22} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{11}}{22} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{66}}{22} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{11}}{22} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{1155}}{154} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 852 | symmetry | $\frac{\sqrt{105}x^6}{32} - \frac{\sqrt{105}ix^5y}{16} + \frac{\sqrt{105}x^4y^2}{32} - \frac{\sqrt{105}x^4z^2}{2} - \frac{\sqrt{105}ix^3y^3}{8} + \sqrt{105}ix^3yz^2 - \frac{\sqrt{105}x^2y^4}{32} + \frac{\sqrt{105}x^2z^4}{2} - \frac{\sqrt{105}ixy^5}{16} + \sqrt{105}ixy^3z^2 - \sqrt{105}ixyz^4 - \frac{\sqrt{105}y^6}{32} + \frac{\sqrt{105}y^4z^2}{2} - \frac{\sqrt{105}y^2z^4}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_6^{(a)}(E^{(a)},4)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{231}}{231} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{66}}{33} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{22}}{11} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}}{33} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{33}}{33} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}}{462} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{33}}{33} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{22}}{22} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{165}}{33} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{66}}{66} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{22}}{11} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{66}}{66} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{66}}{33} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{22}}{22} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{231}}{231} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}}{462} & 0 & 0 \end{bmatrix}$ |
| 853 | symmetry | $\frac{3\sqrt{77}x^5z}{16}+\frac{15\sqrt{77}ix^4yz}{16}-\frac{15\sqrt{77}x^3y^2z}{8}-\frac{15\sqrt{77}ix^2y^3z}{8}+\frac{15\sqrt{77}xy^4z}{16}+\frac{3\sqrt{77}iy^5z}{16}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_6^{(a)}(E^{(b)},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{14} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{14} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{14} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 854 | symmetry | $\frac{5\sqrt{42}x^5z}{16} - \frac{5\sqrt{42}ix^4yz}{16} + \frac{5\sqrt{42}x^3y^2z}{8} - \frac{5\sqrt{42}x^3z^3}{4} - \frac{5\sqrt{42}ix^2y^3z}{8} + \frac{5\sqrt{42}ix^2yz^3}{4} + \frac{5\sqrt{42}xy^4z}{16} - \frac{5\sqrt{42}xy^2z^3}{4} + \frac{\sqrt{42}xz^5}{2} - \frac{5\sqrt{42}iy^5z}{16} + \frac{5\sqrt{42}iy^3z^3}{4} - \frac{\sqrt{42}iyz^5}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_6^{(a)}(E^{(b)},2)$ | $\begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{462}}{924} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}}{132} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{55}}{22} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{33}}{66} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{66}}{132} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{22}}{44} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{77}}{154} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{22}}{44} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{33}}{33} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{5\sqrt{66}}{132} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}}{66} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{5\sqrt{33}}{66} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{55}}{22} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{165}}{66} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{330}}{132} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{33}}{33} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{462}}{924} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{77}}{154} & 0 \end{pmatrix}$ |
| 855 | symmetry | $-\frac{3\sqrt{14}x^6}{32}+\frac{3\sqrt{14}ix^5y}{8}+\frac{15\sqrt{14}x^4y^2}{32}+\frac{15\sqrt{14}x^4z^2}{16}-\frac{15\sqrt{14}ix^3yz^2}{4}+\frac{15\sqrt{14}x^2y^4}{32}-\frac{45\sqrt{14}x^2y^2z^2}{8}-\frac{3\sqrt{14}ixy^5}{8}+\frac{15\sqrt{14}ixy^3z^2}{4}-\frac{3\sqrt{14}y^6}{32}+\frac{15\sqrt{14}y^4z^2}{16}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_6^{(a)}(E^{(b)},3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{385}}{77} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{110}}{22} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{66}}{22} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{1155}}{154} & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{66}}{22} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{11}}{22} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{110}}{22} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{11}}{22} & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{385}}{77} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{1155}}{154} & 0 & 0 & 0 \end{bmatrix}$ |
| 856 | symmetry | $\frac{\sqrt{105}x^6}{32}+\frac{\sqrt{105}ix^5y}{16}+\frac{\sqrt{105}x^4y^2}{32}-\frac{\sqrt{105}x^4z^2}{2}+\frac{\sqrt{105}ix^3y^3}{8}-\sqrt{105}ix^3yz^2-\frac{\sqrt{105}x^2y^4}{32}+\frac{\sqrt{105}x^2z^4}{2}+\frac{\sqrt{105}ixy^5}{16}-\sqrt{105}ixy^3z^2+\sqrt{105}ixyz^4-\frac{\sqrt{105}y^6}{32}+\frac{\sqrt{105}y^4z^2}{2}-\frac{\sqrt{105}y^2z^4}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_6^{(a)}(E^{(b)},4)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{33}}{33} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{165}}{33} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{22}}{11} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{66}}{33} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{231}}{231} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{231}}{231} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}}{462} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{66}}{33} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{22}}{22} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{22}}{11} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{66}}{66} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{165}}{33} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{66}}{66} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{33}}{33} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{22}}{22} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}}{462} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 857 | symmetry | $-\frac{x^2}{2}-\frac{y^2}{2}+z^2$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_2^{(1,-1;a)}(A)$ | $\begin{pmatrix} -\frac{5\sqrt{21}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{15\sqrt{14}}{196} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{21}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{210}}{196} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{4\sqrt{21}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{7}}{98} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{4\sqrt{21}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{7}}{98} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{210}}{196} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{21}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{15\sqrt{14}}{196} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{15\sqrt{14}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{98} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{3\sqrt{210}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{21}}{98} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{3\sqrt{7}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{21}}{98} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{3\sqrt{7}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{21}}{98} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{3\sqrt{210}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{21}}{98} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{15\sqrt{14}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{98} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{14} \end{pmatrix}$ |
| 858 | symmetry | $\frac{\sqrt{6}xz}{2} + \frac{\sqrt{6}iyz}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_2^{(1,-1;a)}(E^{(a)},1)$ | $\begin{pmatrix} 0 & -\frac{\sqrt{70}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{15\sqrt{7}}{196} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{2\sqrt{7}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{11\sqrt{21}}{196} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{2\sqrt{7}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{70}}{196} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{196} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{3}}{28} \\ \frac{5\sqrt{3}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{105}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{42}}{49} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{3\sqrt{70}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{98} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{42}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{11\sqrt{21}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{98} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{15\sqrt{7}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{42}}{49} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}}{14} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$ |
| 859 | symmetry | $\frac{\sqrt{6}x^2}{4} - \frac{\sqrt{6}ixy}{2} - \frac{\sqrt{6}y^2}{4}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_2^{(1,-1;a)}(E^{(a)},2)$ | $\begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{35}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{49} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{3\sqrt{7}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{70}}{98} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{3\sqrt{7}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{21}}{49} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{35}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{98} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{105}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{70}}{98} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{2\sqrt{21}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{49} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{3\sqrt{70}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{49} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{210}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{70}}{98} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{14} & 0 & 0 \end{pmatrix}$ |
| 860 | symmetry | $\frac{\sqrt{6}xz}{2} - \frac{\sqrt{6}iyz}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_2^{(1,-1;a)}(E^{(b)},1)$ | $\begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{3}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{70}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{196} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{2\sqrt{7}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{70}}{196} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{2\sqrt{7}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{11\sqrt{21}}{196} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{15\sqrt{7}}{196} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{15\sqrt{7}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{42}}{49} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{11\sqrt{21}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{98} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{42}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{70}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{98} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{42}}{49} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{3}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}}{14} & 0 \end{pmatrix}$ |
| 861 | symmetry | $\frac{\sqrt{6}x^2}{4} + \frac{\sqrt{6}ixy}{2} - \frac{\sqrt{6}y^2}{4}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_2^{(1,-1;a)}(E^{(b)},2)$ | $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{35}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{98} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{7}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{21}}{49} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{3\sqrt{7}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{70}}{98} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{49} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{14} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{15}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{14} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{210}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{70}}{98} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{3\sqrt{70}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{49} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{2\sqrt{21}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{49} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{70}}{98} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{14} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 862 | symmetry | $\frac{3x^4}{8}+\frac{3x^2y^2}{4}-3x^2z^2+\frac{3y^4}{8}-3y^2z^2+z^4$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_4^{(1,-1;a)}(A,1)$ | $\begin{bmatrix} \frac{\sqrt{21}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{14}}{98} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{21}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{210}}{147} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{2\sqrt{21}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{7}}{98} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{2\sqrt{21}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{7}}{98} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{210}}{147} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{14}}{98} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{5\sqrt{14}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{13\sqrt{21}}{147} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{2\sqrt{210}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{49} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{5\sqrt{7}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{21}}{49} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{5\sqrt{7}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{21}}{49} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{2\sqrt{210}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{49} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{14}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{13\sqrt{21}}{147} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{21} \end{bmatrix}$ |
| 863 | symmetry | $\frac{\sqrt{70}x^3z}{4} - \frac{3\sqrt{70}xy^2z}{4}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_4^{(1,-1;a)}(A,2)$ | $\begin{bmatrix} 0 & 0 & 0 & \frac{\sqrt{3}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{12} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{21} & -\frac{3\sqrt{14}}{56} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{56} & 0 \\ \frac{\sqrt{3}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{56} & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{14}}{56} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{12} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{3}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{3\sqrt{14}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{21} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{2}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{6}}{21} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{5}{28} & -\frac{\sqrt{42}}{21} & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{6}}{21} & 0 \\ \frac{5}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{6}}{21} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{21} \\ 0 & \frac{\sqrt{3}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{2}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{6}}{21} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{14}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{21} & 0 & 0 & 0 \end{bmatrix}$ |
| 864 | symmetry | $\frac{3\sqrt{70}x^2yz}{4} - \frac{\sqrt{70}y^3z}{4}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_4^{(1,-1;a)}(A,3)$ | $\begin{pmatrix} 0 & 0 & 0 & -\frac{\sqrt{3}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5i}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{12} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{21} & -\frac{3\sqrt{14}i}{56} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{56} & 0 \\ \frac{\sqrt{3}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{56} & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{14}i}{56} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{12} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{3}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5i}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{3\sqrt{14}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{21} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{2}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{6}i}{21} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{5i}{28} & -\frac{\sqrt{42}i}{21} & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{6}i}{21} & 0 \\ \frac{5i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{6}i}{21} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{21} \\ 0 & \frac{\sqrt{3}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{2}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{6}i}{21} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{14}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{21} & 0 & 0 & 0 \end{pmatrix}$ |
| 865 | symmetry | $-\frac{3\sqrt{5}x^3z}{4} - \frac{3\sqrt{5}ix^2yz}{4} - \frac{3\sqrt{5}xy^2z}{4} + \sqrt{5}xz^3 - \frac{3\sqrt{5}iy^3z}{4} + \sqrt{5}iyz^3$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_4^{(1,-1;a)}(E^{(a)},1)$ | $\begin{pmatrix} 0 & \frac{2\sqrt{21}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{210}}{294} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{210}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{98} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{588} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{21}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{13\sqrt{14}}{196} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{28} \\ -\frac{\sqrt{10}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{15}}{21} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{13\sqrt{14}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{35}}{49} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{21}}{588} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{6\sqrt{7}}{49} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{35}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{6\sqrt{7}}{49} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{210}}{294} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{35}}{49} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{15}}{21} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$ |
| 866 | symmetry | $\frac{\sqrt{70}x^4}{16} + \frac{\sqrt{70}ix^3y}{4} - \frac{3\sqrt{70}x^2y^2}{8} - \frac{\sqrt{70}ixy^3}{4} + \frac{\sqrt{70}y^4}{16}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_4^{(1,-1;a)}(E^{(a)},2)$ | $\begin{matrix} 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{21} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{21} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{21} & 0 & 0 & 0 & 0 & 0 & \frac{2}{7} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{21} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{2}{7} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{7} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{21} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{7} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{21} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{matrix}$ |
| 867 | symmetry | $-\frac{\sqrt{10}x^4}{8}+\frac{\sqrt{10}ix^3y}{4}+\frac{3\sqrt{10}x^2z^2}{4}+\frac{\sqrt{10}ixy^3}{4}-\frac{3\sqrt{10}ixyz^2}{2}+\frac{\sqrt{10}y^4}{8}-\frac{3\sqrt{10}y^2z^2}{4}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_4^{(1,-1;a)}(E^{(a)},3)$ | $\begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{21}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{14}}{196} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{105}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{17\sqrt{42}}{588} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{105}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{98} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{21}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{7}}{49} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{7} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{5\sqrt{7}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{147} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{35}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{4\sqrt{14}}{49} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{17\sqrt{42}}{588} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{4\sqrt{14}}{49} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{9\sqrt{14}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{147} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{3}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{7} & 0 & 0 \end{pmatrix}$ |
| 868 | symmetry | $-\frac{3\sqrt{5}x^3z}{4}+\frac{3\sqrt{5}ix^2yz}{4}-\frac{3\sqrt{5}xy^2z}{4}+\sqrt{5}xz^3+\frac{3\sqrt{5}iy^3z}{4}-\sqrt{5}iyz^3$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_4^{(1,-1;a)}(E^{(b)},1)$ | $\begin{matrix} 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{2\sqrt{21}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{13\sqrt{14}}{196} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{210}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{588} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{210}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{98} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{2\sqrt{21}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{210}}{294} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{15}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{5\sqrt{210}}{294} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{35}}{49} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{70}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{6\sqrt{7}}{49} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{35}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{21}}{588} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{6\sqrt{7}}{49} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{13\sqrt{14}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{35}}{49} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{15}}{21} & 0 \end{matrix}$ |
| 869 | symmetry | $\frac{\sqrt{70}x^4}{16} - \frac{\sqrt{70}ix^3y}{4} - \frac{3\sqrt{70}x^2y^2}{8} + \frac{\sqrt{70}ixy^3}{4} + \frac{\sqrt{70}y^4}{16}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_4^{(1,-1;a)}(E^{(b)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{6}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2}{7} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{6}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{21} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{15}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{7} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{2}{7} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{7} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{21} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 870 | symmetry | $-\frac{\sqrt{10}x^4}{8} - \frac{\sqrt{10}ix^3y}{4} + \frac{3\sqrt{10}x^2z^2}{4} - \frac{\sqrt{10}ixy^3}{4} + \frac{3\sqrt{10}ixyz^2}{2} + \frac{\sqrt{10}y^4}{8} - \frac{3\sqrt{10}y^2z^2}{4}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_4^{(1,-1;a)}(E^{(b)},3)$ | $\begin{bmatrix} 0 & 0 & \frac{\sqrt{21}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{7}}{49} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{105}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{98} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{17\sqrt{42}}{588} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{14}}{196} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3}{14} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{3}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{7} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{9\sqrt{14}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{147} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{17\sqrt{42}}{588} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{4\sqrt{14}}{49} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{4\sqrt{14}}{49} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{7}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{147} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}}{7} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 871 | symmetry | $-\frac{5x^6}{16}-\frac{15x^4y^2}{16}+\frac{45x^4z^2}{8}-\frac{15x^2y^4}{16}+\frac{45x^2y^2z^2}{4}-\frac{15x^2z^4}{2}-\frac{5y^6}{16}+\frac{45y^4z^2}{8}-\frac{15y^2z^4}{2}+z^6$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_6^{(1,-1;a)}(A,1)$ | $\begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{462}}{924} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{770}}{308} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{231}}{462} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{231}}{462} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{770}}{308} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{462}}{924} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{77}}{154} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{462}}{924} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{77}}{154} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{770}}{308} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{77}}{154} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{5\sqrt{231}}{462} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{77}}{154} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{5\sqrt{231}}{462} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{77}}{154} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{770}}{308} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{77}}{154} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{462}}{924} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{77}}{154} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{77}}{154} \end{pmatrix}$ |
| 872 | symmetry | $\frac{\sqrt{462}x^6}{32} - \frac{15\sqrt{462}x^4y^2}{32} + \frac{15\sqrt{462}x^2y^4}{32} - \frac{\sqrt{462}y^6}{32}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_6^{(1,-1;a)}(A,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{14} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{14} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{14} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{14} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 873 | symmetry | $\frac{3\sqrt{462}x^5y}{16} - \frac{5\sqrt{462}x^3y^3}{8} + \frac{3\sqrt{462}xy^5}{16}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_6^{(1,-1;a)}(A,3)$ | $\begin{matrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{14} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{14} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 \end{matrix}$ |
| 874 | symmetry | $-\frac{3\sqrt{210}x^5z}{16}+\frac{3\sqrt{210}x^3y^2z}{8}+\frac{\sqrt{210}x^3z^3}{2}+\frac{9\sqrt{210}xy^4z}{16}-\frac{3\sqrt{210}xy^2z^3}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_6^{(1,-1;a)}(A,4)$ | $\begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{11}}{44} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{33}}{44} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{154}}{308} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{22}}{44} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{22}}{44} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{154}}{308} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{33}}{44} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{11}}{44} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{154}}{308} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{462}}{77} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{22}}{44} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{66}}{22} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{33}}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{11}}{44} & \frac{\sqrt{462}}{77} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{66}}{22} & 0 \\ -\frac{\sqrt{11}}{44} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{66}}{22} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{462}}{77} \\ 0 & \frac{\sqrt{33}}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{22}}{44} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{66}}{22} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{154}}{308} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{462}}{77} & 0 & 0 & 0 \end{pmatrix}$ |
| 875 | symmetry | $-\frac{9\sqrt{210}x^4yz}{16} - \frac{3\sqrt{210}x^2y^3z}{8} + \frac{3\sqrt{210}x^2yz^3}{2} + \frac{3\sqrt{210}y^5z}{16} - \frac{\sqrt{210}y^3z^3}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_6^{(1,-1;a)}(A,5)$ | $\begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{11}i}{44} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{33}i}{44} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{154}i}{308} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{22}i}{44} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{22}i}{44} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{154}i}{308} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{33}i}{44} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{11}i}{44} & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{154}i}{308} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{462}i}{77} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{22}i}{44} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{66}i}{22} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{33}i}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{11}i}{44} & \frac{\sqrt{462}i}{77} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{66}i}{22} & 0 \\ -\frac{\sqrt{11}i}{44} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{66}i}{22} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{462}i}{77} \\ 0 & \frac{\sqrt{33}i}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{22}i}{44} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{66}i}{22} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{154}i}{308} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{462}i}{77} & 0 & 0 & 0 \end{pmatrix}$ |
| 876 | symmetry | $\frac{3\sqrt{77}x^5z}{16} - \frac{15\sqrt{77}ix^4yz}{16} - \frac{15\sqrt{77}x^3y^2z}{8} + \frac{15\sqrt{77}ix^2y^3z}{8} + \frac{15\sqrt{77}xy^4z}{16} - \frac{3\sqrt{77}iy^5z}{16}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_6^{(1,-1;a)}(E^{(a)},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{84} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{12} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{7} & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{6}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{210}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{7} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 877 | symmetry | $\frac{5\sqrt{42}x^5z}{16}+\frac{5\sqrt{42}ix^4yz}{16}+\frac{5\sqrt{42}x^3y^2z}{8}-\frac{5\sqrt{42}x^3z^3}{4}+\frac{5\sqrt{42}ix^2y^3z}{8}-\frac{5\sqrt{42}ix^2yz^3}{4}+\frac{5\sqrt{42}xy^4z}{16}-\frac{5\sqrt{42}xy^2z^3}{4}+\frac{\sqrt{42}xz^5}{2}+\frac{5\sqrt{42}iy^5z}{16}-\frac{5\sqrt{42}iy^3z^3}{4}+\frac{\sqrt{42}iyz^5}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_6^{(1,-1;a)}(E^{(a)},2)$ | $\begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{33}}{132} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{11}}{132} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{22}}{132} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{330}}{132} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{55}}{132} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{77}}{924} \\ \frac{\sqrt{77}}{924} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{462}}{154} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{55}}{132} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{22}}{11} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{330}}{132} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{110}}{22} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{5\sqrt{22}}{132} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{5\sqrt{11}}{132} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{110}}{22} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{33}}{132} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{22}}{11} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{462}}{154} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$ |
| 878 | symmetry | $-\frac{3\sqrt{14}x^6}{32}-\frac{3\sqrt{14}ix^5y}{8}+\frac{15\sqrt{14}x^4y^2}{32}+\frac{15\sqrt{14}x^4z^2}{16}+\frac{15\sqrt{14}ix^3yz^2}{4}+\frac{15\sqrt{14}x^2y^4}{32}-\frac{45\sqrt{14}x^2y^2z^2}{8}+\frac{3\sqrt{14}ixy^5}{8}-\frac{15\sqrt{14}ixy^3z^2}{4}-\frac{3\sqrt{14}y^6}{32}+\frac{15\sqrt{14}y^4z^2}{16}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_6^{(1,-1;a)}(E^{(a)},3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{11}}{22} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{165}}{66} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}}{462} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{2310}}{462} & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{770}}{154} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{165}}{66} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{66}}{22} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{11}}{22} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{66}}{22} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{770}}{154} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 879 | symmetry | $\frac{\sqrt{105}x^6}{32} - \frac{\sqrt{105}ix^5y}{16} + \frac{\sqrt{105}x^4y^2}{32} - \frac{\sqrt{105}x^4z^2}{2} - \frac{\sqrt{105}ix^3y^3}{8} + \sqrt{105}ix^3yz^2 - \frac{\sqrt{105}x^2y^4}{32} + \frac{\sqrt{105}x^2z^4}{2} - \frac{\sqrt{105}ixy^5}{16} + \sqrt{105}ixy^3z^2 - \sqrt{105}ixyz^4 - \frac{\sqrt{105}y^6}{32} + \frac{\sqrt{105}y^4z^2}{2} - \frac{\sqrt{105}y^2z^4}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_6^{(1,-1;a)}(E^{(a)},4)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{154}}{462} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{11}}{33} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{33}}{33} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{110}}{66} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{22}}{66} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{385}}{77} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{22}}{66} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{33}}{11} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{110}}{66} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{11}}{11} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{33}}{33} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{11}}{11} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{11}}{33} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{33}}{11} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{154}}{462} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{385}}{77} & 0 & 0 \end{bmatrix}$ |
| 880 | symmetry | $\frac{3\sqrt{77}x^5z}{16}+\frac{15\sqrt{77}ix^4yz}{16}-\frac{15\sqrt{77}x^3y^2z}{8}-\frac{15\sqrt{77}ix^2y^3z}{8}+\frac{15\sqrt{77}xy^4z}{16}+\frac{3\sqrt{77}iy^5z}{16}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_6^{(1,-1;a)}(E^{(b)},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{12} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{84} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{84} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{7} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{7} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 881 | symmetry | $\frac{5\sqrt{42}x^5z}{16} - \frac{5\sqrt{42}ix^4yz}{16} + \frac{5\sqrt{42}x^3y^2z}{8} - \frac{5\sqrt{42}x^3z^3}{4} - \frac{5\sqrt{42}ix^2y^3z}{8} + \frac{5\sqrt{42}ix^2yz^3}{4} + \frac{5\sqrt{42}xy^4z}{16} - \frac{5\sqrt{42}xy^2z^3}{4} + \frac{\sqrt{42}xz^5}{2} - \frac{5\sqrt{42}iy^5z}{16} + \frac{5\sqrt{42}iy^3z^3}{4} - \frac{\sqrt{42}iyz^5}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_6^{(1,-1;a)}(E^{(b)},2)$ | $\begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{77}}{924} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{55}}{132} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{330}}{132} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{22}}{132} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{11}}{132} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{33}}{132} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{462}}{154} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{33}}{132} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{22}}{11} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{5\sqrt{11}}{132} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{110}}{22} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{5\sqrt{22}}{132} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{330}}{132} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{110}}{22} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{55}}{132} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{22}}{11} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{77}}{924} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{462}}{154} & 0 \end{pmatrix}$ |
| 882 | symmetry | $-\frac{3\sqrt{14}x^6}{32}+\frac{3\sqrt{14}ix^5y}{8}+\frac{15\sqrt{14}x^4y^2}{32}+\frac{15\sqrt{14}x^4z^2}{16}-\frac{15\sqrt{14}ix^3yz^2}{4}+\frac{15\sqrt{14}x^2y^4}{32}-\frac{45\sqrt{14}x^2y^2z^2}{8}-\frac{3\sqrt{14}ixy^5}{8}+\frac{15\sqrt{14}ixy^3z^2}{4}-\frac{3\sqrt{14}y^6}{32}+\frac{15\sqrt{14}y^4z^2}{16}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_6^{(1,-1;a)}(E^{(b)},3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}}{462} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}}{66} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{11}}{22} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{770}}{154} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{11}}{22} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{66}}{22} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{165}}{66} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{66}}{22} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{2310}}{462} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{770}}{154} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 883 | symmetry | $\frac{\sqrt{105}x^6}{32}+\frac{\sqrt{105}ix^5y}{16}+\frac{\sqrt{105}x^4y^2}{32}-\frac{\sqrt{105}x^4z^2}{2}+\frac{\sqrt{105}ix^3y^3}{8}-\sqrt{105}ix^3yz^2-\frac{\sqrt{105}x^2y^4}{32}+\frac{\sqrt{105}x^2z^4}{2}+\frac{\sqrt{105}ixy^5}{16}-\sqrt{105}ixy^3z^2+\sqrt{105}ixyz^4-\frac{\sqrt{105}y^6}{32}+\frac{\sqrt{105}y^4z^2}{2}-\frac{\sqrt{105}y^2z^4}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 884 | $\mathbb{Q}_6^{(1,-1;a)}(E^{(b)},4)$ | $\begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{22}}{66} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{110}}{66} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{33}}{33} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{11}}{33} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{154}}{462} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{154}}{462} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{385}}{77} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{11}}{33} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{33}}{11} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{33}}{33} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{11}}{11} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{110}}{66} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{11}}{11} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{22}}{66} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{33}}{11} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{385}}{77} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$ |
| | symmetry | 1 |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_0^{(1,1;a)}(A)$ | $\begin{bmatrix} -\frac{\sqrt{42}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{42}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{42}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{42}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{28} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{28} \end{bmatrix}$ |
| 885 | symmetry | $-\frac{x^2}{2} - \frac{y^2}{2} + z^2$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_2^{(1,1;a)}(A)$ | $\begin{pmatrix} \frac{15\sqrt{7}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{42}}{147} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{3\sqrt{7}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{49} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{6\sqrt{7}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{21}}{147} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{6\sqrt{7}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{21}}{147} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{3\sqrt{7}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{49} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{15\sqrt{7}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{42}}{147} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{5\sqrt{42}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{98} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{70}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{7}}{98} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{2\sqrt{21}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{7}}{98} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{2\sqrt{21}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{7}}{98} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{7}}{98} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{42}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{98} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{14} \end{pmatrix}$ |
| 886 | symmetry | $\frac{\sqrt{6}xz}{2} + \frac{\sqrt{6}iyz}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_2^{(1,1;a)}(E^{(a)},1)$ | $\begin{bmatrix} 0 & \frac{3\sqrt{210}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{21}}{147} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{3\sqrt{21}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{11\sqrt{7}}{147} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{21} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{3\sqrt{21}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{147} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{210}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{147} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5}{21} \\ \frac{5}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{35}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{14}}{49} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{210}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{98} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{14}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{11\sqrt{7}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{98} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{21}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{14}}{49} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{14} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 887 | symmetry | $\frac{\sqrt{6}x^2}{4} - \frac{\sqrt{6}ixy}{2} - \frac{\sqrt{6}y^2}{4}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_2^{(1,1;a)}(E^{(a)},2)$ | $\begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{5}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{3\sqrt{105}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{4\sqrt{70}}{147} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{9\sqrt{21}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{210}}{147} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{9\sqrt{21}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{8\sqrt{7}}{147} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{3\sqrt{105}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{35}}{147} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{2\sqrt{35}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{98} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{8\sqrt{7}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{49} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{2\sqrt{210}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{49} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{4\sqrt{70}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{98} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{2\sqrt{5}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{14} & 0 & 0 \end{pmatrix}$ |
| 888 | symmetry | $\frac{\sqrt{6}xz}{2} - \frac{\sqrt{6}iyz}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_2^{(1,1;a)}(E^{(b)},1)$ | $\begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{5}{21} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{3\sqrt{210}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{147} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{3\sqrt{21}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{147} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{21} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{21}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{11\sqrt{7}}{147} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{3\sqrt{210}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{21}}{147} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{5\sqrt{21}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{14}}{49} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{11\sqrt{7}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{98} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{14}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{210}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{98} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{14}}{49} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{5}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{14} & 0 \end{pmatrix}$ |
| 889 | symmetry | $\frac{\sqrt{6}x^2}{4} + \frac{\sqrt{6}ixy}{2} - \frac{\sqrt{6}y^2}{4}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_2^{(1,1;a)}(E^{(b)},2)$ | $\begin{bmatrix} 0 & 0 & \frac{3\sqrt{105}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{35}}{147} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{9\sqrt{21}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{8\sqrt{7}}{147} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{9\sqrt{21}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{210}}{147} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{105}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{4\sqrt{70}}{147} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{5}}{21} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{2\sqrt{5}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{14} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{4\sqrt{70}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{98} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{2\sqrt{210}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{49} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{8\sqrt{7}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{49} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{35}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{98} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{14} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 890 | symmetry | $\frac{3x^4}{8}+\frac{3x^2y^2}{4}-3x^2z^2+\frac{3y^4}{8}-3y^2z^2+z^4$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 891 | $\mathbb{Q}_4^{(1,1;a)}(A,1)$ | $\begin{pmatrix} -\frac{\sqrt{2310}}{294} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{4\sqrt{385}}{539} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{2310}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{16\sqrt{231}}{1617} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{2310}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{770}}{539} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{2310}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{770}}{539} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{2310}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{16\sqrt{231}}{1617} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}}{294} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{4\sqrt{385}}{539} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}}{924} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{4\sqrt{385}}{539} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{13\sqrt{2310}}{6468} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{16\sqrt{231}}{1617} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}}{2156} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{2\sqrt{770}}{539} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2310}}{2156} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{2\sqrt{770}}{539} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2310}}{2156} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{16\sqrt{231}}{1617} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}}{2156} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{4\sqrt{385}}{539} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{13\sqrt{2310}}{6468} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}}{924} \end{pmatrix}$ |
| | symmetry | $\frac{\sqrt{70}x^3z}{4} - \frac{3\sqrt{70}xy^2z}{4}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_4^{(1,1;a)}(A,2)$ | $\begin{pmatrix} 0 & 0 & 0 & -\frac{\sqrt{330}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{110}}{77} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{330}}{165} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{330}}{42} & -\frac{3\sqrt{385}}{385} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{55}}{385} & 0 \\ -\frac{\sqrt{330}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{55}}{385} & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{385}}{385} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{330}}{165} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{330}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{110}}{77} & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{3\sqrt{385}}{385} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{1155}}{462} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{55}}{385} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{165}}{231} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{330}}{165} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{110}}{77} & \frac{\sqrt{1155}}{462} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}}{231} & 0 \\ \frac{\sqrt{110}}{77} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{165}}{231} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{1155}}{462} \\ 0 & \frac{\sqrt{330}}{165} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{55}}{385} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}}{231} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{385}}{385} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{1155}}{462} & 0 & 0 & 0 \end{pmatrix}$ |
| 892 | symmetry | $\frac{3\sqrt{70}x^2yz}{4} - \frac{\sqrt{70}y^3z}{4}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_4^{(1,1;a)}(A,3)$ | $\begin{bmatrix} 0 & 0 & 0 & \frac{\sqrt{330}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{110}i}{77} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}i}{165} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}i}{42} & -\frac{3\sqrt{385}i}{385} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{55}i}{385} & 0 \\ -\frac{\sqrt{330}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{55}i}{385} & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{385}i}{385} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{330}i}{165} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{330}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{110}i}{77} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{3\sqrt{385}i}{385} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{1155}i}{462} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{55}i}{385} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}i}{231} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{330}i}{165} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{110}i}{77} & \frac{\sqrt{1155}i}{462} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{165}i}{231} & 0 \\ \frac{\sqrt{110}i}{77} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{165}i}{231} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{1155}i}{462} \\ 0 & \frac{\sqrt{330}i}{165} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{55}i}{385} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}i}{231} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{385}i}{385} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{1155}i}{462} & 0 & 0 & 0 \end{bmatrix}$ |
| 893 | symmetry | $-\frac{3\sqrt{5}x^3z}{4} - \frac{3\sqrt{5}ix^2yz}{4} - \frac{3\sqrt{5}xy^2z}{4} + \sqrt{5}xz^3 - \frac{3\sqrt{5}iy^3z}{4} + \sqrt{5}iyz^3$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_4^{(1,1;a)}(E^{(a)},1)$ | $\begin{pmatrix} 0 & -\frac{\sqrt{2310}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{20\sqrt{231}}{1617} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{5\sqrt{231}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{4\sqrt{77}}{539} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{154}}{77} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{5\sqrt{231}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}}{8085} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{26\sqrt{385}}{2695} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{11}}{77} \\ -\frac{2\sqrt{11}}{77} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{66}}{462} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{26\sqrt{385}}{2695} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{154}}{1078} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{2310}}{8085} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{770}}{1078} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{154}}{77} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{4\sqrt{77}}{539} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{770}}{1078} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{20\sqrt{231}}{1617} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{154}}{1078} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{66}}{462} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$ |
| 894 | symmetry | $\frac{\sqrt{70}x^4}{16}+\frac{\sqrt{70}ix^3y}{4}-\frac{3\sqrt{70}x^2y^2}{8}-\frac{\sqrt{70}ixy^3}{4}+\frac{\sqrt{70}y^4}{16}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_4^{(1,1;a)}(E^{(a)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{\sqrt{165}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{4\sqrt{66}}{231} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{8\sqrt{110}}{385} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{4\sqrt{385}}{385} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{4\sqrt{385}}{385} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{1155}}{462} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{8\sqrt{110}}{385} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{11}}{154} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{4\sqrt{66}}{231} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{11}}{154} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{1155}}{462} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 895 | symmetry | $-\frac{\sqrt{10}x^4}{8}+\frac{\sqrt{10}ix^3y}{4}+\frac{3\sqrt{10}x^2z^2}{4}+\frac{\sqrt{10}ixy^3}{4}-\frac{3\sqrt{10}ixyz^2}{2}+\frac{\sqrt{10}y^4}{8}-\frac{3\sqrt{10}y^2z^2}{4}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_4^{(1,1;a)}(E^{(a)},3)$ | $\begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{6\sqrt{110}}{385} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{2310}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{18\sqrt{385}}{2695} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{5\sqrt{462}}{294} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{34\sqrt{1155}}{8085} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{5\sqrt{462}}{294} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{154}}{539} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{2310}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{4\sqrt{770}}{539} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{11}}{154} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{4\sqrt{770}}{539} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{1155}}{3234} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{2\sqrt{154}}{539} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{385}}{539} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{34\sqrt{1155}}{8085} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{385}}{539} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{18\sqrt{385}}{2695} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{1155}}{3234} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{6\sqrt{110}}{385} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{11}}{154} & 0 & 0 \end{pmatrix}$ |
| 896 | symmetry | $-\frac{3\sqrt{5}x^3z}{4}+\frac{3\sqrt{5}ix^2yz}{4}-\frac{3\sqrt{5}xy^2z}{4}+\sqrt{5}xz^3+\frac{3\sqrt{5}iy^3z}{4}-\sqrt{5}iyz^3$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_4^{(1,1;a)}(E^{(b)},1)$ | $\begin{matrix} 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{11}}{77} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{2310}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{26\sqrt{385}}{2695} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{5\sqrt{231}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}}{8085} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{154}}{77} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{5\sqrt{231}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{4\sqrt{77}}{539} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{2310}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{20\sqrt{231}}{1617} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{66}}{462} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{20\sqrt{231}}{1617} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{154}}{1078} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{4\sqrt{77}}{539} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{770}}{1078} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{154}}{77} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{2310}}{8085} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{770}}{1078} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{26\sqrt{385}}{2695} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{154}}{1078} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{11}}{77} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{66}}{462} & 0 \end{matrix}$ |
| 897 | symmetry | $\frac{\sqrt{70}x^4}{16} - \frac{\sqrt{70}ix^3y}{4} - \frac{3\sqrt{70}x^2y^2}{8} + \frac{\sqrt{70}ixy^3}{4} + \frac{\sqrt{70}y^4}{16}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{Q}_4^{(1,1;a)}(E^{(b)},2)$ | $\begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{4\sqrt{385}}{385} & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{165}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{8\sqrt{110}}{385} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{165}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{4\sqrt{66}}{231} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{1155}}{462} & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{4\sqrt{66}}{231} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{11}}{154} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{8\sqrt{110}}{385} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{11}}{154} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{4\sqrt{385}}{385} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{1155}}{462} & 0 & 0 & 0 \end{pmatrix}$ |
| 898 | symmetry | $-\frac{\sqrt{10}x^4}{8} - \frac{\sqrt{10}ix^3y}{4} + \frac{3\sqrt{10}x^2z^2}{4} - \frac{\sqrt{10}ixy^3}{4} + \frac{3\sqrt{10}ixyz^2}{2} + \frac{\sqrt{10}y^4}{8} - \frac{3\sqrt{10}y^2z^2}{4}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 899 | $\mathbb{Q}_4^{(1,1;a)}(E^{(b)},3)$ | $\begin{pmatrix} 0 & 0 & -\frac{\sqrt{2310}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{4\sqrt{770}}{539} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{5\sqrt{462}}{294} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{154}}{539} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{5\sqrt{462}}{294} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{34\sqrt{1155}}{8085} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{18\sqrt{385}}{2695} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{6\sqrt{110}}{385} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{6\sqrt{110}}{385} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{11}}{154} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{18\sqrt{385}}{2695} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{1155}}{3234} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{34\sqrt{1155}}{8085} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{385}}{539} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{2\sqrt{154}}{539} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{385}}{539} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{4\sqrt{770}}{539} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{1155}}{3234} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{11}}{154} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_1^{(1,0;a)}(A)$ | $\begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{28} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{42}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{70}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{21}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{21}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$ |
| 900 | symmetry | $\frac{\sqrt{2}x}{2} + \frac{\sqrt{2}iy}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_1^{(1,0;a)}(E^{(a)})$ | $\begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{28} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{4} \\ -\frac{\sqrt{3}i}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{105}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{70}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{42}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$ |
| 901 | symmetry | $\frac{\sqrt{2}x}{2} - \frac{\sqrt{2}iy}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_1^{(1,0;a)}(E^{(b)})$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{7}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{21}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{42}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{70}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{105}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 902 | symmetry | $-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_3^{(1,0;a)}(A,1)$ | $\begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{6} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{6}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$ |
| 903 | symmetry | $\frac{3\sqrt{10}x^2y}{4} - \frac{\sqrt{10}y^3}{4}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_3^{(1,0;a)}(A,2)$ | $\begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{12} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{4} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{24} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{8} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{8} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{24} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{4} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{12} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{42}}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{6}}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{3}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{1}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{6}}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{42}}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$ |
| 904 | symmetry | $\frac{\sqrt{10}x^3}{4} - \frac{3\sqrt{10}xy^2}{4}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_3^{(1,0;a)}(A,3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{12} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{4} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{24} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{8} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{8} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{24} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{4} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{12} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{42}i}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{6}i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{i}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{3}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{i}{4} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{6}i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{42}i}{24} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 905 | symmetry | $-\frac{\sqrt{3}x^3}{4} - \frac{\sqrt{3}ix^2y}{4} - \frac{\sqrt{3}xy^2}{4} + \sqrt{3}xz^2 - \frac{\sqrt{3}iy^3}{4} + \sqrt{3}iyz^2$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_3^{(1,0;a)}(E^{(a)},1)$ | $\begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{6} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{6} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{12} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{12} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10}i}{12} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{12} \\ \frac{\sqrt{14}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{10}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{15}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$ |
| 906 | symmetry | $\frac{\sqrt{30}x^2z}{4} - \frac{\sqrt{30}ixyz}{2} - \frac{\sqrt{30}y^2z}{4}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_3^{(1,0;a)}(E^{(a)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{12} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{6} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{3} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{i}{3} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{5}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{6}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{2}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{7}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 907 | symmetry | $-\frac{\sqrt{3}x^3}{4}+\frac{\sqrt{3}ix^2y}{4}-\frac{\sqrt{3}xy^2}{4}+\sqrt{3}xz^2+\frac{\sqrt{3}iy^3}{4}-\sqrt{3}iyz^2$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_3^{(1,0;a)}(E^{(b)},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{12} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{12} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{6} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{6} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{6}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{2}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{15}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{10}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 908 | symmetry | $\frac{\sqrt{30}x^2z}{4}+\frac{\sqrt{30}ixyz}{2}-\frac{\sqrt{30}y^2z}{4}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_3^{(1,0;a)}(E^{(b)},2)$ | $\begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{3} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{6} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{12} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{12} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{6} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{7}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{2}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{6}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{i}{3} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$ |
| 909 | symmetry | $\frac{15x^4z}{8}+\frac{15x^2y^2z}{4}-5x^2z^3+\frac{15y^4z}{8}-5y^2z^3+z^5$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_5^{(1,0;a)}(A,1)$ | $\begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{14}i}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{42} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{42} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{14}i}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{84} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{210}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{3\sqrt{14}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{105}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{105}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{3\sqrt{14}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$ |
| 910 | symmetry | $-\frac{3\sqrt{70}x^4y}{16} - \frac{\sqrt{70}x^2y^3}{8} + \frac{3\sqrt{70}x^2yz^2}{2} + \frac{\sqrt{70}y^5}{16} - \frac{\sqrt{70}y^3z^2}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_5^{(1,0;a)}(A,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{12} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{20} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{60} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{20} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{20} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{60} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{20} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{12} & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{210}}{60} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{30}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{15}}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{5}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{30}}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{210}}{60} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 911 | symmetry | $-\frac{\sqrt{70}x^5}{16}+\frac{\sqrt{70}x^3y^2}{8}+\frac{\sqrt{70}x^3z^2}{2}+\frac{3\sqrt{70}xy^4}{16}-\frac{3\sqrt{70}xy^2z^2}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_5^{(1,0;a)}(A,3)$ | $\begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{12} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{20} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{60} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{20} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{20} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{60} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{20} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{12} & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{210}i}{60} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{30}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{15}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{5}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{30}i}{20} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{210}i}{60} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$ |
| 912 | symmetry | $\frac{3\sqrt{7}x^5}{16} - \frac{15\sqrt{7}ix^4y}{16} - \frac{15\sqrt{7}x^3y^2}{8} + \frac{15\sqrt{7}ix^2y^3}{8} + \frac{15\sqrt{7}xy^4}{16} - \frac{3\sqrt{7}iy^5}{16}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_5^{(1,0;a)}(E^{(a)},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{12} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{30}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{42}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 913 | symmetry | $\frac{\sqrt{30}x^5}{16}+\frac{\sqrt{30}ix^4y}{16}+\frac{\sqrt{30}x^3y^2}{8}-\frac{3\sqrt{30}x^3z^2}{4}+\frac{\sqrt{30}ix^2y^3}{8}-\frac{3\sqrt{30}ix^2yz^2}{4}+\frac{\sqrt{30}xy^4}{16}-\frac{3\sqrt{30}xy^2z^2}{4}+\frac{\sqrt{30}xz^4}{2}+\frac{\sqrt{30}iy^5}{16}-\frac{3\sqrt{30}iy^3z^2}{4}+\frac{\sqrt{30}iyz^4}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_5^{(1,0;a)}(E^{(a)},2)$ | $\begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{21}i}{84} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{13\sqrt{7}i}{84} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}i}{84} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{11\sqrt{210}i}{420} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{23\sqrt{35}i}{420} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{12} \\ -\frac{i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{23\sqrt{35}i}{420} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{11\sqrt{210}i}{420} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{14}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{13\sqrt{7}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{21}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$ |
| 914 | symmetry | $\frac{3\sqrt{70}x^4z}{16}+\frac{3\sqrt{70}ix^3yz}{4}-\frac{9\sqrt{70}x^2y^2z}{8}-\frac{3\sqrt{70}ixy^3z}{4}+\frac{3\sqrt{70}y^4z}{16}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_5^{(1,0;a)}(E^{(a)},3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{2} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{30} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{30} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{210}i}{30} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{30} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{i}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 915 | symmetry | $-\frac{\sqrt{210}x^4z}{8}+\frac{\sqrt{210}ix^3yz}{4}+\frac{\sqrt{210}x^2z^3}{4}+\frac{\sqrt{210}ixy^3z}{4}-\frac{\sqrt{210}ixyz^3}{2}+\frac{\sqrt{210}y^4z}{8}-\frac{\sqrt{210}y^2z^3}{4}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_5^{(1,0;a)}(E^{(a)},4)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{30} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{10}i}{15} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{30} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{3} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{6} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{5}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{i}{3} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{30}i}{30} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{2\sqrt{10}i}{15} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{30} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 916 | symmetry | $\frac{3\sqrt{7}x^5}{16}+\frac{15\sqrt{7}ix^4y}{16}-\frac{15\sqrt{7}x^3y^2}{8}-\frac{15\sqrt{7}ix^2y^3}{8}+\frac{15\sqrt{7}xy^4}{16}+\frac{3\sqrt{7}iy^5}{16}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_5^{(1,0;a)}(E^{(b)},1)$ | $\begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{12} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{12} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$ |
| 917 | symmetry | $\frac{\sqrt{30}x^5}{16} - \frac{\sqrt{30}ix^4y}{16} + \frac{\sqrt{30}x^3y^2}{8} - \frac{3\sqrt{30}x^3z^2}{4} - \frac{\sqrt{30}ix^2y^3}{8} + \frac{3\sqrt{30}ix^2yz^2}{4} + \frac{\sqrt{30}xy^4}{16} - \frac{3\sqrt{30}xy^2z^2}{4} + \frac{\sqrt{30}xz^4}{2} - \frac{\sqrt{30}iy^5}{16} + \frac{3\sqrt{30}iy^3z^2}{4} - \frac{\sqrt{30}iyz^4}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_5^{(1,0;a)}(E^{(b)},2)$ | $\begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{23\sqrt{35}i}{420} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{11\sqrt{210}i}{420} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{84} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{13\sqrt{7}i}{84} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{21}i}{84} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{5\sqrt{21}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{13\sqrt{7}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{14}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{11\sqrt{210}i}{420} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{23\sqrt{35}i}{420} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$ |
| 918 | symmetry | $\frac{3\sqrt{70}x^4z}{16} - \frac{3\sqrt{70}ix^3yz}{4} - \frac{9\sqrt{70}x^2y^2z}{8} + \frac{3\sqrt{70}ixy^3z}{4} + \frac{3\sqrt{70}y^4z}{16}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_5^{(1,0;a)}(E^{(b)},3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{30} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{30} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{2} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{i}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{15}i}{30} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{210}i}{30} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 919 | symmetry | $-\frac{\sqrt{210}x^4z}{8} - \frac{\sqrt{210}ix^3yz}{4} + \frac{\sqrt{210}x^2z^3}{4} - \frac{\sqrt{210}ixy^3z}{4} + \frac{\sqrt{210}ixyz^3}{2} + \frac{\sqrt{210}y^4z}{8} - \frac{\sqrt{210}y^2z^3}{4}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{G}_5^{(1,0;a)}(E^{(b)},4)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{6} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{3} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{30} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{10}i}{15} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{30} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{35}i}{30} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{2\sqrt{10}i}{15} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{30}i}{30} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{i}{3} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 920 | symmetry | $-\frac{x^2}{2}-\frac{y^2}{2}+z^2$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_2^{(1,0;a)}(A)$ | $\begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{42}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{42} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{42}i}{84} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{5\sqrt{42}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{70}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{21}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{42}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$ |
| 921 | symmetry | $\frac{\sqrt{6}xz}{2} + \frac{\sqrt{6}iyz}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_2^{(1,0;a)}(E^{(a)},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{21}i}{84} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{11\sqrt{7}i}{84} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{12} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{84} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{84} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5i}{12} \\ \frac{5i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{35}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{210}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{14}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{11\sqrt{7}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{21}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 922 | symmetry | $\frac{\sqrt{6}x^2}{4} - \frac{\sqrt{6}ixy}{2} - \frac{\sqrt{6}y^2}{4}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_2^{(1,0;a)}(E^{(a)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{42} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{7}i}{21} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{42} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{35}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{2\sqrt{7}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{210}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{70}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 923 | symmetry | $\frac{\sqrt{6}xz}{2} - \frac{\sqrt{6}iyz}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_2^{(1,0;a)}(E^{(b)},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{84} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{12} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{11\sqrt{7}i}{84} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{21}i}{84} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{5\sqrt{21}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{11\sqrt{7}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{14}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{210}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{5i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 924 | symmetry | $\frac{\sqrt{6}x^2}{4}+\frac{\sqrt{6}ixy}{2}-\frac{\sqrt{6}y^2}{4}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_2^{(1,0;a)}(E^{(b)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{42} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{7}i}{21} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{42} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{21} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{6} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{5}i}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{70}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{210}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{2\sqrt{7}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 925 | symmetry | $\frac{3x^4}{8}+\frac{3x^2y^2}{4}-3x^2z^2+\frac{3y^4}{8}-3y^2z^2+z^4$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_4^{(1,0;a)}(A,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}i}{154} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{154}i}{77} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{1155}i}{154} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{1155}i}{154} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{154}i}{77} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}i}{154} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{2310}i}{154} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{2\sqrt{154}i}{77} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{1155}i}{154} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{1155}i}{154} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{2\sqrt{154}i}{77} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}i}{154} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 926 | symmetry | $\frac{\sqrt{70}x^3z}{4} - \frac{3\sqrt{70}xy^2z}{4}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_4^{(1,0;a)}(A,2)$ | $\begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}i}{44} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{7\sqrt{55}i}{220} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2310}i}{440} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{330}i}{440} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{330}i}{440} & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2310}i}{440} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{7\sqrt{55}i}{220} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}i}{44} & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{3\sqrt{2310}i}{440} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{330}i}{440} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{7\sqrt{55}i}{220} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{165}i}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{165}i}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{7\sqrt{55}i}{220} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{330}i}{440} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{2310}i}{440} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$ |
| 927 | symmetry | $\frac{3\sqrt{70}x^2yz}{4} - \frac{\sqrt{70}y^3z}{4}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_4^{(1,0;a)}(A,3)$ | $\begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}}{44} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{7\sqrt{55}}{220} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2310}}{440} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{330}}{440} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}}{440} & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2310}}{440} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{7\sqrt{55}}{220} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{165}}{44} & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{3\sqrt{2310}}{440} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{330}}{440} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{7\sqrt{55}}{220} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{165}}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{165}}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{7\sqrt{55}}{220} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{330}}{440} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{3\sqrt{2310}}{440} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$ |
| 928 | symmetry | $-\frac{3\sqrt{5}x^3z}{4} - \frac{3\sqrt{5}ix^2yz}{4} - \frac{3\sqrt{5}xy^2z}{4} + \sqrt{5}xz^3 - \frac{3\sqrt{5}iy^3z}{4} + \sqrt{5}iyz^3$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_4^{(1,0;a)}(E^{(a)},1)$ | $\begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{154}i}{154} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{462}i}{154} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{231}i}{44} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{385}i}{1540} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{13\sqrt{2310}i}{1540} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{66}i}{44} \\ -\frac{\sqrt{66}i}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{13\sqrt{2310}i}{1540} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{385}i}{1540} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{231}i}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{462}i}{154} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{154}i}{154} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$ |
| 929 | symmetry | $\frac{\sqrt{70}x^4}{16}+\frac{\sqrt{70}ix^3y}{4}-\frac{3\sqrt{70}x^2y^2}{8}-\frac{\sqrt{70}ixy^3}{4}+\frac{\sqrt{70}y^4}{16}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_4^{(1,0;a)}(E^{(a)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{11}i}{11} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{165}i}{55} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}i}{110} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{2310}i}{110} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{2\sqrt{165}i}{55} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{11}i}{11} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 930 | symmetry | $-\frac{\sqrt{10}x^4}{8}+\frac{\sqrt{10}ix^3y}{4}+\frac{3\sqrt{10}x^2z^2}{4}+\frac{\sqrt{10}ixy^3}{4}-\frac{3\sqrt{10}ixyz^2}{2}+\frac{\sqrt{10}y^4}{8}-\frac{3\sqrt{10}y^2z^2}{4}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_4^{(1,0;a)}(E^{(a)},3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{165}i}{110} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{2310}i}{1540} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{17\sqrt{770}i}{1540} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{231}i}{154} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{1155}i}{77} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{1155}i}{77} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{231}i}{154} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{17\sqrt{770}i}{1540} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{9\sqrt{2310}i}{1540} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{3\sqrt{165}i}{110} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 931 | symmetry | $-\frac{3\sqrt{5}x^3z}{4}+\frac{3\sqrt{5}ix^2yz}{4}-\frac{3\sqrt{5}xy^2z}{4}+\sqrt{5}xz^3+\frac{3\sqrt{5}iy^3z}{4}-\sqrt{5}iyz^3$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_4^{(1,0;a)}(E^{(b)},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{66}i}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{13\sqrt{2310}i}{1540} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{385}i}{1540} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{231}i}{44} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{462}i}{154} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{154}i}{154} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{5\sqrt{154}i}{154} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{462}i}{154} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{231}i}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{385}i}{1540} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{13\sqrt{2310}i}{1540} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{66}i}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 932 | symmetry | $\frac{\sqrt{70}x^4}{16} - \frac{\sqrt{70}ix^3y}{4} - \frac{3\sqrt{70}x^2y^2}{8} + \frac{\sqrt{70}ixy^3}{4} + \frac{\sqrt{70}y^4}{16}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_4^{(1,0;a)}(E^{(b)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}i}{110} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{165}i}{55} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{11}i}{11} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{11}i}{11} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{2\sqrt{165}i}{55} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{2310}i}{110} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 933 | symmetry | $-\frac{\sqrt{10}x^4}{8}-\frac{\sqrt{10}ix^3y}{4}+\frac{3\sqrt{10}x^2z^2}{4}-\frac{\sqrt{10}ixy^3}{4}+\frac{3\sqrt{10}ixyz^2}{2}+\frac{\sqrt{10}y^4}{8}-\frac{3\sqrt{10}y^2z^2}{4}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_4^{(1,0;a)}(E^{(b)},3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{1155}i}{77} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{231}i}{154} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{17\sqrt{770}i}{1540} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{2310}i}{1540} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{165}i}{110} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{3\sqrt{165}i}{110} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{9\sqrt{2310}i}{1540} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{17\sqrt{770}i}{1540} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{231}i}{154} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{1155}i}{77} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 934 | symmetry | $-\frac{5x^6}{16}-\frac{15x^4y^2}{16}+\frac{45x^4z^2}{8}-\frac{15x^2y^4}{16}+\frac{45x^2y^2z^2}{4}-\frac{15x^2z^4}{2}-\frac{5y^6}{16}+\frac{45y^4z^2}{8}-\frac{15y^2z^4}{2}+z^6$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_6^{(1,0;a)}(A,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{66}i}{132} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{110}i}{44} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{33}i}{66} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{33}i}{66} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{110}i}{44} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{66}i}{132} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{66}i}{132} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{110}i}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{5\sqrt{33}i}{66} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{5\sqrt{33}i}{66} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{110}i}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{66}i}{132} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 935 | symmetry | $\frac{\sqrt{462}x^6}{32} - \frac{15\sqrt{462}x^4y^2}{32} + \frac{15\sqrt{462}x^2y^4}{32} - \frac{\sqrt{462}y^6}{32}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_6^{(1,0;a)}(A,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{2} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{2} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{i}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{i}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 936 | symmetry | $\frac{3\sqrt{462}x^5y}{16} - \frac{5\sqrt{462}x^3y^3}{8} + \frac{3\sqrt{462}xy^5}{16}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_6^{(1,0;a)}(A,3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 937 | symmetry | $-\frac{3\sqrt{210}x^5z}{16}+\frac{3\sqrt{210}x^3y^2z}{8}+\frac{\sqrt{210}x^3z^3}{2}+\frac{9\sqrt{210}xy^4z}{16}-\frac{3\sqrt{210}xy^2z^3}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_6^{(1,0;a)}(A,4)$ | $\begin{matrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{77}i}{44} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{231}i}{44} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{22}i}{44} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{154}i}{44} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{154}i}{44} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{22}i}{44} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{231}i}{44} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{77}i}{44} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{22}i}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{154}i}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{231}i}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{77}i}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{77}i}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{231}i}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{154}i}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{22}i}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{matrix}$ |
| 938 | symmetry | $-\frac{9\sqrt{210}x^4yz}{16} - \frac{3\sqrt{210}x^2y^3z}{8} + \frac{3\sqrt{210}x^2yz^3}{2} + \frac{3\sqrt{210}y^5z}{16} - \frac{\sqrt{210}y^3z^3}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_6^{(1,0;a)}(A,5)$ | $\begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{77}}{44} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{231}}{44} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{22}}{44} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{154}}{44} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{154}}{44} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{22}}{44} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{231}}{44} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{77}}{44} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{22}}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{154}}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{231}}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{77}}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{77}}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{231}}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{154}}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{22}}{44} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$ |
| 939 | symmetry | $\frac{3\sqrt{77}x^5z}{16} - \frac{15\sqrt{77}ix^4yz}{16} - \frac{15\sqrt{77}x^3y^2z}{8} + \frac{15\sqrt{77}ix^2y^3z}{8} + \frac{15\sqrt{77}xy^4z}{16} - \frac{3\sqrt{77}iy^5z}{16}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_6^{(1,0;a)}(E^{(a)},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{42}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{30}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 940 | symmetry | $\frac{5\sqrt{42}x^5z}{16}+\frac{5\sqrt{42}ix^4yz}{16}+\frac{5\sqrt{42}x^3y^2z}{8}-\frac{5\sqrt{42}x^3z^3}{4}+\frac{5\sqrt{42}ix^2y^3z}{8}-\frac{5\sqrt{42}ix^2yz^3}{4}+\frac{5\sqrt{42}xy^4z}{16}-\frac{5\sqrt{42}xy^2z^3}{4}+\frac{\sqrt{42}xz^5}{2}+\frac{5\sqrt{42}iy^5z}{16}-\frac{5\sqrt{42}iy^3z^3}{4}+\frac{\sqrt{42}iyz^5}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_6^{(1,0;a)}(E^{(a)},2)$ | $\begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{231}i}{132} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{77}i}{132} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{154}i}{132} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}i}{132} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{385}i}{132} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{11}i}{132} \\ \frac{\sqrt{11}i}{132} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{385}i}{132} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{2310}i}{132} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{5\sqrt{154}i}{132} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{5\sqrt{77}i}{132} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{231}i}{132} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$ |
| 941 | symmetry | $-\frac{3\sqrt{14}x^6}{32}-\frac{3\sqrt{14}ix^5y}{8}+\frac{15\sqrt{14}x^4y^2}{32}+\frac{15\sqrt{14}x^4z^2}{16}+\frac{15\sqrt{14}ix^3yz^2}{4}+\frac{15\sqrt{14}x^2y^4}{32}-\frac{45\sqrt{14}x^2y^2z^2}{8}+\frac{3\sqrt{14}ixy^5}{8}-\frac{15\sqrt{14}ixy^3z^2}{4}-\frac{3\sqrt{14}y^6}{32}+\frac{15\sqrt{14}y^4z^2}{16}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_6^{(1,0;a)}(E^{(a)},3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{77}i}{22} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{1155}i}{66} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{330}i}{66} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{330}i}{66} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{1155}i}{66} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{77}i}{22} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 942 | symmetry | $\frac{\sqrt{105}x^6}{32} - \frac{\sqrt{105}ix^5y}{16} + \frac{\sqrt{105}x^4y^2}{32} - \frac{\sqrt{105}x^4z^2}{2} - \frac{\sqrt{105}ix^3y^3}{8} + \sqrt{105}ix^3yz^2 - \frac{\sqrt{105}x^2y^4}{32} + \frac{\sqrt{105}x^2z^4}{2} - \frac{\sqrt{105}ixy^5}{16} + \sqrt{105}ixy^3z^2 - \sqrt{105}ixyz^4 - \frac{\sqrt{105}y^6}{32} + \frac{\sqrt{105}y^4z^2}{2} - \frac{\sqrt{105}y^2z^4}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_6^{(1,0;a)}(E^{(a)},4)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{22}i}{66} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{77}i}{33} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{231}i}{33} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{770}i}{66} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{154}i}{66} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{154}i}{66} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{770}i}{66} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{231}i}{33} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{77}i}{33} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{22}i}{66} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 943 | symmetry | $\frac{3\sqrt{77}x^5z}{16}+\frac{15\sqrt{77}ix^4yz}{16}-\frac{15\sqrt{77}x^3y^2z}{8}-\frac{15\sqrt{77}ix^2y^3z}{8}+\frac{15\sqrt{77}xy^4z}{16}+\frac{3\sqrt{77}iy^5z}{16}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_6^{(1,0;a)}(E^{(b)},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{12} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{12} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{12} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 944 | symmetry | $\frac{5\sqrt{42}x^5z}{16} - \frac{5\sqrt{42}ix^4yz}{16} + \frac{5\sqrt{42}x^3y^2z}{8} - \frac{5\sqrt{42}x^3z^3}{4} - \frac{5\sqrt{42}ix^2y^3z}{8} + \frac{5\sqrt{42}ix^2yz^3}{4} + \frac{5\sqrt{42}xy^4z}{16} - \frac{5\sqrt{42}xy^2z^3}{4} + \frac{\sqrt{42}xz^5}{2} - \frac{5\sqrt{42}iy^5z}{16} + \frac{5\sqrt{42}iy^3z^3}{4} - \frac{\sqrt{42}iyz^5}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_6^{(1,0;a)}(E^{(b)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{11}i}{132} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{385}i}{132} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}i}{132} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{154}i}{132} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{77}i}{132} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{231}i}{132} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{231}i}{132} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{5\sqrt{77}i}{132} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{5\sqrt{154}i}{132} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{2310}i}{132} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{385}i}{132} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{11}i}{132} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 945 | symmetry | $-\frac{3\sqrt{14}x^6}{32}+\frac{3\sqrt{14}ix^5y}{8}+\frac{15\sqrt{14}x^4y^2}{32}+\frac{15\sqrt{14}x^4z^2}{16}-\frac{15\sqrt{14}ix^3yz^2}{4}+\frac{15\sqrt{14}x^2y^4}{32}-\frac{45\sqrt{14}x^2y^2z^2}{8}-\frac{3\sqrt{14}ixy^5}{8}+\frac{15\sqrt{14}ixy^3z^2}{4}-\frac{3\sqrt{14}y^6}{32}+\frac{15\sqrt{14}y^4z^2}{16}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_6^{(1,0;a)}(E^{(b)},3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}i}{66} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{1155}i}{66} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{77}i}{22} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{77}i}{22} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{1155}i}{66} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{330}i}{66} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 946 | symmetry | $\frac{\sqrt{105}x^6}{32}+\frac{\sqrt{105}ix^5y}{16}+\frac{\sqrt{105}x^4y^2}{32}-\frac{\sqrt{105}x^4z^2}{2}+\frac{\sqrt{105}ix^3y^3}{8}-\sqrt{105}ix^3yz^2-\frac{\sqrt{105}x^2y^4}{32}+\frac{\sqrt{105}x^2z^4}{2}+\frac{\sqrt{105}ixy^5}{16}-\sqrt{105}ixy^3z^2+\sqrt{105}ixyz^4-\frac{\sqrt{105}y^6}{32}+\frac{\sqrt{105}y^4z^2}{2}-\frac{\sqrt{105}y^2z^4}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{T}_6^{(1,0;a)}(E^{(b)},4)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{154}i}{66} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{770}i}{66} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{231}i}{33} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{77}i}{33} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{22}i}{66} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{22}i}{66} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{77}i}{33} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{231}i}{33} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{770}i}{66} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{154}i}{66} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 947 | symmetry | $z$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_1^{(a)}(A)$ | $\begin{pmatrix} \frac{5\sqrt{14}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{98} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{3\sqrt{14}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{98} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{14}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{98} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{14}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{98} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{3\sqrt{14}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{98} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{14}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{98} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{14}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{21}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{15\sqrt{14}}{196} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{35}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{14}}{196} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{42}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{14}}{196} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{42}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{14}}{196} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{9\sqrt{14}}{196} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{15\sqrt{14}}{196} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{14}}{28} \end{pmatrix}$ |
| 948 | symmetry | $\frac{\sqrt{2}x}{2} + \frac{\sqrt{2}iy}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_1^{(a)}(E^{(a)})$ | $\begin{bmatrix} 0 & \frac{2\sqrt{35}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{196} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{4\sqrt{14}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{196} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{6\sqrt{7}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{98} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{4\sqrt{14}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{98} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{35}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{196} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{28} \\ -\frac{\sqrt{6}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{210}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{21}}{49} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{35}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{105}}{98} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{21}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{6\sqrt{7}}{49} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{105}}{98} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{21}}{49} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3}{14} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 949 | symmetry | $\frac{\sqrt{2}x}{2} - \frac{\sqrt{2}iy}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_1^{(a)}(E^{(b)})$ | $\begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{2\sqrt{35}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{196} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{4\sqrt{14}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{98} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{6\sqrt{7}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{98} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{4\sqrt{14}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{196} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{2\sqrt{35}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{196} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{3}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{14}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{21}}{49} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{42}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{105}}{98} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{21}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{6\sqrt{7}}{49} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{35}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{105}}{98} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{21}}{49} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3}{14} & 0 \end{pmatrix}$ |
| 950 | symmetry | $-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_3^{(a)}(A,1)$ | $\begin{bmatrix} -\frac{5\sqrt{3}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{7} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{2\sqrt{3}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{7} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{2\sqrt{3}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{7} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{3}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{7} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{2}}{7} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{3}}{42} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{1}{7} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{7} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{7} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{3}}{42} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{6} \end{bmatrix}$ |
| 951 | symmetry | $\frac{3\sqrt{10}x^2y}{4} - \frac{\sqrt{10}y^3}{4}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_3^{(a)}(A,2)$ | $\begin{pmatrix} 0 & 0 & 0 & \frac{5\sqrt{3}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{14} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{3}i}{42} & \frac{\sqrt{14}i}{28} & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}i}{28} & 0 \\ -\frac{5\sqrt{3}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}i}{28} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}i}{28} \\ 0 & -\frac{\sqrt{30}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{14} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{5\sqrt{3}i}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{14}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{42} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{2}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{6}i}{21} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}i}{21} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{i}{14} & -\frac{\sqrt{42}i}{42} & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{6}i}{21} & 0 \\ -\frac{i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{6}i}{21} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}i}{42} \\ 0 & -\frac{\sqrt{3}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}i}{21} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{3\sqrt{2}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{6}i}{21} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{14}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}i}{42} & 0 & 0 & 0 \end{pmatrix}$ |
| 952 | symmetry | $\frac{\sqrt{10}x^3}{4} - \frac{3\sqrt{10}xy^2}{4}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_3^{(a)}(A,3)$ | $\begin{bmatrix} 0 & 0 & 0 & -\frac{5\sqrt{3}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{14} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{3}}{42} & \frac{\sqrt{14}}{28} & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}}{28} & 0 \\ -\frac{5\sqrt{3}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}}{28} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{28} \\ 0 & -\frac{\sqrt{30}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{14} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{5\sqrt{3}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{14}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{42} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{3\sqrt{2}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{6}}{21} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{21} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{14} & -\frac{\sqrt{42}}{42} & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{6}}{21} & 0 \\ -\frac{1}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{6}}{21} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{42} \\ 0 & -\frac{\sqrt{3}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{21} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{3\sqrt{2}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{6}}{21} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{14}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{42} & 0 & 0 & 0 \end{bmatrix}$ |
| 953 | symmetry | $-\frac{\sqrt{3}x^3}{4} - \frac{\sqrt{3}ix^2y}{4} - \frac{\sqrt{3}xy^2}{4} + \sqrt{3}xz^2 - \frac{\sqrt{3}iy^3}{4} + \sqrt{3}iyz^2$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_3^{(a)}(E^{(a)},1)$ | $\begin{bmatrix} 0 & -\frac{\sqrt{5}}{7} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{7} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{2}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{21} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{2}{7} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{42} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{14} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{7} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{42} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{42} \\ \frac{\sqrt{42}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{7} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{30}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{21} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{5}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{21} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{3}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2}{7} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{21} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{7} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{21} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{7} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 954 | symmetry | $\frac{\sqrt{30}x^2z}{4} - \frac{\sqrt{30}ixyz}{2} - \frac{\sqrt{30}y^2z}{4}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_3^{(a)}(E^{(a)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{5}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{42} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{5}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{14} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{5}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{21} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{5}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{3}}{21} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{2\sqrt{3}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{15}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{21} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{2}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{21} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{6}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{42} & 0 & 0 \end{bmatrix}$ |
| 955 | symmetry | $-\frac{\sqrt{3}x^3}{4}+\frac{\sqrt{3}ix^2y}{4}-\frac{\sqrt{3}xy^2}{4}+\sqrt{3}xz^2+\frac{\sqrt{3}iy^3}{4}-\sqrt{3}iyz^2$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_3^{(a)}(E^{(b)},1)$ | $\begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{5}}{7} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{42} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{2}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{14} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{2}{7} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{42} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{2}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{21} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{7} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{7} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{7} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{2}}{7} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{21} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{6}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{21} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{3}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2}{7} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{5}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{21} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{21} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{7} & 0 \end{pmatrix}$ |
| 956 | symmetry | $\frac{\sqrt{30}x^2z}{4}+\frac{\sqrt{30}ixyz}{2}-\frac{\sqrt{30}y^2z}{4}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_3^{(a)}(E^{(b)},2)$ | $\begin{bmatrix} 0 & 0 & -\frac{5}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{3}}{21} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{5}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{21} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{14} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{5}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{42} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{21} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{21}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{42} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{6}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2}}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{2}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{21} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{21} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{3}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2}}{14} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{42} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 957 | symmetry | $\frac{15x^4z}{8}+\frac{15x^2y^2z}{4}-5x^2z^3+\frac{15y^4z}{8}-5y^2z^3+z^5$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_5^{(a)}(A,1)$ | $\begin{pmatrix} \frac{\sqrt{42}}{294} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{7}}{98} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{5\sqrt{42}}{294} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{105}}{98} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{5\sqrt{42}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{14}}{98} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{5\sqrt{42}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{14}}{98} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{5\sqrt{42}}{294} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{105}}{98} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{294} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{7}}{98} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{5\sqrt{7}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{23\sqrt{42}}{588} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{3\sqrt{105}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{17\sqrt{42}}{588} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{5\sqrt{14}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{42}}{196} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{5\sqrt{14}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{42}}{196} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{3\sqrt{105}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{17\sqrt{42}}{588} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{7}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{23\sqrt{42}}{588} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{84} \end{pmatrix}$ |
| 958 | symmetry | $-\frac{3\sqrt{70}x^4y}{16} - \frac{\sqrt{70}x^2y^3}{8} + \frac{3\sqrt{70}x^2yz^2}{2} + \frac{\sqrt{70}y^5}{16} - \frac{\sqrt{70}y^3z^2}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_5^{(a)}(A,2)$ | $\begin{pmatrix} 0 & 0 & 0 & -\frac{\sqrt{6}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{2}i}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{21} & -\frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & \frac{3i}{14} & 0 \\ \frac{\sqrt{6}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3i}{14} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}i}{14} \\ 0 & -\frac{\sqrt{15}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{6}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{2}i}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{21} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{3i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{42} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}i}{21} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{2}i}{28} & \frac{\sqrt{21}i}{21} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}i}{42} & 0 \\ \frac{5\sqrt{2}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{42} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}i}{21} \\ 0 & -\frac{\sqrt{6}i}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{21} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{3i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{42} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{7}i}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}i}{21} & 0 & 0 & 0 \end{pmatrix}$ |
| 959 | symmetry | $-\frac{\sqrt{70}x^5}{16}+\frac{\sqrt{70}x^3y^2}{8}+\frac{\sqrt{70}x^3z^2}{2}+\frac{3\sqrt{70}xy^4}{16}-\frac{3\sqrt{70}xy^2z^2}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_5^{(a)}(A,3)$ | $\begin{bmatrix} 0 & 0 & 0 & \frac{\sqrt{6}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{2}}{28} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{21} & -\frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & -\frac{3}{14} & 0 \\ \frac{\sqrt{6}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3}{14} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{14} \\ 0 & -\frac{\sqrt{15}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{6}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{2}}{28} & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{21} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{3}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{42} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{21} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{2}}{28} & \frac{\sqrt{21}}{21} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{42} & 0 \\ \frac{5\sqrt{2}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{42} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{21} \\ 0 & -\frac{\sqrt{6}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}}{21} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{3}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{42} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{7}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{21} & 0 & 0 & 0 \end{bmatrix}$ |
| 960 | symmetry | $\frac{3\sqrt{7}x^5}{16} - \frac{15\sqrt{7}ix^4y}{16} - \frac{15\sqrt{7}x^3y^2}{8} + \frac{15\sqrt{7}ix^2y^3}{8} + \frac{15\sqrt{7}xy^4}{16} - \frac{3\sqrt{7}iy^5}{16}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_5^{(a)}(E^{(a)},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{6}}{7} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{5}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{7} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{35}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 961 | symmetry | $\frac{\sqrt{30}x^5}{16}+\frac{\sqrt{30}ix^4y}{16}+\frac{\sqrt{30}x^3y^2}{8}-\frac{3\sqrt{30}x^3z^2}{4}+\frac{\sqrt{30}ix^2y^3}{8}-\frac{3\sqrt{30}ix^2yz^2}{4}+\frac{\sqrt{30}xy^4}{16}-\frac{3\sqrt{30}xy^2z^2}{4}+\frac{\sqrt{30}xz^4}{2}+\frac{\sqrt{30}iy^5}{16}-\frac{3\sqrt{30}iy^3z^2}{4}+\frac{\sqrt{30}iyz^4}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_5^{(a)}(E^{(a)},2)$ | $\begin{pmatrix} 0 & \frac{\sqrt{7}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{70}}{196} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{70}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{13\sqrt{210}}{588} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{2\sqrt{35}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{294} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{11\sqrt{7}}{98} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{23\sqrt{42}}{588} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{84} \\ -\frac{\sqrt{30}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{23\sqrt{42}}{588} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{4\sqrt{105}}{147} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{11\sqrt{7}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{294} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{105}}{294} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{35}}{49} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{13\sqrt{210}}{588} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{294} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{70}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{4\sqrt{105}}{147} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{14} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$ |
| 962 | symmetry | $\frac{3\sqrt{70}x^4z}{16}+\frac{3\sqrt{70}ix^3yz}{4}-\frac{9\sqrt{70}x^2y^2z}{8}-\frac{3\sqrt{70}ixy^3z}{4}+\frac{3\sqrt{70}y^4z}{16}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_5^{(a)}(E^{(a)},3)$ | $\begin{matrix} 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{7} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{14} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{7} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{14} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{7} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{7}}{7} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{14} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{14} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{14} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{matrix}$ |
| 963 | symmetry | $-\frac{\sqrt{210}x^4z}{8}+\frac{\sqrt{210}ix^3yz}{4}+\frac{\sqrt{210}x^2z^3}{4}+\frac{\sqrt{210}ixy^3z}{4}-\frac{\sqrt{210}ixyz^3}{2}+\frac{\sqrt{210}y^4z}{8}-\frac{\sqrt{210}y^2z^3}{4}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_5^{(a)}(E^{(a)},4)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{2}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{4\sqrt{3}}{21} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{10}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{7} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{10}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{21} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{2}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{6}}{42} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{5\sqrt{6}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{30}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{3}}{21} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{1}{7} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{3}}{21} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{4\sqrt{3}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3}{14} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{42} & 0 & 0 \end{bmatrix}$ |
| 964 | symmetry | $\frac{3\sqrt{7}x^5}{16}+\frac{15\sqrt{7}ix^4y}{16}-\frac{15\sqrt{7}x^3y^2}{8}-\frac{15\sqrt{7}ix^2y^3}{8}+\frac{15\sqrt{7}xy^4}{16}+\frac{3\sqrt{7}iy^5}{16}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_5^{(a)}(E^{(b)},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{7} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5}{14} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{14} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{5}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{7} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 965 | symmetry | $\frac{\sqrt{30}x^5}{16} - \frac{\sqrt{30}ix^4y}{16} + \frac{\sqrt{30}x^3y^2}{8} - \frac{3\sqrt{30}x^3z^2}{4} - \frac{\sqrt{30}ix^2y^3}{8} + \frac{3\sqrt{30}ix^2yz^2}{4} + \frac{\sqrt{30}xy^4}{16} - \frac{3\sqrt{30}xy^2z^2}{4} + \frac{\sqrt{30}xz^4}{2} - \frac{\sqrt{30}iy^5}{16} + \frac{3\sqrt{30}iy^3z^2}{4} - \frac{\sqrt{30}iyz^4}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_5^{(a)}(E^{(b)},2)$ | $\begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{7}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{23\sqrt{42}}{588} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{70}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{11\sqrt{7}}{98} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{2\sqrt{35}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{294} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{70}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{13\sqrt{210}}{588} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{7}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{70}}{196} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{5\sqrt{70}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{4\sqrt{105}}{147} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{13\sqrt{210}}{588} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{294} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{105}}{294} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{35}}{49} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{11\sqrt{7}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{294} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{23\sqrt{42}}{588} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{4\sqrt{105}}{147} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{84} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{14} & 0 \end{pmatrix}$ |
| 966 | symmetry | $\frac{3\sqrt{70}x^4z}{16} - \frac{3\sqrt{70}ix^3yz}{4} - \frac{9\sqrt{70}x^2y^2z}{8} + \frac{3\sqrt{70}ixy^3z}{4} + \frac{3\sqrt{70}y^4z}{16}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_5^{(a)}(E^{(b)},3)$ | $\begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{7}}{7} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{3}}{7} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{3}}{7} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{14} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{21}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{30}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{2}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{14} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{7}}{7} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{21}}{14} & 0 & 0 & 0 & 0 \end{pmatrix}$ |
| 967 | symmetry | $-\frac{\sqrt{210}x^4z}{8} - \frac{\sqrt{210}ix^3yz}{4} + \frac{\sqrt{210}x^2z^3}{4} - \frac{\sqrt{210}ixy^3z}{4} + \frac{\sqrt{210}ixyz^3}{2} + \frac{\sqrt{210}y^4z}{8} - \frac{\sqrt{210}y^2z^3}{4}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| 968 | symmetry | $z$ |

$\mathbb{M}_5^{(a)}(E^{(b)},4)$

$$\begin{bmatrix}
0 & 0 & \frac{\sqrt{2}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{6}}{42} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{\sqrt{10}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{21} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{\sqrt{10}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{7} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{4\sqrt{3}}{21} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{42} \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & -\frac{\sqrt{42}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{42} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{4\sqrt{3}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3}{14} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{1}{7} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{3}}{21} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{3}}{21} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{6}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3}{14} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{42} \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
\end{bmatrix}$$

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_1^{(1,-1;a)}(A)$ | $\begin{pmatrix} -\frac{5\sqrt{14}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{21}}{49} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{3\sqrt{14}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{35}}{49} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{14}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{42}}{49} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{14}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{42}}{49} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{3\sqrt{14}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{35}}{49} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{14}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{21}}{49} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{2\sqrt{21}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{14}}{98} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{2\sqrt{35}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{14}}{98} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{2\sqrt{42}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{98} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{2\sqrt{42}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{98} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{2\sqrt{35}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{14}}{98} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{21}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{14}}{98} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{14} \end{pmatrix}$ |
| 969 | symmetry | $\frac{\sqrt{2}x}{2} + \frac{\sqrt{2}iy}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_1^{(1,-1;a)}(E^{(a)})$ | $\begin{pmatrix} 0 & -\frac{\sqrt{35}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{14}}{49} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{2\sqrt{14}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{49} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{7}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{21}}{49} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{2\sqrt{14}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{35}}{49} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{49} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{7} \\ \frac{\sqrt{6}}{7} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{7} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{210}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{21}}{49} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{2\sqrt{35}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{49} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{2\sqrt{21}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{4\sqrt{7}}{49} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{49} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{21}}{49} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{7} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$ |
| 970 | symmetry | $\frac{\sqrt{2}x}{2} - \frac{\sqrt{2}iy}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_1^{(1,-1;a)}(E^{(b)})$ | $\begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{7} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{35}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{49} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{2\sqrt{14}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{35}}{49} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{3\sqrt{7}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{21}}{49} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{2\sqrt{14}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{49} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{14}}{49} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{7} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{14}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{21}}{49} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{42}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{49} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{2\sqrt{21}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{4\sqrt{7}}{49} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{2\sqrt{35}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{49} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{21}}{49} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{7} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{7} & 0 \end{pmatrix}$ |
| 971 | symmetry | $-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_3^{(1,-1;a)}(A,1)$ | $\begin{pmatrix} \frac{\sqrt{70}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{4\sqrt{105}}{147} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{70}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{2\sqrt{70}}{245} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{210}}{147} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{2\sqrt{70}}{245} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{210}}{147} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{70} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{4\sqrt{105}}{147} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{4\sqrt{105}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{70}}{196} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{28} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{2\sqrt{210}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{70}}{196} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{2\sqrt{210}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{70}}{196} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{28} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{4\sqrt{105}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{70}}{196} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{28} \end{pmatrix}$ |
| 972 | symmetry | $\frac{3\sqrt{10}x^2y}{4} - \frac{\sqrt{10}y^3}{4}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_3^{(1,-1;a)}(A,2)$ | $\begin{pmatrix} 0 & 0 & 0 & -\frac{\sqrt{70}i}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{147} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{2\sqrt{7}i}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{49} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{98} & -\frac{\sqrt{15}i}{21} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{49} & 0 \\ \frac{\sqrt{70}i}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}i}{49} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{15}i}{21} \\ 0 & \frac{2\sqrt{7}i}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}i}{49} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{70}i}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}i}{147} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{15}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{105}i}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{35}i}{49} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{70}i}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{7}i}{49} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}i}{147} & -\frac{\sqrt{5}i}{14} & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{35}i}{49} & 0 \\ \frac{\sqrt{210}i}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{35}i}{49} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{5}i}{14} \\ 0 & \frac{\sqrt{70}i}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{7}i}{49} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{105}i}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{35}i}{49} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{15}i}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}i}{14} & 0 & 0 & 0 \end{pmatrix}$ |
| 973 | symmetry | $\frac{\sqrt{10}x^3}{4} - \frac{3\sqrt{10}xy^2}{4}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_3^{(1,-1;a)}(A,3)$ | $\begin{pmatrix} 0 & 0 & 0 & \frac{\sqrt{70}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{147} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{2\sqrt{7}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{49} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{98} & -\frac{\sqrt{15}}{21} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{49} & 0 \\ \frac{\sqrt{70}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{49} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{15}}{21} \\ 0 & \frac{2\sqrt{7}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{49} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{70}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{147} & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{15}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{14} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{105}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{35}}{49} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{7}}{49} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{147} & -\frac{\sqrt{5}}{14} & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{35}}{49} & 0 \\ \frac{\sqrt{210}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{35}}{49} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{14} \\ 0 & \frac{\sqrt{70}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{7}}{49} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{105}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{35}}{49} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{15}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{14} & 0 & 0 & 0 \end{pmatrix}$ |
| 974 | symmetry | $-\frac{\sqrt{3}x^3}{4} - \frac{\sqrt{3}ix^2y}{4} - \frac{\sqrt{3}xy^2}{4} + \sqrt{3}xz^2 - \frac{\sqrt{3}iy^3}{4} + \sqrt{3}iyz^2$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_3^{(1,-1;a)}(E^{(a)},1)$ | $\begin{pmatrix} 0 & \frac{\sqrt{42}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{4\sqrt{105}}{147} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{105}}{245} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{4\sqrt{35}}{147} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{2\sqrt{210}}{245} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{147} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{245} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{42}}{147} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{10\sqrt{7}}{147} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{5}}{21} \\ -\frac{2\sqrt{5}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{14} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{10\sqrt{7}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{98} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{5\sqrt{42}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{14}}{98} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{70}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{49} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{4\sqrt{35}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{14}}{98} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{4\sqrt{105}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{98} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{14} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$ |
| 975 | symmetry | $\frac{\sqrt{30}x^2z}{4} - \frac{\sqrt{30}ixyz}{2} - \frac{\sqrt{30}y^2z}{4}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_3^{(1,-1;a)}(E^{(a)},2)$ | $\begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{10}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{210}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{35}}{147} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{42}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{105}}{147} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{42}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{10\sqrt{14}}{147} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{210}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{4\sqrt{70}}{147} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{4\sqrt{70}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{105}}{98} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{10\sqrt{14}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{49} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{2\sqrt{105}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{49} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{2\sqrt{35}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{105}}{98} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{2\sqrt{10}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5}{14} & 0 & 0 \end{pmatrix}$ |
| 976 | symmetry | $-\frac{\sqrt{3}x^3}{4}+\frac{\sqrt{3}ix^2y}{4}-\frac{\sqrt{3}xy^2}{4}+\sqrt{3}xz^2+\frac{\sqrt{3}iy^3}{4}-\sqrt{3}iyz^2$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_3^{(1,-1;a)}(E^{(b)},1)$ | $\begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{5}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{42}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{10\sqrt{7}}{147} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{105}}{245} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{42}}{147} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{2\sqrt{210}}{245} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{70}}{147} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{105}}{245} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{4\sqrt{35}}{147} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{42}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{4\sqrt{105}}{147} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{4\sqrt{105}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{98} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{4\sqrt{35}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{14}}{98} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{70}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{210}}{49} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{5\sqrt{42}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{14}}{98} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{10\sqrt{7}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{98} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{5}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{14} & 0 \end{pmatrix}$ |
| 977 | symmetry | $\frac{\sqrt{30}x^2z}{4}+\frac{\sqrt{30}ixyz}{2}-\frac{\sqrt{30}y^2z}{4}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_3^{(1,-1;a)}(E^{(b)},2)$ | $\begin{bmatrix} 0 & 0 & \frac{\sqrt{210}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{4\sqrt{70}}{147} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{42}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{10\sqrt{14}}{147} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{42}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{105}}{147} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{210}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{35}}{147} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{10}}{21} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{2\sqrt{10}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5}{14} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{2\sqrt{35}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{105}}{98} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{2\sqrt{105}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{35}}{49} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{10\sqrt{14}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{35}}{49} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{4\sqrt{70}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{105}}{98} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5}{14} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 978 | symmetry | $\frac{15x^4z}{8}+\frac{15x^2y^2z}{4}-5x^2z^3+\frac{15y^4z}{8}-5y^2z^3+z^5$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_5^{(1,-1;a)}(A,1)$ | $\begin{bmatrix} -\frac{\sqrt{385}}{3234} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}}{539} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{5\sqrt{385}}{3234} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{154}}{539} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{5\sqrt{385}}{1617} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{1155}}{539} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{5\sqrt{385}}{1617} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{1155}}{539} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{5\sqrt{385}}{3234} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{9\sqrt{154}}{539} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{385}}{3234} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}}{539} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{385}}{154} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{2310}}{539} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{23\sqrt{385}}{1078} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{9\sqrt{154}}{539} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{17\sqrt{385}}{1078} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{2\sqrt{1155}}{539} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{15\sqrt{385}}{1078} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{2\sqrt{1155}}{539} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{15\sqrt{385}}{1078} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{9\sqrt{154}}{539} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{17\sqrt{385}}{1078} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}}{539} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{23\sqrt{385}}{1078} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{385}}{154} \end{bmatrix}$ |
| 979 | symmetry | $-\frac{3\sqrt{70}x^4y}{16} - \frac{\sqrt{70}x^2y^3}{8} + \frac{3\sqrt{70}x^2yz^2}{2} + \frac{\sqrt{70}y^5}{16} - \frac{\sqrt{70}y^3z^2}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_5^{(1,-1;a)}(A,2)$ | $\begin{pmatrix} 0 & 0 & 0 & \frac{\sqrt{55}i}{231} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{165}i}{77} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{5\sqrt{22}i}{462} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{55}i}{385} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{55}i}{231} & \frac{\sqrt{2310}i}{385} & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{330}i}{385} & 0 \\ -\frac{\sqrt{55}i}{231} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{330}i}{385} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}i}{385} \\ 0 & \frac{5\sqrt{22}i}{462} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{55}i}{385} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{55}i}{231} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{165}i}{77} & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{2310}i}{385} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{770}i}{77} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{3\sqrt{330}i}{385} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{110}i}{154} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{3\sqrt{55}i}{385} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{22}i}{77} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}i}{77} & \frac{\sqrt{770}i}{77} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{110}i}{154} & 0 \\ -\frac{\sqrt{165}i}{77} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{110}i}{154} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{770}i}{77} \\ 0 & \frac{3\sqrt{55}i}{385} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{22}i}{77} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{3\sqrt{330}i}{385} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{110}i}{154} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{2310}i}{385} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{770}i}{77} & 0 & 0 & 0 \end{pmatrix}$ |
| 980 | symmetry | $-\frac{\sqrt{70}x^5}{16}+\frac{\sqrt{70}x^3y^2}{8}+\frac{\sqrt{70}x^3z^2}{2}+\frac{3\sqrt{70}xy^4}{16}-\frac{3\sqrt{70}xy^2z^2}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_5^{(1,-1;a)}(A,3)$ | $\begin{pmatrix} 0 & 0 & 0 & -\frac{\sqrt{55}}{231} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{165}}{77} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{5\sqrt{22}}{462} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{55}}{385} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{55}}{231} & \frac{\sqrt{2310}}{385} & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{330}}{385} & 0 \\ -\frac{\sqrt{55}}{231} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{330}}{385} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}}{385} \\ 0 & \frac{5\sqrt{22}}{462} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{55}}{385} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{55}}{231} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{165}}{77} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{2310}}{385} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{770}}{77} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{330}}{385} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{110}}{154} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{3\sqrt{55}}{385} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{22}}{77} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{165}}{77} & \frac{\sqrt{770}}{77} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{110}}{154} & 0 \\ -\frac{\sqrt{165}}{77} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{110}}{154} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{770}}{77} \\ 0 & \frac{3\sqrt{55}}{385} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{22}}{77} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{3\sqrt{330}}{385} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{110}}{154} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{2310}}{385} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{770}}{77} & 0 & 0 & 0 \end{pmatrix}$ |
| 981 | symmetry | $\frac{3\sqrt{7}x^5}{16} - \frac{15\sqrt{7}ix^4y}{16} - \frac{15\sqrt{7}x^3y^2}{8} + \frac{15\sqrt{7}ix^2y^3}{8} + \frac{15\sqrt{7}xy^4}{16} - \frac{3\sqrt{7}iy^5}{16}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_5^{(1,-1;a)}(E^{(a)},1)$ | $\begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{462}}{77} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{55}}{77} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{330}}{77} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{1155}}{77} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{330}}{77} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{6\sqrt{55}}{77} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{462}}{77} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{1155}}{77} & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$ |
| 982 | symmetry | $\frac{\sqrt{30}x^5}{16}+\frac{\sqrt{30}ix^4y}{16}+\frac{\sqrt{30}x^3y^2}{8}-\frac{3\sqrt{30}x^3z^2}{4}+\frac{\sqrt{30}ix^2y^3}{8}-\frac{3\sqrt{30}ix^2yz^2}{4}+\frac{\sqrt{30}xy^4}{16}-\frac{3\sqrt{30}xy^2z^2}{4}+\frac{\sqrt{30}xz^4}{2}+\frac{\sqrt{30}iy^5}{16}-\frac{3\sqrt{30}iy^3z^2}{4}+\frac{\sqrt{30}iyz^4}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_5^{(1,-1;a)}(E^{(a)},2)$ | $\begin{pmatrix} 0 & -\frac{\sqrt{2310}}{3234} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{231}}{539} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{5\sqrt{231}}{1617} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{13\sqrt{77}}{539} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{5\sqrt{462}}{1617} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{154}}{539} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{5\sqrt{231}}{1617} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}}{245} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}}{3234} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{23\sqrt{385}}{2695} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{11}}{77} \\ \frac{\sqrt{11}}{77} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{66}}{154} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{23\sqrt{385}}{2695} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{20\sqrt{154}}{539} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{2310}}{245} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{770}}{1078} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{154}}{539} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{10\sqrt{462}}{539} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{13\sqrt{77}}{539} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{770}}{1078} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{231}}{539} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{20\sqrt{154}}{539} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{66}}{154} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$ |
| 983 | symmetry | $\frac{3\sqrt{70}x^4z}{16}+\frac{3\sqrt{70}ix^3yz}{4}-\frac{9\sqrt{70}x^2y^2z}{8}-\frac{3\sqrt{70}ixy^3z}{4}+\frac{3\sqrt{70}y^4z}{16}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_5^{(1,-1;a)}(E^{(a)},3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & -\frac{\sqrt{110}}{154} & 0 & 0 & 0 & 0 & 0 & -\frac{6\sqrt{11}}{77} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{110}}{154} & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{165}}{385} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{2310}}{385} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{2\sqrt{2310}}{385} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{770}}{154} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{2\sqrt{165}}{385} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{66}}{154} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{6\sqrt{11}}{77} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{66}}{154} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{770}}{154} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 984 | symmetry | $-\frac{\sqrt{210}x^4z}{8}+\frac{\sqrt{210}ix^3yz}{4}+\frac{\sqrt{210}x^2z^3}{4}+\frac{\sqrt{210}ixy^3z}{4}-\frac{\sqrt{210}ixyz^3}{2}+\frac{\sqrt{210}y^4z}{8}-\frac{\sqrt{210}y^2z^3}{4}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_5^{(1,-1;a)}(E^{(a)},4)$ | $\begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{385}}{385} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{165}}{462} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{8\sqrt{110}}{385} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{5\sqrt{33}}{462} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{330}}{385} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{5\sqrt{33}}{462} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{4\sqrt{11}}{77} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{165}}{462} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{55}}{77} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{154}}{154} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{2\sqrt{55}}{77} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{330}}{154} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{4\sqrt{11}}{77} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{110}}{77} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{2\sqrt{330}}{385} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{110}}{77} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{8\sqrt{110}}{385} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{330}}{154} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{2\sqrt{385}}{385} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{154}}{154} & 0 & 0 \end{pmatrix}$ |
| 985 | symmetry | $\frac{3\sqrt{7}x^5}{16}+\frac{15\sqrt{7}ix^4y}{16}-\frac{15\sqrt{7}x^3y^2}{8}-\frac{15\sqrt{7}ix^2y^3}{8}+\frac{15\sqrt{7}xy^4}{16}+\frac{3\sqrt{7}iy^5}{16}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_5^{(1,-1;a)}(E^{(b)},1)$ | $\begin{pmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{55}}{77} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{330}}{77} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{462}}{77} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{462}}{77} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{1155}}{77} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{330}}{77} & 0 & 0 & 0 & 0 & 0 & \frac{6\sqrt{55}}{77} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{1155}}{77} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$ |
| 986 | symmetry | $\frac{\sqrt{30}x^5}{16} - \frac{\sqrt{30}ix^4y}{16} + \frac{\sqrt{30}x^3y^2}{8} - \frac{3\sqrt{30}x^3z^2}{4} - \frac{\sqrt{30}ix^2y^3}{8} + \frac{3\sqrt{30}ix^2yz^2}{4} + \frac{\sqrt{30}xy^4}{16} - \frac{3\sqrt{30}xy^2z^2}{4} + \frac{\sqrt{30}xz^4}{2} - \frac{\sqrt{30}iy^5}{16} + \frac{3\sqrt{30}iy^3z^2}{4} - \frac{\sqrt{30}iyz^4}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_5^{(1,-1;a)}(E^{(b)},2)$ | $\begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{11}}{77} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{2310}}{3234} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{23\sqrt{385}}{2695} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{5\sqrt{231}}{1617} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2310}}{245} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{5\sqrt{462}}{1617} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{154}}{539} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{5\sqrt{231}}{1617} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{13\sqrt{77}}{539} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{2310}}{3234} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{231}}{539} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{66}}{154} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{5\sqrt{231}}{539} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{20\sqrt{154}}{539} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{13\sqrt{77}}{539} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{770}}{1078} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{154}}{539} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{10\sqrt{462}}{539} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{2310}}{245} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{770}}{1078} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{23\sqrt{385}}{2695} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{20\sqrt{154}}{539} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{11}}{77} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{66}}{154} & 0 \end{pmatrix}$ |
| 987 | symmetry | $\frac{3\sqrt{70}x^4z}{16} - \frac{3\sqrt{70}ix^3yz}{4} - \frac{9\sqrt{70}x^2y^2z}{8} + \frac{3\sqrt{70}ixy^3z}{4} + \frac{3\sqrt{70}y^4z}{16}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_5^{(1,-1;a)}(E^{(b)},3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{2310}}{385} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{110}}{154} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{165}}{385} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{110}}{154} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{6\sqrt{11}}{77} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{770}}{154} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{6\sqrt{11}}{77} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{66}}{154} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{2\sqrt{165}}{385} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{66}}{154} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{2\sqrt{2310}}{385} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{770}}{154} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 988 | symmetry | $-\frac{\sqrt{210}x^4z}{8} - \frac{\sqrt{210}ix^3yz}{4} + \frac{\sqrt{210}x^2z^3}{4} - \frac{\sqrt{210}ixy^3z}{4} + \frac{\sqrt{210}ixyz^3}{2} + \frac{\sqrt{210}y^4z}{8} - \frac{\sqrt{210}y^2z^3}{4}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_5^{(1,-1;a)}(E^{(b)},4)$ | $\begin{bmatrix} 0 & 0 & -\frac{\sqrt{165}}{462} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{55}}{77} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{5\sqrt{33}}{462} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{4\sqrt{11}}{77} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{5\sqrt{33}}{462} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{330}}{385} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{165}}{462} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{8\sqrt{110}}{385} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{385}}{385} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{2\sqrt{385}}{385} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{154}}{154} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{8\sqrt{110}}{385} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{330}}{154} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{2\sqrt{330}}{385} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{110}}{77} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{4\sqrt{11}}{77} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{110}}{77} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{55}}{77} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{330}}{154} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{154}}{154} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 989 | symmetry | $\frac{3\sqrt{6006}x^5yz}{16} - \frac{5\sqrt{6006}x^3y^3z}{8} + \frac{3\sqrt{6006}xy^5z}{16}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_7^{(1,-1;a)}(A,1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{2} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{2} & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 990 | symmetry | $-\frac{35x^6z}{16} - \frac{105x^4y^2z}{16} + \frac{105x^4z^3}{8} - \frac{105x^2y^4z}{16} + \frac{105x^2y^2z^3}{4} - \frac{21x^2z^5}{2} - \frac{35y^6z}{16} + \frac{105y^4z^3}{8} - \frac{21y^2z^5}{2} + z^7$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_7^{(1,-1;a)}(A,2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{858}}{1716} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{7\sqrt{858}}{1716} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{7\sqrt{858}}{572} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{35\sqrt{858}}{1716} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{35\sqrt{858}}{1716} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{7\sqrt{858}}{572} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{7\sqrt{858}}{1716} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{858}}{1716} \end{bmatrix}$ |
| 991 | symmetry | $\frac{\sqrt{6006}x^6z}{32} - \frac{15\sqrt{6006}x^4y^2z}{32} + \frac{15\sqrt{6006}x^2y^4z}{32} - \frac{\sqrt{6006}y^6z}{32}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_7^{(1,-1;a)}(A,3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{1}{2} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 992 | symmetry | $\frac{9\sqrt{21}x^6y}{32}+\frac{15\sqrt{21}x^4y^3}{32}-\frac{45\sqrt{21}x^4yz^2}{8}+\frac{3\sqrt{21}x^2y^5}{32}-\frac{15\sqrt{21}x^2y^3z^2}{4}+\frac{15\sqrt{21}x^2yz^4}{2}-\frac{3\sqrt{21}y^7}{32}+\frac{15\sqrt{21}y^5z^2}{8}-\frac{5\sqrt{21}y^3z^4}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_7^{(1,-1;a)}(A,4)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{1430}i}{286} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10010}i}{286} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2002}i}{286} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{1430}i}{286} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10010}i}{286} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10010}i}{286} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{1430}i}{286} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2002}i}{286} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10010}i}{286} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{1430}i}{286} & 0 & 0 & 0 \end{bmatrix}$ |
| 993 | symmetry | $\frac{3\sqrt{21}x^7}{32} - \frac{3\sqrt{21}x^5y^2}{32} - \frac{15\sqrt{21}x^5z^2}{8} - \frac{15\sqrt{21}x^3y^4}{32} + \frac{15\sqrt{21}x^3y^2z^2}{4} + \frac{5\sqrt{21}x^3z^4}{2} - \frac{9\sqrt{21}xy^6}{32} + \frac{45\sqrt{21}xy^4z^2}{8} - \frac{15\sqrt{21}xy^2z^4}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_7^{(1,-1;a)}(A,5)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{1430}}{286} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10010}}{286} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2002}}{286} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{1430}}{286} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10010}}{286} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10010}}{286} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{1430}}{286} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2002}}{286} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10010}}{286} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{1430}}{286} & 0 & 0 & 0 \end{bmatrix}$ |
| 994 | symmetry | $\frac{\sqrt{858}x^7}{64} + \frac{7\sqrt{858}ix^6y}{64} - \frac{21\sqrt{858}x^5y^2}{64} - \frac{35\sqrt{858}ix^4y^3}{64} + \frac{35\sqrt{858}x^3y^4}{64} + \frac{21\sqrt{858}ix^2y^5}{64} - \frac{7\sqrt{858}xy^6}{64} - \frac{\sqrt{858}iy^7}{64}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_7^{(1,-1;a)}(E^{(a)},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 995 | symmetry | $-\frac{\sqrt{462}x^7}{64}+\frac{5\sqrt{462}ix^6y}{64}+\frac{9\sqrt{462}x^5y^2}{64}+\frac{3\sqrt{462}x^5z^2}{16}-\frac{5\sqrt{462}ix^4y^3}{64}-\frac{15\sqrt{462}ix^4yz^2}{16}+\frac{5\sqrt{462}x^3y^4}{64}-\frac{15\sqrt{462}x^3y^2z^2}{8}-\frac{9\sqrt{462}ix^2y^5}{64}+\frac{15\sqrt{462}ix^2y^3z^2}{8}-\frac{5\sqrt{462}xy^6}{64}+\frac{15\sqrt{462}xy^4z^2}{16}+\frac{\sqrt{462}iy^7}{64}-\frac{3\sqrt{462}iy^5z^2}{16}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_7^{(1,-1;a)}(E^{(a)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{39}}{13} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{91}}{13} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{39}}{13} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 996 | symmetry | $-\frac{5\sqrt{14}x^7}{64}-\frac{5\sqrt{14}ix^6y}{64}-\frac{15\sqrt{14}x^5y^2}{64}+\frac{15\sqrt{14}x^5z^2}{8}-\frac{15\sqrt{14}ix^4y^3}{64}+\frac{15\sqrt{14}ix^4yz^2}{8}-\frac{15\sqrt{14}x^3y^4}{64}+\frac{15\sqrt{14}x^3y^2z^2}{4}-\frac{15\sqrt{14}x^3z^4}{4}-\frac{15\sqrt{14}ix^2y^5}{64}+\frac{15\sqrt{14}ix^2y^3z^2}{4}-\frac{15\sqrt{14}ix^2yz^4}{4}-\frac{5\sqrt{14}xy^6}{64}+\frac{15\sqrt{14}xy^4z^2}{8}-\frac{15\sqrt{14}xy^2z^4}{4}+\sqrt{14}xz^6-\frac{5\sqrt{14}iy^7}{64}+\frac{15\sqrt{14}iy^5z^2}{8}-\frac{15\sqrt{14}iy^3z^4}{4}+\sqrt{14}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_7^{(1,-1;a)}(E^{(a)},3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{429}}{429} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{1001}}{143} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5005}}{143} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{3003}}{429} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5005}}{143} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{1001}}{143} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{429}}{429} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 997 | symmetry | $-\frac{3\sqrt{462}x^6z}{32}-\frac{3\sqrt{462}ix^5yz}{8}+\frac{15\sqrt{462}x^4y^2z}{32}+\frac{5\sqrt{462}x^4z^3}{16}+\frac{5\sqrt{462}ix^3yz^3}{4}+\frac{15\sqrt{462}x^2y^4z}{32}-\frac{15\sqrt{462}x^2y^2z^3}{8}+\frac{3\sqrt{462}ixy^5z}{8}-\frac{5\sqrt{462}ixy^3z^3}{4}-\frac{3\sqrt{462}y^6z}{32}+\frac{5\sqrt{462}y^4z^3}{16}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_7^{(1,-1;a)}(E^{(a)},4)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{65}}{26} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{273}}{26} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{273}}{26} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{65}}{26} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 998 | symmetry | $\frac{15\sqrt{21}x^6z}{32} - \frac{15\sqrt{21}ix^5yz}{16} + \frac{15\sqrt{21}x^4y^2z}{32} - \frac{5\sqrt{21}x^4z^3}{2} - \frac{15\sqrt{21}ix^3y^3z}{8} + 5\sqrt{21}ix^3yz^3 - \frac{15\sqrt{21}x^2y^4z}{32} + \frac{3\sqrt{21}x^2z^5}{2} - \frac{15\sqrt{21}ixy^5z}{16} + 5\sqrt{21}ixy^3z^3 - 3\sqrt{21}ixyz^5 - \frac{15\sqrt{21}y^6z}{32} + \frac{5\sqrt{21}y^4z^3}{2} - \frac{3\sqrt{21}y^2z^5}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_7^{(1,-1;a)}(E^{(a)},5)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{858}}{286} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10010}}{286} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30030}}{286} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30030}}{286} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10010}}{286} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{858}}{286} & 0 & 0 \end{bmatrix}$ |
| 999 | symmetry | $\frac{\sqrt{858}x^7}{64} - \frac{7\sqrt{858}ix^6y}{64} - \frac{21\sqrt{858}x^5y^2}{64} + \frac{35\sqrt{858}ix^4y^3}{64} + \frac{35\sqrt{858}x^3y^4}{64} - \frac{21\sqrt{858}ix^2y^5}{64} - \frac{7\sqrt{858}xy^6}{64} + \frac{\sqrt{858}iy^7}{64}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_7^{(1,-1;a)}(E^{(b)},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 1000 | symmetry | $-\frac{\sqrt{462}x^7}{64} - \frac{5\sqrt{462}ix^6y}{64} + \frac{9\sqrt{462}x^5y^2}{64} + \frac{3\sqrt{462}x^5z^2}{16} + \frac{5\sqrt{462}ix^4y^3}{64} + \frac{15\sqrt{462}ix^4yz^2}{16} + \frac{5\sqrt{462}x^3y^4}{64} - \frac{15\sqrt{462}x^3y^2z^2}{8} + \frac{9\sqrt{462}ix^2y^5}{64} - \frac{15\sqrt{462}ix^2y^3z^2}{8} - \frac{5\sqrt{462}xy^6}{64} + \frac{15\sqrt{462}xy^4z^2}{16} - \frac{\sqrt{462}iy^7}{64} + \frac{3\sqrt{462}iy^5z^2}{16}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_7^{(1,-1;a)}(E^{(b)},2)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{39}}{13} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{91}}{13} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{39}}{13} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 1001 | symmetry | $-\frac{5\sqrt{14}x^7}{64}+\frac{5\sqrt{14}ix^6y}{64}-\frac{15\sqrt{14}x^5y^2}{64}+\frac{15\sqrt{14}x^5z^2}{8}+\frac{15\sqrt{14}ix^4y^3}{64}-\frac{15\sqrt{14}ix^4yz^2}{8}-\frac{15\sqrt{14}x^3y^4}{64}+\frac{15\sqrt{14}x^3y^2z^2}{4}-\frac{15\sqrt{14}x^3z^4}{4}+\frac{15\sqrt{14}ix^2y^5}{64}-\frac{15\sqrt{14}ix^2y^3z^2}{4}+\frac{15\sqrt{14}ix^2yz^4}{4}-\frac{5\sqrt{14}xy^6}{64}+\frac{15\sqrt{14}xy^4z^2}{8}-\frac{15\sqrt{14}xy^2z^4}{4}+\sqrt{14}xz^6+\frac{5\sqrt{14}iy^7}{64}-\frac{15\sqrt{14}iy^5z^2}{8}+\frac{15\sqrt{14}iy^3z^4}{4}-\sqrt{14}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_7^{(1,-1;a)}(E^{(b)},3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{429}}{429} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{1001}}{143} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5005}}{143} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{3003}}{429} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{5005}}{143} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{1001}}{143} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{429}}{429} & 0 \end{bmatrix}$ |
| 1002 | symmetry | $-\frac{3\sqrt{462}x^6z}{32}+\frac{3\sqrt{462}ix^5yz}{8}+\frac{15\sqrt{462}x^4y^2z}{32}+\frac{5\sqrt{462}x^4z^3}{16}-\frac{5\sqrt{462}ix^3yz^3}{4}+\frac{15\sqrt{462}x^2y^4z}{32}-\frac{15\sqrt{462}x^2y^2z^3}{8}-\frac{3\sqrt{462}ixy^5z}{8}+\frac{5\sqrt{462}ixy^3z^3}{4}-\frac{3\sqrt{462}y^6z}{32}+\frac{5\sqrt{462}y^4z^3}{16}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_7^{(1,-1;a)}(E^{(b)},4)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{65}}{26} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{273}}{26} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{273}}{26} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{65}}{26} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 1003 | symmetry | $\frac{15\sqrt{21}x^6z}{32}+\frac{15\sqrt{21}ix^5yz}{16}+\frac{15\sqrt{21}x^4y^2z}{32}-\frac{5\sqrt{21}x^4z^3}{2}+\frac{15\sqrt{21}ix^3y^3z}{8}-5\sqrt{21}ix^3yz^3-\frac{15\sqrt{21}x^2y^4z}{32}+\frac{3\sqrt{21}x^2z^5}{2}+\frac{15\sqrt{21}ixy^5z}{16}-5\sqrt{21}ixy^3z^3+3\sqrt{21}ixyz^5-\frac{15\sqrt{21}y^6z}{32}+\frac{5\sqrt{21}y^4z^3}{2}-\frac{3\sqrt{21}y^2z^5}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_7^{(1,-1;a)}(E^{(b)},5)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{858}}{286} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10010}}{286} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30030}}{286} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30030}}{286} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10010}}{286} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{858}}{286} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 1004 | symmetry | $z$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_1^{(1,1;a)}(A)$ | $\begin{bmatrix} \frac{2\sqrt{105}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{70}}{196} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{6\sqrt{105}}{245} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{42}}{196} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{2\sqrt{105}}{245} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{35}}{98} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{2\sqrt{105}}{245} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{35}}{98} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{6\sqrt{105}}{245} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{42}}{196} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{105}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{70}}{196} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{3\sqrt{70}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{105}}{294} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{5\sqrt{42}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{98} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{3\sqrt{35}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{294} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{35}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{294} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{5\sqrt{42}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{98} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{70}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{105}}{294} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{42} \end{bmatrix}$ |
| 1005 | symmetry | $\frac{\sqrt{2}x}{2} + \frac{\sqrt{2}iy}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_1^{(1,1;a)}(E^{(a)})$ | $\begin{bmatrix} 0 & \frac{2\sqrt{42}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{105}}{196} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{8\sqrt{105}}{245} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{35}}{196} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{6\sqrt{210}}{245} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{70}}{196} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{8\sqrt{105}}{245} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{42}}{196} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{42}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{15\sqrt{7}}{196} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{5}}{28} \\ \frac{3\sqrt{5}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{42} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{15\sqrt{7}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{49} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{5\sqrt{42}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{14}}{98} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{3\sqrt{70}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{210}}{147} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{3\sqrt{35}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{14}}{98} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{49} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{42} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 1006 | symmetry | $\frac{\sqrt{2}x}{2} - \frac{\sqrt{2}iy}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_1^{(1,1;a)}(E^{(b)})$ | $\begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{5}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{2\sqrt{42}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{15\sqrt{7}}{196} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{8\sqrt{105}}{245} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{42}}{196} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{6\sqrt{210}}{245} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{70}}{196} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{8\sqrt{105}}{245} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{35}}{196} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{2\sqrt{42}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{105}}{196} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{42} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{105}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{49} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{3\sqrt{35}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{14}}{98} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{3\sqrt{70}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{210}}{147} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{5\sqrt{42}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{14}}{98} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{15\sqrt{7}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{70}}{49} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{5}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{42} & 0 \end{pmatrix}$ |
| 1007 | symmetry | $-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_3^{(1,1;a)}(A,1)$ | $\begin{pmatrix} -\frac{5\sqrt{77}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{462}}{98} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{77}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{4\sqrt{77}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{231}}{98} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{4\sqrt{77}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{231}}{98} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{77}}{21} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{77}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{462}}{98} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{77}}{77} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{462}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{77}}{539} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{77}}{77} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{231}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{77}}{539} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{231}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{77}}{539} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{77}}{77} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{462}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{77}}{539} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{77}}{77} \end{pmatrix}$ |
| 1008 | symmetry | $\frac{3\sqrt{10}x^2y}{4} - \frac{\sqrt{10}y^3}{4}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_3^{(1,1;a)}(A,2)$ | $\begin{pmatrix} 0 & 0 & 0 & \frac{5\sqrt{77}i}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{231}i}{196} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{2\sqrt{770}i}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{77}i}{196} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{77}i}{147} & -\frac{\sqrt{66}i}{56} & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{462}i}{392} & 0 \\ -\frac{5\sqrt{77}i}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{462}i}{392} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{66}i}{56} \\ 0 & -\frac{2\sqrt{770}i}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{77}i}{196} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{5\sqrt{77}i}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{231}i}{196} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{66}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{22}i}{77} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{3\sqrt{462}i}{392} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{4\sqrt{154}i}{539} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{3\sqrt{77}i}{196} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{770}i}{539} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{231}i}{196} & \frac{\sqrt{22}i}{77} & 0 & 0 & 0 & 0 & 0 & -\frac{4\sqrt{154}i}{539} & 0 \\ \frac{\sqrt{231}i}{196} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{4\sqrt{154}i}{539} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{22}i}{77} \\ 0 & \frac{3\sqrt{77}i}{196} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{770}i}{539} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{3\sqrt{462}i}{392} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{4\sqrt{154}i}{539} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{66}i}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{22}i}{77} & 0 & 0 & 0 \end{pmatrix}$ |
| 1009 | symmetry | $\frac{\sqrt{10}x^3}{4} - \frac{3\sqrt{10}xy^2}{4}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_3^{(1,1;a)}(A,3)$ | $\begin{pmatrix} 0 & 0 & 0 & -\frac{5\sqrt{77}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{231}}{196} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{2\sqrt{770}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{77}}{196} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{77}}{147} & -\frac{\sqrt{66}}{56} & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{462}}{392} & 0 \\ -\frac{5\sqrt{77}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{462}}{392} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{66}}{56} \\ 0 & -\frac{2\sqrt{770}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{77}}{196} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{5\sqrt{77}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{231}}{196} & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{66}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{22}}{77} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{3\sqrt{462}}{392} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{4\sqrt{154}}{539} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{3\sqrt{77}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{770}}{539} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{231}}{196} & \frac{\sqrt{22}}{77} & 0 & 0 & 0 & 0 & 0 & \frac{4\sqrt{154}}{539} & 0 \\ \frac{\sqrt{231}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{4\sqrt{154}}{539} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{22}}{77} \\ 0 & \frac{3\sqrt{77}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{770}}{539} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{3\sqrt{462}}{392} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{4\sqrt{154}}{539} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{66}}{56} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{22}}{77} & 0 & 0 & 0 \end{pmatrix}$ |
| 1010 | symmetry | $-\frac{\sqrt{3}x^3}{4} - \frac{\sqrt{3}ix^2y}{4} - \frac{\sqrt{3}xy^2}{4} + \sqrt{3}xz^2 - \frac{\sqrt{3}iy^3}{4} + \sqrt{3}iyz^2$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_3^{(1,1;a)}(E^{(a)},1)$ | $\begin{pmatrix} 0 & -\frac{2\sqrt{1155}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{462}}{98} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{462}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{154}}{98} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{4\sqrt{231}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{77}}{196} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{462}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{1155}}{196} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{1155}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{770}}{196} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{22}}{28} \\ -\frac{\sqrt{22}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{33}}{77} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{770}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{77}}{539} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{1155}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{385}}{539} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{77}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{4\sqrt{231}}{539} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{154}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{385}}{539} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{462}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{77}}{539} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{33}}{77} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$ |
| 1011 | symmetry | $\frac{\sqrt{30}x^2z}{4} - \frac{\sqrt{30}ixyz}{2} - \frac{\sqrt{30}y^2z}{4}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_3^{(1,1;a)}(E^{(a)},2)$ | $\begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{11}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{5\sqrt{231}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{154}}{196} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{1155}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{462}}{196} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{1155}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{385}}{98} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{5\sqrt{231}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{77}}{49} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{110}}{77} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{77}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{462}}{539} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{385}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{154}}{539} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{462}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{154}}{539} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{154}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{462}}{539} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{11}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{110}}{77} & 0 & 0 \end{pmatrix}$ |
| 1012 | symmetry | $-\frac{\sqrt{3}x^3}{4}+\frac{\sqrt{3}ix^2y}{4}-\frac{\sqrt{3}xy^2}{4}+\sqrt{3}xz^2+\frac{\sqrt{3}iy^3}{4}-\sqrt{3}iyz^2$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_3^{(1,1;a)}(E^{(b)},1)$ | $\begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{22}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{2\sqrt{1155}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{770}}{196} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{462}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{1155}}{196} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{4\sqrt{231}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{77}}{196} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{462}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{154}}{98} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{2\sqrt{1155}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{462}}{98} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{33}}{77} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{462}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{77}}{539} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{154}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{385}}{539} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{77}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{4\sqrt{231}}{539} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{1155}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{385}}{539} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{770}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{77}}{539} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{22}}{28} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{33}}{77} & 0 \end{pmatrix}$ |
| 1013 | symmetry | $\frac{\sqrt{30}x^2z}{4}+\frac{\sqrt{30}ixyz}{2}-\frac{\sqrt{30}y^2z}{4}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_3^{(1,1;a)}(E^{(b)},2)$ | $\begin{bmatrix} 0 & 0 & -\frac{5\sqrt{231}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{77}}{49} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{1155}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{385}}{98} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{1155}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{462}}{196} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{231}}{147} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{154}}{196} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{11}}{14} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{11}}{14} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{110}}{77} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{154}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{462}}{539} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{462}}{196} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{154}}{539} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{385}}{98} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{154}}{539} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{77}}{49} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{462}}{539} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{110}}{77} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 1014 | symmetry | $\frac{15x^4z}{8}+\frac{15x^2y^2z}{4}-5x^2z^3+\frac{15y^4z}{8}-5y^2z^3+z^5$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_5^{(1,1;a)}(A,1)$ | $\begin{bmatrix} \frac{\sqrt{1001}}{539} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{6006}}{6468} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{5\sqrt{1001}}{539} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{10010}}{2156} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{10\sqrt{1001}}{539} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{3003}}{3234} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{10\sqrt{1001}}{539} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{3003}}{3234} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{5\sqrt{1001}}{539} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{10010}}{2156} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{1001}}{539} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{6006}}{6468} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{1001}}{2002} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{5\sqrt{6006}}{6468} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{23\sqrt{1001}}{14014} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{3\sqrt{10010}}{2156} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{17\sqrt{1001}}{14014} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{5\sqrt{3003}}{3234} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{15\sqrt{1001}}{14014} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{5\sqrt{3003}}{3234} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{15\sqrt{1001}}{14014} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{3\sqrt{10010}}{2156} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{17\sqrt{1001}}{14014} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{6006}}{6468} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{23\sqrt{1001}}{14014} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{1001}}{2002} \end{bmatrix}$ |
| 1015 | symmetry | $-\frac{3\sqrt{70}x^4y}{16} - \frac{\sqrt{70}x^2y^3}{8} + \frac{3\sqrt{70}x^2yz^2}{2} + \frac{\sqrt{70}y^5}{16} - \frac{\sqrt{70}y^3z^2}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_5^{(1,1;a)}(A,2)$ | $\begin{pmatrix} 0 & 0 & 0 & -\frac{2\sqrt{143}i}{77} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{429}i}{924} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{1430}i}{77} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{143}i}{308} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{143}i}{77} & \frac{\sqrt{6006}i}{924} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{858}i}{308} & 0 \\ \frac{2\sqrt{143}i}{77} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{858}i}{308} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6006}i}{924} \\ 0 & -\frac{\sqrt{1430}i}{77} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{143}i}{308} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{2\sqrt{143}i}{77} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{429}i}{924} & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{6006}i}{924} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2002}i}{1001} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{858}i}{308} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{286}i}{2002} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{143}i}{308} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{1430}i}{1001} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{429}i}{924} & -\frac{\sqrt{2002}i}{1001} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{286}i}{2002} & 0 \\ -\frac{5\sqrt{429}i}{924} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{286}i}{2002} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2002}i}{1001} \\ 0 & \frac{\sqrt{143}i}{308} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{1430}i}{1001} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{858}i}{308} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{286}i}{2002} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{6006}i}{924} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2002}i}{1001} & 0 & 0 & 0 \end{pmatrix}$ |
| 1016 | symmetry | $-\frac{\sqrt{70}x^5}{16}+\frac{\sqrt{70}x^3y^2}{8}+\frac{\sqrt{70}x^3z^2}{2}+\frac{3\sqrt{70}xy^4}{16}-\frac{3\sqrt{70}xy^2z^2}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_5^{(1,1;a)}(A,3)$ | $\begin{pmatrix} 0 & 0 & 0 & \frac{2\sqrt{143}}{77} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{429}}{924} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{1430}}{77} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{143}}{308} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{143}}{77} & \frac{\sqrt{6006}}{924} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{858}}{308} & 0 \\ \frac{2\sqrt{143}}{77} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{858}}{308} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6006}}{924} \\ 0 & -\frac{\sqrt{1430}}{77} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{143}}{308} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{2\sqrt{143}}{77} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{429}}{924} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{6006}}{924} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2002}}{1001} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{858}}{308} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{286}}{2002} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{143}}{308} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{1430}}{1001} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{429}}{924} & -\frac{\sqrt{2002}}{1001} & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{286}}{2002} & 0 \\ -\frac{5\sqrt{429}}{924} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{286}}{2002} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2002}}{1001} \\ 0 & \frac{\sqrt{143}}{308} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{1430}}{1001} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{858}}{308} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{286}}{2002} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{6006}}{924} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2002}}{1001} & 0 & 0 & 0 \end{pmatrix}$ |
| 1017 | symmetry | $\frac{3\sqrt{7}x^5}{16} - \frac{15\sqrt{7}ix^4y}{16} - \frac{15\sqrt{7}x^3y^2}{8} + \frac{15\sqrt{7}ix^2y^3}{8} + \frac{15\sqrt{7}xy^4}{16} - \frac{3\sqrt{7}iy^5}{16}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_5^{(1,1;a)}(E^{(a)},1)$ | $\begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{30030}}{924} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{6\sqrt{143}}{77} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{858}}{924} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3003}}{1001} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{5\sqrt{858}}{924} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{6\sqrt{143}}{1001} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{30030}}{924} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3003}}{1001} & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$ |
| 1018 | symmetry | $\frac{\sqrt{30}x^5}{16}+\frac{\sqrt{30}ix^4y}{16}+\frac{\sqrt{30}x^3y^2}{8}-\frac{3\sqrt{30}x^3z^2}{4}+\frac{\sqrt{30}ix^2y^3}{8}-\frac{3\sqrt{30}ix^2yz^2}{4}+\frac{\sqrt{30}xy^4}{16}-\frac{3\sqrt{30}xy^2z^2}{4}+\frac{\sqrt{30}xz^4}{2}+\frac{\sqrt{30}iy^5}{16}-\frac{3\sqrt{30}iy^3z^2}{4}+\frac{\sqrt{30}iyz^4}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_5^{(1,1;a)}(E^{(a)},2)$ | $\begin{pmatrix} 0 & \frac{\sqrt{6006}}{539} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{15015}}{6468} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{2\sqrt{15015}}{539} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{13\sqrt{5005}}{6468} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{2\sqrt{30030}}{539} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10010}}{6468} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{2\sqrt{15015}}{539} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6006}}{588} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6006}}{539} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{23\sqrt{1001}}{6468} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{715}}{924} \\ \frac{\sqrt{715}}{924} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{4290}}{2002} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{23\sqrt{1001}}{6468} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{4\sqrt{10010}}{7007} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{6006}}{588} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2002}}{14014} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{10010}}{6468} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{30030}}{7007} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{13\sqrt{5005}}{6468} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2002}}{14014} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{15015}}{6468} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{4\sqrt{10010}}{7007} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{4290}}{2002} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$ |
| 1019 | symmetry | $\frac{3\sqrt{70}x^4z}{16}+\frac{3\sqrt{70}ix^3yz}{4}-\frac{9\sqrt{70}x^2y^2z}{8}-\frac{3\sqrt{70}ixy^3z}{4}+\frac{3\sqrt{70}y^4z}{16}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_5^{(1,1;a)}(E^{(a)},3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & \frac{3\sqrt{286}}{77} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{715}}{154} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{286}}{77} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{429}}{462} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6006}}{462} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{6006}}{462} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2002}}{2002} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{429}}{462} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{4290}}{2002} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{715}}{154} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{4290}}{2002} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2002}}{2002} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 1020 | symmetry | $-\frac{\sqrt{210}x^4z}{8}+\frac{\sqrt{210}ix^3yz}{4}+\frac{\sqrt{210}x^2z^3}{4}+\frac{\sqrt{210}ixy^3z}{4}-\frac{\sqrt{210}ixyz^3}{2}+\frac{\sqrt{210}y^4z}{8}-\frac{\sqrt{210}y^2z^3}{4}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_5^{(1,1;a)}(E^{(a)},4)$ | $\begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{1001}}{462} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{429}}{77} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{286}}{231} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{2145}}{77} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{858}}{462} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{2145}}{77} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{715}}{231} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{429}}{77} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{143}}{462} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10010}}{2002} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{5\sqrt{143}}{462} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{858}}{2002} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{715}}{231} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{286}}{1001} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{858}}{462} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{286}}{1001} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{2\sqrt{286}}{231} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{858}}{2002} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{1001}}{462} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10010}}{2002} & 0 & 0 \end{pmatrix}$ |
| 1021 | symmetry | $\frac{3\sqrt{7}x^5}{16}+\frac{15\sqrt{7}ix^4y}{16}-\frac{15\sqrt{7}x^3y^2}{8}-\frac{15\sqrt{7}ix^2y^3}{8}+\frac{15\sqrt{7}xy^4}{16}+\frac{3\sqrt{7}iy^5}{16}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_5^{(1,1;a)}(E^{(b)},1)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{6\sqrt{143}}{77} & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{858}}{924} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{30030}}{924} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{30030}}{924} & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3003}}{1001} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{858}}{924} & 0 & 0 & 0 & 0 & 0 & -\frac{6\sqrt{143}}{1001} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{3003}}{1001} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 1022 | symmetry | $\frac{\sqrt{30}x^5}{16} - \frac{\sqrt{30}ix^4y}{16} + \frac{\sqrt{30}x^3y^2}{8} - \frac{3\sqrt{30}x^3z^2}{4} - \frac{\sqrt{30}ix^2y^3}{8} + \frac{3\sqrt{30}ix^2yz^2}{4} + \frac{\sqrt{30}xy^4}{16} - \frac{3\sqrt{30}xy^2z^2}{4} + \frac{\sqrt{30}xz^4}{2} - \frac{\sqrt{30}iy^5}{16} + \frac{3\sqrt{30}iy^3z^2}{4} - \frac{\sqrt{30}iyz^4}{2}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_5^{(1,1;a)}(E^{(b)},2)$ | $\begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{715}}{924} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{6006}}{539} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{23\sqrt{1001}}{6468} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{2\sqrt{15015}}{539} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6006}}{588} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{2\sqrt{30030}}{539} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10010}}{6468} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{2\sqrt{15015}}{539} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{13\sqrt{5005}}{6468} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{6006}}{539} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{5\sqrt{15015}}{6468} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{4290}}{2002} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{5\sqrt{15015}}{6468} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{4\sqrt{10010}}{7007} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{13\sqrt{5005}}{6468} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2002}}{14014} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{10010}}{6468} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{30030}}{7007} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{6006}}{588} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2002}}{14014} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{23\sqrt{1001}}{6468} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{4\sqrt{10010}}{7007} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{715}}{924} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{4290}}{2002} & 0 \end{pmatrix}$ |
| 1023 | symmetry | $\frac{3\sqrt{70}x^4z}{16} - \frac{3\sqrt{70}ix^3yz}{4} - \frac{9\sqrt{70}x^2y^2z}{8} + \frac{3\sqrt{70}ixy^3z}{4} + \frac{3\sqrt{70}y^4z}{16}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_5^{(1,1;a)}(E^{(b)},3)$ | $\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6006}}{462} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{3\sqrt{286}}{77} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{429}}{462} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{3\sqrt{286}}{77} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{715}}{154} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{2002}}{2002} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{715}}{154} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{4290}}{2002} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{429}}{462} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{4290}}{2002} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{6006}}{462} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{2002}}{2002} & 0 & 0 & 0 & 0 \end{bmatrix}$ |
| 1024 | symmetry | $-\frac{\sqrt{210}x^4z}{8}-\frac{\sqrt{210}ix^3yz}{4}+\frac{\sqrt{210}x^2z^3}{4}-\frac{\sqrt{210}ixy^3z}{4}+\frac{\sqrt{210}ixyz^3}{2}+\frac{\sqrt{210}y^4z}{8}-\frac{\sqrt{210}y^2z^3}{4}$ |

*continued ...*

Table 10

| No. | multipole | matrix |
|---|---|---|
| | $\mathbb{M}_5^{(1,1;a)}(E^{(b)},4)$ | $\begin{pmatrix} 0 & 0 & \frac{\sqrt{429}}{77} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{143}}{462} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{2145}}{77} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{715}}{231} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{2145}}{77} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{858}}{462} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{429}}{77} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{286}}{231} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{1001}}{462} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{1001}}{462} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{10010}}{2002} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{2\sqrt{286}}{231} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{3\sqrt{858}}{2002} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{858}}{462} & 0 & 0 & 0 & 0 & 0 & 0 & \frac{2\sqrt{286}}{1001} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{715}}{231} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{2\sqrt{286}}{1001} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{5\sqrt{143}}{462} & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{3\sqrt{858}}{2002} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{10010}}{2002} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$ |